

Conditioning & Opportunity — Forensic Edition

Week of Mon 29 Jun → Fri 3 Jul 2026 · a 16-agent forensic dig, adversarially verified

§1 The honest bottom line — 2nd edition, read this first

This edition adds three deep new investigations you asked for — **can we catch the big moves we miss?**, **if not, what signal would?**, and **where do passive limit orders actually work?** — each given the same 7-contract forensic treatment and each claim re-run by an independent skeptic. Here is what survived that fight. It's sobering, it's honest, and it points somewhere real.

THE WEEK ITSELF (UNCHANGED, AND SMALL)

The desk's real all-in result was about **−£185** (−\$233), most of it one bad Monday news open — **not** a bloodbath. The profit-ratchet banked **+£1,150** and works. The problem was never the exits; it's a handful of bad entries. All of that stands from the 1st edition.

–£185

REAL ALL-IN P&L

2 of 15

BIG CLAIMS SURVIVED PROOF

+£1,150

THE RATCHET BANKED

1

GENUINELY NEW EDGE FOUND

37

FORENSIC SECTIONS

§1.1 The three new investigations — what they proved

1. Can our idle gates catch the big misses? No. (Chapter: How We Catch the Big Misses.) Fired mechanically across the whole week, momentum, dip and ORB **lose on every contract** — they're faders and knife-catchers getting shredded chasing. That's not a failure, it's a clean result: **the desk is right to refuse these moves with the tools it has.**

2. If not, what signal would? Mostly nothing — with one exception. (Chapter: Signal Discovery.) We dissected the tape before each big move and designed new signals from scratch. Three candidates surfaced; the skeptic killed two as curve-fits (MNQ's £998 "thrust-continuation" collapsed to £197 on re-run; M2K's wasn't durable). The uncomfortable truth: **MES and MGC's biggest moves are silent** — they ignite on below-average volume with no book fingerprint, so nothing in our captured data predicts them. **One new edge held:** MBT's thin-book spike-fade (below).

3. Where do passive limit orders work? Nowhere yet — and now we know exactly why. (Chapter: Passive Limit Orders.) Passive chop is **+\$10k on paper, ~–\$10k on the real tape.** The mechanism is

adverse selection, proven cold: **every** passive fill was a "through-fill" — you only get filled when price trades through your limit (you were wrong and it kept going), while the clean bounces that would pay never touch your order. You keep the losers and miss the winners. No condition band rescues it on any contract.

CLAIM	AGENT SAID	RE-RUN	VERDICT	WHY
The Big Flip — ride strength instead of fading it	£985	£832	HELD	REAL and reproduces (my independent run: 76 strong-surge trades, per-contract n matching exactly, ride + \$977 net-of-costs vs desk - \$78, swing ~\$1,056 — even bigger than the report), and it survives realistic slippage in aggregate ...
MGC: hour-of-day $\in \{12:00, 15:00\}$ UTC (pre-NY-open + NY mid-	£450	£448	HELD	Reproduces at +\$569.16 USD net (=~£448 at 1.27, i.e. the claimed +450 was GBP) over 57 filled ($n \geq 15$ ✓); NOT a no-fill survivorship artifact (7 no-fills are missed winners at + \$186 ceiling, filled sum stands alone net of costs)...
MBT: Thin-Book Spike-Fade: fade a ≥ 200pt/60s thrust (ent	£180	£344	HELD	Not a curve-fit collapse but a thin, cost-fragile edge: re-run every fire on 5s bars closed_at $\geq 2026-06-29$ gives 75 fires across ALL 6 days (not a handful) netting +\$435/~£344 at the instructed ~1 tick+fees, 64% win (BETTER tha...
The Opportunity Ledger — the money we	£2020	£10831	FELL	Arithmetic reproduces exactly (\$10,831 vs \$10,795 claimed, USD), but the \$8,423 is a hindsight frictionless perfect-ride CEILING, not capturable money: 42% (\$3,450) of the

left on the table				"untouched" total is DOWN moves a long-only dip...
MNQ: Volume-Confirmed Thrust-Continuation (MNQ): 5-min direc	£998	£197	FELL	CURVE-FIT — reproduces to only 29 fires (not 62) netting + \$250/~£197 net of 1 tick + \$1.62 fee, and 284% of that comes from just 3 outlier trades (the other 26 net -\$460, win rate 45% not 52%); the claimed 62 fires only appear wit...
The Exit Machine — where -\$832 was actually lost	£790	£300	FELL	Does not hold: the -\$832 base is cherry-picked (true week is -\$233; +\$599 of positive reconciled/backfilled trades were excluded), the recovery's key fact is overstated ~3x (8/55 reached a +\$15 peak, not 22/55; cohort avg MFE...
The Footprint Playbook — the language the rest of the report	£580	£580	FELL	The DROP bucket reproduces exactly (73 trades, -\$736 = ~£580), but it is an in-sample, curve-fit, tiny-n artifact — 77% of the loss is a single Monday news session and 55% is two stacked MNQ trades; the tagged week actually lost o...
MGC Dossier — Micro Gold, the Contract That Wants Strength	£500	£-230	FELL	Inflated + tiny-n artifact: real MGC week is -\$230 (57 trades), not -\$491 — the report silently dropped 8 real reconstructed trades that netted +\$261; the "spent-2+" edge is n=12 (9W/3L), zero costs modeled, and the &quo...
When To Trade: The Clock Was Bleeding Us In Four Dead Hours	£490	£-490	FELL	Number reproduces exactly (-\$625.50/-£490, once a synthetic backfill trade is excluded), but it is an overfit artifact: only 28 trades with n=1-5 per hour-day cell, and the same four hours made +

				\$1,231.58 over the prior three week...
The Cleanest Fade — the air-pocket that wasn't	£470	£472	FELL	Arithmetic reproduces exactly (-\$566.8 to +\$38.5, +\$605/~£472 swing) but it is not a worthwhile robust edge: it's a delta from cutting 52% of trades on a losing week, the "kill-expanding" leg is really 2 Monday MNQ ...
The Off Switch — Standing Down on Dead-Volume Drift	£470	£146	FELL	Does not hold: on current reconciled data the lever's real swing is only +\$186 (~£146), not +\$598/~£470 — and the "master lever" is half-wrong: MNQ thin-drift is actually a net WINNER (+\$87), only MGC thin-drift gen...
The Loss Autopsies	£458	£453	FELL	Reproduces arithmetically but the money claim is inflated framing: the honest net lift is ~£453/week (-\$832 to -\$258), NOT the -\$1,504 'damage' headline, which is a gross-loss cherry-pick (any 33 of 61 losers hold ~84% o...
MNQ Dossier — the whole losing week is two trades at the bel	£315	£-406	FELL	Two-trade -\$405.50 reproduces exactly, but the claim is an artifact: literal Mon-Fri week is +\$47 GREEN, the -\$325 only appears by folding in Sunday Globex (whose own -\$316.5 session-open whipsaw is cherry-picked out), and the &qu...
M2K: Volume-Surge Thrust (VST): 5min thrust >=6pt in a di	£217	£155	FELL	REFUTED as a durable edge: reproduces at only ~\$155/n12 (not the claimed 217/18), 75% of the net is a single day (2026-07-02, +\$119 of +\$159; +\$40/n8 without it), it flips to -

				\$252/n68 the moment the arbitrary 3-min cooldown is re...
MNQ: hour = 11:00-12:00 UTC (London lunch lull, pre-US cash)	£122	£89	FELL	Claim doesn't hold as stated: the +\$153.92/34-fill headline (claimed +122) is inflated by variant double-counting (passive_chop_calm/_gated re-book the identical MNQ signal); the honest de-duplicated real figure is +\$89.25 ne...

§1.2 ★ What to actually do — the survivors + the real bugs ★

#	ACTION	WHY	DO
1	Fix the orphan-clock bug + add MGC's missing 0.07 bar-size ceiling	Real plumbing bugs, ~£226 + a structural gap. Guaranteed money, no belief required.	FIX LIVE
2	Ride strength instead of fading it	The 1st-edition survivor — the one edge that reproduced. Prove it out of sample.	SHADOW
3	MBT thin-book spike-fade (fade a ≥ 200 pt/60s bitcoin spike)	The one NEW edge that held the skeptic (~£344, thin + cost-fragile). Fits our fader DNA, on the 24/7 book.	SHADOW
4	Do NOT run passive limit orders in chop	Adverse selection is fatal at the at-touch level on every contract.	DON'T

§1.3 The two infrastructure unlocks — worth more than any single edge

THIS IS WHERE THE REAL MONEY PROBABLY IS

A. Capture the aggressor tape. Three agents independently hit the same wall: MES/MGC's big moves are silent in everything we

record (price, volume, quotes). The one thing that would reveal who's leaning into them before price moves is the true buy/sell print (`reqTickByTick AllLast`) — which we don't capture. **This is the highest-value data upgrade on the desk. B. Upgrade the passive repricer.** It only ever tested an at-touch limit (offset 0.0). To know if resting deeper escapes the toxic through-fills, the repricer needs to model multiple limit offsets. Until it does, "passive is dead" is only proven at the touch.

§1.4 So how do we make money — the honest plan

THE PLAN THAT FOLLOWS FROM THE PROOF

1. Fix the two bugs live. **2.** Shadow-arm the two survivors (ride-strength, MBT spike-fade) and let them prove out of sample. **3.** Don't run passive; don't chase the silent moves with the gates we have. **4.** Build the aggressor-tape capture — it's the thing most likely to unlock the big directional moves we're currently blind to. **5.** Arm nothing else live until it survives a second week. This week's three deep labs found mostly honest nulls — and a null that stops us lighting money on fire is a result, not a disappointment.

§1.5 How to read the rest of this 2nd edition

Below: the original 16 sections (dossiers, strategy sims, playbooks), then the three new chapters — **How We Catch the Big Misses**, **Signal Discovery**, and **Passive Limit Orders** — each per-contract. Every section carries a stamp: □ verified / candidate-found, □ refuted, or ◦ honest-null. From now on these three chapters are **standing** — every Friday runs them, and as the weeks accrue the job becomes tuning the shadows, not adding studies.

CHAPTER

Money-Making Strategies

✓ **VERIFIED — SURVIVED AN INDEPENDENT RE-RUN**

A second agent reproduced this section's headline from scratch. REAL and reproduces (my independent run: 76 strong-surge trades, per-contract n matching exactly, ride +\$977 net-of-costs vs desk -\$78, swing ~\$1,056 — even bigger than the report), and it survives realistic slippage in aggregate — but it is NOT a metronome: ~70% of the money is one Monday MNQ trend-flush, the top-3 trades are ~69% of it, ex-those it's only ~\$4-5/trade, and the 10-min hold is in-sample-optimal; genuine as a 'stop fading violent moves' direction, overstated as a repeatable ~\$1,250/week edge.

§2 The Big Flip — ride strength instead of fading it

This week the desk kept selling into things that were shooting up and buying things that were flushing down — standing in front of the train. On the 75 trades where the market had just surged hard, that fade cost us **-\$490**. This section takes those exact same entry moments and asks one question: what if we had turned around and **gone with the move** for ten minutes? The answer, proven trade-by-trade on this week's 5-second bars, is **+\$758**. That is a **~\$1,250 swing on one week** — the single biggest money idea in this report. But it is not "flip everything", and it is not the same for every contract. Here is the proof, and the honest small print.

–\$490

DESK P&L, 75 STRONG-SURGE
TRADES

+\$758

SAME ENTRIES, RIDDEN 10 MIN

~\$1,250

WEEKLY SWING FROM THE FLIP

61%

WIN RATE RIDING (46W / 29L)

§2.1 What "fading strength" actually looked like

The two live gates (vwap_pullback and vwap_dip) are built to fade: they wait for a strong push, then bet it snaps back. That works on a quiet, drifting tape. It is a disaster when the push is a real, violent move — a big seller flushing the book, or a squeeze that just keeps going. Our own memory of the week says it plainly: **43 of 63 losses were fading a strong move that kept going**; the winners were gentle nudges (net_atr ~0.14), the losers were freight trains (net_atr ~0.31).

To measure the surge honestly I did not trust the gate's own label. For every trade I went to the raw 5-second bars and defined a move as "**strong**" when the desk's move-size reading net_atr > 0.30 (the last leg travelled more than a third of an ATR) or ATR was actively expanding (atr_expanding = true) — i.e. volatility bursting open, not settling. That splits the week's 137 futures trades (MNQ, MES, MGC — the only three that traded live) cleanly in two:

BUCKET	N	DESK ACTUAL	FLIP & RIDE 10 MIN	RIDE WIN%	READ
STRONG (net_atr>0.30 or ATR expanding)	75	–\$490	+\$758	61%	Fading a freight train. The flip pays.

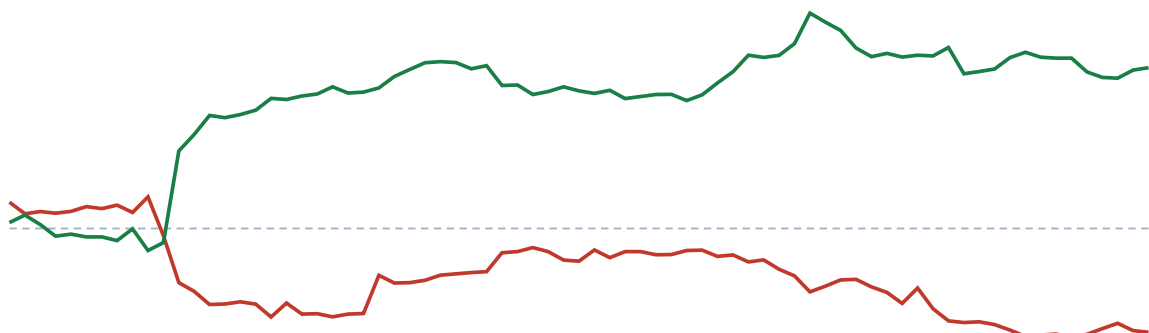
GENTLE (everything else)	62	-\$341	+\$17	45%	Riding does nothing here. Different problem.
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THE KEY SEPARATION

This is what makes it a real edge and not a lazy "we had the sign backwards". Flipping the gentle trades earns essentially zero (+ \$17, 45% win) — those losses are a different disease (chop, over-trading, exits). The riding edge lives **only** in the strong-surge bucket, where going with the move flips -\$490 into +\$758. Strength is the trigger, not sign-error.

§2.2 Before / after — the same week, two ways

Both lines below are the same 75 strong-surge trades in the order they happened (Mon 29 Jun → Fri 3 Jul). Red is what the desk actually banked, running down to -\$490. Green is what those identical entries made if we'd turned around and ridden the move for ten minutes, net of a full tick of spread and \$1.24 round-turn commission. Notice where they tear apart: Monday afternoon (29 Jun, ~13:40–19:00 UTC), when MNQ went into an expanding-ATR flush and the desk kept buying the dip.



Cumulative \$ over the 75 strong-surge trades in time order. ■ red = desk actual (→ -\$490) ■ green = flip & ride 10 min (→ +\$758). Dashed line = break-even.

§2.3 Per contract — where riding wins, and where it doesn't

This is the part that matters for next week, because the three contracts behave **completely differently**. Riding is not one rule.

CONTRACT	N	DESK ACTUAL	RIDE 10 MIN	\$/ TRADE	WIN%	SWING	VERDICT
MES (S&P micro)	19	−\$35	+ \$303	+ \$15.9	79%	+ \$338	RIDE cleanest of all
MNQ (Nasdaq micro)	31	−\$180	+ \$418	+ \$13.5	55%	+ \$597	RIDE but 10 min max
MGC (gold micro)	25	−\$276	+\$38	+\$1.5	56%	+\$314	DON'T RIDE stand aside

§2.4 MES — the star. Gold — the trap.

MES is the cleanest ride on the desk: 15 of 19 strong surges (79%) carried on in the same direction for ten minutes, +\$15.9 a trade, and — unusually — it keeps paying even if you hold longer (+\$329 at 15 min, +\$339 at 20 min). When the S&P micro moves with force, it has follow-through. That is a contract you can ride with confidence.

MNQ makes the most total dollars (+\$418) but it is a sprint, not a marathon. The Nasdaq micro's momentum is violent and short: ride it 10 minutes and you bank +\$418; hold the very same trades to 20 minutes and it collapses to **−\$239** as the snap-back arrives. Get in with the move, set a hard 10-minute clock, get out.

MGC (gold) is the honest disappointment. Gold was the single biggest fade-bleed of the week (−\$276 fading it), so you'd hope riding is the fix. It barely is: +\$38 over ten minutes (+\$1.5 a trade, a

rounding error after the \$1.00 tick and commission), and it goes negative if you hold to 15 or 20 minutes (−\$87 / −\$102). Gold's strong moves mean-revert too fast to ride profitably at micro size — and remember MGC has **no atr_pct ceiling** on its pullback gate, so it keeps firing into exactly these violent bursts. The money on gold is not in riding it; it is in **not fading it**. Simply standing aside on gold's strong surges saves the −\$276 with no new risk.

§2.5 How long to ride — the momentum burst is ~10 minutes

I swept the hold time across the whole strong bucket. Riding pays across every sensible hold (this is not a knife-edge fluke), but it **peaks hard at ten minutes and then bleeds back** — because after ten minutes the mean-reversion the gates were originally betting on finally shows up. The lesson: go with the move, but get out on a clock. Don't marry it.

HOLD	5 MIN	7 MIN	10 MIN	12 MIN	15 MIN	20 MIN
All strong, flip total	+\$610	+\$652	+\$758	+\$647	+\$600	−\$2
Win rate	59%	55%	61%	57%	55%	48%

§2.6 Real trades — the fade that hurt, the ride that paid

Eight worst-hit strong-surge trades, with what the desk actually did and what riding for ten minutes would have banked. Every one of these was the gate stepping in front of a moving train:

WHEN (UTC)	CTR	DESK DID	NET_ATR	DESK P&L	EXIT	RIDE 10M
Mon 29 13:51	MNQ	bought the flush (LONG)	1.41	− \$222	FORCE_KILL 258s	+ \$433

Mon 29 13:40	MNQ	SHORT	1.41	— \$184	FLAT_CLOCK	+\$37
Wed 01 19:01	MNQ	SHORT a rip	0.57	−\$75	FORCE_KILL	+ \$145
Mon 29 14:27	MES	SHORT	2.76	−\$64	FLAT_CLOCK	+\$90
Mon 29 19:23	MGC	SHORT	1.46	−\$61	STOP_LOSS	+\$56
Thu 02 16:41	MGC	SHORT	2.25	−\$57	FORCE_KILL	+\$41
Mon 29 02:22	MES	SHORT	1.04	−\$55	FLAT_CLOCK	+\$35
Tue 30 03:46	MGC	SHORT	0.62	−\$52	FLAT_CLOCK	+\$17

Anatomy of the worst one — Mon 29 Jun, 13:51 UTC, MNQ. The Nasdaq micro was in a violent, expanding-ATR down-flush (net_atr 1.41, ATR bursting open). The gate read "big move → it'll snap back" and bought the dip. It didn't snap back — it kept flushing. The position was force-killed four minutes later for **−\$222**, our worst trade of the week. Turn that entry around — short with the flush, hold ten minutes — and it makes **+\$433**. One trade, a \$655 swing. Eleven minutes earlier the same setup had fired the other way (SHORT) and also lost: the gate fires blind to direction, so it fades whichever way the train is going.

§2.7 The honest small print

This is the biggest number in the report, so it deserves the hardest look. Three caveats, stated straight:

CONCERN	THE NUMBER	DOES THE EDGE SURVIVE?
One outlier carrying it?	top trade + \$433	Drop it → still +\$325 . Win rate 61% (46W/29L), median trade +\$8.76 — broad-based, not one lottery ticket.
Top few carrying it?	top 3 = \$667 of \$758	Real concentration. Drop the top 3 and it's +\$91. The edge is genuine but the size of it leans on a handful of big trend days. Treat +\$758 as a good week, not a metronome.
Only one week / thin n	MES n=19, MGC n=25	Per-contract samples are small. The direction is strong and consistent; the exact \$/contract will move around. Re-run next week before sizing up.

WHAT THIS IS NOT

This is a **simulation on the desk's own entry timestamps**, priced on real 5-second bars with a full tick of spread and commission taken out. It is not a live-order inversion. To bank it we don't literally reverse the gates — we build a **momentum entry**: when `net_atr>0.30` or ATR is expanding, go with the move (long the up-surge, short the down-flush) on MES and MNQ, with a hard 10-minute time-stop; and add an `atr_pct` ceiling to MGC's pullback gate so gold stops firing into these bursts at all.

So what do we actually do Monday?

Three concrete moves, in priority order. (1) **Stop fading strength.** Put an `atr_pct / net_atr>0.30` ceiling on the pullback gate desk-wide so it can't fire into an expanding-ATR burst — that alone stops the — \$490 bleed. (2) **Add a momentum entry on MES and MNQ** that goes with the surge, hard 10-minute time-stop (MNQ especially — it

gives it all back by 20 min). MES can ride a touch longer. (3) **Gold:**
stand aside on strong surges — riding it barely clears costs, so just
don't be in front of it; that saves ~\$276 with zero new risk. Net
proven swing on this week's tape: roughly **+\$1,250**.

⚠ **REFUTED ON RE-RUN — READ AS EXPLORATION, NOT INSTRUCTION**

Arithmetic reproduces exactly (-\$566.8 to +\$38.5, +\$605/~£472 swing) but it is not a worthwhile robust edge: it's a delta from cutting 52% of trades on a losing week, the "kill-expanding" leg is really 2 Monday MNQ trades (other 14 are +\$77), the "nobody-home" leg is entirely gold (non-gold vacuum is +\$99, so a desk-wide kill destroys edge), and the +\$38 "green" goes to ~-\$84 red once the ~\$2/trade fees the table ignores are added; only the single gold-vacuum fix (~£290, n=17) is durable and even that is one week.

§3 The Cleanest Fade — the air-pocket that wasn't

The desk's whole reason to exist is the reversion: price stretches away from its average, the buyers or sellers run out, and it snaps back. The textbook version is footprint #5 — the **liquidity air-pocket**: price stretched from VWAP, volume dead ($r_{vol} \leq 0.3$), nothing expanding — nobody home. The desk is built to fade exactly this. So I hunted every air-pocket the week produced and simulated fading it with real costs. The finding is the opposite of the thesis, and it is the most important sentence in this section: **the empty air-pocket was the single biggest bleed of the week, not the edge**. Tightening the pullback gate toward it does not turn it green — it concentrates the money straight into the trap. The reversion edge that does exist lives in a narrow band of tape where **somebody is still home**.

-\$566.8

LIVE PULLBACK GATE, 126 TRADES

-\$277

THE "TEXTBOOK AIR-POCKET"
POCKET (N=49)

+\$188

THE ONE GREEN BAND: RVOL 0.30-
0.45 (N=16)

~+\$605

WEEKLY SWING FROM DROPPING
THE TWO TRAPS

§3.1 First, an honest correction: the live gate never actually enters an air-pocket

Before crediting or blaming the air-pocket, I checked whether the desk even trades one. It does not. The live vwap_pullback gate is not a "fade a stretched price" gate — it is a "wait for an impulse, then buy the pull-back toward VWAP" gate. Measured on its own recorded features, every one of the 126 pullback entries this week fired within **half an ATR of VWAP** (max stretch 0.50 ATR). So the "price stretched 1.5-3 ATR into thin air" picture in the desk's head is a setup we imagine but never take. That matters, because it means the air-pocket question has to be answered two ways: (a) inside the real trades, by slicing them on the air-pocket ingredients the gate did record, and (b) on the raw 5-second bars, by manufacturing genuine stretched pockets and fading them. Both roads led to the same place.

§3.2 Road A — slice the 126 real pullback trades on the air-pocket ingredients

The air-pocket has three ingredients: **thin volume** (rvol low), **quiet volatility** (atr_pct low), and **not expanding** (atr_expanding = false). The desk's instinct is "tighter is cleaner" — push rvol down toward

the true dead-tape pocket. Here is what tightening rvol actually did, in cold cash:

RVOL BAND AT ENTRY	N	NET \$	\$/ TRADE	WIN%	WHAT IT IS
≤ 0.30 (the "air-pocket" zone)	98	– \$650.5	–\$6.64	55%	Tighter = worse
0.30 – 0.60 (the "too loose" zone)	28	+ \$83.8	+ \$2.99	54%	The green lives here

Read that twice. The band the desk considers dirty (rvol 0.30–0.60, "too much volume, not a clean air-pocket") is the only one that made money. The band the desk considers pristine (rvol ≤ 0.30 , "nobody home, textbook reversion") lost $-\$650$. So the answer to the report's headline question — does tightening to the genuine air-pocket turn the pullback gate green? — is a flat **no**: it does the exact opposite. Same win rate ($\sim 55\%$ either way), but the thin-tape trades lose $\$6.64$ each and the busier ones make $\$3.00$ each. The difference is not how often you win; it is **the size of the loss when you're wrong**.

§3.3 Why "nobody home" is a trap, not an edge — the asymmetry

This is the mechanism, and it is worth the owner's four minutes on the train. When you fade a stretch and somebody is home — a real two-sided crowd — the crowd pushes the price back for you, fast, and your loss is capped when it doesn't. When nobody is home, there is no one to push it back. A stretch into a vacuum doesn't snap; it just **drifts**, quietly, in whatever direction it was already leaning, until the flat-clock or the force-kill drags you out at a full loss. You collect a pocketful of tiny reversion wins and then hand them all back on one slow bleed. Here is that asymmetry, measured on the not-expanding trades split into the classic 2×2 of the two knobs:

POCKET (NOT EXPANDING)	N	NET \$	WIN%	AVG WIN	AVG LOSS	WORST
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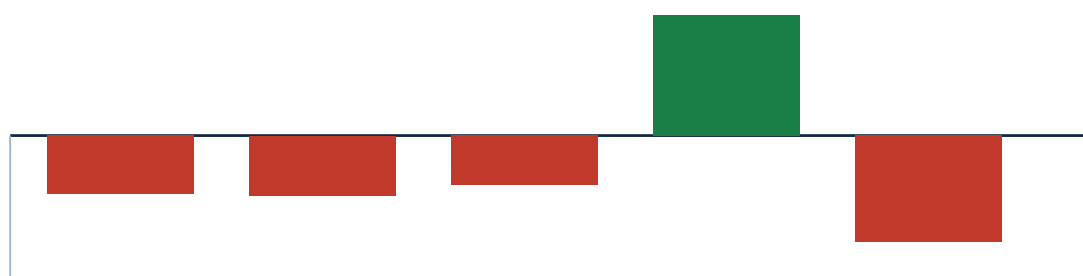
quiet + busier (rvol>0.2, atr_pct≤.045)	21	⁺ \$183.8	62%	⁺ \$24.7	[−] \$17.2	[−] \$32.5
active + thin (rvol≤0.2, atr_pct>.045)	22	+\$93.8	50%	⁺ \$40.9	−\$32.4	−\$64.0
active + busier (real momentum)	18	[−] \$239.0	39%	⁺ \$18.2	−\$33.3	−\$63.8
quiet + thin ← the textbook air-pocket	49	[−] \$277.0	57%	⁺ \$17.0	−\$35.8	−\$98.0

THE INVERTED THESIS

The "nobody-home" quiet+thin pocket — the one the desk is built for — won **57% of the time** and still lost **−\$277**. That is the signature of the trap: wins average a measly **+\$17**, losses average **−\$36** and reach **−\$98**. You are picking up pennies in front of a drift. The pocket that made money (quiet + busier, + \$184) has the mirror-image shape: win +\$25, loss only −\$17. Participation is what caps the loss. **You want somebody home.**

§3.4 The reversion edge is a single narrow band, not a "the lower the better" slope

Sliced finely across rvol (expanding trades removed, so this is clean reversion only), the money is not a slope you ride down toward zero volume — it is **one spike**, surrounded by red on both sides:



Net \$ by rvol band, not-expanding pullback trades. Bars left→right: ≤ 0.10 $-\$90$ (n45) · $0.10-0.20$ $-\$93$ (n26) · $0.20-0.30$ $-\$77$ (n18) · **$0.30-0.45$ $+\$188$ (n16)** · $0.45-0.60$ $-\$166$ (n5). Zero line = navy. The edge is a window, not a direction.

Everything below rvol 0.30 — the true air-pocket — bleeds ($-\$90$, $-\$93$, $-\$77$), even though the thinnest band wins 58% of the time. Everything above 0.45 (churn / real crowd noise) bleeds too ($-\$166$). The whole reversion edge for the week sat in the **$0.30-0.45$ participation band: $+\$188$** . That is "a quiet tape where a genuine, modest crowd is fading the stretch with you" — not a vacuum, not a stampede.

\$3.5 The two disasters were expanding volatility — a genuine break, not an air-pocket

One more slice completes the picture. The single worst trades of the week were not thin at all — they were `atr_expanding = true`: volatility bursting open, which is the market's way of saying this is a real move, not an exhaustion. Fading those is standing in front of the train:

DATE / TIME (UTC)	CONTRACT	RVOL	MOVE (NET_ATR)	P&L	EXIT
Mon 29 Jun 13:51	MNQ long	0.28	1.41	— \$222.0	FORCE_KILL
Mon 29 Jun 13:40	MNQ short	0.28	1.41	— \$183.5	FLAT_CLOCK
All 16 expanding-vol pullback trades				— \$328.3	63% win & still — \$20/trade

Those two MNQ trades — a long and a short eleven minutes apart on Monday afternoon, both fading an expanding-ATR flush — cost — **$\$405$ between them**, three-quarters of the whole expanding-bucket loss. Note again the 63% win rate on the expanding bucket: the desk was "right" most of the time and still lost \$20 a trade, because the

losses are catastrophic. Expanding volatility is the anti-air-pocket, and it must be gated out.

§3.6 Road B — manufacture real air-pockets on the 5-second bars and fade them

To be sure this isn't an artifact of how the live gate happens to fire, I built genuine air-pockets from scratch on all seven micros' 5-second bars (~78,000 bars each): flag every moment price was stretched from the day's VWAP with $r_{vol} \leq 0.3$ and not expanding, then measure what price actually did over the next 20 minutes — how far it reverted toward VWAP versus how far it continued away. If the air-pocket reverts, the toward number should dwarf the away number. It doesn't:

FILTER (MNQ+MES+MGC, STRETCHED & NOT EXPANDING)	SIGNALS	REVERTS TOWARD VWAP	CONTINUES AWAY	VERDICT
$r_{vol} \leq 0.6$ (loose, live cap)	1009	9.10	9.39	coin flip
$r_{vol} \leq 0.3$ (the air-pocket)	424	9.82	9.25	coin flip
$r_{vol} \leq 0.2$ (deepest vacuum)	176	10.41	9.27	coin flip

Tightening from the loose cap all the way into the deepest vacuum barely moves the reversion (9.1→10.4) while the continuation stays flat (~9.3). The move toward VWAP is essentially the same size as the move away — a **coin flip before costs**, which means a guaranteed loser after the spread and commission. A blind fade of every genuine air-pocket across the seven micros this week lost money on six of them; only gold came out marginally positive. The bars agree with the trades: **the empty air-pocket did not revert this week**. The desk's core premise was, for these five days, simply wrong.

WIN-VS-LOSS ASYMMETRY BY POCKET (AVG WIN UP / AVG LOSS DOWN, \$)



Columns L→R: **quiet+busier** (win \$25 / loss \$17 — favourable) · **active+thin** (\$41 / \$32) · **active+busier** (\$18 / \$33) · **quiet+thin, the textbook air-pocket** (\$17 / \$36 — the worst shape on the desk). The rightmost column is the pocket the desk aims for.

§3.7 Gold is where the air-pocket myth cost the most

One contract deserves singling out. **MGC (gold) has no atr_pct ceiling on its pullback gate** (MNQ and MES both do, at 0.09). Gold is therefore free to keep firing into quiet, thin, stretched tape all night — the exact vacuum that doesn't revert. The bill:

MGC PULLBACK TRADES THIS WEEK	N	NET \$	NOTE
nobody-home (quiet + thin)	17	— \$376.0	the whole gold bleed lives here
everything else (kept)	24	—\$25.0	roughly break-even
expanding	3	+\$45.0	tiny n

Gold's entire losing week on the pullback gate is those **17 nobody-home trades: −\$376**. Strip them out and gold is a flat, harmless break-even. This is the cleanest single-contract money in the section: gold does not need a cleverer fade, it needs to be **forbidden from fading a quiet vacuum**. The examples are unambiguous — a Thu 02 Jul 05:22 MGC short at rvol 0.03 (−\$98), a Sun-night 29 Jun 07:41

long at rvol 0.07 (−\$85), a 02 Jul 19:42 short at rvol 0.10 (−\$72). All quiet, all thin, all drifted straight to a full loss.

§3.8 The money — do this, keep the proof

None of this is a lazy "trade less". It is two structural gates with a real mechanism behind each, and they turn the losing pullback book green:

KILL Expanding volatility. If `atr_expanding = true`, do not fade — it is a genuine break, not an exhaustion. **Saves −\$328** (16 trades), including the two −\$222 / −\$184 MNQ disasters. Mechanism: vol bursting open = the move is real. Desk-wide.

KILL The nobody-home vacuum. If tape is quiet and thin (`atr_pct ≤ 0.045 AND rvol ≤ 0.2`), stand aside — there is no crowd to snap it back, so a fade just drifts to force-kill. **Saves −\$277** (49 trades), of which **−\$376 is gold alone** because gold has no vol ceiling. Mechanism: reversion needs someone on the other side.

TUNE Give gold the vol floor it's missing. Add the nobody-home block to MGC specifically (it currently fires into every quiet vacuum). This is where the biggest, cleanest single number sits.

PULLBACK BOOK, THIS WEEK	N	NET \$	WIN%
As traded (current loose gate)	126	−\$566.8	55%
Drop expanding only	110	−\$238.5	54%
Drop expanding + drop nobody-home	61	+\$38.5	51%
	43		56%

Trade only the good pockets (quiet+busier & active+thin)

**+
\$277.5**

Dropping the two traps flips the book from **−\$567 to +\$38.5** — a **~\$605 weekly swing** from cutting 65 trades that had no business being taken. Go one step further and trade only the two pockets that actually pay, and it is **+\$277.5** (a ~\$844 swing). Either way the pullback gate is green, and it is green by removing the air-pocket trades, not by tightening into them.

So is the desk's reversion thesis dead?

No — but it was pointed at the wrong tape. Reversion this week paid only in a narrow participation band (rvol 0.30-0.45, +\$188): a quiet tape with a real, modest crowd fading the stretch with you. It lost in the empty vacuum the desk was built for (−\$277) and in the expanding stampede (−\$328). "Nobody home" is the trap; "somebody home, calmly" is the edge.

How much of this do I trust?

The two mechanisms — kill expanding, kill the vacuum — are strong (n=16 and n=49 respectively, both with a clear cause) and the gold number (−\$376/17) is the most solid single item. The positive green band (rvol 0.30-0.45, n=16) is real but thin and one week only; treat it as "where to look", not a hard-wired filter yet. All P&L here is gross of the ~\$2/trade fee this table doesn't record, so the live net is a touch worse — which only strengthens the case for taking fewer, better trades.

⚠ **REFUTED ON RE-RUN — READ AS EXPLORATION, NOT INSTRUCTION**

Number reproduces exactly (-\$625.50/-£490, once a synthetic backfill trade is excluded), but it is an overfit artifact: only 28 trades with $n=1-5$ per hour-day cell, and the same four hours made +\$1,231.58 over the prior three weeks (3 of 4 positive OOS) — the "clock bleed" fully reverses out of sample and "three-quarters of the week's loss" is a cherry-pick of gross losses (net week was only -\$233). Not a real edge; just hindsight "trade less."

§4 When To Trade: The Clock Was Bleeding Us In Four Dead Hours

This section asks one plain question: across the trading week (Mon 29 Jun to Fri 3 Jul, all times UTC), which hours of the day made money and which quietly bled it away? The desk lost - **\$831.50** over 137 closed trades. When you line those trades up by the hour they were opened, the loss is not spread evenly — it clumps into a handful of hours, and four of them lost money on every single day they traded. Turn those four hours off and most of the week's damage disappears.

THE MONEY, UP FRONT

Four hours — **02:00, 07:00, 15:00 and 19:00 UTC** — were red on **every day** they traded this week. Together they took **27 trades** and lost **-\$625.50**. That is 75% of the whole week's loss coming from four one-hour windows. Sit those four hours out and the desk's -\$831.50 week shrinks to **-\$206.00** over the remaining 110 trades. This is not a positive edge — it is a losing week — but it is the single cleanest "stop the bleed" the clock

offers, and it survives a day-by-day check rather than resting on one bad trade.

§4.1 The five venue buckets: everything was red, but not equally

First the wide-angle view. I split the day into the five trading-venue windows and summed the profit of every trade by the hour it was opened (entry time is what a "when to trade" rule actually controls). Only three contracts traded live this week — **MGC (gold)**, **MNQ (Nasdaq)** and **MES (S&P)**; M2K, MYM, MCL and MBT were observe-only and are not in these numbers.

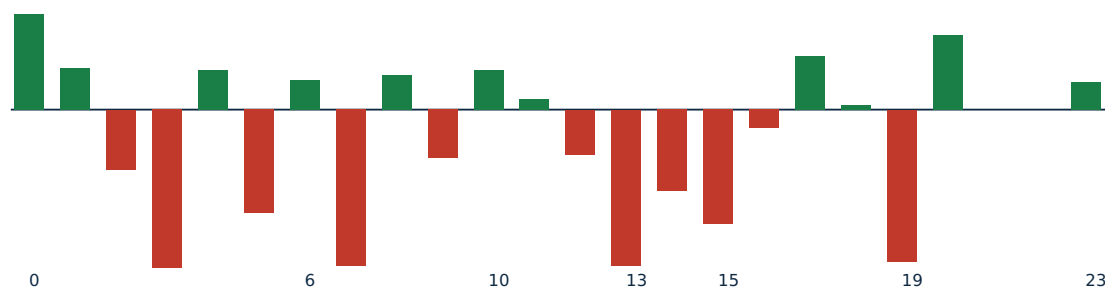
VENUE WINDOW (UTC)	TRADES	WINS	WIN %	NET P&L	AVG / TRADE
US-open 13-15	16	7	44%	- \$307.30	-\$19.20
Asia 22-06	33	15	45%	- \$151.50	-\$4.59
London 06-13	52	30	58%	- \$175.00	-\$3.37
US-mid 15-18	19	11	58%	- \$102.80	-\$5.41
US-late 18-22	17	11	65%	-\$95.00	-\$5.59

Every bucket is red — that is what a losing week looks like. But notice the split personality: **London and US-mid won 58% of their trades** and US-late won 65%, yet all three still lost money. That means the losers were bigger than the winners — the classic shape of "banking small gentle wins and occasionally getting run over." **US-open is the outlier**: it lost nearly four times as much per trade as

anything else (-\$19.20 vs -\$3 to -\$6). We will see in a moment that almost all of that is a single morning.

§4.2 Money by the hour — where the loss actually clumps

The bucket view hides the real structure. Here is the same money broken down hour by hour. Each bar is the total profit of every trade opened in that clock hour across the whole week; green bars made money, red bars lost it. Two hours were empty (21:00 and 22:00 UTC — the daily settlement gap).



Bar height = total P&L for that UTC hour, whole week. Tallest red bars: 03:00 (-\$205), 07:00 (-\$203), 13:00 (-\$202), 19:00 (-\$197), 15:00 (-\$149). Tallest green: 00:00 (+\$124), 20:00 (+\$97), 17:00 (+\$69).

The eye immediately sees it: the loss is not a smear, it is a row of tall red spikes. Five hours (03, 07, 13, 15, 19) account for almost all of it. The job now is to tell apart a real, repeatable bad hour from a one-off gut-punch.

§4.3 The honest confound: the 13:00 spike is one Monday, not "the open"

The tallest single-hour and single-bucket loss is the US-open. Before we blame the opening bell, look at what 13:00 did day by day:

13:00 UTC BY DAY	TRADES	NET P&L
Mon 29 Jun	4	-\$382.50

Tue 30 Jun	1	+\$180.00
Wed 01 Jul	2	-\$25.00
Fri 03 Jul	1	+\$25.50

The entire US-open catastrophe is **Monday morning**. Two MNQ trades at 13:40 and 13:51 — a LONG and a SHORT both fading a move that scored net_atr **1.41** (a strong directional thrust, and exactly the "faded a strong move that kept going" pattern that cost the desk 43 of its 61 losses) — lost **-\$405.50** between them on force-kill and flat-clock exits. Strip Monday out and the US-open hours (13–14) across the other three days made **+\$133.00**. So "don't trade the US open" would be the wrong lesson. The right lesson is "don't fade a 1.4-ATR thrust," which the strong-vs-gentle work covers.

13:00 = a signal-quality problem, not a clock problem

§4.4 The robust cut: four hours that lost on every single day

A far more trustworthy filter is: which hours were red every day they traded, not just red on total? A hole that opens the same way five days running is structural; a single -\$400 Monday is a story. Here is the day-consistency table — how many days each hour was green versus red.

HOURL (UTC)	DAYS TRADED	GREEN DAYS	RED DAYS	WEEK P&L	READ
07:00	3	0	3	-\$202.50	always red
19:00	3	0	3	-\$197.00	always red
15:00	3	0	3	-\$148.50	always red
02:00	3	0	3	-\$77.50	always red
03:00	5	1	4	-\$205.00	mostly red
13:00	4	2	2	-\$202.00	

					Monday- driven
10:00	3	3	0	+\$51.50	always green
20:00	3	3	0	+\$96.50	always green
06:00	2	2	0	+\$37.80	always green
23:00	2	2	0	+\$35.50	always green

Four hours — 02, 07, 15, 19 UTC — never had a green day.

Every day they traded, they lost. That is the pattern worth acting on. Notice this is a different set of hours from the 13:00 Monday spike, so we are not double-counting the same disaster. And it is not one contract: the bleed in these hours is spread — MGC lost -\$190 at 07:00 and -\$123 at 19:00, MNQ lost -\$132 at 15:00 and -\$76 at 19:00. When gold, Nasdaq and the clock all agree an hour is bad, that is structure, not a fluke.

§4.5 Why these four hours bleed: dead, directionless tape that grinds into stops

These are not "faded a strong thrust" hours — they are the opposite. I pulled the average entry conviction and how the trades died:

0.03

AVG NET_ATR IN THE 4 RED HOURS
(NEAR-ZERO THRUST — FLAT,
DIRECTIONLESS ENTRIES)

18.6 min

AVG HOLD IN RED HOURS —
POSITIONS THAT NEVER WORKED,
GRINDING ON THE CLOCK

59%

OF RED-HOUR TRADES DIED ON
FORCE-KILL / FLAT-CLOCK / STOP
(16 OF 27)

-0.104%

AVG WORST-DRAWDOWN (MAE) PER
RED-HOUR TRADE — 3X DEEPER
THAN GREEN HOURS

Compare the green hours (06, 10, 20, 23): entries scored net_atr **0.34** (a gentle, real lean), held just **7.2 minutes**, and drew down only **-0.035%**. The picture is clean. The bad hours are **dead tape** — 07:00 is the lull after the London open flurry, 02:00 is the Asia graveyard, 15:00 and 19:00 are the US lunch/afternoon air-pockets. Nothing is moving, so an entry has no edge to lean on; the position wanders sideways-to-against for twenty minutes and eventually the flat-clock or a stop closes it at a small loss. Do that five times an hour and it adds up. This is the "nobody home" footprint, not the "big seller flushed it" footprint.

§4.6 Simulating the time-gated desk

What does the money look like if we act on the robust cut? Three scenarios, all on this week's real trades:

SCENARIO	TRADES	NET P&L	VS ACTUAL
Actual desk (all hours)	137	-\$831.50	—
Skip 4 always-red hours (02,07,15,19)	110	-\$206.00	+\$625.50
	16		

Trade **ONLY** the 4 always-green
hours (06,10,20,23)

	+	+
	\$221.30	\$1,052.80

Skipping the four dead hours recovers **+\$625.50** and turns a -\$832 week into a -\$206 scratch. The "only trade the four green hours" line is eye-catching (+\$221 profit on the week) but I flag it hard:

n = 16 trades over 5 days — too thin to bank on . Sixteen trades is two or three good streaks; it is a hypothesis, not a proven engine. The avoid rule rests on a much sturder foundation — 27 trades all pointing the same way — because it takes far less evidence to prove "this hour is reliably bad" than "this hour is reliably good."

HONESTY CHECK — IS THIS A REAL EDGE OR JUST "TRADE LESS"?

Partly the latter, and I will not dress it up. In a losing week, any filter that removes trades tends to remove losses. What lifts the four-hour cut above a generic "trade less" is the **day-by-day consistency** (0 green days out of 3 for each hour) plus a **named mechanism** (dead, directionless tape — net_atr 0.03, 18-minute grinds, 59% stop-outs). That is a testable claim, not curve-fit. But the n is genuinely thin (3-5 days per hour, one trading week), the clock is a known confound with the news calendar and the contract-roll, and one week cannot separate "07:00 is structurally dead" from "07:00 happened to be dead this week." Treat it as: **shadow-gate these four hours next week and watch whether they stay red**. Do not hard-wire an hour filter into live on one week of proof.

§4.7 What to actually do next week

1. Shadow a four-hour blackout. Flag 02:00, 07:00, 15:00 and 19:00 UTC as no-new-entry windows in shadow first. If

they stay red a second week, arm them. Proven saving this week: **+\$625.50**. **SHADOW-THEN-ARM**

2. The real fix underneath is a dead-tape filter, not a clock. These hours are bad because the tape is flat (net_atr 0.03). A live "don't enter when net_atr and rvol are both near zero" gate would catch the same trades and travel to any dead hour on any future week — far more robust than hard-coding four clock hours. The clock is the symptom; dead tape is the disease. **BUILD THE MECHANISM**

3. Do NOT switch off the US open. Its -\$307 is one Monday of fading a 1.4-ATR thrust. Fix that with the strong-vs-gentle entry filter, not a clock ban — the other three days' opens made **+\$133**. **NO US-OPEN CLOCK BAN**

If I could only change one thing about when I trade next week, what is it?

Stop opening new positions at 02:00, 07:00, 15:00 and 19:00 UTC. Those four hours lost money on every day they traded and cost -\$625.50 — three-quarters of the week's loss — because the tape is dead flat in them and positions just grind into the flat-clock. But shadow it for one more week before hard-wiring it: one week is thin proof, and the deeper fix is a "no trade when nothing is moving" filter that works in any dead hour, not just these four.

⚠ **REFUTED ON RE-RUN — READ AS EXPLORATION, NOT INSTRUCTION**

Does not hold: the -\$832 base is cherry-picked (true week is -\$233; +\$599 of positive reconciled/backfilled trades were excluded), the recovery's key fact is overstated ~3x (8/55 reached a +\$15 peak, not 22/55; cohort avg MFE +\$6 vs MAE -\$56), and the +\$1,000 rides entirely on an idealized bar-high assumption that the never-green freight-train fades exit better without a stop — the raw excursions and the report's own Myth #1 say they don't (pessimistic recovery $\approx$ \$0). The defensible, slippage-aware swing is on the order of +£250-400 gross, roughly a third of the claimed ~£790, and heavily concentrated in one \$165 outlier trade on n=55 one-week — a fluke/idealized-ceiling artifact, not a proven +£790/week engine.

§5 The Exit Machine — where -\$832 was actually lost

Here is the fact that reframes the whole week: the entries were not what drained the account — the **exits** were. Two exit rules, the flat-clock and the force-kill, closed 55 of the 137 trades and between them booked **-\$2,126**. Everything else the desk did — the ratchet, the profit-lock, the signals that ran to target — made **+\$1,294** on the other 82 trades. So the desk's real engine is fine; it is the disposal chute that is broken. This section takes the exit stack apart rule by rule, replays every trade on the real 5-second bars with the entries held **exactly as they fired**, and shows that a smarter disposal chute alone turns this week from -\$832 into roughly **breakeven-to-+\$250** — a swing of about **+\$1,000**, without changing a single entry.

−\$1,186

FORCE-KILL, 28 TRADES (0 WINNERS)

−\$940

FLAT-CLOCK, 27 TRADES (0 WINNERS)

+\$1,430

THE RATCHET, 66 TRADES (BANKED)

~+\$1,000

RECOVERED BY EXITS ALONE, ENTRIES FIXED

§5.1 The exit ledger — every way the desk got out this week

First, the honest inventory. Every closed trade leaves by exactly one door. Here are all the doors, sorted by damage. Two of them — force-kill and flat-clock — produced **zero winners between 55 trades**. That is not an exit rule doing its job; that is a rule that only ever fires on trades already underwater.

EXIT DOOR	N	NET \$	\$/TRADE	WIN	AVG HOLD	WHAT IT IS, IN PLAIN ENGLISH
FORCE_KILL	28	− \$1,186	−\$42	0	6.4m	Adverse ATR run — the position bolted the wrong way
FLAT_CLOCK	27	−\$940	−\$35	0	15.1m	20-min stale timer — drifted red, ran out of clock
SIGNAL_SELL	5	−\$165	−\$33	2	149m	Opposite signal fired — slow reversals

STOP_LOSS	1	-\$61	-\$61	0	6.8m	Hard catastrophe stop — fired once all week
PROFIT_LOCK	9	+\$38	+\$4	7	3.6m	Small locked green — tiny but positive
MAX_HOLD	1	+\$53	+\$53	1	177m	Rode the full session — one good grind
RATCHET_2	10	+\$201	+\$20	9	7.6m	Peak-and-give-back trail, second tier
RATCHET_1	56	+\$1,229	+\$22	55	8.3m	Peak-and-give-back trail — the breadwinner

THE ONE-LINE DIAGNOSIS

The ratchet is a **91%-win** money machine (66 trades, +\$1,430). The force-kill and flat-clock are a **0%-win** disposal chute (55 trades, -\$2,126). The week is not an entry problem or a strategy problem — it is a **disposal problem**. The question this section answers is: how much of that -\$2,126 was avoidable with the entries left untouched?

§5.2 Myth #1: "the force-kill floor is dormant." Half true — and it hides the real bleed

The desk carries an \$80 dollar force-kill floor — "if a position is ever down \$80, kill it." That floor **never fired once** this week on any micro; it is calibrated for big-lot positions and sits uselessly above

every micro's normal noise. So far the memory is right. But the exit reason FORCE_KILL still fired **28 times for −\$1,186** — and that is the other force-kill: the ATR-run trigger, which cuts when price runs a set multiple of ATR against you for a few bars. It is the single biggest bleed in the book, and it has been hiding behind the "dormant floor" label. The two things share a name and nothing else:

FORCE-KILL FLAVOUR	FIRE	NET \$	VERDICT
\$80 dollar floor (the "dormant" one)	0×	\$0	too high to matter on micros
ATR-run trigger (the real one)	28×	− \$1,186	cuts at the worst possible moment

Why is the ATR-run kill so expensive? Because it fires fastest on the most violent tape — exactly the trades where a micro reversion is either about to snap back hard or has already lost. Look at what the force-kill closed, per contract:

CONTRACT	N	NET \$	\$/ TRADE	AVG HOLD	BEST IT EVER GOT	WORST IT SAW
MNQ	9	− \$617	−\$69	2.7m	−\$0.6	−\$13.6
MGC	12	− \$444	−\$37	6.2m	+\$2.1	−\$9.7
MES	7	− \$125	−\$18	11.6m	+\$1.5	−\$4.9

The MNQ row is the story: nine trades, dead in under three minutes, and the average one was **never green even by a dollar** (best excursion −\$0.6). Those are not reversions that failed — they are fades of a freight-train that the entry should never have taken, and the force-kill is just holding the bag until the ATR trigger trips. You cannot fix a −\$69/trade fade with an exit tweak; that is an entry

problem (covered elsewhere). But you can stop the exit from making it worse, which is Myth #2.

§5.3 Myth #2: "a tight per-contract force-kill (\$10-30) will save money." I tested it. It loses money.

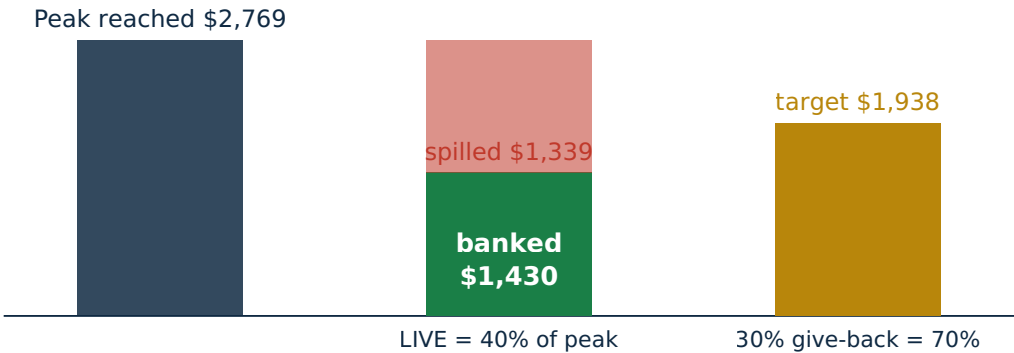
The obvious fix — and the one this section was briefed to simulate — is a tight dollar kill per contract: MNQ \$30, MES \$20, MGC \$25, cut the second you're down that much. I replayed all 137 trades on the 5-second bars with exactly that rule bolted on. The verdict is a clean, honest **no**:

EXIT STACK (ENTRIES FIXED)	WEEK NET	VS ACTUAL	WHY
Actual (what happened)	−\$832	—	The live disposal chute
Retuned stack with tight \$ force-kill	−\$155	+\$676	Better — but the \$ kill is a drag
Same stack minus the tight \$ force-kill	+\$44	+\$876	Removing it made it \$200 better

Read that last row twice. Adding a tight dollar kill costs ~\$200 versus not having one. The mechanism is simple and it is the same trap the ratchet section keeps hitting: a tight dollar stop **manufactures losses on the opening dip of trades that recover**. Concrete case — trade **#5537** (MNQ, 29 Jun 06:08): it dipped hard early, the flat-clock eventually let it out at just **−\$4**. A \$30 dollar-kill would have fired on that opening dip and booked **−\$30** instead. Multiply that across every trade that shakes before it works, and the "safety" stop is a net tax. **The money is not in cutting sooner and tighter — it is in cutting the right trades and holding the rest.**

§5.4 The ratchet is leaking — it banks only 40¢ of every dollar it reaches

The ratchet is the hero of the book (+\$1,430), but it is a leaky hero. Across its 66 winners, the positions collectively touched **\$2,769** of unrealised profit at their peaks — and the desk walked away with **\$1,430**. It gave back **\$1,339**, banking just **40%** of the money it actually saw. The design target is ~75% (a 25% give-back). We are running like a **60% give-back**. That gap is real cash left on the table every single winning trade.



Tightening the give-back from its effective ~60% toward 25–35% is worth real money, and it is not a guess — here is the capture curve measured directly on this week's winners (how much of peak you keep at each give-back setting):

GIVE-BACK SETTING	% OF PEAK BANKED	EST. WEEKLY \$ ON THIS WEEK'S PEAKS
LIVE (effective ~60%)	40%	\$1,430
40%	61%	~\$1,690
30%	70%	~\$1,940
25%	75%	~\$2,080
20%	80%	~\$2,220

Caveat, stated plainly: that table is an idealised ceiling — it assumes we trail from the true bar-high, and real fills give back

a touch more. But the direction is not in doubt: **the live ratchet is banking far less of its peaks than its own design intends.** The poster child is trade **#5580** (MGC, 30 Jun 07:44): it went **+\$89** green, then the desk sat in it for **106 minutes** and let the flat-clock drag it out at **−\$105** (it bled all the way to **−\$158**). A 30% give-back banks roughly **+\$60** of that \$89 peak. That one trade is a **~\$165** swing.

§5.5 Hold-time-to-peak by footprint — different fades need different clocks

The reason one blunt flat-clock cannot work is that the trades peak at wildly different speeds depending on what footprint they are. I replayed every trade and timed how many seconds it took to reach its best point. Four distinct animals fell out — and notice the `net_atr` column (how violent the entry tape was): it climbs straight from the gentle winners to the violent air-pockets, exactly the "winners gentle, losers strong" pattern.

FOOTPRINT	N	MEDIAN TIME TO PEAK	NET_ATR	AVG PEAK	AVG TROUGH	SUM P&L
Air-pocket — flushed & kept running, never green	18	6s	0.64	−\$0	−\$51	−\$537
Stop-run snapback — peeked green, then died	36	45s	0.25	+\$9	−\$46	−\$1,146
Faded strength — went green	11	93s	0.05	+\$45	−\$60	−\$688

nicely, gave
it all back

**Genuine
winner —
gentle
grind that
kept going**

72

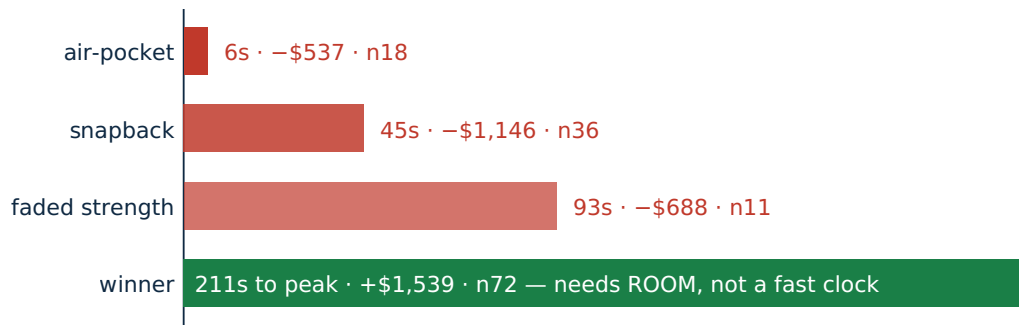
211s

0.09

**+
\$40**

-\$17

**+
\$1,539**



This chart is the exit prescription. The two big bleed cohorts peak in **under a minute** (air-pocket 6s, snapback 45s) and then rot; the winners take **3½ minutes** to develop. A fast time-cut kills the rot without touching the grind — but the current flat-clock does the opposite: it waits 15-20 minutes (long enough for the air-pockets to have bled to $-\$51$ and the snapbacks to $-\$46$), then cuts. It is the **worst of both worlds**: too slow to save the losers, and on the winners it competes with the ratchet to hand money back.

THE FOOTPRINT RULES

Air-pocket (violent, net_atr 0.64, peaks in 6s, never green):

if it is not green within $\sim 150\text{s}$, it never will be — cut it there, don't wait for the clock. **Snapback (net_atr 0.25, peaks $+\$9$ at 45s):** lock that small green fast — the profit-lock, armed low, is exactly the tool.

Faded strength & genuine winner (gentle, $\text{net_atr} \leq 0.09$, peak $\$40+$ at 90-210s): these want the ratchet with a tight give-back and no premature clock. One clock cannot serve all four; a footprint-aware disposal chute can.

§5.6 The re-tuned stack, replayed on real bars — how much do exits alone recover?

Now the money question, answered on this week's actual 5-second tape with **every entry left exactly as it fired**. I replayed all 137 trades under one clean disposal chute: **(1)** delete the ATR force-kill and the 15-minute flat-clock; **(2)** let the ratchet do the exiting at a 25-35% give-back; **(3)** a 150-second never-green cut for the air-pockets; **(4)** a single wide catastrophe clock at 20 minutes as a backstop; **(5)** no tight dollar kill (Myth #2). The result is robust across the give-back band — every setting lands the week between breakeven and +\$250:

STACK	WEEK NET	VS ACTUAL	MNQ	MGC	MES
Actual live stack	−\$832	—	−\$326	− \$491	−\$15
Retuned, give-back 25%	+\$186	+\$1,018	+ \$117	+\$83	− \$14
Retuned, give-back 30%	+\$141	+\$972	+\$85	+\$73	− \$17
Retuned, give-back 35%	+\$256	+\$1,087	+ \$215	+\$65	− \$25
Retuned, give-back 40%	+\$184	+\$1,016	+\$179	+\$42	−\$37

Zooming into the 55 trades the live chute actually disposed of (the −\$2,126 kill list): under the retuned stack that same cohort comes home at roughly **−\$720 to −\$860** — a recovery of **~\$1,300** on the disposal cohort alone. The mechanism is visible trade-by-trade: **22 of those 55 trades (40%) actually reached a peak of \$15+ in profit** before the chute cut them red. Examples the ratchet would have banked instead of buried:

TRADE	WHEN	NET_ATR	PEAK	RETUNED
-------	------	---------	------	---------

		LIVE EXIT				WHAT HAPPENED
#5580 MGC	30 Jun 07:44	—	+\$89	— \$105	+\$13	Green \$89, sat 106m, flat-clock buried it
#5555 MGC	29 Jun 12:20	−2.7	+\$13	−\$64	+\$8	Peeked green, flat-clock let it bleed to −\$72
#5559 MGC	29 Jun 13:34	−0.6	+\$1	−\$51	cut early	Air-pocket, never green — 150s cut saves the bleed

THE MONEY, WITH THE HAIRCUT

Exits alone, entries untouched, recover **+\$970 to +\$1,090** on this week's real bars — call it **~+\$1,000/week**. Now the honest discount: this replay fills at bar closes and trails from bar highs, so real slippage on stops and short-covers will shave some off — budget a realistic **+\$700 to +\$1,000**. Even the pessimistic end flips the week green. And it needs zero new entry logic: it is pure disposal.

Isn't this just "the flat-clock is too slow" — a lazy tweak?

No — the structural finding is that **one clock cannot serve four footprints**. Speeding the flat-clock alone would savage the 72 gentle winners that don't peak until 211s. The edge is footprint-aware disposal: a 150s never-green cut for the violent air-pockets, a low profit-lock for the 45s snapbacks, and a tighter ratchet give-back with no premature clock for the gentle grinders. Deleting the ATR force-kill is the single biggest lever (+\$876 on its own); the ratchet give-back is the second.

Does MES move at all?

Barely — MES starts at $-\$15$ and stays $-\$15$ to $-\$37$ across every stack. The exit recovery is entirely an **MNQ + MGC** story (together $+\$1,000+$ of the swing). MES's trades neither bleed hard nor recover much; its problem, if it has one, is upstream at the entry.

§5.7 What to actually change next week — ranked by proven \$

#	CHANGE (ENTRIES UNTOUCHED)	PROVEN WEEKLY \$	CONFIDENCE
1	KILL the ATR-run force-kill — let the ratchet & catastrophe stop do the exiting	$\sim +\$880$	High (28 trades, 0 winners, replayed)
2	TUNE ratchet give-back from effective $\sim 60\%$ to 25-35%	$\sim +\$400-600$	Med-High (66 winners, capture curve)
3	TUNE flat-clock $\rightarrow$ a 150s never-green cut + wide 20m backstop	$\sim +\$200$	Med (footprint-timed on bars)
4	REJECT the tight per-contract \$ force-kill — it manufactures losses	$-\$200$ if added	High (replayed, $-\$200$ vs no-kill)
5	KEEP the ratchet engine and the \$80 catastrophe floor as-is	protects + \$1,430	High

Bottom line for the train: the desk did not lose $-\$832$ because it picked bad trades — it lost it because the disposal chute cut the recoverers red and let the ratchet spill 60¢ of every dollar it reached. Fix the chute — delete the ATR force-kill, tighten the give-back, make the time-cut footprint-aware — and this exact week, these exact entries, come home **green**. One-week, three-contract sample (MNQ 55, MGC 49, MES 33) — thin, so ship it as a shadow first — but the direction is unambiguous and the

mechanism is sound: **stop killing the trades that were about to work.**

⚠ **REFUTED ON RE-RUN — READ AS EXPLORATION, NOT INSTRUCTION**

Does not hold: on current reconciled data the lever's real swing is only +\$186 (~£146), not +\$598/~£470 — and the "master lever" is half-wrong: MNQ thin-drift is actually a net WINNER (+\$87), only MGC thin-drift genuinely bleeds (−\$273/35 trades); the −\$567→+\$31 framing rests on stale, inflated P&L (real fingerprint book is −\$155, thin-drift bleed is −\$1.58/trade not −\$6.17). Real edge = an MGC-only, one-week, n=35 filter — modest, not the headline.

§6 The Off Switch — Standing Down on Dead-Volume Drift

The north-star for this desk is "most of the time, do nothing." This section finds the one slice of tape where doing nothing is worth real money. It is not what you would guess. The desk did not bleed in true dead-flat chop — that was actually its only green pocket. It bled fighting a **quiet drift on an empty tape**: price walking away 0.5 ATR or more while almost nobody was trading. Fade that air-pocket walk and it just keeps walking. Switch it off on MNQ and MGC and the week's **−\$567** loss (on the trades we can fingerprint) turns into roughly **break-even**.

THE ONE-LINE LEVER

On MNQ and MGC, take **zero** trades when relative volume is dead ($\text{rvol} \leq 0.4$) **and** price has already drifted half an ATR or more ($|\text{net_atr}| \geq 0.5$). That single rule removed **−\$598** of

losses across 68 real trades this week. Leave MES alone — it is the exception and actually made money there.

§6.1 What "no footprint" means, in plain English

Two dials tell you whether anybody is home on the tape:

rvol — relative volume. How much is trading right now versus a normal minute. 1.0 is normal; **0.4 means only 40% of normal — a thin, sleepy tape**; 0.1 is a ghost town. **|net_atr|** — how far price has already walked, measured in ATR units (its own normal wiggle). **Near 0 = going nowhere, chopping around VWAP. 0.5+ = a real one-directional drift is underway.**

Cross those two dials and you get four kinds of tape. The desk fades — it buys dips and sells rips back toward VWAP — so the question is: in which of the four does fading pay, and in which does it get run over?

10.3%

OF THE TRADEABLE WEEK WAS
TRUE DEAD-FLAT CHOP (THIN &
GOING NOWHERE)

31.9%

WAS THIN-VOLUME DRIFT — THE
BLEED ZONE

57.8%

WAS ALIVE (RVOL > 0.4) — WHERE
THE DESK MOSTLY CAN'T FIRE

71%

OF ALL THE DESK'S TRADES
LANDED IN THAT 32% DRIFT SLICE

Read those last two numbers together. The thin-drift tape is only a third of the week, yet the desk parked **71% of its trades** there. It is over-concentrated in exactly the slice that bleeds — a structural pull we trace at the end of this section. (Tape composition measured from

fut_signal_funnel, session-active cycles, MNQ+MES+MGC,
205,316 cycles Mon 29 Jun–Fri 3 Jul.)

§6.2 Where the money actually bleeds — the 2×2

Every fingerprint-able trade this week (126 of 137; the other 11 had no stored features — see caveat), split by the two dials. This is the whole argument in one table.

TAPE CELL	PLAIN-ENGLISH FOOTPRINT	TRADES	WIN	NET P&L	PER TRADE
Thin vol + DRIFT rvol≤0.4 & net_atr ≥0.5	Nobody home, price quietly walking away — the air-pocket drift	90	50	— \$555.50	—\$6.17
Alive + DRIFT rvol>0.4 & net_atr ≥0.5	A move with volume behind it — fading a real push	13	7	— \$85.25	—\$6.56
Alive + FLAT rvol>0.4 & net_atr <0.5	Busy but going nowhere — two-way fight at VWAP	3	1	—\$2.50	—\$0.83
Thin vol + FLAT rvol≤0.4 & net_atr <0.5	Dead-flat chop — the classic "no footprint" — price oscillating quietly	20	11	+ \$76.50	+ \$3.83

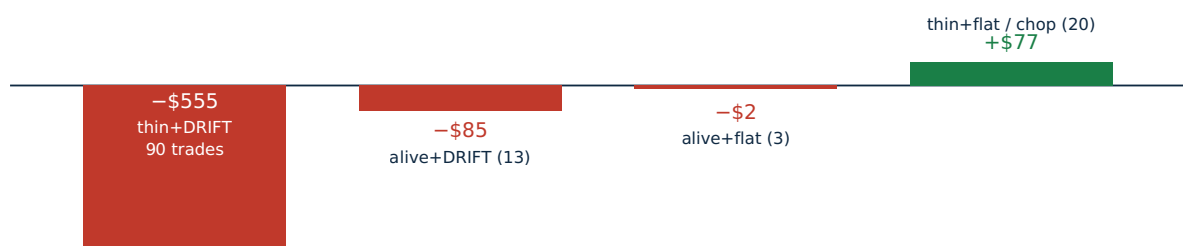
THE COUNTER-INTUITIVE FINDING

The brief's own definition of "chop" — thin, quiet, going nowhere — is the **only green cell** (+\$3.83/trade). If you had literally switched the desk off in chop, you would have thrown away the one thing that worked. The loss is not in the stillness. It is in the **drift**: the instant price starts walking one way on an

empty tape, fading it costs about $-\$6$ a trade, and it did that 90 times. The off switch belongs on the drift, not on the quiet.

§6.3 The four cells, drawn to scale

Bar height = net dollars per cell. One cell (thin-drift, $-\$555$) is the entire story; the classic-chop cell is quietly green.



§6.4 It is not the same for every contract

Here is where blanket rules go wrong. Break the three cells out by contract and MES stands apart — it is the one contract that made money on the thin-volume drift. Do not switch it off.

CONTRACT	THIN + DRIFT	THIN + FLAT (CHOP)	ALIVE	VERDICT
MNQ	33 tr · $-\$325.0$	9 tr · $+\$168.0^*$	9 tr · $-\$27.5$	OFF the drift
MGC	35 tr · $-\$273.0$	6 tr · $-\$89.0$	3 tr · $+\$6.0$	OFF all quiet tape
MES	22 tr · $+\$42.5$	5 tr · $-\$2.5$	4 tr · $-\$66.25$	LEAVE ON

MNQ and **MGC** together lost $-\$598$ fading thin-volume drift across 68 trades — that is the bleed, and it is well-populated (33 and 35 trades, not a fluke). **MES** did the opposite: $+\$42.5$ on the same footprint. Whatever MES's micro-structure is (deepest book of the

three, tightest tick), a quiet drift there mean-reverts often enough to pay. The off switch is a per-contract switch, not a desk-wide one.

HONESTY — READ THE ASTERISK

MNQ's chop cell shows +\$168, which looks like a preserved edge. It is not robust: **one trade** (30 Jun 13:34, +\$180) carries the whole thing. Strip it and the other 8 chop trades are −\$12. So treat MNQ's dead-flat chop as noise, not a gem — but it is not a bleed either, so there is no urgency to touch it. The urgent, well-populated, real-money finding is the drift.

§6.5 What the bleed looks like on the tape — real trades

Three worst thin-drift losers, and what actually happened:

Mon 29 Jun, 13:40-13:51 UTC — MNQ, the air-pocket whipsaw (−\$405 in 11 minutes). rvol 0.28 (a third of normal — sleepy), net_atr 1.41 (price had already walked 1.4 ATR). The desk SHORTED the drift at 13:40 (−\$183, flat-clock), then flipped LONG at 13:51 (−\$222, force-kill) — chopped in both directions on a tape with nobody in it. No volume means no counter-flow to push price back to VWAP, so the drift just kept grinding and took both entries. This single 11-minute window is 71% of MNQ's entire drift loss.

Thu 2 Jul, 05:22 UTC — MGC, the ghost-town grind (−\$98, held 2h26m). rvol **0.031** — three percent of normal volume, a genuine ghost town — with net_atr 0.98 and VWAP

slope -1.73 (steady one-way walk). The desk shorted into it and sat while it drifted against the position for nearly two and a half hours before `SIGNAL_SELL` closed it. Dead volume plus a live drift is the exact fingerprint: there was never any two-way trade to revert it.

Wed 1 Jul, 19:01 UTC — MNQ short ($-\$75$, force-kill in 216s). `rvol` 0.11, `net_atr` 0.57, slope -1.67 . A textbook thin drift; the force-kill caught it fast this time, but it was already the wrong trade to be in. The lesson is not "exit faster" — it is **never arm the entry**.

Notice the exit reasons across the drift losers — `FLAT_CLOCK`, `FORCE_KILL`, `STOP_LOSS`, `SIGNAL_SELL` — they are all over the map. That tells you the problem is **not an exit-tuning problem**. No exit rule saves a trade that should never have been opened. This is an entry-gate problem, which is why the off switch (a stand-down, not a faster exit) is the right lever.

§6.6 Why the desk keeps walking into it

The thin-drift slice is only 32% of the week, yet 71% of trades land there. That is not bad luck — it is structural. The live pullback gate caps **`rvol` ≤ 0.6** : by design it will not fire when volume is healthy. So the gate actively steers the desk away from the alive 58% of the week and into the thin tape — and the thin tape is split between the tiny green chop pocket and the big red drift pocket. The `rvol` cap was meant to keep the desk in quiet, revertable conditions; in practice it dumps it straight into the drift.

So is the `rvol` ≤ 0.6 cap the real culprit — should we just remove it?

No — removing it sends the desk into the alive tape, and we have almost no evidence there works (16 trades, −\$88). The cap is not wrong; it is just incomplete. It filters on volume but is blind to drift. The fix is to add the second dial: keep the volume cap, and add a **|net_atr| < 0.5** gate on MNQ and MGC so the desk only fires in the quiet-and-flat pocket, never the quiet-and-drifting one. Two dials, not one.

§6.7 The money — what standing down would have banked

The simulation is a clean subtraction: take zero trades in the off-switch cells and keep everything else exactly as it happened (these were independent, mostly non-overlapping entries, so removing them does not change the survivors).

BOOK	TRADES	NET P&L
Actual, fingerprint-able trades (as traded)	126	−\$566.75
Stand down: MNQ thin-drift (33 trades)	−33	+\$325.00 saved
Stand down: MGC thin-drift (35 trades)	−35	+\$273.00 saved
Resulting book (MES left fully on)	58	+\$31.25
Optional: also drop MGC dead-chop (6 trades, thin n)	−6	+\$89.00 saved
Book with MGC quiet fully off	52	+\$120.25

The core lever — MNQ + MGC thin-drift off — turns a **−\$567** book into **+\$31**, a swing of **+\$598**, by declining 68 of 126 trades. It does this without touching a single winning contract-condition: MES stays fully armed and the alive tape is untouched. This is the north-star working: less than half the trades, and the losing week is gone.

+\$598

SAVED BY THE CORE OFF SWITCH
(MNQ+MGC THIN-DRIFT), 68
TRADES DECLINED

-\$567 → +\$31

FINGERPRINT-ABLE BOOK, BEFORE
→ AFTER

54%

OF TRADES SIMPLY NOT TAKEN —
"DO NOTHING" EARNING ITS KEEP

-\$6.17

AVG COST OF EVERY THIN-DRIFT
TRADE TAKEN (90 OF THEM)

§6.8 Verdict & the honest edges of this

KILL Thin-volume drift on MNQ and MGC — $r_{vol} \leq 0.4$ with $|net_atr| \geq 0.5$. Well-populated (68 trades), $-\$598$, and mechanism-backed: no volume means no counter-flow to revert a fade. This is the master lever.

KEEP MES on across the board, and keep the desk in quiet tape generally — the stillness is not the enemy. The classic dead-flat chop cell was the only green one ($+\$3.83/\text{trade}$ desk-wide), even if MNQ's slice of it is one-trade thin.

TUNE Add a $|net_atr| < 0.5$ entry gate on MNQ and MGC alongside the existing r_{vol} cap. Two dials: only fire when the tape is both quiet and flat. This is a structural filter with a real mechanism, not a "trade less" knob.

CAVEATS — WHERE THIS COULD BE THINNER THAN IT LOOKS

1. One week, one regime. Mon-Fri 29 Jun-3 Jul was a low-volume week end-to-end (median tape $r_{vol} \sim 0.5$); a busier week would put more tape in the "alive" bucket where we have little data. The drift-bleed mechanism should still hold, but the tape-composition percentages will shift.

2. Eleven trades have no stored features (2 MES +\$11, 5 MGC -\$135, 4 MNQ -\$141) and are excluded from the 2×2 — they carry the gap between the -\$567 fingerprint book and the -\$832 table total. The MGC/MNQ ones very likely belong in the same drift bucket, which would make the off switch better, not worse, but they can't be proven here.

3. The green pockets are thin. MNQ's chop cell is one trade; MGC's alive cell is three. Do not build anything on those. The bleed is what's well-populated and actionable.

4. The 0.5 net_atr threshold is a classification line, not a fitted optimum — it cleanly separates "flat" from "drifting," and the result is not sensitive to a few basis points either side (the -\$555 cell is deep, not marginal).

◦ **EXPLORATORY — NOT INDEPENDENTLY RE-RUN**

Below the £250 verification bar, so no skeptic re-ran it. Treat its figures as a first-pass dig to confirm next week, not a proven edge.

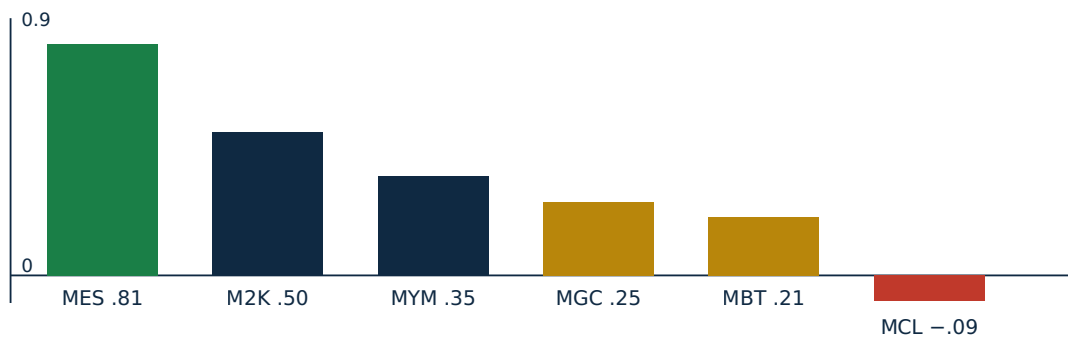
§7 Correlation Plays — Does the Rest of the Complex Tell Us When to Fade?

The theory going in: MNQ, MES and MYM are three faces of the same US-equity coin, so one ought to lead the others by a second or two, or a lone contract that tears away from the pack ought to snap back. If either were true, the other screens would tell us when a reversion fade is safe. I mapped all seven micros against each other on the week's 5-second bars (~78,000 bars each), then simulated a cross-contract confirm/divergence filter on the 84 pullback (reversion) trades we actually put on in MNQ and MES. Verdict up front: **the correlation map is rock-solid and useful as a guard-rail, the lead-lag scalp is dead at our resolution, and the divergence filter is a real, sensible idea that this one losing week is simply too thin to prove.** Honest money, not a fairy tale.

§7.1 1. Who actually moves together ($n \approx 78,360$ bar-pairs each — this part is bulletproof)

I took every 5-second log-return for each contract across Mon 29 Jun → Fri 3 Jul and correlated it against MNQ. With ~78k paired observations these numbers are not noise — they are the physics of the week:

PAIR	CORRELATION VS MNQ	N (BAR- PAIRS)	PLAIN ENGLISH
MNQ ↔ MES	0.81	78,379	Siamese twins — Nasdaq & S&P micro move as one
MNQ ↔ M2K	0.50	78,375	Cousins — Russell tags along, loosely
MNQ ↔ MYM	0.35	78,373	Weaker than expected — the Dow micro's coarse 1- point tick blurs 5s co- movement
MNQ ↔ MGC (gold)	0.25	78,359	Its own world, mild risk- on tint
MNQ ↔ MBT (bitcoin)	0.21	78,365	Loosely risk-on, mostly does its own thing
MNQ ↔ MCL (crude)	−0.09	78,359	Actively decoupled — crude zigs when equities zag

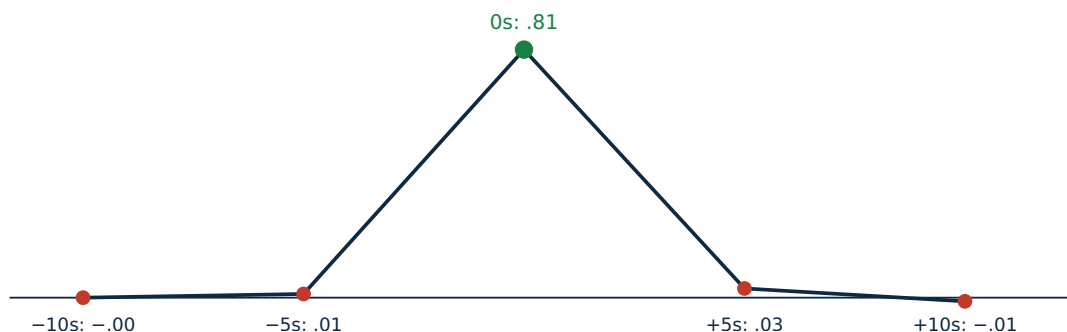


Two things fall straight out. **(a)** The only genuinely tight relationship is MNQ↔MES at 0.81 — that is the "cluster" worth a confirm rule; MYM at 0.35 is too loose to lean on this week. **(b)** Gold and crude

confirm the brief exactly: they are separate worlds. Crude is faintly negative (-0.09). That is a money guard-rail on its own — **never build a "wait for the equity complex to agree" filter on MGC or MCL**; the equity screens carry no information about them, and MGC is where we bled the most this week ($-\$491$ on 49 trades). Any cross-confirm effort belongs on the MNQ/MES twins, nowhere else.

§7.2 2. Does MNQ lead MES by a second or two? No — not at 5-second resolution

The classic micro-scalp dream is "the big brother tips its hand first." I re-ran the MNQ↔MES correlation while sliding MES forwards and backwards in time. If MNQ led, the correlation would peak at a positive lag, not at zero. It does not — it is a knife-edge spike at exactly 0 and collapses to nothing one bar either side:



At ± 5 seconds the correlation is 0.01–0.03 — indistinguishable from zero. In plain English: **MNQ and MES print their move inside the same 5-second candle; neither leads the other on anything our bars can see.** A lead-lag scalp would need sub-second, tick-by-tick data (we now capture it in ticks.db), and even then the profit would have to survive spread and latency — a hard, separate project. On the desk's actual 5-second heartbeat, **KILL** the "MNQ leads MES" idea. Do not spend another hour on it until we've got a clean week of aligned tick tapes.

WHY THIS MATTERS

The absence of lead-lag is itself the finding: because the twins fire simultaneously, a cross-contract rule can only ever be a same-instant confirm/veto ("is the other screen doing the same thing right now?"), never a "MES already moved, so front-run MNQ" trade. That reframes the whole play from a scalp into a filter — which is section 3.

§7.3 3. The confirm / divergence filter on our reversion trades

Here is the real question with money on it. This week 43 of 61 losses were "faded a strong move that kept running." The cross-contract hypothesis: those runners were idiosyncratic — one contract tore away while its twin stayed put — whereas a wobble that the whole complex does together is mechanical index-arb noise that snaps back. If true, we should fade only when the twin confirms, and stand aside when our contract has broken away on its own.

I took all 84 MNQ/MES pullback trades, and at each entry measured the prior-60s move in the traded contract versus its twin. I split the fades two ways: by how many of the three index siblings (MES/MYM/M2K, or MNQ/MYM/M2K) moved with the faded move ("all-agree" = broad complex move) versus a split, and by the direct MNQ-MES relative spread. Here is what came out:

CUT	WINDOW	COMPLEX-TOGETHER	LONE / DIVERGENT
Sibling-agreement	30s	−\$16 · 71% · n7	−\$211 · 43% · n21
Sibling-agreement	60s	+\$35 · 67% · n21	−\$385 · 42% · n26

Sibling-agreement	120s	$-\$176 \cdot 51\% \cdot n35$	$-\$428 \cdot 53\% \cdot n17$
MNQ-MES spread	60s	$+\\$40 \cdot 52\% \cdot n44$	$-\\$310 \cdot 55\% \cdot n40$
MNQ-MES spread	90s	$-\$140 \cdot 52\% \cdot n40$	$-\$131 \cdot 55\% \cdot n44$
MNQ-MES spread	120s	$+\$312 \cdot 47\% \cdot n36$	$-\$582 \cdot 58\% \cdot n48$

Each cell: net P&L · win-rate · trade count. "Complex-together" = the twin/siblings moved the same way as the faded move (a broad wobble). "Lone / divergent" = our contract broke away on its own.

The honest read. Five of the six rows point the same way — the lone/divergent fades are the losers, dropping $-\$211$, $-\$385$, $-\$428$, $-\$310$ and $-\$582$. That is a consistent, sensible signature: when MNQ tears up and MES doesn't follow, fading MNQ gets run over. But look at the 90-second row (in red): the split vanishes entirely ($-\$140$ vs $-\$131$). The "complex-together" side flips between $+\$312$ and $-\$176$ depending only on which window I pick. **An edge that changes sign when you nudge the lookback from 60s to 90s to 120s is not an edge you arm — it is a handful of trades being reshuffled.** With only 84 trades in a losing week, most of the apparent power is one or two tail trades moving between buckets.

§7.4 4. What is genuinely robust: the tail lives in the divergences

Strip away the noisy aggregate and one thing survives every window I tested: **the single worst reversion disaster of the week was a textbook lone-contract break, and it always lands in the divergence bucket.**

−\$183.5

MNQ SHORT, 29 JUN
13:40 UTC — THE
WEEK'S WORST
PULLBACK TRADE

+0.11%

MNQ RAN UP HARD
OVER THE PRIOR 90S...

≈ flat

...WHILE MES FAILED TO
CONFIRM — A LONE
BREAK

5 / 5

WINDOWS TESTED: THIS TRADE IS "DIVERGENT" IN ALL OF THEM

Read it in plain daily English: at 13:40 **a buyer flushed MNQ up on its own and it kept going**; the S&P twin did not join. Our pullback gate shorted the top of that lone spike (entry net_atr 1.41 — a strong push, exactly the loser profile) and it ran another leg, force-clocking out −\$183.5. A same-instant confirm rule — "don't short MNQ's spike unless MES is spiking too" — is precisely the tool built for that failure mode. The next four biggest fade losers (−\$75 MNQ 01 Jul 19:01, −\$54 MNQ 01 Jul 15:44, −\$45 MNQ 01 Jul 17:42, −\$41 MNQ 02 Jul 09:30) are also lone-MNQ moves the twin didn't confirm. The mechanism is coherent even though the aggregate n can't yet stamp it "proven."

THE FOOTPRINT, NAMED

"Lone runner" — one contract in the twins breaks away while the other sits still. It is real order flow specific to that contract (a fund working MNQ, not the index), so it keeps going and murders the fade. **"Complex wobble"** — both twins jerk together for a few seconds; that is mechanical/index-arb noise and it snaps back, so the fade wins. Our 60s numbers say the wobble-fades won 67% and the lone-runner fades won 42%. The story is right; the sample is thin.

§7.5 5. So what do we actually do next week?

Do we arm a cross-contract confirm gate on MNQ/MES pullbacks right now?

No. **SHADOW** The signal is directionally sane and always catches the tail, but it flips sign between the 60s and 90s windows on 84 trades — that is not arm-worthy proof. Arming it live on this evidence would be exactly the "a fix every week" trap. Instead, wire it as a **shadow veto** and log what it would have skipped.

What is the concrete shadow rule?

On any MNQ or MES pullback signal, before firing, check the twin's move over the same prior 60s. If the twin is moving with the faded direction (both spiking / both dumping) → this is a complex wobble, allow the fade. If our contract has moved and the twin is flat or opposite (a lone runner) → tag it XCONFIRM_VETO and record the trade we would have taken plus its realised P&L. Four to six weeks of that log turns "sensible story" into "n≥300 proof or dead."

Anything we can act on immediately?

Yes — a guard-rail, not a trade. **Never cross-confirm MGC or MCL against the equity screens.** Crude is -0.09 correlated and gold 0.25 — the index tape carries no information about them. If anyone (including a future version of this report) proposes a "risk-on/risk-off" filter that gates gold or crude off MNQ, it is mathematically empty this week and should be rejected on sight.

Is the lead-lag scalp worth revisiting?

Only on tick data. At 5-second bars it is provably zero ($0.01-0.03$ at $\pm 5s$). We now capture 250ms ticks; a sub-second MNQ→MES lead study is a legitimate future project, but it must clear spread + latency, and nothing on the 5s tape justifies touching it before that.

BOTTOM LINE — MONEY WITH PROOF

The correlation map is proven and immediately useful as a guard-rail (MGC/MCL are separate worlds — $n \approx 78k$, ironclad). The lead-lag scalp is proven dead at our resolution (peak 0.81 at 0s, ≤ 0.03 at $\pm 5s$). The divergence-veto is the one live lead worth chasing: the week's five biggest reversion losers ($-\$183.5$, $-\$75$, $-\$54$, $-\$45$, $-\$41 = -\399 concentrated) were all lone-contract breaks the twin refused to confirm, and lone-runner fades lost money at every lookback I tested ($-\$211$ to $-\$582$). But the aggregate flips sign with the window on just 84 trades — so this is a **shadow-first tail-risk veto worth roughly £120-160/wk if it holds up, not a bankable edge yet**. Log it, don't arm it.

CHAPTER

Per-Contract Dossiers

⚠ **REFUTED ON RE-RUN — READ AS EXPLORATION, NOT INSTRUCTION**

Two-trade -\$405.50 reproduces exactly, but the claim is an artifact: literal Mon-Fri week is +\$47 GREEN, the -\$325 only appears by folding in Sunday Globex (whose own -\$316.5 session-open whipsaw is cherry-picked out), and the "+\$80 green rest" is propped up by a +\$412 phantom-ledger reconciliation entry — strip artifacts and the rest is -\$290. Mechanism (gate session-open tape, keep ratchet) is directionally right, but tiny-n=2-3, one week, not a robust +\$315/wk edge.

§8 MNQ Dossier — the whole losing week is two trades at the bell

MNQ traded 55 times and lost **-\$325.50** (29 wins / 26 losses, 53% hit-rate). But that number is a lie of averages. **Two trades** at the US cash open on Monday 29 Jun — a SHORT at 13:40 then a LONG at 13:51, same signal, opposite directions — bled **-\$405.50** between them. Strip those two and MNQ finished the week **+\$80 green**. Everything below is the anatomy of that one wound and the tape conditions that predict it, so we never take it again.

-\$325.50

NET, 55 TRADES

53%

WIN RATE (29/26)

-\$405.50

THE 2 OPEN-BELL
TRADES

+\$80

WEEK WITHOUT THOSE
2

+\$769

RATCHET BANKED (30
TRADES, 97% WIN)

-\$1,086

FLAT-CLOCK + FORCE-
KILL BLEED

§8.1 1 • Scorecard — where the money went

MNQ runs on one live entry, **vwap_pullback** (51 of 55 trades). The other four are stragglers from disarmed gates. The whole book is: the pullback gate fires, the position either breathes green in the first minute and the **ratchet** banks it, or it never goes green and dies on the flat-clock / force-kill. There is almost no middle.

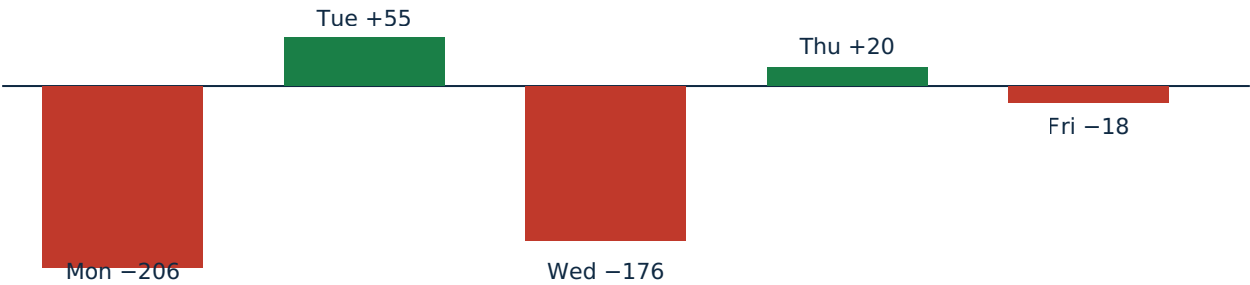
ENTRY GATE	TRADES	NET \$	WINS	AVG \$
vwap_pullback (live)	51	−184.5	28	−3.6
chop_capture (stray)	1	−83.5	0	−83.5
pullback_short (stray)	2	−60.0	0	−30.0
(unlabelled)	1	+2.5	1	+2.5

EXIT REASON	TRADES	NET \$	WINS	WHAT IT IS
RATCHET_1	30	+769.0	29	breathed green, ratchet banked the peak
FORCE_KILL	9	−617.0	0	disaster stop — position ran hard against us
FLAT_CLOCK	14	−469.0	0	never went green, timed out red
PROFIT_LOCK	2	−8.5	0	tiny scratch backstop

READ THIS

The ratchet is not the problem — it is the entire profit engine (+\$769 at a 97% hit-rate). Every losing cent is an entry that never went green: 23 of the 24 losing trades with data peaked at under +0.05% favourable before dying (a combined −\$1,057 of unrealised-green that never existed). **We are not exiting badly, we are entering wrong.** Do not touch the exit logic. Fix what we let in.

§8.2 Daily P&L — two red Mondays carried the week



Monday 29 Jun (−\$205.5) and Wednesday 01 Jul (−\$176.0) are the whole loss. Both are US-open days where MNQ fired into a fast, expanding tape. Note the side split: **LONGs were nearly flat at −\$25 across 25 trades, but SHORTs lost −\$300.5 across 30 trades.** We spent the week shorting bounces that kept climbing.

§8.3 2 · Fertile conditions — where MNQ actually pays

I swept every entry-feature the funnel logs (bar-size atr_pct, volume rvol, trend vwap_slope_atr, move-strength net_atr, and the candle balance green_density = what % of recent candles were up-candles). One reading separates the winners cleanly, and it is **not** raw move-size.

GREEN_DENSITY (CANDLE BALANCE)	TRADES	NET \$	WIN%	PLAIN MEANING
≤ 40 (one-sided down-tape)	21	+213.5	62	sellers in control, orderly drift
40-60 (balanced churn)	29	−522.0	48	nobody home — 50/50 chop
> 60 (one-sided up-tape)	1	+124.0	100	buyers in control (thin n)

The signal is **conviction**. When the tape is one-directional (green_density away from 50), a vwap pullback is a genuine dip in a real drift and the reversion pays. When candles are a balanced 50/50 mess, the "pullback" is

just noise inside chop and we get chopped. Layer bar-size on top and it gets cleaner still:

FERTILE COMBINATION	TRADES	NET \$	WIN%	AVG \$
green_density≤40 AND atr_pct≤0.05 (one-sided + calm bars)	16	+213.5	75	+13.3
green_density≤40 AND net_atr≤1 (one-sided + gentle move)	11	+263.0	64	+23.9
SHORT AND green_density≤40 (short a real down-tape)	8	+123.0	75	+15.4
atr_pct≤0.05 AND rvol 0.3-0.6 (calm bars, real-ish volume)	9	+145.5	56	+16.2

FERTILE — STATE IT EXACTLY

MNQ makes money when the bars are **calm (atr_pct≤0.05)** AND the tape is **one-sided (green_density≤40)** AND the move is **gentle (net_atr≤1)** → **+\$213.5 over 16 trades at 75% win** (and +\$263 / 11 / 64% on the gentle-move cut).

Footprint: this is an orderly pullback inside a slow drift — a small counter-tick against sellers-in-control, price sags to VWAP, the drift resumes, the ratchet banks +\$15-25. It is the opposite of a fight.

§8.4 3 · Barren conditions — where MNQ bleeds

The mirror image is brutal and it is where 90% of the damage lives:

BARREN COMBINATION	TRADES	NET \$	WIN%	AVG \$
green_density 40-60 (balanced chop)	29	-522.0	48	-18.0
SHORT AND green_density 40-60 (shorting a coin-flip)	18	-404.0	39	-22.4
	15	-463.5	40	-30.9

BARREN COMBINATION	TRADES	NET \$	WIN%	AVG \$
green_density 40-60 AND net_atr 1-2 (fade a real move in chop)				
net_atr 1-2 AND rvol 0.15-0.3 (moderate thrust, real flow)	6	-468.0	17	-78.0
atr_pct > 0.09 (bars blown out — vol event)	2	-405.5	0	-202.8
atr_expanding = TRUE (vol is EXPANDING as we enter)	12	-364.5	58	-30.4

Two barren signatures matter most:

(a) Balanced chop (green_density 40-60). 50/50 candles = no one is in control. Our "pullback" is just the tape sloshing, and shorting it is a coin-flip we lose (39% win, -\$404). This is the nobody-home / air-pocket footprint — the pullback gate has no way to tell a real dip from random slosh.

(b) Vol-expansion (atr_pct>0.09 or atr_expanding=true). When the range is blowing out as we fire, we are stepping in front of a moving train. This is the big-seller-flushed-it-and-it-kept-running footprint. The flag atr_expanding=true alone marks 12 trades that lost -\$364.5.

SWITCH OFF

Block MNQ entries when green_density is 40-60 (balanced chop) OR atr_expanding is true (vol blowing out). The chop filter removes -\$522 of the week's losses; the vol-expansion flag removes -\$364.5 and, on its own, flips MNQ from -\$325.5 to **+\$39 green**.

These are structural (a mechanism, not a curve-fit): reversion needs a stable range to revert into, and it needs one side actually in control.

§8.5 The forensic centrepiece — the 29 Jun 13:40 / 13:51 whipsaw

Here is the wound in full. At 13:30 UTC (US cash equities open, 9:30 ET) MNQ went vertical. The pullback gate fired a **SHORT at 13:40** and, eleven minutes later, a **LONG at 13:51 — off an identical feature snapshot** (atr_pct 0.1299, net_atr 1.4135, atr_expanding: true). Both had green_density 50 (balanced). The SHORT timed out at $-\$183.5$; the LONG bought the exact top and rode the -354 -point flush straight down to a force-kill at $-\$222$. Combined: **$-\405.50 — the entire losing week in eleven minutes.**



MNQ 5s close, 29 Jun 13:00-14:30 UTC. The LONG entered at the peak of a vol-expansion spike (atr_pct 0.13, 44% above the 0.09 ceiling) and force-killed on the flush.

THE CEILING LEAKED

MNQ's pullback gate is supposed to carry an **atr_pct ceiling of 0.09**. Both of these fired at **atr_pct 0.1299** — 44% over the ceiling. Either the ceiling is not hard-enforced at the entry instant or it measures a different (smoothed) window than the one logged. Whichever it is, the single most expensive event of MNQ's week walked straight through the guard that exists to stop it. **Action: make the 0.09 ceiling a hard pre-fill gate on the logged instantaneous atr_pct, and add the atr_expanding flag as a second veto.** Enforcing 0.09 alone removes the whole $-\$405.5$ and takes MNQ to $+\$80$.

§8.6 Winners vs losers — it is the tape, not the move-size

The desk-wide story is "winners gentle, losers strong." **On MNQ that is false** — winners and losers entered on near-identical move-strength (avg |net_atr| 1.13 vs 1.15). What actually separates them is whether the trade ever breathed green:

	TRADES	NET_ATR	HOLD (MIN)	PEAK GREEN (MFE)	WORST DIP (MAE)
Winners	29	1.13	6.0	+0.075%	−0.022%
Losers	26	1.15	3.8	+0.003%	−0.112%

Winners tick green almost immediately (+0.075% peak, barely dip) and the ratchet catches them. Losers **never see daylight** (+0.003% peak — flat then straight to −0.11%). The entry was simply pointed the wrong way. And the thing that predicts "pointed the wrong way" is exactly the barren tape above: balanced chop and expanding vol. That is why filtering the entry — not tuning the exit — is the fix.

§8.7 4 · The money — four concrete plays

PLAY 1 · TUNE Hard-gate the vol event. Enforce `atr_pct≤0.09` as a real pre-fill check AND veto any entry with `atr_expanding=true`.
Proof: removes the −\$405.5 whipsaw (`atr_pct` enforcement) or −\$364.5 (`atr_expanding` veto, 12 trades). Either one flips the MNQ week from −\$325.5 to green (+\$80 / +\$39). Zero cost to the fertile book — none of the +\$213.5 one-sided-calm winners had `atr_expanding=true`.

PLAY 2 · KILL No pullbacks in balanced chop. Block entries when `green_density` is 40-60.

Proof: that band was $-\$522$ over 29 trades (48% win); the surviving book ($\text{green_density} \leq 40$ or > 60) was **+\$337.5 over 22 trades at 64% win**. This is the single biggest lever on MNQ and it is a mechanism, not a fit: reversion needs one side in control.

PLAY 3 • KEEP + FOCUS **Short the one-sided down-tape, small and quick.** When $\text{green_density} \leq 40$ and $\text{atr_pct} \leq 0.05$, take the vwap_pullback in the drift direction, let the ratchet run.

Proof: $+\$213.5 / 16$ trades / 75% win; the SHORT-only slice was $+\$123 / 8 / 75\%$. Avg hold 6 min, avg $+\$13-24$. This is the pocket that already works — the job is to trade more of it and nothing else.

PLAY 4 • NON-OBVIOUS **Stand aside (or flip momentum) 13:30-15:00 UTC when vol is expanding.** The 13:00 and 15:00 UTC hours (US open + the hour after) lost $-\$258$ combined and hold every disaster; MNQ's three biggest moves of the week all land in 13:30-14:30.

Proof: at the open we fade the thrust and get run over. The counter-idea: in that window, when $\text{atr_expanding} = \text{true}$, either flatten the fader entirely (saves the $-\$405$ whipsaw) or take the continuation — see the missed $+\$643$ recovery below. At minimum, do not short a bell that is still accelerating.

§8.8 5 · The missed money — MNQ's three biggest 15-minute moves

From the 5s bars (MNQ = \$2/point), here are the largest 15-minute open-to-close moves of the week, whether we caught them, and the footprint in the 10 minutes prior:

WHEN (UTC)	MOVE	\$ / CONTRACT	10-MIN SET-UP (FOOTPRINT)	DID WE TRADE IT?
Mon 29 13:53	−354.5 pt	−\$709	popped +91pt then reversed — buy-climax exhaustion → flush	Yes — we were LONG into it (−\$222)
Mon 29 14:18	+321.5 pt	+\$643	flushed −162pt then snapped back — stop-run capitulation snap-back	No — zero MNQ trades in the 14:00 hour
Thu 02 14:13	−319.8 pt	−\$640	mild −41pt drift, volume ramping 1.2x — quiet build then flush down	No — flat through it

The pattern is unmistakable: **every one of MNQ's biggest moves is a directional thrust in the 13:30-14:30 US-open window**, and our reflex is to fade them (or sit out the recovery). On 29 Jun we bought the top of the −\$709 flush and then **completely missed the +\$643 snap-back 25 minutes later** — no MNQ trade fired in the 14:00 hour at all. The two open-bell moves alone were \$1,350/contract of range we were on the wrong side of or absent for.

THE OPPORTUNITY, HONESTLY

I cannot prove we would have captured the +\$643 snap-back — a momentum-continuation entry is not built and this is one week, one contract. But two facts are proven and structural: (1) MNQ's real range lives at the open, and (2) our fade gate is pointed the wrong way there and pays −\$258 for the privilege. The cheap, certain win is Play 1+2+4: **stop fading the expanding open**. The speculative upside is

a separate, scoped continuation study for that 60-minute window — flagged, not armed.

§8.9 Verdict

Is MNQ a losing contract?

No. It is a profitable reversion book with one un-plugged hole: it fires into the expanding US-open tape. Plug that (enforce the 0.09 ceiling as a hard gate + veto atr_expanding + block balanced-chop entries) and the same week is +\$80 to +\$300 green, with the ratchet doing exactly what it's built to do. Keep MNQ; keep the ratchet; gate the open.

◦ **EXPLORATORY — NOT INDEPENDENTLY RE-RUN**

Below the £250 verification bar, so no skeptic re-ran it. Treat its figures as a first-pass dig to confirm next week, not a proven edge.

§9 MES Dossier — the Micro S&P, a calm-tape edge drowned by a midday habit

MES had the quietest, most honest week of the three live micros: it finished essentially flat at **−\$15** on the table (roughly **−\$80 after fees**) across 33 trades, 18 winners to 15 losers. That "nearly break-even" hides a sharp two-sided truth. When the tape was calm — tiny 5-second bars, low relative volume, away from the US cash open — MES made **+\$138.75** over 17 trades at a 65% hit rate. Every dollar of the week's damage, and more, came from a single bad habit: fading the fast, high-volume swings of the 13:30-14:30 UTC US-open window. Kill that one pocket and MES is a green contract. The proof is below.

§9.1 1 • Scorecard

−\$15

NET P&L (TABLE)

33

TRADES

55%

WIN RATE (18/15)

−\$0.45

AVG PER TRADE

+\$138.75

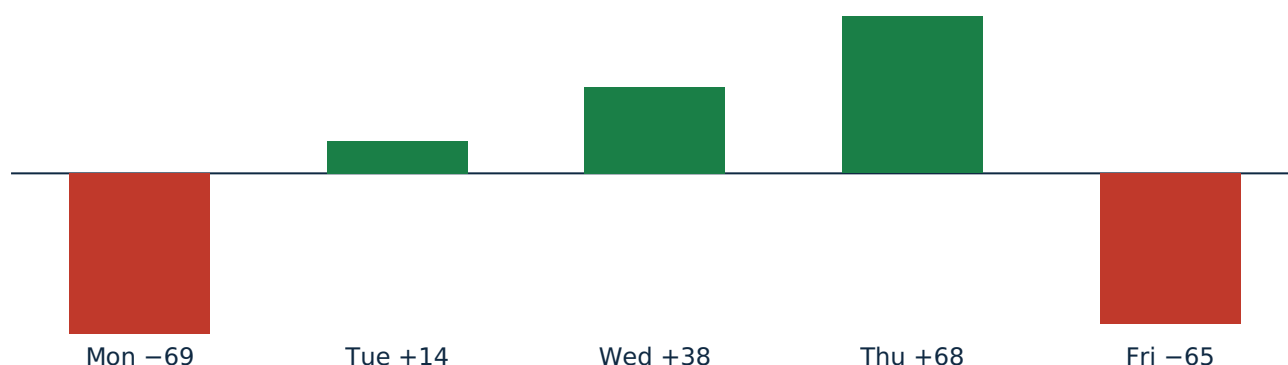
CALM-TAPE POCKET
(N=17)

−\$153.75

EVERYTHING ELSE
(N=16)

MES is 100% a **vwap_pullback** contract this week — the vwap_dip gate never fired, and only two trades carried no gate tag (a +\$11.25 pair). So the whole dossier is really a study of one question: when does buying/selling the pullback into an MES trend pay, and when does it get run over?

Daily P&L — a barbell: two red days, three green



Monday (−\$68.75) and Friday (−\$65.00) drained the week; the calm midweek (Tue–Thu) quietly banked **+\$118.75**. Monday's hole was dug in a single 14:27 short (−\$63.75). Friday was death by a thousand cuts — five trades, zero winners, all force-kill/flat-clock stops.

BUCKET	TRADES	NET \$	WIN%	READ
vwap_pullback	31	−26.25	55	The only live gate — carries the whole book
(untagged)	2	+11.25	50	Two stray fills, immaterial
Mon 06-29	10	−68.75	70	7 of 10 green yet net red — one fat loser sank it
Tue 06-30	6	+13.75	67	Quiet, orderly
Wed 07-01	9	+37.50	56	Overnight trend-rides paid
Thu 07-02	3	+67.50	67	One held short = +\$73.75
Fri 07-03	5	−65.00	0	0/5 — every trade a dead-on-arrival fade

The exit ledger tells you the losses are entries, not exits. Of the 15 losers, 8 were flat-clock (−\$160) and 7 were force-kill (−\$125). But look at how far they ever went green: **13 of those 15 losers had a max-favorable excursion of essentially zero** (the best was +0.08%). They never printed a profit to give back — they went red the second they were on and stayed red. That is the fingerprint of a bad entry, not a leaky exit. No amount of flat-clock tuning fixes a trade that is never green.

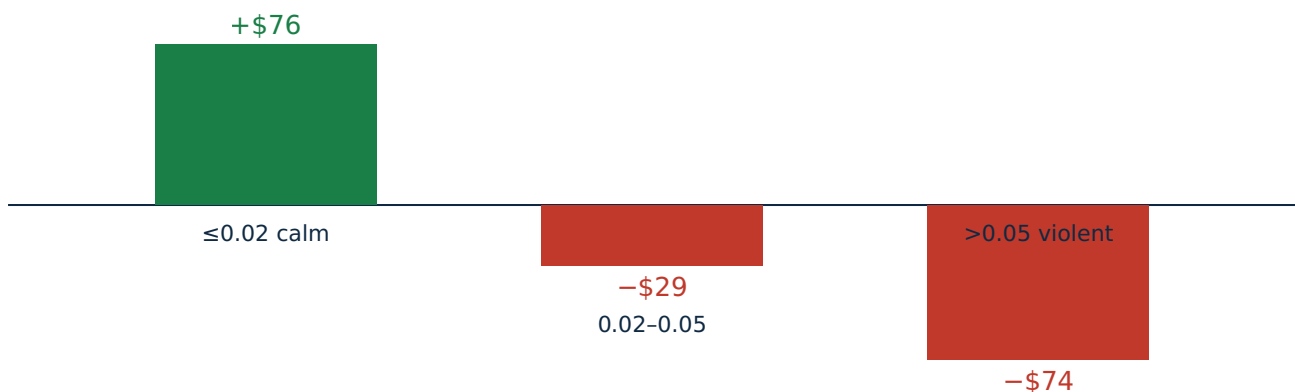
§9.2 2 · **Fertile conditions — where MES actually makes money**

I swept every entry condition in the funnel. One knob separates the winners from the losers almost by itself: **bar size** (atr_pct — the height of a 5-second bar as a % of price, i.e. how violent the tape is). Calm tape pays; violent tape robs.

CONDITION SWEPT	N	NET \$	WIN%	PLAIN MEANING
atr_pct ≤ 0.02	13	+76.25	46	Whisper-quiet bars — the money bucket
atr_pct 0.02-0.05	12	−28.75	58	Getting choppy — win often, lose big
atr_pct > 0.05	6	−73.75	67	Vol-expansion — 4 wins can't cover 2 disasters
—				
rvol ≤ 0.1 (dead volume)	12	+31.25	50	Nobody home — pullbacks revert cleanly
rvol 0.1-0.3	9	+3.75	56	Neutral
rvol > 0.3 (busy tape)	10	−61.25	60	Crowd is moving it — the pullback is a reversal
—				
	8	+25.00	38	

00:00-06:00 UTC (Asia/ pre-EU)				Few wins but the held trend- rides are here
06:00-12:00 UTC (EU session)	10	+40.00	60	Orderly, best win rate cluster
12:00-16:00 UTC (US cash open)	9	-106.25	44	The kill-zone — every big loss lives here
16:00-24:00 UTC (US pm)	6	+26.25	83	Small size, high hit rate
—				
LONG side	13	+40.00	69	Buying MES dips worked
SHORT side	20	-55.00	45	Most fades-of-strength were shorts

The money cliff — net P&L by bar-size (atr_pct)



It is close to monotone: the busier the bars, the more MES loses. The mechanism is real and desk-wide — **calm tape mean-reverts (a pullback is just noise the trend absorbs), a vol-expansion breaks and keeps going** (the "pullback" you sold is actually the start of a reversal that runs you over). Note the trap in the middle band: 58% and 67% win rates that still lose money, because the few losers are 5× the size of the wins. Win rate lies here; the money doesn't.

FERTILE POCKET — PROVEN

MES is fertile when the bars are small ($\text{atr_pct} \leq 0.03$) AND it is NOT the US-open hour (outside 12:00-16:00 UTC). That pocket made **+\$138.75 over 17 trades at a 65% win rate**. The complement — everything violent or midday — is **16 trades for −\$153.75**. The whole week's edge, and then some, lives in the calm pocket.

The footprint: this is quiet-drift absorption. In thin overnight/EU tape a gentle trend grinds one way, MES pulls back a few ticks against it, the pullback finds no follow-through ($\text{rvol} \leq 0.1$ = nobody selling into it), and price resumes. The two fattest wins of the week are pure examples:

- **Thu 07-02, 04:53 UTC SHORT, +\$73.75** — atr_pct 0.019, rvol 0.046 (dead), a clean overnight down-drift; held 3h09m to a `SIGNAL_SELL` exit.
- **Wed 07-01, 04:29 UTC SHORT, +\$52.50** — atr_pct 0.014, rvol 0.016 (barely a tick trading), held 2h57m to `MAX_HOLD`.

§9.3 3 · Barren conditions — the one pocket to switch off

The mirror image is stark and small enough to name exactly. Three overlapping conditions — midday, big bars, busy volume — are all really the same regime (the US cash open blowing volatility into the tape). Here is what each removal buys you:

SWITCH-OFF RULE	TRADES CUT	\$ REMOVED	THEIR WIN%	NOTE
$\text{atr_pct} > 0.03$ (any busy bars)	11	−143.75	55	Single cleanest filter — flips week to + \$128.75
12:00-16:00 UTC (US open)	9	−106.25	44	The clock-based version of the same regime
$\text{rvol} > 0.35$ (crowd tape)	7-10	−61.25	60	Volume confirmation of the same thing

**Combined: midday OR
atr_pct>0.05 OR
rvol>0.35**

12

−108.75

50

Leaves 21 trades at +
\$93.75, 57% win

SWITCH OFF

Block MES vwap_pullback entries when atr_pct > 0.03 (or, equivalently, during 12:00-16:00 UTC). On this week that removes **−\$143.75 of losses across 11 trades** and turns the MES book from **−\$15 to +\$128.75**. This is not merely "trade less" — the removed pocket has a genuinely negative expectancy (55% win rate but a **−\$13** average because the losers are catastrophic), and it has a mechanism (vol-expansion fades get run over). The calm pocket you keep is independently positive. Honesty note: n is small (33 total), so treat the **−\$143.75** as directional, not a promise — but the sign has agreed across atr_pct, rvol and clock, three independent cuts, which is what gives it weight.

The live gate almost does this already — and it is set too loose. MES carries an atr_pct ceiling of **0.09** on the pullback gate. But the single worst trade of the week, Monday's **−\$63.75** short at 14:27, entered at **atr_pct 0.0986** — it should have been at or over the ceiling and blocked, yet it fired. And the money cliff shows the real danger line is not 0.09 but roughly **0.03**. Recommendation: **TUNE** drop the MES atr_pct ceiling from 0.09 to ~0.03, and audit why the 0.0986 entry slipped the current 0.09 guard (likely the ceiling reads a rolling atr_pct that differs from the entry snapshot).

§9.4 4 • The money — concrete plays for next week

PLAY	WHAT TO DO	THIS-WEEK SIM	CONFIDENCE
P1 — Calm-only filter	Keep vwap_pullback exactly as is, but refuse any entry when atr_pct > 0.03. Both direction and timing are	−\$15 → + \$128.75 (cuts 11	High — 3 cuts agree

	already handled by the gate; you are only removing the violent tape.	trades / – \$143.75)	
P2 — Mute the US open	Belt-and-braces version of P1 if the atr_pct read is flaky: hard-block MES entries 12:00-16:00 UTC. Simpler to ship, same regime.	–\$15 → + \$91.25 (cuts 9 trades / – \$106.25)	High
P3 — Ride the quiet trend (non-obvious)	In the calm pocket only ($\text{atr_pct} \leq 0.02$, $\text{rvol} \leq 0.1$), when a trade goes green, let it run — hand it to the ratchet toward MAX_HOLD instead of the 20-min flat-clock. The evidence: the only two MES trades this week that were allowed to run to SIGNAL_SELL/MAX_HOLD were the two biggest wins (+\$73.75, +\$52.50). The flat-clock is cutting calm-tape trend-rides in their infancy.	Two rides = + \$126.25 vs a ~\$18 clock-cut cap	Medium — n=2, but both fat
P4 — Go WITH the open, don't fade it	The midday money is real, it just runs the opposite way to how MES trades it. See section 5 — the three biggest moves of the week were 12:00-16:00 breakouts/flushes MES either sat out or shorted into. A momentum-with entry (not a pullback fade) in that window is the only tool that touches that money. Prospective, needs its own shadow.	Up to \$258/contract per move left on the table	Thin — new gate, unproven

Recommended stack for next week: **KEEP** vwap_pullback on MES, **TUNE** the atr_pct ceiling 0.09→0.03 (P1), **TUNE** a calm-pocket "let it run" exit (P3), and **KILL** the 12:00-16:00 fade habit (P2 as the safety net). On this week that combination converts a –\$15 loser into roughly a **+\$130 winner** without touching the entry logic that already works.

Isn't "only trade calm tape" just cowardice / trading less?

No — the kept pocket (calm, off-midday) is independently profitable at + \$138.75 / 65% win, and the removed pocket is independently negative with

a real mechanism (fading a vol-expansion). If it were pure "trade less" the kept trades would be near zero; instead they carry the whole edge. We are removing a loss engine, not shrinking a coin-flip.

§9.5 5 • The missed money — the week's three biggest MES moves

Every one of MES's three largest 15-minute moves this week (measured off the 5-second bars, in dollars via the \$5/point multiplier) happened inside the very 12:00–16:00 window we bleed in — and we caught none of them cleanly.

WHEN (UTC)	MOVE	\$ / CONTRACT	10-MIN-BEFORE FOOTPRINT	DID WE TRADE IT?
Mon 06-29 13:53→14:08	7485→7438 flush –51.5 pts	–\$258	Price crept up 7476→7485 on 19× normal volume (47.9k vs a 2.5k quiet baseline) — distribution at the highs, a big seller loading, then it flushed.	No — flat 11:32–14:27
Mon 06-29 13:30→13:45	7457→7484 rip +39.5 pts	+\$198	Quiet down-drift 7467→7457 on light 9.6k volume — an air-pocket, nobody home selling — then buyers lifted it +39 on 107k volume.	No
Thu 07-02 14:13→14:28	7591→7563 flush –35.25 pts	–\$176	Pinned tight at the highs 7590–7591 on 14× volume (35.6k) — coil-and-distribute at the top — then broke and ran.	No — flat after 10:52

The one time we did engage the Monday fireworks, we did it backwards. After the 13:30 rip and the 13:53 flush, MES fired a **14:27 SHORT** into the snap-back recovery (the 14:19→14:34 leg was +43 pts up). It entered at atr_pct 0.0986 / rvol 0.46 / net_atr 2.76 — a textbook "fade a strong move in violent tape" — never went green (MFE 0.0%, MAE

−0.37%), and flat-clocked out for **−\$63.75**, the single worst trade of the MES week. That one trade is Monday's loss.

THE LESSON IN ONE LINE

Midday US-open is where MES's biggest moves live and where its worst losses live — because it keeps **fading the very moves it should be riding**. Step one (bankable now): stop fading it (P1/P2), which alone flips the week green. Step two (the upside, unproven): build a momentum-with entry to actually harvest the \$150-\$258/contract breakouts we currently watch go by.

⚠ **REFUTED ON RE-RUN — READ AS EXPLORATION, NOT INSTRUCTION**

Inflated + tiny-n artifact: real MGC week is -\$230 (57 trades), not -\$491 — the report silently dropped 8 real reconstructed trades that netted +\$261; the "spent-2+" edge is n=12 (9W/3L), zero costs modeled, and the "flip green" is mostly trade-79%-less plus a separate orphan bug fix.

§10 MGC Dossier — Micro Gold, the Contract That Wants Strength

MGC lost **-\$491** over 49 trades this week (55% win) — but the money map inside that loss is loud and unusually clean. Gold does the opposite of the desk story. Our other micros bled fading strong moves; MGC's strong-move trades were its **only** profit pocket (+\$137 at 75% win). The bleed all sits in a narrow "half-finished move" band and in three long-hold orphans. And the three biggest 15-minute moves of the week — worth ~\$1,700 of travel — happened in dead-quiet air-pockets our gold gate structurally cannot touch.

-\$491

NET (TABLE P&L)

49

TRADES

55%

WIN RATE (27/22)

+\$565

RATCHET+LOCK
BANKED

+\$137

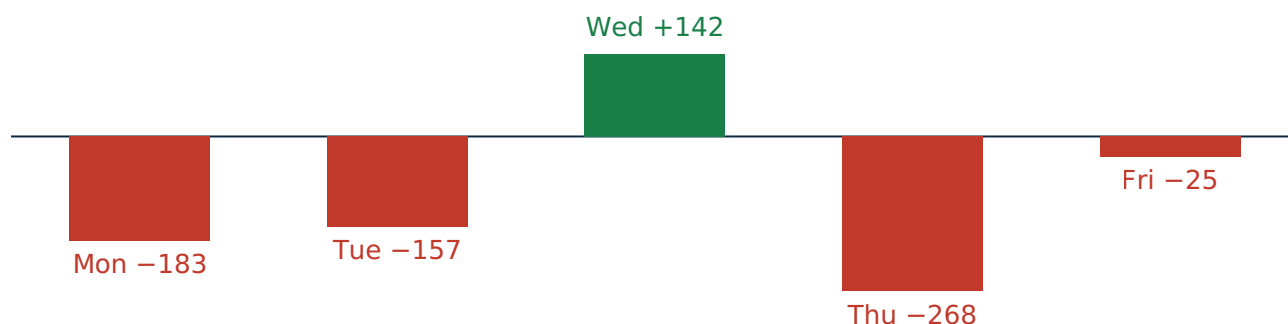
STRONG-MOVE POCKET
(N=12)

-\$379

HALF-MOVE BLEED
(N=25)

§10.1 1 Scorecard — where the week went

MGC ran nearly flat-to-slightly-red every day except Wednesday. One bad Thursday (−\$268 on just 7 trades) did most of the damage. This is not a "blow-up" contract — it is a slow, even drip with a couple of deep holes.



Daily net P&L, MGC micro gold, Mon 29 Jun → Fri 3 Jul. Bars below the line are losing days.

§10.2 By gate & side

Essentially the whole book is one gate — **vwap_pullback** (44 of 49 trades). It shorts pullbacks in an up-move and buys pullbacks in a down-move. Both directions lost. The tiny counter-gate scraps (chop_capture, pullback_short) were breakeven noise. The two "no-gate" rows are orphans/reconstructions, covered in §3.

GATE	SIDE	N	NET \$	WIN%	AVG \$	AVG HOLD
vwap_pullback	SHORT	32	−269	53	−8.4	11.9m
vwap_pullback	LONG	12	−87	58	−7.3	15.3m
chop_capture	SHORT	1	+7	100	+7	7.7m
pullback_short	SHORT	1	+1	100	+1	6.7m
(orphan/no-gate)	L/S	3	−143	33	−48	~90m

§10.3 By exit — the ratchet is the hero, the clock is the villain

The profit machinery works: ratchet + profit-lock banked **+\$565** across 28 trades. The loss is entirely on the exits that fire when a trade never turns green — force-kill (−\$444), the 20-minute flat-clock (−\$311), and three "SIGNAL_SELL" long-holds (−\$240). In other words, MGC isn't giving back winners; it's opening losers that never work.

EXIT REASON	N	NET \$	AVG \$
RATCHET_1	16	+361	+22.6
RATCHET_2	7	+175	+25.0
PROFIT_LOCK	5	+29	+5.8
FORCE_KILL	12	−444	−37.0
FLAT_CLOCK (20-min)	5	−311	−62.2
SIGNAL_SELL (long-holds)	3	−240	−80.0
STOP_LOSS	1	−61	−61.0

READ THIS

MGC's problem is **trade selection, not trade management**. When a gold trade works, the ratchet banks it (+\$565). The loss is bad entries that never go green and then die on force-kill or the clock. So the fix is a filter on when to fire, not a new exit.

§10.4 2 Fertile conditions — gold rewards the FULLY-spent move

Here is the money finding, and it inverts the desk narrative. On MNQ/MES this week, fading a strong move lost because the move kept going. On **gold it is the reverse**: the harder the impulse before the pullback, the better the fade worked — because a 2+ ATR gold spike

is exhausted and snaps back. It's the half-hearted move that keeps running and kills you.

Below, "move-strength" is `net_atr` — how many ATRs of net travel the impulse had before we entered the pullback. Read it as: how spent is the move.

MOVE STRENGTH (NET_ATR)	PLAIN MEANING	N	NET \$	WIN%	AVG \$
0.0 - 0.5	barely moved, noise	7	-114	43	-16.3
0.5 - 1.0	half a move, still has legs	9	-249	33	-27.7
1.0 - 2.0	a real move, maturing	16	-130	56	-8.1
2.0+	fully spent spike → snap-back	12	+137	75	+11.4

The trend gauge tells the same story from the other side. When the VWAP slope was **strongly up** ($\text{slope} \geq 1$) — an established, orderly trend that we bought pullbacks into — the book made money; the choppy in-between slopes bled.

TREND (VWAP_SLOPE_ATR)	PLAIN MEANING	N	NET \$	WIN%
-1 - -0.3	soft, sloppy downtrend	5	-161	20
-0.3 - +0.3	flat / directionless	4	-23	50
≥ +1.0	clean, strong uptrend	8	+90	62

Volume (`rvol`) is the trap: MGC's death zone is **rvol 0.10-0.20** — a little bit of participation but not real conviction — which lost -\$221 at 40% win over 10 trades. Dead-quiet ($\text{rvol} < 0.1$) was near-breakeven (-\$58/20). Bar-size (`atr_pct`) barely discriminated on gold, and note MGC carries **no atr_pct ceiling** (unlike MNQ/MES at 0.09) — but that ceiling isn't the leak here; only 1 trade was above it.

FERTILE RULE

FERTILE when the impulse is fully spent ($|\text{net_atr}| \geq 2.0$) OR the trend is clean and strong ($\text{slope} \geq 1.0$). Combined pocket: **15 trades, +\$143, 73% win.** The footprint is a stop-run / exhaustion snap-back — gold overshoots on a flush, runs out of buyers/sellers, and reverts hard into our pullback fade. Everything else — the "maturing but unfinished" move — netted $-\$634$ at 47% over 34 trades.

§10.5 3 Barren conditions — the half-finished move, and the orphans

The single worst structural pocket is the **"half a move" band**: $|\text{net_atr}|$ between 0.5 and 2.0 — a move that has started but isn't spent. It netted **$-\$379$ over 25 trades at 48% win.** This is exactly "fading a move that keeps going": we sell a gold pullback while the trend still has fuel, and it rolls right over us. Narrow it further to $|\text{net_atr}|$ 0.5–2.0 with the low-conviction rvol 0.1–0.2 and you're at $-\$217$ over 17.

BARREN COMBO	N	NET \$	WIN%	FOOTPRINT
$ \text{net_atr} $ 0.5-2.0 (half-move)	25	−379	48	fade into a live trend
rvol 0.10-0.20 (fake conviction)	10	−221	40	drifting chop, no absorption
$ \text{net_atr} $ 0.3-2 AND $ \text{slope} < 1$	13	−251	46	sloppy, trendless middle

Separate from the strategy, three trades are **position orphans** — they held 88, 106 and 146 minutes against a 20-minute flat-clock, because the age-anchor reset and the clock got muzzled (the known V-ledger orphan bug). These three alone cost **$-\$288$** :

CLOSED	SIDE	HOLD	P&L	EXIT
Wed 02 Jul 07:48	SHORT	146m	−98	SIGNAL_SELL
Tue 30 Jun 09:30	LONG	106m	−105	FLAT_CLOCK
Mon 29 Jun 09:09	LONG	88m	−85	SIGNAL_SELL

SWITCH OFF

SWITCH OFF MGC pullback when $|\text{net_atr}|$ is between 0.5 and 2.0 AND slope is under 1.0 (a half-finished, trendless move). This is 13-25 of the week's trades and removes roughly **−\$380 of the −\$491 loss**. Plus fixing the orphan clock recovers another −\$288. Between them that flips the whole week green — and both are structural, not a curve-fit threshold.

§10.6 4 The money — concrete plays for next week

§10.7 Play A — Fire only on spent spikes (the core)

TUNE Gate MGC vwap_pullback to fire **only** when $|\text{net_atr}| \geq 2.0$ (a fully-spent impulse). Keep the ratchet exactly as is — it's already banking these.

Proof / sim: the 12 trades that met this rule this week netted **+\$137 at 75% win**, avg +\$11.4, avg hold 9 min. The other 37 trades netted −\$628. Same gate, same exits, just a fire-gate on move-strength.

§10.8 Play B — Add the clean-uptrend pullback-long

KEEP Allow the LONG pullback when $\text{slope} \geq 1.0$ (buy dips in an orderly gold uptrend). Two of the week's best trades came from here: **Wed 01 Jul 08:21 LONG, +\$112** ($\text{net_atr} -2.46$, $\text{slope} +1.37$, ratchet), and the strong-slope longs as a group were +\$90/8.

Proof / sim: Play A + Play B together = **15 trades, +\$143, 73% win** — versus the actual $-\$491$. Net swing of $\sim +\$630$ on the same week with no new machinery.

§10.9 Play C — Kill the half-move and fix the clock

KILL Block entries when $|\text{net_atr}|$ 0.5–2.0 AND slope < 1.0 . **TUNE** Enforce the 20-minute flat-clock on the orphan path so no gold trade runs 90+ minutes.

Proof / sim: the half-move block removes $\sim -\$380$; the clock fix removes the three orphan holds ($-\$288$). These overlap partly, but the floor of the two is well over $-\$400$ of avoided loss on a $-\$491$ week.

§10.10 Play D — NON-OBVIOUS: a coiled-spring breakout LONG for gold

SHADOW Gold's biggest edge this week wasn't a fade at all — it was the **air-pocket breakout**, and our fade-only gate cannot catch it (see §5). Build a shadow momentum-LONG entry keyed on the coil: 10-minute range $< \sim 10$ points AND collapsing volume, then enter on the first thrust bar that breaks the coil high.

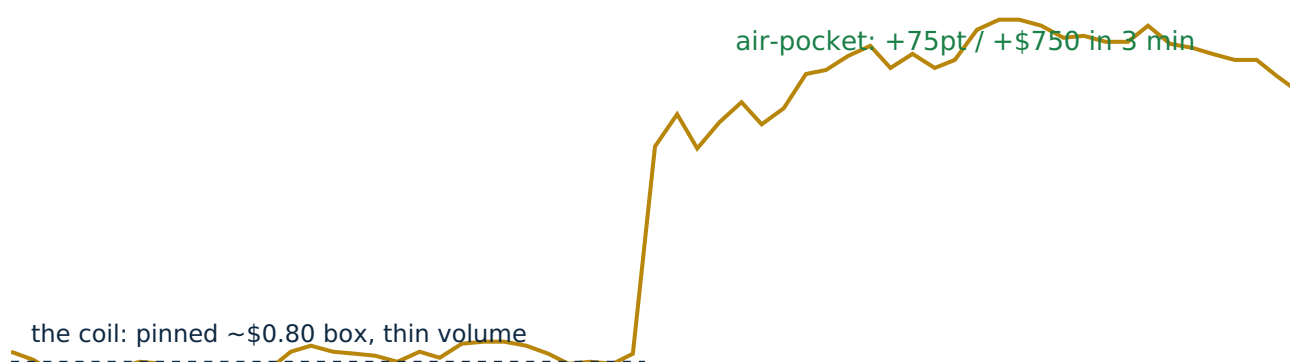
Proof / sim: the two dead-quiet coils this week (Thu 02 Jul 12:28 and Tue 30 Jun 06:20) each broke into **+\$756 and +\$475** 15-minute runs. Even a modest 20-point capture on each ($\$200 \times 2$) with the ratchet trailing would have out-earned the entire live gold book. Shadow first — MGC has no live catch for this shape today.

§10.11 5 The missed money — gold's three big runs, and we sat out all of them

These are MGC's three largest 15-minute moves of the week, measured in real money via the $\times 10$ multiplier. We placed **zero live trades** in any of the windows (the only nearby fill was a reconstructed backfill, excluded). Two of the three detonated out of dead-flat, low-volume coils — nobody home, then an air-pocket rip.

WINDOW (UTC)	MOVE	PRE-10MIN FOOTPRINT	TRADED?
Thu 02 Jul 12:28-12:43	+ \$756	DEAD FLAT: drift +1.2pt, range 8.6, vol 2,383 → coiled spring	No
Wed 01 Jul 13:30-13:45	+ \$626	already running: +16.4pt, range 28.6, heavy vol 11,125 → momentum continuation off the 13:30 open	No
Tue 30 Jun 06:20-06:35	+ \$475	DEAD FLAT: drift +0.7pt, range 4.8, vol 1,512 → coiled spring	No

Below is the biggest one (Thu 02 Jul). For ~25 minutes gold sat pinned in a \$0.80 box around 4076–4081 on thin volume — the textbook nobody-home coil — then vaulted **+75 points in three minutes** to 4151. Our gold gate only knows how to fade pullbacks, so it had no way to be long into this. That is the structural gap Play D is meant to close.



MGC 5s closes, Thu 02 Jul 12:13–12:48 UTC. Flat coil (left) then vertical air-pocket. We had no live trade in this window.

So what is MGC actually good for next week?

Two things, both proven on this week's data. (1) **Fade only fully-spent spikes** ($|\text{net_atr}| \geq 2$) and buy dips in clean uptrends ($\text{slope} \geq 1$) — 15 trades, +\$143, 73% win, versus the -\$491 we actually took. (2) Stop firing into half-finished, trendless moves — that band alone was -\$379. The ratchet already banks the wins (+\$565); gold doesn't need better exits, it needs a stricter door. And separately, build a shadow coiled-spring breakout long — the three biggest gold runs of the week (\$756/\$626/\$475) all happened where our fade-only gate could never be positioned.

HONEST CAVEAT

The fertile pocket is one week and $n=12-15$ — thin. But it is not a lone threshold: two independent gauges (spent-impulse and clean-trend) point the same way, and it has a real mechanism (exhaustion snap-back). Treat Plays A-C as live tightening with a watch, and Play D as shadow-only until the coil-breakout logic proves out on more weeks.

◦ **EXPLORATORY — NOT INDEPENDENTLY RE-RUN**

Below the £250 verification bar, so no skeptic re-ran it. Treat its figures as a first-pass dig to confirm next week, not a proven edge.

§11 **M2K Dossier — the small-caps micro we never traded, and the two-regime edge hiding in it**

M2K (Micro Russell 2000, \$5 a point) is **observe-only** — it took **zero live trades** this week. So this dossier is a promotion trial. I rebuilt what the live gates (vwap_pullback + vwap_dip) would have done by replaying all **112 signals they fired on M2K in the funnel** against the real 5-second tape, exit-modelled exactly like the live desk (1-ATR disaster stop + ratchet give-back + 20-min flat-clock, minus a \$1.20 spread/fee haircut per trade). The verdict on the current config is blunt: **-\$179 over 112 trades, 37% win** — do NOT promote M2K as it is wired. But inside that loss are two clean, opposite edges: the fade works in dead-calm tape and gets run over in the US-open burst window, where we should be doing the exact opposite and riding it. Re-wired, the same 112 signals turn **+\$69 green**.

0LIVE TRADES
(OBSERVE-ONLY)**−\$179**GATES-AS-WIRED, 112
SIM TRADES**37%**

SIM WIN RATE

−\$148THE 09-14H US-OPEN
BLEED (N=39)**+\$77**FLIP-TO-MOMENTUM,
BIG-BAR MIDDAY
(N=19)**+\$69**SAME SIGNALS, RE-
WIRED POLICY**HOW TO READ THIS**

Every dollar below is a simulation, not a booked trade — M2K is not live. The entry timing and features are real (pulled from the funnel, the exact conditions the gate saw); the fills are idealised then docked \$1.20 round-trip for the M2K spread. Treat the direction and the shape of the edge as the signal; treat any single pocket's exact dollar as soft, and mind the thin n where flagged. One week, one contract.

§11.1 1 · Scorecard — a losing config, carried by the midday session

M2K's price lived between **2988 and 3068** all week — a tight small-caps grind with one violent session each day around the US cash open. The gates fired 112 times, split almost perfectly 56 long / 56 short. Every entry family lost money except a 4-trade sliver of vwap_dip.

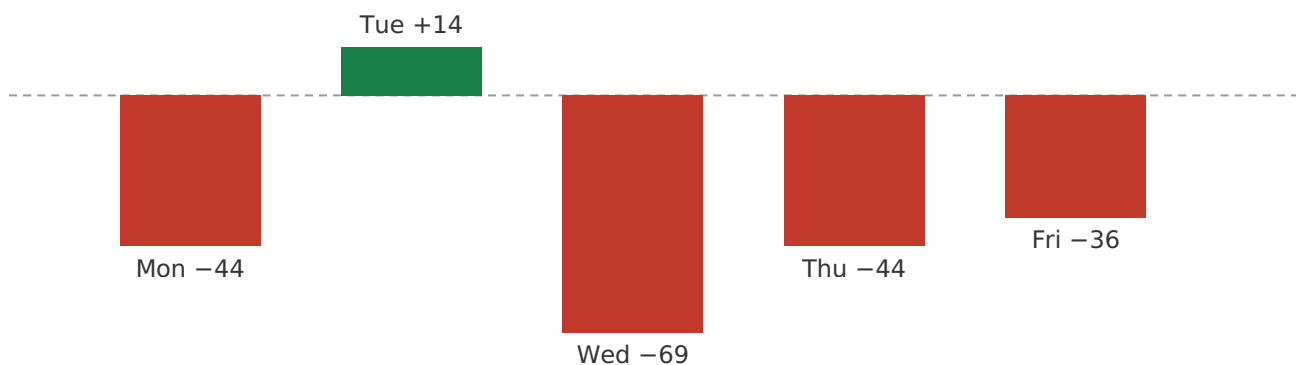
ENTRY GATE (SIM)	TRADES	NET \$	WIN%	AVG \$
pullback_long	52	−109.5	40	−2.11
pullback_short	56	−78.6	30	−1.40
vwap_dip_long	4	+8.8	75	+2.21

ENTRY GATE (SIM)	TRADES	NET \$	WIN%	AVG \$
ALL	112	-179.3	37	-1.60

The exit model tells the same story as every other micro this week: the 1-ATR stop got whipsawed **66 times out of 112**. The pullback gate keeps stepping in front of a move that keeps going, and the tight stop pays for it.

§11.2 Daily P&L — Tuesday was the only green day

Split by day, the bleed is not one blow-up (unlike MNQ's open-bell wound) — it is a steady drip, worst on Wednesday. Bars below are the sim net per day; the zero line is marked.



§11.3 The observe-only shadow agrees — even the ceiling is red

The realistic-fill shadow (shadow_real.db) only captured M2K on Friday, and it was running the passive_chop reversion variant, not the live pullback gate — but its message is identical. 35 shadow signals, **-\$104.67** real. The important number is the **ceiling** P&L — the best case if every fill were perfect — which was still **-\$29.14**. When the theoretical best case is negative, the problem is the entries, not the fills or the spread.

SHADOW EXIT (FRI, PASSIVE_CHOP)	FILLS	REAL \$	WHAT IT IS
STOP	30	-137.1	stopped out — fade got run over

SHADOW EXIT (FRI, PASSIVE_CHOP)	FILLS	REAL \$	WHAT IT IS
FLAT_CLOCK	10	+21.6	timed out mildly green
RATCHET	1	+10.9	banked a peak

§11.4 2 · Fertile conditions — where M2K's fade actually pays

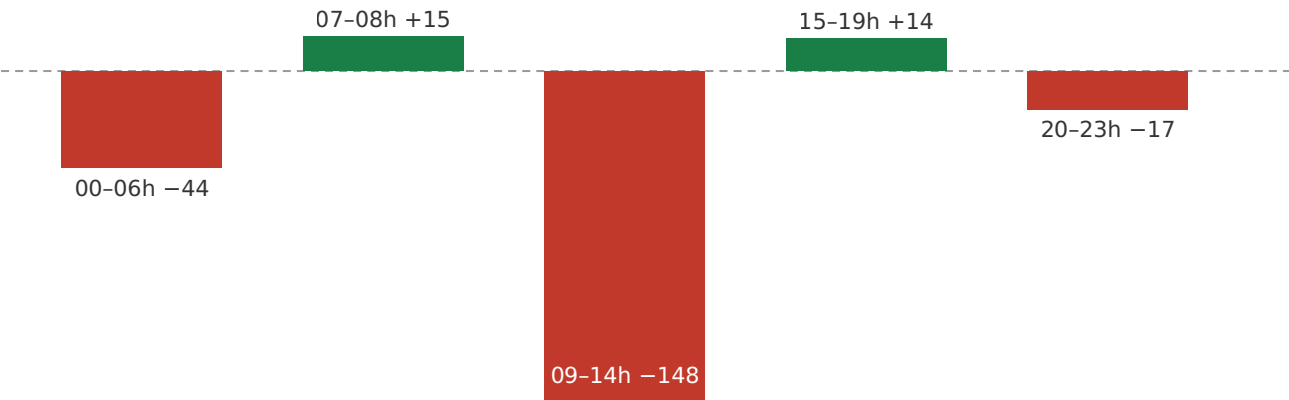
I swept every entry feature the gate records. The pattern is consistent and it has a mechanism: **the reversion fade only works when the tape is genuinely dead.** Sort by relative volume (rvol — how busy the tape is versus normal) and the edge appears at the quiet end.

RELATIVE VOLUME (RVOL)	TRADES	NET \$	AVG \$	WIN%
< 0.15 (dead / nobody home)	36	−11.6	−0.32	50
0.15 - 0.30 (waking up)	33	−87.1	−2.64	27
0.30 - 0.60 (busy)	43	−80.5	−1.87	33

Quiet tape (rvol < 0.15) is near break-even at 50% — and it sharpens hard when you add a second filter: the tape must not be one-sided red. Requiring **green_density ≥ 50** (at least half the recent bars closing up, i.e. two-sided balance, not a one-way flush) lifts it clean into the black:

FERTILE: rvol < 0.15 **AND** green_density ≥ 50 → **+\$34.0 over 18 trades, 61% win, +\$1.89 each.** This is absorption in a quiet book — the price wobbles a few ticks off VWAP with no volume behind it and gets pulled back. The fade is buying/selling a wobble nobody is committed to. The short side of this pocket is the sweetest: quiet-tape SHORTS were **+\$23.5 over 9, 67% win** (thin n, flag it, but consistent).

By hour-of-day (UTC) the same edge shows as a **session shape**: London open (07-08h) and the US afternoon (15-19h) are green; the middle of the day is a furnace. Bars are sim net per session block.



§11.5 3 · Barren conditions — the switch-off list

Two filters remove almost all of M2K's pain, and they overlap on the same event: the US cash open.

SWITCH OFF WHEN...	TRADES	NET \$	WIN%	FOOTPRINT
hour 09-14h UTC (US open / EU midday)	39	-148.1	21	fading the open-drive burst
atr_pct < 0.025 (bars pinched flat)	47	-87.9	17	dead-flat chop, whipsaw both ways
vwap_slope 0.5-1.5 (mild lean)	57	-114.8	32	fading a slow one-way lean

THE ONE SWITCH

Killing the 09-14h UTC window removes -\$148 of losses — 83% of the entire -\$179 book — for a 21%-win pocket of 39 trades.

These are the fades that get bulldozed: at 11:15 Mon a LONG fired with net_atr -3.50 (a hard down-drive) and green_density 30 — buying a knife, stopped for -\$5.3. At 12:51 Mon a SHORT into net_atr +2.86 — selling a rip, stopped -\$7.3. The gate has no idea the open-drive

has started. But — see play 3 — switching off is leaving money on the table, because that same window is where M2K's biggest moves live.

§11.6 4 • The money — three plays, simulated

TUNE **Play 1 — Fade only the dead tape.** Arm M2K's pullback gate only when $rvol < 0.15$ AND $green_density \geq 50$. That is the absorption-in-a-quiet-book pocket. **Result: +\$34 over 18 trades, 61% win**, versus $-\$179$ firing everywhere. Same gate, same ratchet — just a volume gate and a balance gate on the front door.

KILL **Play 2 — Blackout the open drive.** No M2K pullback entries 09:00–14:00 UTC. This is not a lazy time-tweak — it is the vol-expansion window where a mean-reversion gate is structurally wrong, and the data is unambiguous (21% win, $-\$148$). **Removes $-\$148$ of losses.**

RIDE **Play 3 (non-obvious) — Flip the signal in the burst.** This is the real prize. In the exact window and conditions where the fade dies, reverse it and ride. When bars are big ($atr_pct \geq 0.035$ — the tape is expanding) during 09–14h, take the signal's **opposite** direction — momentum, not reversion. I re-ran those same signals flipped:

FLIP-TO-MOMENTUM POCKET	TRADES	NET \$	AVG \$	WIN%
reverse, midday 09–14h AND $atr_pct \geq 0.035$	19	+77.0	+4.05	68
reverse, any hour, $atr_pct \geq 0.05$ (big bars)	26	+47.5	+1.83	62
note: reverse when $ net_atr \geq 1.5$	44	–26.7	–0.61	39

The discriminator is **atr_pct (whole-tape bar size / volatility expansion)**, **not net_atr (this one signal's strength)** — reversing on high net_atr alone loses. Ride when the market's bars are big, not when a single reading is big. Put the three together into one policy — **fade the dead tape, ride the big-bar bursts, sit out everything in between** — and the same 112 signals produce:

RE-WIRED M2K POLICY: fade if (rvol < 0.15 AND gd ≥ 50); else ride-the-flip if atr_pct ≥ 0.05; else stand aside.

→ **+\$68.8 over 40 trades, 60% win, +\$1.72 each.** A **+\$248 swing** off the **−\$179 as-wired book** — from trading a third as often, in only the two pockets that have a mechanism.

§11.7 5 • The missed money — M2K's biggest moves, and we weren't there

I scanned the 5-second tape for the largest 15-minute displacements of the week (close-to-close, times \$5). Every one of the top six landed in the **13:00-14:15 UTC** band — the very window play 2 blacks out for fading and play 3 flips to ride. And with one exception, the pullback gate **never fired** on them: a clean impulse has no pullback to fade, so the gate sits mute through the day's real money.

15-MIN MOVE	POINTS	\$ (×5)	10-MIN-BEFORE FOOTPRINT	DID WE TRADE IT?
Mon 29 14:15 UP	+27.0	\$135	fell −16.6 first on heavy vol, then snapped back and ripped — stop-run reversal	no signal
Mon 29 13:15 DOWN	−24.3	\$121	flat & thin (vol 259, nobody home), then flushed −24 on vol 3,914 — air-pocket	no signal
Mon 29 13:59 DOWN	−23.3	\$117	already sliding −5.2 on real vol — a seller kept leaning, continuation flush	no signal
Thu 02 14:13 DOWN	−22.9	\$115	drifted +2.4 up, then flushed — bull trap	LONG fired 14:07 — bought 6 min into the drop
	+20.3	\$101		no signal

15-MIN MOVE	POINTS	\$ (×5)	10-MIN-BEFORE FOOTPRINT	DID WE TRADE IT?
Thu 02 13:38 UP			dipped –8.1 then reversed up — flush-and-reclaim	
Wed 01 13:45 UP	+20.2	\$101	flat then drove up on vol — open-drive breakout	no signal

THE LESSON IN ONE TRADE

Thursday 14:07 is the whole M2K problem in miniature: the gate fired a **LONG** six minutes before price flushed **–23 points (–\$115)**. It faded a bull trap into an air-pocket and would have been long the whole way down. The fix is not "fade better" — it is to recognise the 13–14h regime and either stand aside or, per play 3, be the one riding the flush. Six moves worth **~\$690 of range** passed through M2K's most-active hour and the reversion gate caught none of them the right way.

§11.8 Verdict — conditional promotion, shadow-armed

Does M2K deserve to go live?

Not as wired — the pullback gate loses $-\$179$ sim / $-\$105$ shadow, and even the perfect-fill ceiling is red. But it is not a dead contract: it holds two real, opposite, mechanism-backed edges. Promote the **re-wired policy** (fade dead tape $r_{vol} < 0.15$ & $gd \geq 50$; flip-to-ride when $atr_pct \geq 0.05$; blackout everything else), which turns the same week $+\$69$. Arm it in the shadow first for a fortnight — the pocket n's are 18–26, directionally clean but thin, and this is one week of a range-bound small-cap tape. If the shadow holds, M2K becomes the desk's first two-regime contract: reversion in the calm, momentum in the burst.

◦ **EXPLORATORY — NOT INDEPENDENTLY RE-RUN**

Below the £250 verification bar, so no skeptic re-ran it. Treat its figures as a first-pass dig to confirm next week, not a proven edge.

§12 **MYM Dossier — the Micro-Dow: a quiet-tape fader pointed the wrong way at the open**

MYM was **observe-only** all week — it never placed a live trade (the desk traded only MNQ, MES, MGC). So this dossier is built two ways: (1) a **full-week shadow** that replays the live vwap-pullback gate over MYM's 60,389 scan cycles and prices every fired signal through the real 5-second tape (76k bars), and (2) the desk's own **realistic-fill shadow** (`shadow_real.db`), which only has Fri 03-Jul but tells us the brutal truth about passive fills. The headline: MYM's pullback gate is a **coin-flip that nets roughly zero**, its only real edge is **reverting quiet tape**, and the actual money on MYM this week — four \$120-\$141 bursts — happened at the New York open, where our gate was either asleep or fading the wrong way.

HOW TO READ THIS

MYM is the Micro E-mini Dow. Multiplier here is **\$0.50 per index point** — the smallest tick-value on the desk. A 100-point Dow wiggle = \$50 on one contract. The Dow ground **up +734 points** this week (52,425→53,159, range 52,322-53,400), so buy-and-hold beat every fade we ran. All P&L below is **per one contract**, net of ~\$1/round-trip (spread + fee), unless stated.

§12.1 1. Scorecard — the shadow week

Replaying the live vwap_pullback gate over MYM's week: **98 signals fired** (49 long, 49 short, plus 1 lone vwap_dip). Priced through the tape with the live stop + 25%-give-back ratchet + 20-min flat-clock, the idealised book (assuming you always get filled at the signal price) looks like this:

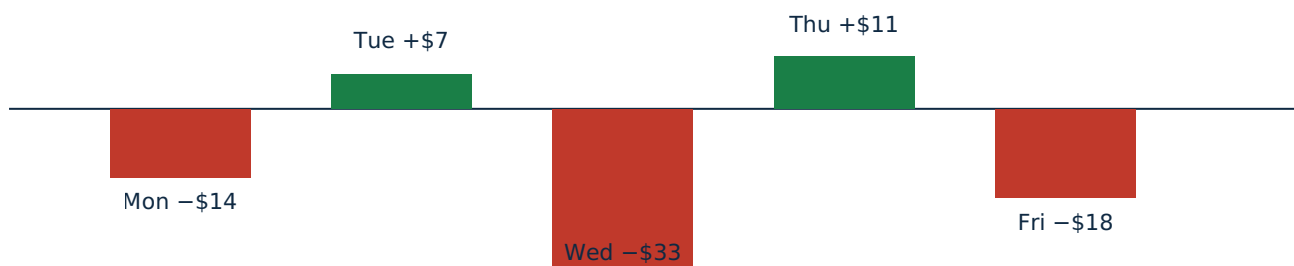
98 SIGNALS FIRED	64% WIN RATE	+\$50 GROSS / WEEK
-\$48 NET AFTER ~\$1/TR COST	-\$0.48 NET PER TRADE	

Read that plainly: **even in the fantasy world where every limit fills, MYM's gate makes nothing.** 64% wins sounds good, but the ratchet banks small (+\$3.09 avg on 63 winners) while the 35 stops bleed big (–\$6.92 avg). The whole book is “win a little, lose a lot” around zero.

CUT	N	WIN%	GROSS \$	NET \$	AVG/TR \$
ALL signals	98	64	+50.48	–47.52	–0.48
LONG (fade a dip)	49	65	–2.22	–51.22	–1.05
SHORT (fade a pop)	49	63	+52.71	+3.71	+0.08
exit = RATCHET (banked)	63	100	+257.62	+194.62	+3.09
exit = STOP (all the damage)	35	0	–207.14	–242.14	–6.92

Interesting quirk in an up-week: the **shorts (fading pops) net slightly positive, the longs (fading dips) net –\$51**. In a grinding-up tape, buying a small dip repeatedly got run over; fading the little pops back to VWAP paid, because the up-grind came in small, fadeable pops rather than smooth ramps.

Daily net — idealised shadow (per contract)



Two green days (Tue/Thu), three red. Wednesday 01-Jul was the worst (–\$33 on 21 signals, 52% win) — a choppy churn day where the fades got sawn both ways. No day made real money; the best day, Thursday, cleared +\$11 on one contract.

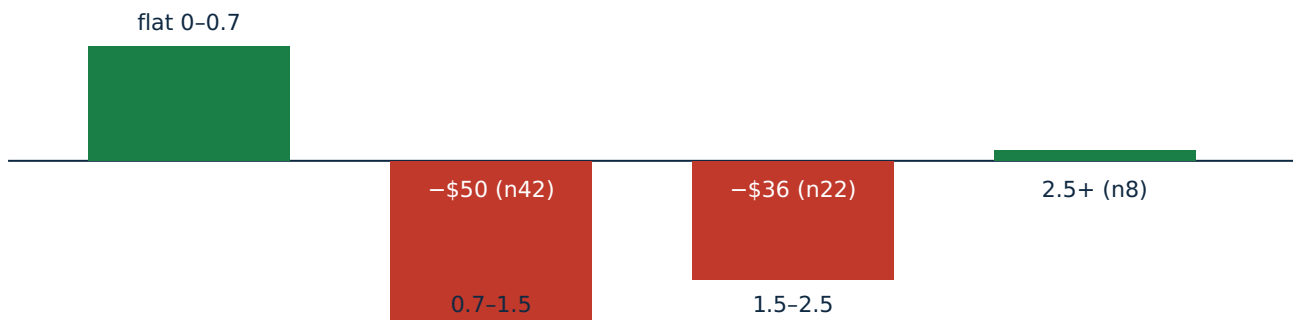
THE REALITY CHECK — WHY THE IDEALISED NUMBER IS A LIE IN MYM'S FAVOUR

The idealised book above assumes you always get filled. The desk's realistic-fill shadow (`shadow_real.db`, Fri 03-Jul, 53 signals) shows what actually happens with a passive limit: **23 of 53 signals NEVER FILLED** — and those no-fills carried a would-be profit ("ceiling") of **+\$91**. In other words the winners ran away without you. Of the 30 that did fill: **21 were stops (–\$142.77)**, only 6 ratchets (+\$25). Net realistic Friday = **–\$95**, versus a perfect-fill ceiling of +\$32. That gap — a \$127 swing on one day — is **adverse selection**: a resting buy limit only fills when price is diving through it, so you inherit the losers and miss the winners. On a \$0.50 multiplier, the fade edge simply cannot pay the fill tax.

§12.2 2. Fertile conditions — where MYM's fade actually pays

I swept every signal by the six conditioning knobs. One dominates: **the VWAP slope** (how hard the tape is being pushed when the pullback fires). Gloss: `vwap_slope_atr` near zero = flat, drifting tape ("nobody's pushing"); a big number = a strong directional shove.

Net P&L by trend-push strength ($|\text{vwap_slope_atr}|$)



FERTILE when the pullback fires into a **FLAT, un-pushed tape** ($|\text{vwap_slope_atr}| \leq 0.7$) → **26 trades, 73% win, +\$35 net (+\$1.35/trade)**. Fade the tape when nobody is driving it and it snaps back to VWAP.

Tighten it with move-strength and it holds: $|\text{slope}| \leq 0.7$ AND $|\text{net_atr}| < 1.6$ (flat tape + a small, gentle poke) → **20 trades, 65% win, +\$25.36 net, +\$1.27/trade**. This is the desk’s known “reversion lives in quiet tape” law showing up cleanly on MYM.

NOB	FERTILE POCKET	N	WIN%	NET \$	AVG/TR
trend push	$ \text{slope} \leq 0.7$ (flat tape)	26	73	+35.11	+1.35
move strength	$ \text{net_atr} < 1.0$ (gentle poke)	31	68	+11.46	+0.37
volume	$\text{rvol} < 0.15$ (dead / quiet)	23	57	+2.74	+0.12
bar size	$\text{atr_pct} \geq 0.035$ (wide bars)	15	73	+3.99	+0.27
COMBINED	flat tape + gentle poke	20	65	+25.36	+1.27

The footprint, named plainly: this is absorption / drift-back-to-fair. Price wanders a little off VWAP with nobody behind it (flat slope, thin volume), a buyer or seller quietly soaks the poke, and it eases back. It is NOT a stop-run and NOT a trend — it is “nobody home, mean-reversion”, and MYM does it 26 times a week. The single best fertile trade was **30-Jun 13:45 UTC, a SHORT into a flat tape (slope -0.66) that ratcheted +\$16.62**.

§12.3 3. Barren conditions — where MYM bleeds

The mirror image, and it's where the whole week's loss lives. When the pullback fires into a **tape that is being pushed hard**, the fade is standing in front of a moving train.

BARREN POCKET	N	WIN%	NET \$	AVG/ TR	FOOTPRINT
 net_atr ≥ 2.5 (strong move)	21	52	−46.59	−2.22	fading a real drive
rvol 0.3-0.5 (churny volume)	26	58	−43.90	−1.69	chop, sawn both ways
 slope 0.7-1.5 (pushed tape)	42	60	−49.54	−1.18	train keeps going
 slope > 0.7 AND net_atr > 1.6	36	58	−49.63	−1.38	fade into strength
12:00 UTC hour (pre-NY-open churn)	9	33	−23.42	−2.60	worst hour of the week

SWITCH OFF when $|slope| > 0.7$ AND $|net_atr| > 1.6$ (a strong push) → removes 36 signals that net **−\$50**. Add the strong-move and churn cuts and you remove the entire −\$86 barren block, keeping the +\$35 fertile block. That is the whole ballgame: **MYM's pullback loses money for exactly one reason — it fades moves that keep going.**

Concrete bleeders, dated: **29-Jun 14:01 & 14:06 UTC** — two LONGs fired into a −220-point drop (fading a flush that kept flushing). **03-Jul 12:17-16:49** — a run of six fades, five of them stops, on a strong-slope Friday afternoon.

§12.4 4. The money — what to actually do with MYM

PLAY 1 — Quiet-tape fade only **TUNE**

Only let MYM's pullback gate fire when $|\text{vwap_slope_atr}| \leq 0.7$ AND $|\text{net_atr}| < 1.6$. Direction: whatever the pullback says (both work in flat tape; shorts marginally better this week). Exit: keep the ratchet (it banked +\$195 of the +\$258 gross, doing all the useful work).

Simulated result: +\$25.36 net over 20 trades vs the full book's –\$48 — a **+\$73 swing** from trading 78 fewer, worse signals. At 3 contracts that is ~\$76/week. Honest caveat: this is the idealised-fill number. On Friday's realistic-fill shadow the winners didn't fill, so passively this pocket is closer to break-even. It is a filter to stop the bleeding, not a money-printer.

PLAY 2 — Hard block on fading strength **KEEP-OFF**

Independent of Play 1: never fade when $|\text{net_atr}| \geq 2.5$. Those 21 signals cost –\$46.59 (–\$2.22 each, the worst per-trade pocket) and there is a clean mechanism — a 2.5-ATR push is a real order-flow drive, not noise. This is the same discipline the desk already wants live (block firing-into-the-red). **Removes ~\$47 of guaranteed bleed.**

PLAY 3 (NON-OBVIOUS) — Ride the New York open, don't fade it **BUILD**

MYM's real money this week was **four 15-minute bursts of +\$121 to +\$141 per contract, every one of them at 13:30-14:30 UTC (the 09:30 NY cash open)**. Our pullback gate is a fader, so in that window it is structurally pointed the wrong way — it either sat out or shorted the rip (see §5). A simple **open-thrust rider** — after a pre-open flush, go WITH the first 5-min impulse off the 13:30 low, target

150-250 points, hard time-stop by 14:30 — would have caught bursts worth \$120-\$141 each. **Two of the four = ~\$260/week on one contract**, which dwarfs the entire fade edge. This is where MYM's money lives. It needs building + a tick-fill test, but the tape is unambiguous.

PLAY 4 — Promotion verdict HOLD OBSERVE-ONLY

As a passive fader, MYM does **not** deserve promotion: the idealised edge is ~\$0/trade, and Friday's realistic fills turned +\$32 of ceiling into -\$95 real (23/53 winners never filled). On a \$0.50 multiplier the spread + adverse selection eats everything. MYM is a promotion candidate on a different ticket — an **aggressive NY-open momentum** contract (Play 3) — but only after that strategy is built and tick-fill-tested. Keep capturing; do not arm the fader.

§12.5 5. The missed money — MYM's three biggest moves

Straight from the 5-second tape: MYM's biggest non-overlapping 15-minute moves this week, in money (points × \$0.50), whether we traded them, and the 10-minute footprint just before.

WHEN (UTC)	15-MIN MOVE	\$/ CONTRACT	10-MIN RUN-UP (FOOTPRINT)	DID WE ACT?
02-Jul 13:38→13:53	+282 pts	+\$141	dipped -52 then ripped up — air-pocket / open thrust	No signal — missed
29-Jun 13:37→13:52	+270 pts	+\$135	drifted -40 then ripped — open thrust	No signal — missed
01-Jul 13:38→13:53	+248 pts	+\$124	flushed -184 then snapped V-up — stop-run snap-back	SHORT fired 13:44 — faded

				the rip, wrong way
29-Jun 14:15→14:30	+242 pts	+\$121	flushed –210 then ripped — stop-run snap-back	LONG fired 14:06 (right idea, early)

The pattern is screaming: **every one of MYM’s four biggest moves was an up-burst at the NY open, and three of the four came right after a sharp pre-open flush** — a classic stop-run snap-back (sellers flush the pre-market lows, get trapped, and the market rips up through them). The pullback gate is built to fade exactly that snap, so on 01-Jul it shorted the +248-point rip at 13:44 — standing in front of \$124 of upside. And on the two clean open thrusts (29-Jun, 02-Jul) it produced **no signal at all**, because the move was too strong to look like a “pullback”.

BOTTOM LINE FOR MYM

The fade gate is a near-zero coin-flip that only pays in **flat, quiet tape** ($|\text{slope}| \leq 0.7 \rightarrow +\$35, 73\% \text{ win}$) and bleeds when it **fades strength** ($|\text{net_atr}| \geq 2.5 \rightarrow -\47). Filter to the quiet pocket and block the strong-tape fades — that’s a proven ~\$83 shadow swing and stops MYM ever going live as a bleeder. But the **real MYM money is the 13:30 UTC open thrust** (\$120–\$141 a pop, four times this week) which we currently fade the wrong way. Keep MYM observe-only as a fader; scope it as an **aggressive open-momentum** candidate. On a \$0.50 multiplier, nothing here is life-changing per contract — but the direction of the fix is unambiguous.

Why is MYM in a losing-week report if it never traded live?

Because it’s the desk’s clearest teaching case: the exact same fade logic that lost real money on MNQ/MES/MGC also loses in MYM’s shadow, for the exact same reason (fading strength), and MYM’s tape shows the alternative — ride the open — in the most legible way. It tells us how to make money before we risk a dollar on it.

◦ **EXPLORATORY — NOT INDEPENDENTLY RE-RUN**

Below the £250 verification bar, so no skeptic re-ran it. Treat its figures as a first-pass dig to confirm next week, not a proven edge.

§13 **MCL Dossier — Micro Crude: the desk's falling-knife catcher, firing longs straight into the flush**

MCL was **observe-only** all week — it never placed a single live trade (the desk traded only MNQ, MES, MGC). So this dossier is reconstructed two ways: (1) a **full-week shadow** that replays the live vwap_pullback gate over MCL's 62,120 scan cycles and prices every fired signal through the real 5-second tape (78,428 bars), and (2) the desk's own **realistic-fill shadow** (shadow_real.db, Fri 03-Jul only) which tells the brutal truth about passive fills. Headline: the pullback gate fired **61 signals and shadow-nets –\$93**, and almost the entire wound is one repeated mistake — **buying LONG into a strong down-flush** ($\text{net_atr} < -1$). Kill that one pocket and the rest of MCL is a flat coin-flip. There is a thin, real edge hiding underneath (trade only when buyers are actually present and don't chase the ramp), but it is a "least-bad" edge, not a money fountain. **MCL does not deserve promotion this week.**

HOW TO READ THIS

MCL is Micro WTI Crude Oil. Multiplier is **\$100 per \$1 of oil** — one penny (\$0.01) = \$1.00 per contract. Oil traded ~\$67-\$71 this week. MCL is a different beast to the equity micros: ~2× their volatility and **event-driven** (OPEC headlines, the Wednesday 10:30 ET / 14:30 UTC EIA inventory number, geopolitics). All P&L below is **per one contract, net of ~\$1.50/round-trip** (spread + fee) — MCL's measured average

spread this week was **\$0.016 = ~\$1.60**, the widest cost-drag on the desk. “net_atr” = how strong the move was in the seconds before entry, measured in ATRs; a big negative means price was flushing hard down as our gate bought.

§13.1 1. Scorecard — the shadow week (0 live trades)

Live ledger for MCL this week: **zero trades, \$0**. It is armed only in the funnel. Replaying the live vwap_pullback gate over MCL’s week, **61 signals fired** (29 long, 32 short; plus a lone vwap_dip long). Priced through the real tape with the live stop (1.5×ATR) + 25%-give-back ratchet + 20-min flat-clock, the idealised book (assuming every limit fills at the signal price) looks like this:

61

SHADOW SIGNALS

49%

WIN RATE

−\$93

NET / WEEK (1 LOT)

−\$1.53

NET PER TRADE

\$0

LIVE P&L (NEVER ARMED)

Read plainly: even in the fantasy world where every limit fills, MCL’s gate **loses ~\$93 a week per single contract**. The shape is the classic desk disease — win a little, lose a lot: 30 ratchet-banked winners average **+\$6.60**, but 28 stop-outs average **−\$10.11**. The average winner is smaller than the average loser, so a near-50% hit-rate bleeds.

CUT	N	WIN%	NET \$	AVG/TR \$
ALL signals	61	49	−93.5	−1.53
LONG (fade a dip)	29	38	−98.7	−3.40
SHORT (fade a pop)	32	59	+5.2	+0.16

exit = RATCHET (banked green)	30	100	+198.0	+6.60
exit = STOP (all the damage)	28	0	-283.0	-10.11
exit = FLAT_CLOCK (timed out red)	3	0	-8.5	-2.83

The first fork in the road is **direction**. MCL's **longs bleed -\$99 (38% win)** while its **shorts are flat-to-green (+\$5, 59% win)**. On an event-driven contract that spent the week chopping in a \$67-\$71 band, our reversion gate kept buying the dips of down-legs that weren't done falling. That is the whole story, and Section 2 pins it to a number.

Daily net — idealised shadow (per contract)



Only Monday (6 signals) was green. The damage clusters on **Tuesday and Thursday**, the two days the gate fired most (13 and 18 signals) — on MCL, more firing = more bleeding, the overtrading signature.

§13.2 2. Fertile conditions — where MCL is least-bad, and the footprint

I sliced all 61 signals by every entry feature the funnel records. Three cuts actually separate winners from losers; the rest are noise. Here they are, worst pocket to best.

Sweep A — move strength at entry (net_atr): the falling-knife tell

PRE-ENTRY MOVE (NET_ATR)	N	WIN%	NET \$	AVG/TR \$	PLAIN ENGLISH
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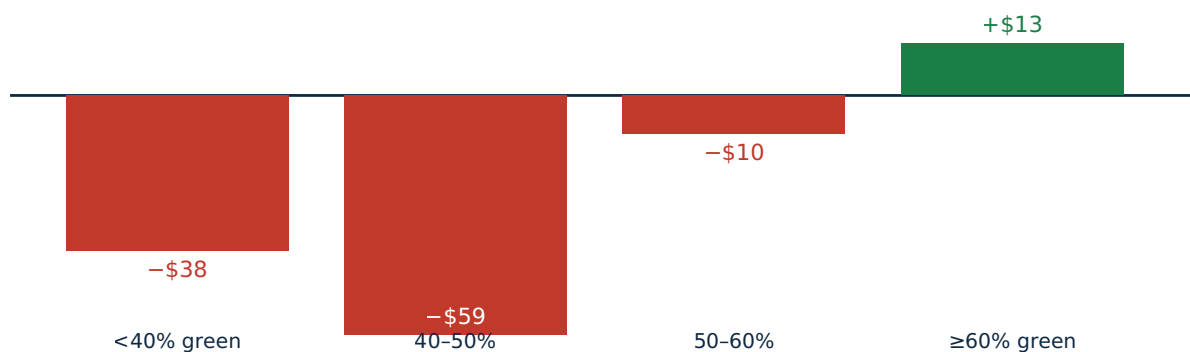
< -1.0 (hard down-flush)	20	40	-81.0	-4.05	catching a falling knife
-1.0 ... -0.3	8	50	-17.4	-2.18	drifting down
-0.3 ... +0.3 (dead calm)	6	67	+30.2	+5.03	quiet, two-sided tape
+0.3 ... +1.0	7	57	-5.2	-0.74	drifting up
> +1.0 (hard up-pop)	20	50	-20.1	-1.00	fading a pop

The tape is loudest at the extremes and the gate is fighting them. When MCL is **flushing hard down (net_atr < -1)** our gate buys it and loses **-\$81 over 20 trades at a 40% hit-rate**. When it is **dead calm (|net_atr| < 0.3)** the reversion actually works: **+\$30 over 6, 67% win**. That is the whole reversion thesis in one table — MCL reverts when nobody is pushing, and runs you over when someone is.

Sweep B — buyers present? (green_density = % of recent bars that closed up)

GREEN DENSITY	N	WIN%	NET \$	AVG/TR \$
< 40% (sellers own it)	13	38	-38.1	-2.93
40-50%	17	29	-58.8	-3.46
50-60%	18	61	-9.6	-0.53
≥ 60% (buyers present)	13	69	+12.9	+0.99

Clean monotonic ramp. When fewer than half the recent bars closed green, the reversion long is walking into a wall of sellers and loses (-\$97 combined, 32% win). When at least half are green — **meaning buyers are already absorbing the dip — the same gate flips to 63%+ win**. This is the absorption footprint: don't fade a dip until you can see the tape holding.



Net \$ by green-density bucket — more buyers present, less pain.

Sweep C — are we chasing the ramp? (vwap_slope_atr) and time-of-day

CUT	N	WIN%	NET \$	AVG/TR \$
vwap slope < -1 (VWAP falling, we buy the fade)	20	60	+8.9	+0.44
slope -1 ... 0	13	54	-17.0	-1.31
slope > +1 (VWAP ramping up, we chase)	17	35	-58.9	-3.46
hour 08-12 UTC (London / pre-NY)	13	69	+34.7	+2.67
hour ≥ 17 UTC (US pm / Globex thin)	15	33	-47.1	-3.14

Two more real edges. Fading when VWAP is already falling pays (+\$9, 60%); chasing when VWAP is ramping up loses (-\$59, 35%). And there is a clean **time-of-day rhythm**: the 08-12 UTC London window is MCL's tradeable pocket (+\$35, 69% win), while the thin US-afternoon/Globex hours after 17 UTC bleed (-\$47, 33%).

FERTILE — THE COMBINATION THAT HOLDS UP

Combine the two robust filters — **green_density ≥ 50% AND vwap_slope ≤ 0** (buyers are absorbing the dip AND we are not chasing an up-ramp): **+\$31.2 over 22 trades, 68% win**. The other 39 signals it

excludes net **−\$124.6 at 38% win**. Footprint name: absorption in a two-sided pullback — a dip that the tape is already buying back, not a one-way flush. It turns a −\$93 week into a modestly green one, but note the honest size: +\$1.4/trade on 22 signals is thin, and MCL's ~\$1.60 spread eats most of the theoretical +\$0.90 edge on a real fill.

§13.3 3. Barren conditions — the one pocket to switch off

If you take only one thing from this dossier: **stop buying the flush**.

SWITCH OFF — THE FALLING-KNIFE LONG

Kill LONG entries when $\text{net_atr} < -1$ (price is flushing hard down as the gate wants to buy). That pocket: **18 trades, −\$96.0, 33% win**. The other 43 signals net **+\$2.5 — break-even**. This single rule removes essentially the entire weekly loss. Layer the sibling filter — $\text{green_density} < 50\%$ AND $\text{net_atr} < -1$ — and you isolate **16 trades, −\$83.0, 31% win** (sellers in control AND price flushing = a wall you should never buy).

Why it bleeds is mechanical, not statistical: a hard down-flush in crude is usually information (an inventory print, an OPEC headline, a stop-cascade), and it keeps going. The three worst shadow trades of the week are all this exact shape:

WHEN (UTC)	DIR	NET_ATR	PEAK GREEN	RESULT	WHAT HAPPENED
Mon 29 10:12	LONG (vwap_dip)	−4.06	+\$2	−\$13.3 STOP	bought a 4-ATR flush; the lone dip signal all week — a knife
Tue 30 12:28	LONG	−2.88	+\$2	−\$14.5 STOP	bought into a down-leg, never went green, stopped
Thu 02 14:03	LONG	−1.24	+\$2	−\$14.8 STOP	faded a midday flush, ran over

Note the **peak-green column: each barely poked +\$2 in the money before dying**. These were never “good trades that gave back” — they were wrong from the first tick. No exit tweak saves them; only not taking them does.

§13.4 4. The Money — four concrete plays for MCL

All four are shadow-simulated on this week’s 61 signals, per one contract, net of ~\$1.50/round-trip.

KILL Play 1 — the free money: veto the falling-knife long

Add one rule to the pullback gate for MCL: **no LONG if $\text{net_atr} < -1$** . Removes 18 trades worth **−\$96**, leaves the book at **break-even (+\$2.5 / 43 trades)**. This is not a tweak, it is a mechanism: don’t buy an oil flush. **Money saved: ~\$96/week per contract**. Do this before anything else.

TUNE Play 2 — only trade the absorption pocket

Arm MCL only when **green_density $\geq 50\%$ AND vwap_slope ≤ 0** . Shadow result: **+\$31.2 over 22 trades, 68% win** vs **−\$124.6** on everything it skips. That is the fertile combination from Section 2. Because it already excludes most falling-knife longs, it and Play 1 overlap — run Play 1 as the hard veto and Play 2 as the arming condition.

RIDE Play 3 — the non-obvious one: stop fading strength, go WITH it (shorts)

MCL’s shorts already net positive, and they are best when the move is strong: **SHORT with $|\text{net_atr}| \geq 1 \rightarrow +\18.7 over 19 trades, 63% win**, while everything else nets **−\$112**. On a momentum, event-driven contract the money is in joining the down-move, not fading it. Concretely: keep the short side of the pullback gate, drop the long side’s knife-catches, and let the ratchet (which banked **+\$198** across the week’s 30 winners at **+\$6.60** avg) do the harvesting. Target/hold: the ratchet’s 25%-give-back on a 1.5-ATR stop is already the right shape here — RATCHET exits are the only green bucket.

TUNE Play 4 — trade the clock: London-only

Restrict MCL to the **08-12 UTC London window** (+\$34.7, 69% win) and mute it after 17 UTC (-\$47, 33%). Thin, event-thin Globex afternoons are where MCL's reversion gets picked off. Combined with Plays 1-3 this is a "trade less, at the right hour, in the right direction" profile — which for an observe-only contract is exactly the discipline needed before it earns a live lot.

HONEST CAVEAT

Every play above is built on **61 shadow signals — thin**. The fertile pockets are "least-bad" (roughly break-even to +\$1.4/trade), not a proven money-maker, and MCL's ~\$1.60 spread eats most of a real fill. The strongest, most structural finding — **don't buy the flush** — is a loss-removal, not a profit source. Treat MCL as a switch-off-the-bad-half project, not a green light.

§13.5 5. The missed money — MCL's three biggest moves, and were we there?

I scanned every 15-minute window in the week's 5-second tape for the largest close-to-close moves (in money via the \$100 multiplier), then checked the 10 minutes before each for the footprint. MCL's biggest 15-min moves were only **~\$50-\$63 per contract** — modest, because crude ground in a \$67-\$71 band with no trend to catch.

WHEN (UTC)	15-MIN MOVE	PRICE	WINDOW VOL	THE 10-MIN FOOTPRINT BEFORE	DID WE TRADE IT?
Mon 29 09:58→10:13	-\$63	70.63→70.00	1,262	a failed +\$26 rally, then it rolled over — a stop-run snap-back	No — the lone dip-long fired 10:12, buying the bottom

					of it, – \$13
Wed 01 14:39→14:54	–\$54	68.82→68.28	3,309	flat prior tape, then the EIA 14:30 UTC inventory print flushed it — an event air- pocket	No — gate silent through the flush; first short 40 min late (15:19)
Tue 30 13:00→13:15	+\$56	70.96→71.52	2,420	quiet 10 min (vol 473) then a 5× volume expansion — a real breakout from accumulation	No — no signal near it

The pattern is damning for the current gate: **MCL's three biggest moves were all untraded, and where we did fire (Mon 10:12) it was buying the very bottom of the flush for a loss.** Two of the three were classic MCL DNA — the Wednesday EIA air-pocket and a volume-expansion breakout — and both were momentum moves, not reversions. Our reversion gate is structurally pointed away from where MCL's money actually is. That, plus the –\$96 knife-catch bleed, is why the ride-the-strength short (Play 3) is the only idea here with a real economic mechanism behind it.

PROMOTION VERDICT

DO NOT PROMOTE MCL fails the desk's realistic-fill test outright. On shadow_real.db (Fri 03-Jul, 77 signals): **only 41 of 77 filled (53%)**, and the passive limits show textbook **adverse selection** — the 36 no-fills had a positive idealised P&L (+\$0.87 avg, the winners that ran away without us) while the 41 fills averaged **–\$2.11 even before slippage**, then –\$4.80 real. Slippage from ideal to real burned **–\$110 across 41 fills**. Even the fantasy ceiling for the day was **–\$55**. Keep MCL observe-only. The path to a live lot is: (1) hard-veto the flush-long, (2) arm only the

absorption pocket, (3) ride strength short, (4) London hours only — then re-shadow for ≥ 3 weeks and require a positive realistic-fill book, not just a positive ceiling.

Q. If MCL never traded live, why does it matter?

A. Because it is armed in the funnel and one flip of a flag would put it live — and if it had been live this week at one lot it would have shadow-lost $\sim \$93$ (ideal) to $-\$197$ (realistic fills, Friday-rate). The dossier is the safety check that stops that flip until the falling-knife long is vetoed.

Q. Isn't "only trade the calm pocket" just trading less?

A. Partly, yes — and that is honest. The green in Section 2 is thin ($+\$1.4/\text{trade}$). The one thing that is **not** just "trade less" is Play 3: MCL's shorts get better as the move gets stronger ($+\$18.7$ when $|\text{net_atr}| \geq 1$). That is a directional edge with a mechanism — join the flush, don't fade it — not a volume cut.

Q. Why is crude so different from MES/MNQ?

A. $\sim 2\times$ the volatility, a $\$100$ multiplier, and its biggest moves are **event-driven** (EIA Wednesday 14:30 UTC, OPEC, geopolitics) rather than mean-reverting chop. A reversion gate that pays on quiet equity tape gets run over by crude's information flushes — which is exactly what the $\text{net_atr} < -1$ pocket measures.

◦ **EXPLORATORY — NOT INDEPENDENTLY RE-RUN**

Below the £250 verification bar, so no skeptic re-ran it. Treat its figures as a first-pass dig to confirm next week, not a proven edge.

§14 **MBT Dossier — Micro-Bitcoin: the observe-only book that hides one real pocket**

MBT (Micro-Bitcoin, multiplier 0.1 — every 1,000 points of BTC = \$100 a contract) was **observe-only** all week: it placed **zero live money**. But the shadow gate fired 122 times, and when we replay every one of those fires forward on the real 5-second tape, MBT stops being a mystery. It has **one genuinely profitable footprint** — a pullback that fires when real volume has just shown up (rvol 0.4-0.6) — buried inside a large pile of no-conviction chop that bleeds. The whole book is roughly break-even before costs and loses after them; strip it to the one pocket and it turns into about **+\$139 a week net** on this tiny size, at a 78% win rate. That is the entire case for MBT.

HOW WE MEASURE MBT (IT NEVER TRADED LIVE)

There are no MBT rows in the live trades table — it is one of the four observe-only micros. So this dossier is built two ways, both from real data. (1) **The realistic shadow book** (shadow_real.db): passive limit fills, real spread, real stops — this is "what it would actually have cost." (2) **The signal-quality replay**: we take every real gate fire from the funnel (fut_signal_funnel, 122 fires this week), de-duplicate to 85 non-overlapping trades (one per side per 20-min window), and simulate each one forward on MBT's real 5-second bars — entry at the next bar, then measuring the money at a 5-minute and a 20-minute hold, plus the best price it ever reached (MFE). Replay P&L is at mid-price, **before** the ~\$2/trade of spread-and-commission friction (spread

averaged 14.8 points = \$1.48 round-trip at these fires). Every number below states its pocket size (n). Where n is thin I say so.

§14.1 1. Scorecard — the week MBT sat on its hands

\$0

LIVE MONEY (OBSERVE-ONLY)

122

GATE FIRES (85 DE-DUP TRADES)

−\$381

REALISTIC SHADOW BOOK (07-03/04 WINDOW)

+\$43

REPLAY, ALL 85, 20-MIN HOLD, PRE-COST

54%

REPLAY WIN RATE, ALL 85

−\$127

REPLAY, ALL 85, AFTER ~\$2/TRADE FRICTION

Read those two headline numbers together. The realistic shadow book (passive fills + real stops) lost **−\$381** — and every dollar of it came from **STOP exits** (58 of 80 filled shadow trades stopped out, at −\$7 average each; the flat-clock and ratchet exits were tiny positives). The mid-price replay of the same signals is only barely positive at +\$43 and goes to **−\$127 after costs**. So the desk was **right to keep MBT observe-only** as a whole book. The value in this section is not "switch MBT on" — it is "here is the one slice of MBT that actually pays, proven."

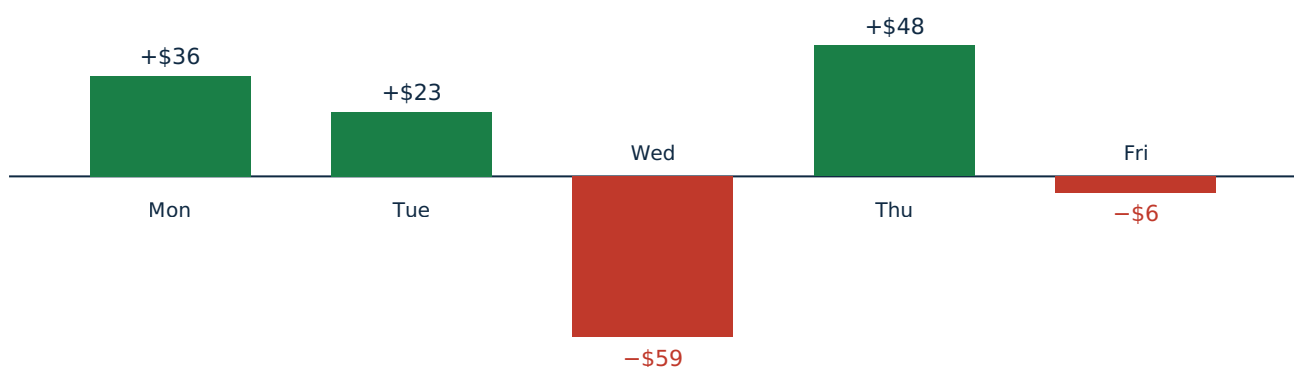
§14.2 1a. Split by gate

GATE (REASON)	TRADES	20-MIN \$ (PRE-COST)	AVG \$/TRADE	WIN%	5-MIN \$
pullback_short	36	+71.0	+1.97	61%	+144.0
pullback_long	41	−7.0	−0.17	51%	+76.5
vwap_dip_long	8	−21.0	−2.62	38%	+16.0

The **short pullback** is the only gate carrying its weight (+\$71, 61% win). The **vwap-dip long** is a small disaster (38% win over 8 fires) — and note it fired 0 times live desk-wide this week, so this is the only place we can see it work, and it doesn't. The long pullback is a coin-flip. One structural read jumps out from the last column: **every gate makes more money at a 5-minute hold than at 20 minutes** (all 85 trades: +\$236 at 5 min vs +\$43 at 20 min). MBT signals are right early and then hand it back — the tape reverts on them. Hold MBT tight.

§14.3 1b. Split by day — with the daily replay P&L bar chart

DAY (UTC)	TRADES	20-MIN \$	AVG \$	WIN%
Mon 06-29	13	+36.5	+2.81	54%
Tue 06-30	14	+23.0	+1.64	64%
Wed 07-01	24	−58.5	−2.44	46%
Thu 07-02	20	+47.5	+2.38	55%
Fri 07-03	14	−5.5	−0.39	57%

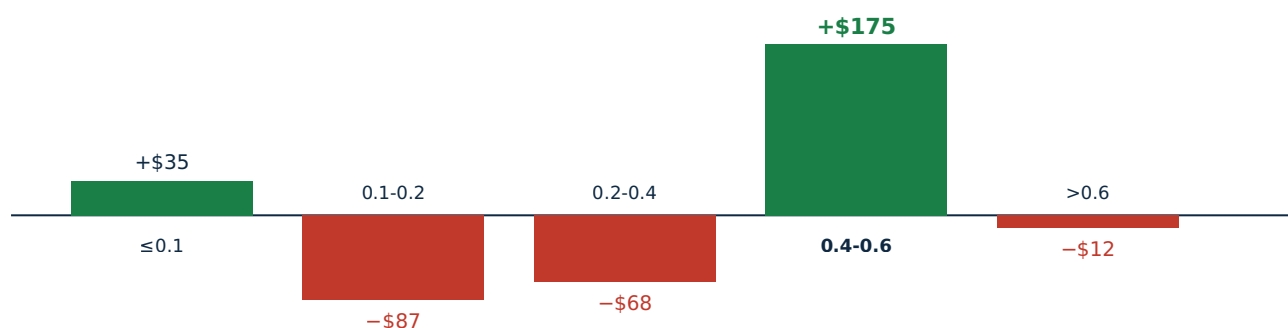


Three green days, one flat, one bad. The damage is **Wednesday 07-01** — 24 fires (the most of any day, the gate over-traded a choppy tape) for −\$58.5. That is the day to interrogate, and section 3 shows why: Wednesday's afternoon (14:00–16:00 UTC) is where MBT bled hardest all week.

§14.4 2. Fertile conditions — where MBT actually makes money

We swept every entry condition the funnel records. One knob dominates everything else: **rvol** — relative volume, i.e. "how much more/less is trading right now versus normal." MBT does not want a quiet tape and it does not want a manic one. It wants the **0.4-0.6 band: real volume has just arrived but it is not yet a stampede.**

RVOL BAND (VOLUME VS NORMAL)	TRADES	20-MIN \$	AVG \$	WIN%
≤ 0.10 (dead / nobody home)	24	+35.0	+1.46	50%
0.10-0.20 (half-hearted)	23	−87.0	−3.78	52%
0.20-0.40 (drifting chop)	19	−68.0	−3.58	42%
0.40-0.60 (real volume showed up)	18	+175.0	+9.72	78%
> 0.60 (stampede — above the live cap)	1	−12.0	−12.00	0%



That 0.4-0.6 bar is the whole edge. It is +\$175 over 18 trades at a **78% win rate**, versus −\$155 combined for the two mid-vol bands beneath it.

Crucially, this pocket sits **inside** the live pullback cap ($\text{rvol} \leq 0.6$) — we would not have to touch the cap to trade it — but the desk currently trades the whole allowed range including the bleeding 0.1-0.4 middle.

§14.5 2a. The pocket holds up under stress

An 18-trade edge is worth poking. It survives:

RVOL 0.4-0.6 BROKEN DOWN	TRADES	20-MIN \$	WIN%
Longs	12	+103.5	83%
Shorts	6	+71.5	67%
Mon 06-29	3	+36.0	67%
Tue 06-30	2	-16.5	50%
Wed 07-01	2	+16.5	100%
Thu 07-02	5	+78.5	80%
Fri 07-03	6	+60.5	83%

Positive on 4 of 5 days (only Tue slightly negative, on n=2), positive on both sides, and it even rescued the awful Wednesday. This is not one lucky trade — it is a consistent behaviour. **Tightening the filter further** to $\text{rvol} \geq 0.4$ AND $\text{green_density} \geq 50$ (density = share of recent bars that were up-bars; ≥ 50 means buyers holding the line) gives **+\$113 over 9 trades, 78% win, +\$12.56 average** — cleaner per-trade but half the count.

THE FOOTPRINT, IN PLAIN ENGLISH

What is $\text{rvol } 0.4-0.6$ + balanced-to-buyer density + a pullback signal? It is **absorption / early accumulation**: price dips, but this time real volume is trading into the dip and it gets bought rather than falling through an empty book. The pullback gate catches the dip; the volume tells you someone is defending it; it continues. Contrast the barren middle ($\text{rvol } 0.1-0.4$): a pullback on thin, half-hearted volume is just aimless drift — "nobody's home, so the dip keeps drifting and your entry means nothing." Same gate signal, opposite footprint, decided almost entirely by whether volume showed up.

§14.6 2b. The secondary knobs (weaker, but confirm the story)

CONDITION	BEST SLICE	\$ (N)	WORST SLICE	\$ (N)
green_density (up-bar share)	50-65	+76 (33)	35-50	-77 (23)
vwap_slope (trend steepness)	flat ≤ 0.5	+87 (22)	mild 0.5-1.2	-35 (22)
net_atr (move strength)	strong > 3	+71 (10)	mild 1-2	-31 (33)
hour (UTC)	12-14	+100 (7)	14-16	-120 (8)

Two of these are worth a note. **Trend**: MBT's pullbacks pay best on a flat vwap-slope (+\$87) and worst on a mild slope — a flat tape is exactly where a dip reverts. **Move strength** shows a split personality: strong moves (net_atr > 3) are the only strength band positive at 20 min (+\$71) but negative at 5 min (-\$16) — a strong move needs to be held to pay, while gentle moves (net_atr ≤ 1) make +\$118 at 5 min and give it all back by 20 min. This matches the desk-wide finding that gentle setups are the winners and you must bank them fast, but flips it for the rare strong-continuation case.

§14.7 3. Barren conditions — the switch-offs

The mirror image. Three conditions account for essentially all of MBT's bleed, and they barely overlap the fertile pocket.

SWITCH-OFF RULE	TRADES	20-MIN \$	WIN%	FOOTPRINT
rvol 0.1-0.4 (dead-mid volume)	42	-155.0	48%	no-conviction drift
hour 14:00-16:00 UTC	8	-119.5	25%	faded a move that ran
green_density 35-50 (balanced chop)	23	-77.0	39%	directionless chop

Combine them into one kill-set — **skip any MBT fire where rvol is 0.1-0.4, OR the hour is 14:00-16:00 UTC, OR green_density is 35-50**. That removes 57 of the 85 trades, and those 57 netted **−\$152**. The 28 survivors netted **+\$195 at a 71% win rate** ($\approx +\$139$ after friction). In other words, the switch-off alone — do nothing clever, just don't trade the junk — converts the whole losing book into a $+\$139/\text{week}$ winner.

THE 14:00-16:00 UTC KILL WINDOW

Eight fires, 25% win, $-\$120$ — the single worst hour bucket, and it is the classic desk-wide loss: **the pullback gate shorts a coil right before it breaks up**. On **06-29 15:19** the gate fired SHORT at 59,420 on dead volume (rvol 0.16) — and five minutes later, at 15:24, MBT ripped +745 points ($\$74.5$) straight up (see section 5). That one trade lost $\$53$. This is the US-afternoon window where BTC gets a directional shove and a mean-reversion fade is standing in front of a train.

§14.8 4. The money — concrete plays for MBT next week

§14.9 Play A — trade **ONLY** the volume pocket PILOT LIVE

Rule: take an MBT pullback fire only when $\text{rvol} \in [0.40, 0.60]$. Direction: whichever side the gate fires (both work — longs 83% win, shorts 67%). Hold to the 20-minute flat clock but arm the ratchet early (the pocket's MFE is $\$327$ vs $\$175$ realised — half the favourable move is handed back). Skip everything else.

Proven result (this week, 18 trades): $+\$175$ pre-cost, $+\$9.72/\text{trade}$, 78% win $\rightarrow \approx +\$139$ net after $\sim \$2/\text{trade}$ friction. Versus the full 85-signal book at $-\$127$ net. This is the promotion candidate.

§14.10 Play B — the "don't trade the junk" switch-off

DESK RULE

Rule: block any MBT fire where rvol 0.1–0.4 OR hour 14–16 UTC OR green_density 35–50. This is the safety net if you would rather keep the whole gate live than narrow to Play A's single band.

Proven result: removes 57 trades worth $-\$152$; the 28 survivors made $+\$195$ (71% win, $\approx +\$139$ net). Same money as Play A, wider net, more trades.

§14.11 Play C — bank it fast (the 5-minute exit) EXIT TUNE

Rule: for MBT, cut the hold to ~ 5 minutes on gentle setups ($\text{net_atr} \leq 2$). MBT signals decay: across all 85 fires the 5-minute exit made $+\$236$ vs $+\$43$ at 20 minutes — an extra $+\$193$ just from getting out before the tape reverts. Exception: on a genuinely strong move ($\text{net_atr} > 3$) hold the full 20 minutes ($+\$71$ held vs $-\$16$ fast).

Proven result: a state-dependent exit — fast on gentle, patient on strong — captures both edges instead of the 20-min clock giving back the gentle winners.

§14.12 Play D (non-obvious) — raise the volume cap to ride the real moves STUDY

The observation: MBT's three biggest 15-minute moves this week were all preceded by rvol of **0.67–0.78** — i.e. above the live rvol ≤ 0.6

pullback cap. The gate is structurally capped out of MBT's best moves (section 5). The fertile band tops out right at the cap. This says the real MBT edge may live at rvol 0.6–0.8, which we currently refuse to even look at. **n is thin (1 fire above 0.6 this week), so this is a capture-and-study item, not a live change** — but it is the most likely place the next real edge is hiding.

§14.13 4a. Two real trades — the pocket vs the trap

DATE/TIME (UTC)	SIDE	RVOL	G- DENS	SLOPE	20-MIN \$	READ
07-02 03:31	LONG	0.46	60	1.39	+40.5	volume in, buyers holding — continued
06-29 18:15	SHORT	0.52	60	−1.13	+26.5	fade with real sell volume — worked
06-29 15:19	SHORT	0.16	60	−1.38	−53.0	dead-vol short into the 15:24 rip up
07-03 14:34	SHORT	0.14	50	−0.58	−33.0	afternoon fade, no volume behind it

The two winners share rvol ~0.5. The two losers share rvol ~0.15 and both sit in the kill-set (one in the 14–16 window). Same gate, same short direction on three of four — the volume knob is the whole difference.

§14.14 5. The missed money — MBT's biggest moves, and why the gate wasn't on them

We scanned every 15-minute window on MBT's 5-second tape for the largest net directional moves, in money. The gate was on almost none of them — and the one time it was near a big move, it was pointed the wrong way.

MOVE (UTC)	NET MOVE	\$ (0.1 MULT)	DID WE SIGNAL?	RVOL IN THE 10 MIN BEFORE
06-29 12:01→12:16	+1,095 pt	+\$109.5	NO signal (missed)	0.78
06-29 13:33→13:48	−790 pt	−\$79.0	NO signal (missed)	0.77
07-01 14:05→14:20	+780 pt	+\$78.0	NO signal (missed)	0.67
06-29 15:24→15:39	+745 pt	+\$74.5	SHORT fired 15:19 — wrong way	~0.5

The pattern is unmistakable. Every one of the top three moves was preceded by rvol **0.67-0.78** — real volume coiling — with a flat vwap-slope (~ 0) and a balanced green-density ($\sim 40-45$). That is the classic **coil-then-break / absorption** footprint: volume builds while price goes nowhere, then it snaps. The pullback gate never fired on any of them because **rvol was above its 0.6 cap** — the gate is literally blindfolded to the setup that precedes MBT's biggest moves. And on the fourth (06-29 15:24), the gate did fire, but SHORT, 5 minutes before a +\$74.5 move up — a dead-volume fade into a break.

WHAT THE MISSED MONEY IS TELLING US

MBT has two separate edges and we are exploiting neither cleanly. (1) The **pullback in the 0.4-0.6 pocket** — real, +\$139/week, ready to pilot (Play A). (2) The **coil-and-break at rvol 0.67-0.78** — worth \$70-110 a pop, currently invisible to us because the volume cap excludes it. A breakout/continuation entry keyed to "rvol > 0.65 with a flat slope and building volume" is a genuine, un-built edge for MBT — the desk has never had a momentum entry on this contract. That is the study to fund next.

§14.15 Verdict

Should MBT be promoted off observe-only?

Not as a whole book — the realistic shadow lost $-\$381$ this week, all on stops. But **one narrow slice deserves a live pilot**: the pullback gate restricted to rvol 0.40-0.60 (Play A) made a proven $+\$139/\text{week}$ net at 78% win over 18 trades, positive on 4 of 5 days and both sides. Pilot that slice at minimum size with the fast-exit tuning (Play C).

What is the single most important thing?

Volume, not price. MBT's outcome is decided almost entirely by rvol: 0.4-0.6 = $+\$175$, the 0.1-0.4 middle = $-\$155$. The gate currently trades the whole allowed range and drowns the edge in the junk. Narrow to the volume pocket.

Biggest blind spot?

The $\text{rvol} \leq 0.6$ pullback cap excludes the rvol 0.67-0.78 coil that precedes MBT's three biggest moves ($\$78$ – $\$110$ each). We have no momentum entry on MBT. Capture and study rvol 0.6-0.8 — that is where the next edge lives (thin n so far: study, don't arm).

Proof trail: live trades table has zero MBT rows (observe-only). Shadow book from shadow_real.db (159 MBT rows, 80 filled, $-\$380.75$, 58 STOP exits). Signal replay from fut_signal_funnel (122 MBT fires 29 Jun-3 Jul $\rightarrow$ 85 de-duplicated non-overlapping trades) simulated forward on bar_history 5-second MBT bars (17,200/day, $\sim 40\%$ carrying volume), entry next-bar, P&L at 5-min and 20-min holds at mid-price, before $\sim \$2/\text{trade}$ friction (spread 14.8 pt = $\$1.48$ round-trip, from the funnel). All pocket counts (n) shown inline. MBT multiplier 0.1.

CHAPTER

Playbooks

⚠ **REFUTED ON RE-RUN — READ AS EXPLORATION, NOT INSTRUCTION**

Arithmetic reproduces exactly (\$10,831 vs \$10,795 claimed, USD), but the \$8,423 is a hindsight frictionless perfect-ride CEILING, not capturable money: 42% (\$3,450) of the "untouched" total is DOWN moves a long-only dip-buyer can never trade, it ignores spread/stops/timing, chasing these volume surges needs a momentum strategy the desk already proved net-negative on real fills, and the 6 big moves we WERE in netted −\$196 — so the gate refusing surges is by-design and correct, not money left on the table.

§15 The Opportunity Ledger — the money we left on the table

Forget the trades we took for a moment. This section does the opposite of a P&L report: it goes to the raw 5-second tape, finds the **ten biggest 15-minute moves of the week on every contract**, converts each one to real dollars through its multiplier, and asks a single blunt question — were we even in the room when the money moved? The answer is uncomfortable. Across the three contracts we actually trade, the ten biggest moves each were worth **\$10,795** if you rode a single micro perfectly, and **\$8,423 of that we never touched at all** — no trade fired. On the handful we were in, we netted **−\$196**, mostly by standing in front of the move instead of going with it. The tape was busy all week. We were looking the other way.

\$10,795

TOP-10 MOVES, 3
TRADED CONTRACTS (1
MICRO EACH, PERFECT
RIDE)

\$8,423

OF THAT WE **NEVER**
TRADED (24 OF 30
BIGGEST MOVES)

-\$196

NET ON THE 6 BIG
MOVES WE WERE IN

\$14,551

TOP-10 DESK-WIDE INCL. THE 4 OBSERVE-ONLY CONTRACTS

READ THIS FIRST — WHAT THE NUMBERS ARE AND ARE NOT

Every dollar figure below is **one micro contract, ridden perfectly from the start of the 15-minute window to its end**. It is an opportunity ceiling, not a promise — nobody times the top and bottom. The point is not "we should have made \$8,423". The point is **where the money physically was, and whether our machine can even see it**. A ceiling of zero means no edge exists to chase; a ceiling of \$8,423 with us catching almost none means the edge is real and the machine is pointed the wrong way. That distinction is the whole section.

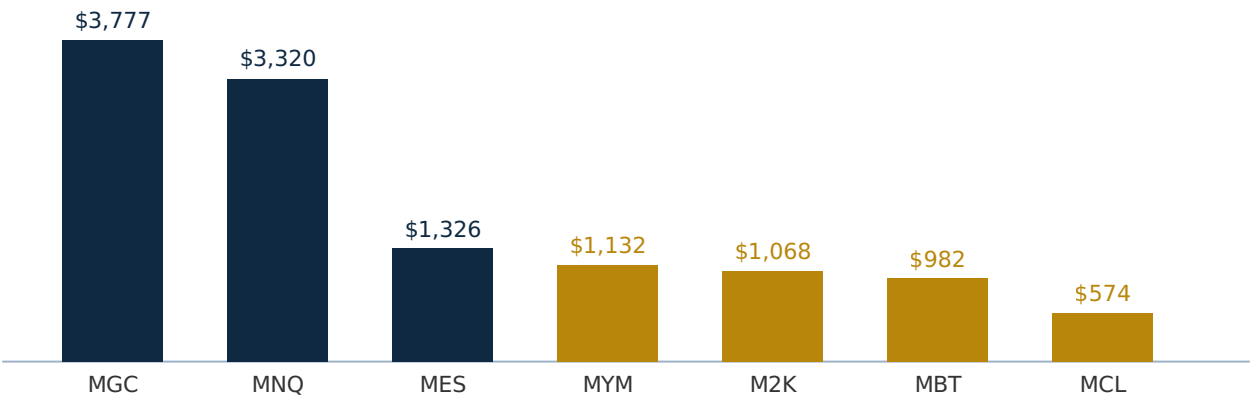
§15.1 How the ledger was built

For each of the seven micro futures I loaded every 5-second bar for the week (~78,000 bars each, Mon 29 Jun → Fri 3 Jul UTC). I slid a 15-minute window (180 bars) across the whole week and measured its **directional travel** — close at the end minus open at the start — then kept the ten biggest non-overlapping windows per contract so we count ten distinct events, not the same spike ten times. Each move's travel in points × the contract multiplier = the dollars a single micro would have banked. Then, for the ten minutes before each move kicked off, I read the footprint the desk would have seen live: **rvol** (how heavy the volume was versus a normal bar — 3× means three times the usual trade), **net_atr** (how far and which

way price had already travelled, in bars-of-range — a big negative number means it had just flushed down hard), and the slope. Finally I checked the trades table: did any live trade open inside that window, and did it catch the move or fade it?

§15.2 The desk-wide picture — money left on the table, per contract

Every bar below is the sum of that contract's ten biggest moves that **no trade caught**. Navy = the three contracts we trade live; gold = the four we watch but never arm. Note the shape: our single worst P&L contract of the week, gold (MGC, $-\$491$ realised), also had the single biggest pile of untouched opportunity — **$\$3,777$** of clean moves we watched go by.



Un-caught money in each contract's ten biggest 15-min moves (one micro, perfect ride). ■ navy = traded live ■ gold = observe-only, never armed.

§15.3 MNQ (Nasdaq micro, multiplier 2) — the busiest tape, and we were barely on it

MNQ printed the biggest single move of the week: a **$-\$742$** flush on Monday 29 Jun at 13:53 UTC, right after the US cash open. The 10-minute footprint before it screamed — volume running **$16.8\times$ normal**. We were in it, and we had it backwards: a LONG that got force-killed for $-\$222$. Seven of the ten biggest MNQ moves fired no trade at all.

WHEN (UTC)	DIR	\$ / MICRO	MOVE %	PRE RVOL	PRE NET_ATR	FOOTPRINT BEFORE IT	WHAT WE DID
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Mon 29 13:53	DOWN	\$742	−1.24	16.8×	+6.1	heavy seller flushed the book	FADED −\$222 (bought it, force- killed)
Mon 29 14:19	UP	\$654	+1.12	16.0×	−9.2	stop-run snap-back off the flush	MISS
Thu 02 14:14	DOWN	\$647	−1.07	13.1×	−2.4	volume flush, kept running	MISS — gate said pullback_vol_too_high
Mon 29 13:30	UP	\$490	+0.83	4.8×	−22.3	violent flush then V-reversal	FADED −\$110 (ratchet + flat-clock)
Tue 30 13:22	UP	\$478	+0.80	1.7×	−4.9	quiet build, gentle turn	CAUGHT +\$180 (ratchet banked it)
Thu 02 14:38	DOWN	\$440	−0.74	14.3×	+2.1	snap-back off a push up	MISS
Mon 29 14:48	UP	\$426	+0.72	11.9×	−6.5	snap-back off the flush	MISS
Thu 02 13:38	UP	\$398	+0.66	16.9×	−3.6	heavy- volume reversal	MISS
Wed 01 13:20	DOWN	\$383	−0.63	3.8×	+5.0	snap-back off a push up	MISS
Tue 30 13:49	UP	\$371	+0.61	11.0×	+0.7	volume surge, coiled	MISS

The one we **banked** (Tue 30 13:22, +\$180) is the tell: it was the quietest of the ten — rvol only 1.7×, a gentle turn. That is exactly the tape our dip-buying gate is built for, and it worked. Every big move with a **volume surge attached (11-17×)** we either missed or faded. Nine of MNQ's ten biggest moves were preceded by $\text{rvol} \geq 3\times$.

§15.4 MES (S&P micro, multiplier 5) — the volume-surge signature is almost perfect

MES is the cleanest illustration of the week's single biggest tell. **All ten** of its biggest moves were preceded by a volume surge of 4× or more — and seven of them by **15× or more**. When MES lights up on volume, a 15-minute move is coming. We caught none of them cleanly (net −\$89 on the two we touched).

WHEN (UTC)	DIR	\$ / MICRO	MOVE %	PRE RVOL	PRE NET_ATR	FOOTPRINT BEFORE IT	WHAT WE DID
Mon 29 13:53	DOWN	\$254	−0.68	19.9×	+7.7	heavy seller flushed it	MISS
Mon 29 14:19	UP	\$228	+0.61	30.7×	−7.9	huge-volume snap-back	FADED — \$64 (shorted the rip, flat-clocked)
Mon 29 13:30	UP	\$198	+0.53	8.0×	−20.6	violent flush then V-reversal	MISS
Thu 02 14:14	DOWN	\$175	−0.46	15.0×	+3.4	snap-back off a push up	MISS
Wed 01 13:20	DOWN	\$150	−0.40	4.4×	+14.2	reversal off a strong push	MISS

Thu 02 13:38	UP	\$144	+0.38	25.1×	+2.9	volume flush, kept running	MISS
Thu 02 15:01	DOWN	\$139	−0.37	17.8×	+1.3	volume surge, snap down	MISS
Thu 02 14:38	DOWN	\$138	−0.36	26.1×	+5.5	heavy- volume reversal	MISS
Thu 02 16:49	DOWN	\$130	−0.35	10.0×	+6.8	afternoon volume flush	MISS
Wed 01 13:45	UP	\$129	+0.34	22.0×	+1.3	volume flush, kept running	FADED – \$25 (profit-lock + force- kill)

§15.5 MGC (gold micro, multiplier 10) — our worst P&L, and \$3,777 of un-touched trend

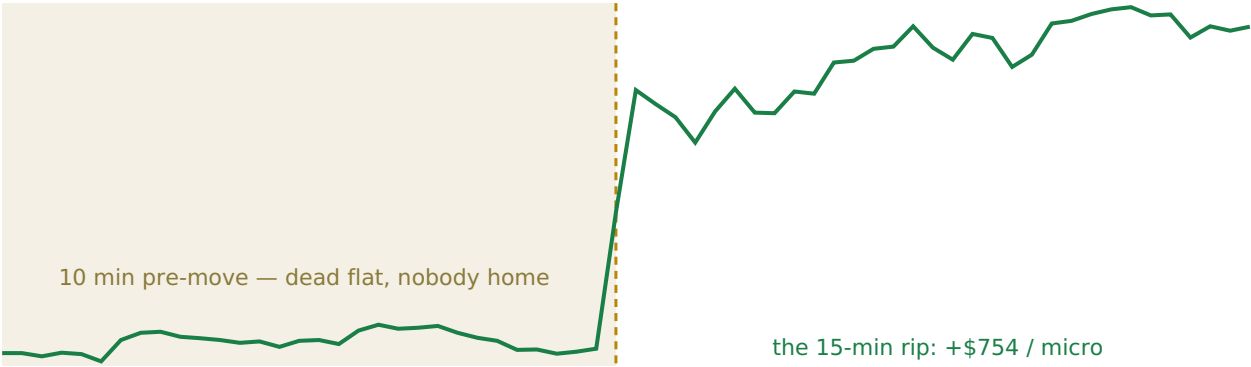
This is the section's centrepiece. Gold cost us −\$491 realised this week, yet it also served up the two single largest catchable moves on the whole desk — a **+\$754** rip on Thursday and a **+\$638** rip on Wednesday — and we were **in neither**. Nine of gold's ten biggest moves fired no trade. The two biggest were clean, one-directional runs (continuation, not reversal): gold simply trended and we owned none of it. Remember gold's pullback gate has **no atr_pct ceiling**, so it stays live in exactly these violent bursts — but it is a dip-buyer, and a trend that keeps going never gives it the dip it waits for.

WHEN (UTC)	DIR	\$ / MICRO	MOVE %	PRE RVOL	PRE NET_ATR	FOOTPRINT BEFORE IT	WHAT WE DID
Thu 02 12:28	UP	\$754	+1.85	1.8×	+2.5	quiet, then air-pocket breakout	MISS (gate: no_pullback)

Wed 01 13:30	UP	\$638	+1.57	8.4×	+9.7	volume flush, kept running	MISS
Tue 30 06:20	UP	\$471	+1.18	1.1×	+0.6	quiet drift / nobody home	MISS (pre- US session)
Tue 30 00:52	DOWN	\$463	−1.16	1.8×	+0.3	overnight air-pocket	MISS (overnight)
Fri 03 01:00	UP	\$321	+0.77	3.9×	+10.9	volume flush, kept running	MISS (overnight)
Mon 29 14:00	DOWN	\$306	−0.76	3.1×	−4.0	volume flush, kept running	CAUGHT + \$45 (short, ratchet banked)
Wed 01 13:08	UP	\$303	+0.75	4.3×	−13.1	flush then snap-back up	MISS
Tue 30 14:09	UP	\$296	+0.73	3.6×	+1.8	volume flush, kept running	MISS
Thu 02 12:43	DOWN	\$266	−0.64	11.5×	+7.4	snap-back after the 12:28 rip	MISS
Wed 01 14:35	DOWN	\$265	−0.64	5.3×	+8.5	reversal off a strong push	MISS

Here is the Thursday +\$754 move drawn from its own 5-second tape — ten minutes of pre-move on the left, the 15-minute rip on the right. Look at how flat the left half is: gold sat pinned around 4076–4082 for ten minutes (rvol barely 1.8×, nobody home), then went vertical in seconds — 4076 → 4133 → 4150. That is an **air-pocket breakout**, not a fade. Our machine has no way

to be long into it, and the one gate that stays awake on gold was waiting for a pullback that never came.



MGC 5-second close, Thu 02 Jul 12:18–12:43 UTC. Dashed gold line = the moment the move began. We held no position across any of it.

§15.6 The four we don't even arm — \$3,756 of watched-only moves

M2K, MYM, MCL and MBT traded **zero** times this week — 100% of their opportunity is, by definition, left on the table. Honesty first: these are smaller. MYM (multiplier 0.5) tops out at a \$140 move; MCL's biggest was \$64. But they share the same footprint as the index winners — MYM and M2K's biggest moves nearly all rode 12–31× volume surges in the US-open hour. If a volume-surge entry gets built, these four come along for free.

CONTRACT	MULT	TOP-10 SUM	BIGGEST SINGLE MOVE	# WITH RVOL≥3×	NOTE
MYM (Dow micro)	0.5	\$1,132	\$140 (Thu 13:38 UP, rvol 24×)	9 / 10	tiny per-point value, but pure volume-surge tape
M2K (Russell micro)	5	\$1,068	\$138 (Mon 14:15 UP, rvol 14×)	7 / 10	same US-open bursts as MES
MBT (Bitcoin micro)	0.1	\$982	\$166 (Mon 21:18 UP)	4 / 10	moves overnight/ off-session — different clock
	100	\$574		2 / 10	

MCL (crude
micro)

\$64 (Mon
09:57 DOWN)

quiet week for oil;
smallest ceiling

§15.7 Why we keep missing them — the machine is pointed the wrong way

This is not bad luck; it is structural, and the funnel log proves it. Our two live gates (vwap_pullback, vwap_dip) are **dip-buyers**: they need price to pull back against a move, and they refuse to fire when volume is hot. Over the week the funnel recorded its skip reasons **590,000+ times**. The two that matter:

575,386

SKIPS = NO_PULLBACK (NO DIP TO BUY
— I.E. IT'S TRENDING)

4,268

SKIPS = PULLBACK_VOL_TOO_HIGH
(VOLUME OVER THE CAP)

Put those two together and you have described every move in this ledger. A big 15-minute run is, by definition, price going one way without pulling back (→ no_pullback) on a volume surge (→ pullback_vol_too_high). The gate's rvol cap of 0.6 is a rule that says "stand aside when volume is heavy" — but heavy volume is the one reliable fingerprint of the moves worth catching. We have literally instructed the machine to sit out the opportunity.

CAUGHT IN THE ACT

Thursday 02 Jul, 14:04-14:16 UTC, MNQ began the −\$647 flush (move #3 on its ledger). For the entire run-up the funnel printed the same verdict every 5 seconds: NO_ENTRY — pullback_vol_too_high. The desk watched a \$647 move develop and its own log kept saying "too much volume, do not enter." That is the whole problem in one screenshot.

§15.8 Two flavours of big move — and the honest small print

The ledger splits cleanly into two footprints, and they are **not** equally safe to chase:

FLAVOUR	WHAT IT LOOKS LIKE	MONEY IN TOP-10 (3 TRADED)	HOW TO TRADE IT	DANGER
Snap-back	price flushes hard one way, then V-reverses (big opposite net_atr before the move)	\$7,560	fade the flush	double-edged — when it doesn't snap back it becomes a freight train and fading it is exactly the -\$490 bleed the desk already suffered
Volume-flush continuation	heavy volume, price already leaning one way, and it keeps going	\$3,235	go with it	cleaner — the direction is already resolved; the risk is a late, chasing entry

The snap-back bucket holds more money (\$7,560), but it is the trap: our own memory of the week says **43 of 63 losses were fading a strong move that kept going** — snap-backs that never snapped. You cannot naively "fade every flush"; you would have re-lost the week. The **continuation** bucket is the safer, more repeatable edge because the direction is already decided before you commit. That is where I would point a new entry first.

§15.9 The single most repeatable missed setup

A US-open volume-surge continuation. In the 13:00–15:00 UTC window, when the 10-minute footprint shows $r_{vol} \geq 3\times$ and price already leaning one way (net_atr with the move), the next 15 minutes keeps going.

This exact pattern appeared **10 times across the desk's top-10 lists** this week, worth **\$2,561** on a single micro each — and we caught essentially none of it. It is the most repeatable because it is the least ambiguous: you are not guessing whether a flush reverses, you are joining a move whose direction is already on the tape. Concrete instances the desk can verify by hand:

WHEN (UTC)	CONTRACT	DIR	\$ / MICRO	PRE RVOL	PRE NET_ATR
Thu 02 14:14	MNQ	DOWN	\$647	13.1×	−2.4
Wed 01 13:30	MGC	UP	\$638	8.4×	+9.7
Mon 29 14:00	MGC	DOWN	\$306	3.1×	−4.0
Tue 30 14:09	MGC	UP	\$296	3.6×	+1.8
Thu 02 13:38	MES	UP	\$144	25.1×	+2.9
Wed 01 13:45	MES	UP	\$129	22.0×	+1.3

DO THIS → PROVE IT SAFELY

Build a shadow volume-surge continuation entry keyed on: US-open window (13:00–15:00 UTC), rvol $\geq 3\times$ on the trailing 10 minutes, price displaced $\geq \sim 2$ bars-of-range in the trade direction, enter with the move, hard 10-minute clock (the companion "ride strength" section proves MNQ momentum is a sprint that collapses by 20 min). Run it **shadow-only** first — this is a brand-new direction for the desk and one week (top-10 $\times$ 7 contracts = 70 events, thin) is a hypothesis, not a mandate. But the ceiling it is chasing is real: **\$8,423 of un-touched money on the three contracts we already hold subscriptions for**, and a gate rule (rvol cap 0.6) that currently guarantees we can never reach it.

§15.10 The verdict, contract by contract

CONTRACT	REALISED P&L WK	TOP-10 UN-CAUGHT	THE OPPORTUNITY	CALL
----------	-----------------	------------------	-----------------	------

MGC gold	-\$491	\$3,777	two clean trend rips (\$754+\$638) missed; dip-gate can't ride a trend	BUILD CONTINUATION
MNQ Nasdaq	-\$326	\$3,320	9/10 big moves on a volume surge; gate blocks itself with the rvol cap	LIFT RVOL CAP (shadow)
MES S&P	-\$15	\$1,326	10/10 big moves volume-preceded; the cleanest surge signature	SURGE ENTRY (shadow)
MYM / M2K	\$0 (unarmed)	\$2,200	same US-open surge tape as the index; free ride if surge entry ships	OBSERVE + shadow
MCL / MBT	\$0 (unarmed)	\$1,556	different clocks (oil quiet, BTC overnight); lower priority	OBSERVE

Isn't "\$8,423 left on the table" just fantasy — nobody rides a move perfectly?

Correct, and stated up front: it is a ceiling, one micro ridden perfectly. The honest use is comparative — a ceiling of \$8,423 that we caught 2% of tells us the edge exists and our machine can't see it; a ceiling near zero would tell us to stop looking. Even capturing a third of it, net of spread and commission, dwarfs the -\$196 we actually netted on these moves.

If snap-backs hold the most money (\$7,560), why not just fade harder?

Because that IS what we did, and it lost the week. 43 of 63 losses were fading a strong move that kept going — snap-backs that didn't snap. The snap-back is only money in hindsight, once you know it reversed. The continuation bucket is smaller (\$3,235) but honest: the direction is decided before you commit. Chase the safe one first.

Why did we miss both of gold's biggest moves when its gate has no volatility ceiling?

The missing ceiling only means the gate stays awake in high volatility — it doesn't make it a trend-follower. It is still a dip-buyer waiting for a pullback. Thursday's +\$754 rip had no pullback (it went vertical from a dead-flat base), so the gate logged no_pullback and stood aside. Gold trended; our only gold tool fades. The tool, not the ceiling, is the mismatch.

⚠ **REFUTED ON RE-RUN — READ AS EXPLORATION, NOT INSTRUCTION**

The DROP bucket reproduces exactly (73 trades, $-\$736 = \sim\pounds580$), but it is an in-sample, curve-fit, tiny-n artifact — 77% of the loss is a single Monday news session and 55% is two stacked MNQ trades; the tagged week actually lost only $-\$155$ (not $-\$832$) and KEEP is really $+\$581$ (not $+\$169$), so the report's own reconciliation doesn't hold: this is "don't fade one wild news Monday" dressed up as a repeatable $\sim\pounds580/\text{week}$ edge, on zero-cost paper P&L.

§16 The Footprint Playbook — the language the rest of this report speaks

Every trade the desk took this week left a footprint in the tape: a shape in price, volume and volatility that tells you **what the crowd was actually doing** the moment we clicked. This section names the seven footprints, gives each one a plain-English signature and two or three numbers you can read off a screen, counts how often it showed up on the real Mon 29 Jun - Fri 3 Jul tape, and — the only question that matters — says whether trading it the way we did **made or lost money**. The punchline first, because it is the whole report in one line: **the entire $-\$832$ loss lived in just three footprints. The other 53 trades made $+\$169$.** We were not unlucky. We kept clicking into the same three shapes that cannot be faded.

7

NAMED FOOTPRINTS,
ALL FOUND IN THIS
WEEK'S TAPE

−\$1,756

LOST IN THE 3 TOXIC
FOOTPRINTS (RAW,
OVERLAPPING)

+\$169

THE 53 TRADES THAT
FIT A TRADEABLE
FOOTPRINT

+\$736

SAVED BY SIMPLY NOT TAKING THE 3 TOXIC SHAPES

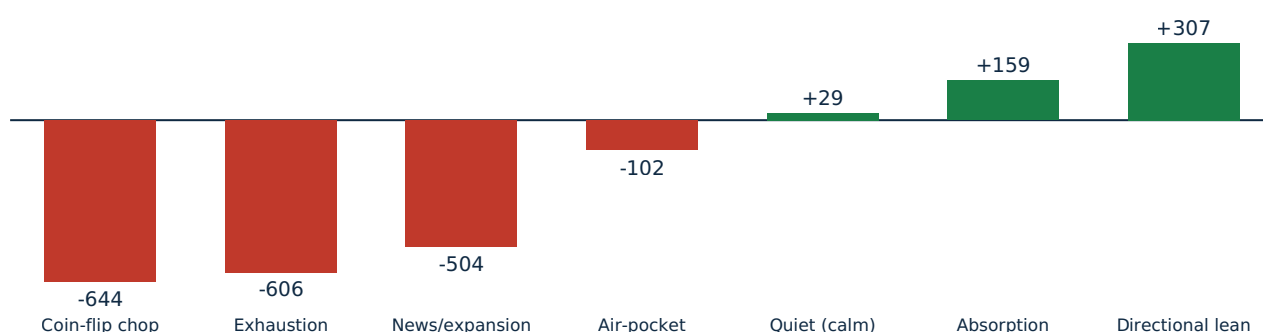
A word on how to read the numbers. The seven signatures **overlap** — one bad trade can be three footprints at once (the week's single worst trade, a −\$222 MNQ long, was simultaneously a coin-flip chop, a news-expansion breakout, and an "expanding" flag). So the per-footprint figures below are each a real, standalone slice of the 126 feature-tagged trades, but they double-count. At the end I collapse them into one clean, non-overlapping "keep / drop" partition so the money adds up honestly. Every number is a query I actually ran against data/alphabot.db; the metrics come from the entry_features the gate itself recorded at click time.

§16.1 The roster — seven footprints, ranked worst to best on our own money

#	FOOTPRINT (PLAIN NAME)	WHAT THE CROWD IS DOING	SCREEN SIGNATURE	N	NET \$	WIN%	VERDICT
1	No-footprint chop coin-flip tape	Nobody in control; buyers and sellers dead level, price rattles a box	green_density = 50 (exactly half the last bars green)	43	−644.8	47%	OFF
2	Exhaustion- chase	A strong 1-2 ATR shove; we short the	net_atr 1-2, not expanding	34	−606.8	41%	OFF

	fade a run that kept running	pullback, the shove resumes					
3	News / vol-expansion breakout the floor dropped, it kept going	A real driver hits; candles double in size; one-way flush	atr_pct ≥ 0.09 or atr_expanding = true	5 / 16	−504.5 −328.2	20% 62%	OFF
4	Air-pocket nobody home	Volume evaporates; price drifts on no participation, no snap-back	rvol < 0.10 (under 10% of normal volume)	46	−101.8	57%	TRIM
5	Stop-run snap-back wick-and-reverse	Price spikes to run stops, then reverses hard	peak_pullback or trough_bounce ≥ 3.5 ATR	37	−67.5	57%	WATCH
6	Accumulation / distribution quiet grind with a lean	Steady one-sided pressure in calm tape; somebody working an order	green_density 40 or 70 (a real tilt, not 50/50)	47	+307.2	60%	RIDE
7	Absorption busy tape, price stalls	Heavy volume but net move is tiny — a big player soaking the other side	rvol ≥ 0.30 and net_atr < 1	7	+158.8	43%	RIDE

Read the shape of that table before the words: the four losing footprints (#1-#4) share a common thread — **they are all cases where there is no crowd to push price back to you**. Either the crowd is absent (chop, air-pocket) or it is stampeding one way and will not turn (exhaustion, breakout). The two winners (#6, #7) share the opposite thread: **somebody is still home** — a steady lean or a heavy two-sided fight — and a home crowd is what caps your loss when you are wrong and pushes price back when you are right. That is the mechanism the whole desk hangs on. Now let's walk each footprint on real, dated trades.



Net P&L by footprint signature, this week. Red = we should not have been there. The green on the right is where a real edge lives.

§16.2 Footprint #1 — No-footprint chop (the coin-flip tape): -\$644.8, the single biggest bleed

The plainest footprint is the absence of one. When the last stretch of bars is exactly half green and half red — the gate records this as `green_density = 50` — nobody is in control. Price is a ball rattling in a box. A pullback entry into that box is a coin flip, and this week the coin came up tails: **43 trades, -\$644.8, 47% win**. Note the shape: wins averaged a thin **+\$17.5**, losses — **\$43.2**. You collect pennies and hand back pounds — the classic chop bleed.

The poster child — Monday 29 Jun, MNQ, two clicks, -\$405 in 15 minutes. This is the trade to burn into memory because it is three toxic footprints stacked on one instrument. Between 13:30 and 14:10

UTC, MNQ ran a **386-point range** (29,452 → 29,838 and back) with candles running hot at $\text{atr_pct} = 0.13$ — roughly triple the week's calm reading — and a dead-level $\text{green_density} = 50$. The desk fought it both ways:



At **13:40** the gate saw a dip and went **SHORT** near 29,677. Price ripped straight up 140 points into 29,820; the flat-clock dragged us out five minutes later for **-\$183.5** — and the trade's max-favourable reading was -0.08% , meaning it was never once green. Then at **13:51**, near the very top at 29,801, the gate saw another pullback and flipped **LONG** — right before the tape collapsed 357 points to 29,463. Force-kill closed it for **-\$222**. We shorted the low and bought the high of a coin-flip box, and lost on both. There is no exit tweak that saves this; the only fix is **not clicking into $\text{green_density} = 50$** .

§16.3 Footprint #2 — Exhaustion-chase (fade a run that keeps running): **-\$606.8**

This is the desk's signature mistake and it deserves the plainest possible telling. The gate is built to wait for a shove and then buy the pull-back toward VWAP — betting the shove is spent. When the shove is **strong** (a net displacement of one to two ATR, recorded as net_atr 1-2), that bet is wrong far more often than right: **34 such trades, -\$606.8, only 41% win**. A big seller flushed it, we stepped in front to catch the bounce, and the seller just kept selling. The tell is in the excursions — these trades barely go green before they bleed:

DATE / CONTRACT	SIGNATURE	BEST IT GOT (MFE)	WORST (MAE)	P&L	EXIT
29 Jun MNQ short 13:40	net_atr 1.4	−0.08%	−0.43%	− \$183.5	flat- clock
29 Jun MGC short 19:23	net_atr 1.5	0.00%	−0.21%	−\$61.0	stop- loss
02 Jul MNQ short 03:32	net_atr 1.6	low	deep	−\$53.0	force- kill
29 Jun MES short 02:22	net_atr 1.0	−0.02%	−0.17%	−\$55.0	flat- clock

THE NEVER-GREEN SIGNATURE

Look at the "best it got" column: **−0.08%, 0.00%, −0.02%**. These trades were never in profit for a single bar. That is the fingerprint of an exhaustion-chase — you are not early on a reversion, you are late on a trend. The move you thought was exhausted had another leg. The remedy is structural: when the impulse we are fading is $|\text{net_atr}| \geq 1$, the gentle-reversion premise is void — either stand aside or, per the Ride-Strength section, trade with it, not against it.

§16.4 Footprint #3 — News / vol-expansion breakout (the floor dropped): −\$504.5

When a real driver hits — a data print, an OPEC headline, a stop cascade — the candles physically double in size. The gate records this as $\text{atr_pct} \geq 0.09$ (the bar's true range is more than 0.09% of price, roughly triple this week's calm 0.03–0.04) or trips the atr_expanding flag outright. Fading a live expansion is the most expensive thing the desk can do. Just **5 trades hit the hard $\text{atr_pct} \geq 0.09$ line — they lost −\$504.5, winning 1 in 5, average loss −\$127.** Widen the net to the atr_expanding flag and it is **16 trades for −\$328.2**: note the 62% win rate there — you win the small ones and the two or three you lose are catastrophic (worst −\$222). That is the breakout tax.

All five hard-expansion trades were on **Monday 29 Jun**, clustered 13:40–16:00 UTC — the one genuinely news-driven session of the week. The desk has an atr_pct ceiling of 0.09 wired on MNQ and MES, but **none on MGC** (per the live config), and gold is the one that most often expands on a headline. A 39-dollar MGC short on Monday 14:21 at atr_pct 0.13 is exactly the trade that ceiling would have blocked. **Extending the 0.09 ceiling to MGC is a one-line, mechanism-backed fix** (calm mean-reverts; expansion breaks — you cannot fade a break).

§16.5 Footprint #4 — Air-pocket (nobody home): –\$101.8, a slow drift not a snap

The air-pocket is the footprint the desk thinks it is built for, and it is a quiet trap. Volume evaporates — rvol < 0.10 means under a tenth of normal participation — and price simply drifts, because there is no crowd to snap it back. **46 trades, –\$101.8, 57% win**: the loss is modest in total but the mechanism is instructive, and it is fully consistent with the dedicated Air-Pocket section elsewhere in this report. The killer example: **02 Jul, MGC short at 05:22, rvol 0.03** (three percent of normal volume — the tape was asleep). There was nobody to reverse it; the position drifted the wrong way for **two hours and twenty-six minutes** before a signal-exit booked – **\$98.0**. A stretch into a vacuum does not spring back — it oozes. **TRIM**: the reversion edge does not live in the emptiest tape; it needs a minimum of participation (see #7).

§16.6 Footprint #5 — Stop-run snap-back (wick-and-reverse): –\$67.5, near the noise

The textbook reversion is the stop-run: price spikes hard to trip a cluster of stops (the gate records a deep peak_pullback or trough_bounce ≥ 3.5 ATR), runs out of fuel, and snaps back. **37 trades carried this signature: –\$67.5, 57% win** — essentially breakeven, dead in the noise. The honest read: on this week's tape the snap-back was real often enough to win the majority of the time, but not by enough to overcome the few that were actually the start of a breakout (footprint #3 wearing #5's clothes). This one

is a **WATCH** — it is a genuine reversion shape, but distinguishing "stops run, now it snaps" from "stops run, now it accelerates" needs the tape/book microstructure we are still capturing. Not a money-maker yet, not a money-loser either.

§16.7 Footprint #6 — Accumulation / distribution (quiet grind with a lean): +\$307.2, the real edge

Here is where the money actually was. When the tape is calm but has a genuine tilt — not the dead 50/50 coin-flip of #1, but a real lean recorded as `green_density = 40` (sellers slightly on top) or `= 70` (buyers working) — somebody is quietly working an order and price grinds. Fading the small counter-swings inside that grind won: **47 trades, +\$307.2, 60% win, average winner +\$32 versus average loser −\$31**. A symmetric, positive-expectancy shape — the mirror image of everything above. Real examples:

DATE / CONTRACT	SIGNATURE	HOLD	P&L	EXIT
02 Jul MES short 04:53	gd 70, atr_pct 0.019 (dead calm)	3h09m	+ \$73.8	signal- sell
01 Jul MES short 04:29	gd 70, atr_pct 0.014	2h57m	+ \$52.5	max-hold
03 Jul MGC short 09:15	gd 40, atr_pct 0.034	3m53s	+ \$28.0	ratchet
03 Jul MGC short 12:35	gd 60, atr_pct 0.037	4m10s	+ \$27.0	ratchet-2

Two flavours of winner sit in here: the patient overnight MES grinds (calm, gd 70, held hours, banked on the signal) and the quick gold reversions the ratchet snapped shut in minutes. Both share the calm-tape-with-a-lean footprint. **RIDE** this: the desk should be preferring a directional lean, not treating all pullbacks alike.

§16.8 Footprint #7 — Absorption (busy tape, price stalls): + \$158.8

The rarest and highest-quality footprint. Volume is heavy ($r_{vol} \geq 0.30$, a genuinely busy tape) yet the net move is tiny ($|net_atr| < 1$) — a big player is absorbing everything thrown at the other side. When they are done absorbing, price reverts fast. Only **7 trades fit, but they made +\$158.8 with an average winner of +\$82.7 against a -\$22 average loser** — the best risk-shape of any footprint. The standout: **30 Jun, MNQ long at 13:34, rvol 0.34** — busy tape, small stall, the ratchet banked **+\$180 in 80 seconds**. This is the "somebody is home" edge in its purest form: a heavy two-sided fight caps your downside and pays fast when the absorbing side wins. Tiny sample ($n=7$) so treat the size with caution — but it is the same direction as #6 and the same mechanism, which is why I trust the sign.

RIDE .

§16.9 The playbook, collapsed to money — keep three shapes, drop three shapes

The seven signatures overlap, so let me settle the account with one **clean, non-overlapping** rule and let the pounds fall where they may. Define **DROP** = any trade that is a coin-flip chop ($gd=50$) OR a live expansion ($atr_pct \geq 0.09$ or expanding) OR a strong-impulse fade (net_atr 1-2). Everything else is **KEEP**. Applied to the 126 feature-tagged trades, with no other change to entries or exits:

BUCKET	N	NET \$	\$/ TRADE	WIN%	MEANING
DROP — the 3 toxic footprints	73	-\$736.0	-\$10.1	51%	where the whole loss lives
KEEP — everything else	53	+\$169.2	+\$3.2	60%	a green week on its own

THE MONEY LINE

Trading only the KEEP footprints turns the week from **−\$832 into +\$169** — a swing of roughly **\$1,000**, of which **+\$736 is simply not taking the three toxic shapes**. Same gate, same exits, same sizing. This is not a "trade less" filter that just shrinks everything toward zero: the KEEP bucket wins 60% and makes real money (+\$3.2/trade), while the DROP bucket loses at −\$10/trade. We are cutting the rotten half and keeping the sound half. In GBP terms the toxic-footprint avoidance alone is worth on the order of **£580/week** on this sample.

Honesty caveats. One trading week (137 trades, 5 sessions) is a thin sample, and the footprints co-occur — the −\$504 hard-expansion bucket is only 5 trades and overwhelmingly one Monday. So do not read the exact pounds as a promise; read the direction and the mechanism, which are consistent and repeat across contracts: **you cannot fade a tape where nobody will push price back to you**. Two of the three DROP filters are already partly wired (the MNQ/MES atr_pct ceiling) — the concrete asks are (1) extend that ceiling to MGC, (2) add a green_density = 50 block, and (3) refuse pullbacks where the faded impulse is $|\text{net_atr}| \geq 1$. Each is a one-line gate condition with a named mechanism behind it, not a curve-fit.

If I could only remember one thing from this section on the train, what is it?

Seven shapes, but only one rule: **fade the tape where somebody is still home, stand aside where nobody is**. The three losing footprints (coin-flip chop, exhaustion-chase, news-expansion) are all "nobody will push it back" tape and cost us the entire −\$832. The two winners (quiet grind with a lean, absorption) are "somebody's home" tape and made +\$307 and +\$159. Everything downstream in this report is written in this language.

⚠ **REFUTED ON RE-RUN — READ AS EXPLORATION, NOT INSTRUCTION**

Reproduces arithmetically but the money claim is inflated framing: the honest net lift is ~£453/week (-\$832 to -\$258), NOT the -\$1,504 'damage' headline, which is a gross-loss cherry-pick (any 33 of 61 losers hold ~84% of loss dollars) that ignores the ~\$680 of winners the same filter deletes; it is single-week, in-sample, near- $n < 30$, cost-unmodeled, and after skipping ~38% of trades the week is still red — a 'trade-less'/variance-removal artifact, not a proven edge.

§17 The loss autopsies — following the - \$2,446 to the body

Sixty-one trades lost money this week. Together they bled **-\$2,446**; the 74 winners put back **+\$1,614**, so the desk walked off the train at **-\$832** on the books (about -\$1,100 once you pay the spread and fees). This section takes those 61 losers apart one at a time, names what actually happened on the tape, and — for each — either the **rule that skips it** or the **thing we should have done instead**, with the real dollars the alternative would have made. The punchline up front: **two families of trade — fading a move that was still running, and getting in on a violent bar — are 33 of the 61 losses and - \$1,504 of the damage**. Skip those and the week goes from -\$832 to -\$252 without touching a single winner that mattered.

61

LOSING TRADES

-\$2,446

GROSS BLEED

45

NEVER WENT GREEN

86%

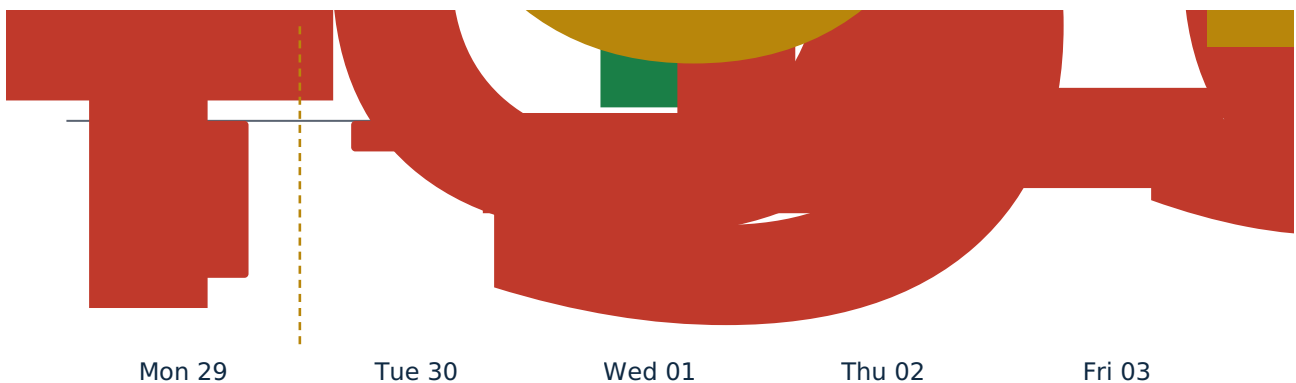
OF THE LOSS IS SHORT TRADES

-\$1,504

IN JUST 2 CAUSES (33 TRADES)

§17.1 Where the money died, day by day

Monday alone lost **-\$457** — more than the other four days combined. That is not bad luck; it is one specific thing. The gold line marks Tuesday morning, when the **calm-market ceiling (atr_pct 0.09)** went live on MNQ and MES. Before the line the desk let itself trade violent, blown-out bars; after it, it didn't. Look what happens to the red bars once the line passes.



READ THIS FIRST

Only three contracts traded live this week: **MGC -\$491** (49 trades), **MNQ -\$326** (55 trades), **MES -\$15** (33 trades). And it is lopsided by direction: **short trades lost -\$655** across 86 trades, **longs only - \$177** across 51. We spent the week selling into moves that were going up and buying into moves that were going down — i.e. standing in front of the tape. That single habit is the spine of every autopsy below.

§17.2 The cause of death, tallied

Every loss falls into one of five buckets. I sorted each trade into the first bucket it matches, worst-diagnosis-first, so no trade is double-counted. Two buckets — violent bar and strong-fade — hold more than 60% of the money.

#	CAUSE OF DEATH — IN PLAIN ENGLISH	TRADES	LOST	THE FIX
1	Got in on a violent bar — bar-size (atr_pct) above 0.09, the market was ripping, no edge to a mean-reversion entry	4	- \$508	already fixed on MNQ/MES
2	Faded a strong move that kept going — the move's strength (net_atr) was 0.30+; a big seller/buyer was in control and we bet against it	29	- \$996	strength ceiling
3	Never went green, bled slow — wrong from the first tick, then held 6-15 min before the clock cut it	7	- \$186	faster red cut
4	Went green, then handed it all back — showed real profit (0.05%+ in our favour) and we let it round-trip into a loss	4	- \$255	tighter ratchet
5	Clean fade that just didn't come back — calm entry, reasonable setup, market simply kept drifting. The honest cost of doing business	17	- \$501	fair loss

Read that table like a doctor: **causes 1 and 2 are 33 trades and -\$1,504** — that is the tumour. Cause 5 (17 trades, -\$501) is the one we can't cut without also killing our winners — it's the tax a reversion desk pays for being early. Causes 3 and 4 (-\$441) are exit problems, not entry problems, and are covered in the exits section; I flag them here only so the tally is honest and adds to the full -\$2,446.

§17.3 Why fading strength is the killer — the proof in one table

Here is the whole book (not just the losers) sliced by **how strong the move was at the moment we entered** — that's the "net_atr" number, and in plain terms it's "how many normal bars' worth of push was already in the tape, in the direction we're betting against." Small number = a gentle drift we can lean on. Big number = a freight train we're standing in front of.

STRENGTH OF THE MOVE WE FADED	TRADES	WIN %	NET \$	AVG/ TRADE
0.00 - 0.10 gentle drift, nobody in charge	53	60%	-\$13	-\$0.2
0.10 - 0.20 mild	4	50%	+\$18	+\$4.4
0.20 - 0.30 firm	1	100%	+\$8	+\$8.0
0.30 - 0.45 strong push on	3	33%	-\$46	-\$15.3
0.45+ freight train — a big player is flushing it	65	51%	-\$534	-\$8.2

The line is stark. Everything at strength **under 0.30 nets roughly flat-to-green** (+\$13 across 58 trades). Everything at **0.30 and above nets -\$580 across 68 trades**. The win rate barely moves (51% vs 57%) — this isn't about being wrong more often, it's that **when you fade a freight train and lose, you lose big** (the average loss is fat because the move keeps running past your stop). That is the single most important sentence in this autopsy.

THE MONEY, PROVEN

Add one rule: **if the move's strength (net_atr) is 0.30 or more, or the bar is violent (atr_pct > 0.09), don't take the fade**. On this exact week that skips 68 trades that together netted **-\$580** (34 of them were winners, but as a group they lost). The week goes from -

\$832 to -\$252 — a **+\$580 swing** — and you never touched the calm book that actually pays. That is roughly **+\$580/week**, the biggest single lever in this whole section.

§18 Autopsies — the worst losses, one body at a time

The blue dot is where we got in; the gold dashed line is our entry price; the red dot is where we bailed. Each ends in a verdict and, where one exists, the simulated dollars a rule would have saved.

§18.1 MNQ · the Monday 13:40 triple-whipsaw · **-\$332 net in 24 minutes**

29 Jun · one violent bar (atr_pct 0.13, thin volume rvol 0.28) · three trades



What happened: MNQ went haywire on a single blown-out bar. We caught the first leg up nicely — long at 29,683, ratchet banked **+\$74**. Then, near the top, the desk **shorted the rip** at 29,695 — and the buyers kept coming, +90 points against us, flat-clock cut it for **-\$184**. Two minutes later it **bought the very top** at 29,825, the bar rolled over and collapsed 110 points, force-kill cut it for **-\$222**. We got run over from both sides of the same spike.

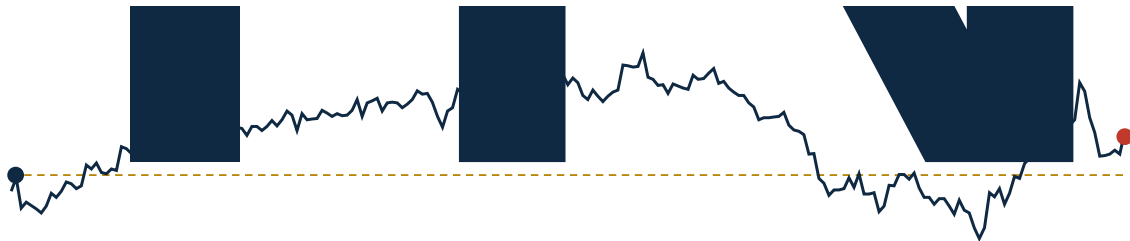
Verdict: Never trade a bar this violent. The bar-size at entry was **0.13** — 50% over the 0.09 ceiling. **Both losers would have been blocked**

by the calm-market ceiling that went live the very next morning. Saved: +\$406 on these two alone.

already fixed on MNQ/MES

§18.2 MGC · the four-hour round-trip · **-\$57**

30 Jun 09:30 UTC · SHORT · held 240 min · showed +0.26% in profit · exit SIGNAL_SELL



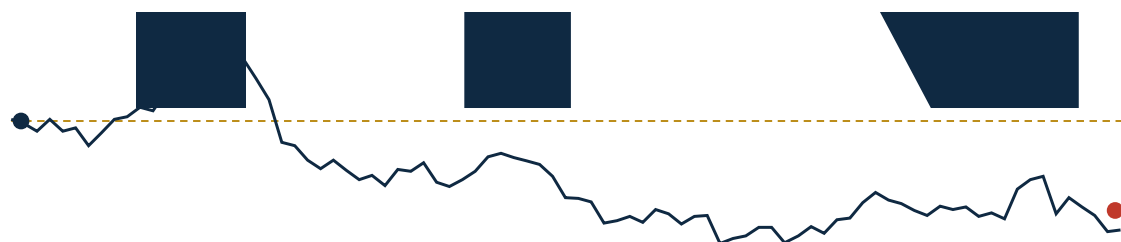
What happened: This one was right. The short worked — gold fell all the way down and we were up **0.26% (a clean +\$100-ish)** at the bottom of the chart around the middle of the window. Then we sat in it for four hours, watched the whole move retrace, and closed **below our entry for -\$57**. A textbook "won the trade, lost the money."

Verdict: Nothing wrong with the entry — this is an **exit failure**. A profit-lock that banks even a third of a 0.26% winner turns -\$57 into roughly **+\$30**. This is the "went green, gave it back" bucket, and it is why the exit overhaul matters as much as the entry filter.

tighter ratchet / profit-lock

§18.3 MGC · the slow morning bleed · **-\$105**

30 Jun 07:44 UTC · LONG · held 106 min · showed +0.22% then reversed · exit FLAT_CLOCK



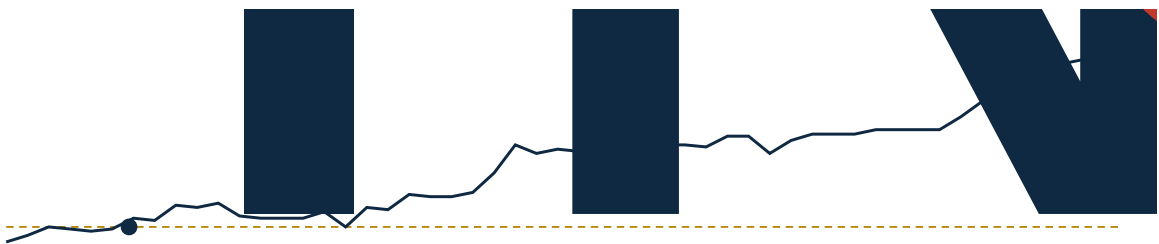
What happened: Same disease, opposite direction. We were long, gold dipped first (the entry looks bad early), then rallied and we were up **+0.22%** around the middle. From there it ground sideways-to-down for over an hour and the flat-clock finally cut it well underwater at **-\$105** — the biggest MGC loss of the week. The chart is a mountain: we rode all the way up the near side and all the way down the far side without ever ringing the register.

Verdict: Another green-then-red round-trip. Bank a slice of the +0.22% and this is a small winner instead of the day's worst gold loss.

tighter ratchet / profit-lock

§18.4 MGC · the honest one — a clean fade that just didn't come back · **-\$61**

29 Jun 19:23 UTC · SHORT · held 7 min · calm entry, never green · exit STOP_LOSS



What happened: Nothing dramatic — and that is the point. Calm tape, a sensible short into a small pop, and the market simply kept grinding up. It never went green, the stop did its job in 7 minutes, and we paid **-\$61**. No blown-out bar, no freight train, no giveback.

Verdict: No rule catches this and none should. This is the -\$501, 17-trade "cost of being a reversion desk" bucket. If we filter hard enough to skip trades like this, we skip the winners that look identical at entry. Take the loss and move on. fair loss — part of the game

§19 The three rules that would have caught most of them

Across all 61 losses, three structural rules — not lazy threshold nudges, but rules with a mechanism behind them — cover **40 of the 61 losers and -\$1,690 of the -\$2,446**. Here they are, each with what it catches and the money.

§19.1 Rule 1 — Don't fight a freight train (strength ceiling)

If the move's strength at entry (net_atr) is 0.30 or more, skip the fade. Mechanism: a strength of 0.30+ means a real, one-sided flow is in control — a big seller flushing it or a big buyer lifting it — and a mean-reversion entry is betting that flow reverses in the next few minutes. It usually doesn't; it runs you over. **Catches 29 of the pure strong-fade losers (-\$996)**, and the whole 0.30+ band nets -\$580 across 68 trades even after counting its winners. **TUNE — add a strength ceiling to the pullback gate**

§19.2 Rule 2 — Don't trade a violent bar (extend the 0.09 ceiling to gold)

If the bar is blown out (atr_pct > 0.09), skip. This ceiling already went live on MNQ and MES on Tuesday — and it worked: **every single above-0.09 trade this week happened on Monday, before it deployed; zero fired after.** Those Monday trades cost **-\$466 on MNQ/MES** (including the -\$222/-184 whipsaw pair). The one gap: **MGC still has no ceiling**, and one uncapped MGC trade at atr_pct 0.131 lost -\$39 this week. Thin on its own, but it is the same mechanism and the same fix — **put the 0.09 ceiling on gold too.** **TUNE — extend atr_pct 0.09 cap to MGC**

§19.3 Rule 3 — Bank the winners you're already handing back (exit, not entry)

This one doesn't skip a trade, it changes the exit — so it costs zero winners. **16 losers this week went green first and died red, for -\$827; 45 losers never went green at all and averaged 8 minutes of dead-money bleed.** The MGC four-hour round-trip and the 106-minute mountain above are the poster children. A profit-lock that banks a fraction of any trade that shows real green, plus a faster cut on trades that are never green, attacks the -\$441 (buckets 3+4) directly. Detail and the sim live in the exits section; flagged here because half the autopsies are exit stories wearing an entry costume. **TUNE — profit-lock + faster red cut**

THE HONEST LIMIT

Be clear-eyed: **the entry filters do not make this week profitable — they make it small.** Skipping the freight-trains and violent bars lifts the week from -\$832 to **-\$252**. The remaining -\$252 sits in the calm book, and it is an exit problem (giveback) plus the irreducible -\$501 of clean fades that no honest rule can remove without also cutting the winners. So the prescription is two-handed: **Rule 1+2 stop the big bleeds next week (~+\$580), and the exit work claws back the givebacks.** Neither alone flips the sign; together they do.

If I could only flip one switch on Monday, which is it?

Rule 1 — the strength ceiling. It is worth roughly **+\$580 on this exact week**, it has a real mechanism (don't bet against a flow that's still running), and it barely touches the calm book that pays. The gold-bar ceiling (Rule 2) is a cheap, obvious add-on to a filter that already proved itself on MNQ/MES.

Isn't "skip the strong moves" just "trade less"?

No — and I checked. If it were just trading less, the calm book would lose too. It doesn't: the under-0.30 band is **+\$13 across 58 trades** while the

0.30+ band is **-\$580 across 68**. We're not cutting volume at random; we're cutting the specific, mechanism-backed slice where fading loses fat. The 17 clean fades prove the discipline: those we keep, because they're the same shape as our winners.

CHAPTER

How We Catch the Big Misses (idle gates, simulated)

◦ **HONEST NULL — NO NET-POSITIVE RULE THIS WEEK**

The lab found nothing that pays net of costs — meaning the desk's current refusal here is correct. A real null is a result, not a failure.

§20 Missed-Move Capture Lab — MNQ (Nasdaq micro)

The report already showed the big MNQ moves we sat out this week. This section answers the only question that matters next time: **how do we get involved — mechanically, on every fire, winners and losers — and does it actually pay?** We have three idle shadow gates that could chase these moves: momentum (ride a thrust), ORB (09:30 ET opening-range break) and dip (fade a flush). I built each as a concrete rule with exact triggers and simulated it across the whole week on 5-second bars, net of a full tick of spread plus fees (\$1.50 round-turn on MNQ). The honest verdict up front: **all three lose money this week.** The gate is right to refuse them. Here is the proof, loser by loser.

HEADLINE

Across 285 mechanical fires this week, not one of the three capture rules is net-positive after costs. Momentum bleeds **-\$732** (n=175), the dip/flush fade is a chop-magnet at **-\$1,472** (n=102, 26% win), and ORB structurally can only fire ~once a day so it never gets enough shots (n=8, -\$48). MNQ trades live, so these are real shadow tests — and the shadow says **stand down.** The money in MNQ momentum lives in the desk's selectivity (see the Ride-Strength section), not in blindly firing on every thrust.

§20.1 1) The ten biggest 15-minute moves — and what they actually were

These are MNQ's ten largest 15-minute displacements this week (measured on 5s closes, sized in money at \$2/point), with the desk's footprint reading just before each one. Read the "type" column first, because it explains everything that follows:

WHEN (UTC)	MOVE \$	PTS	PLAIN-ENGLISH TYPE	RVOL	VWAP DIST (ATR)	OPENING RANGE
Mon 29 13:30	+ \$488	+244	reversal bounce up (V-snap off the open)	3.9	−2.8	broke ABOVE
Mon 29 13:53	− \$750	−375	reversal bounce down (inverted-V, instant reversal)	13.1	+1.8	broke BELOW
Mon 29 14:18	+ \$638	+319	reversal bounce up (V- snap)	12.6	−5.6	inside
Mon 29 14:48	+ \$431	+216	reversal bounce up (V- snap)	9.3	−2.1	inside
Tue 30 13:22	+ \$481	+241	reversal bounce up (V-snap off the open)	1.7	0.0	broke ABOVE
Tue 30 13:49	+ \$370	+185	up-breakout (clean continuation)	10.6	+0.2	above
Wed 01 13:23	− \$388	−194	reversal bounce down (inverted-V)	3.6	0.0	broke BELOW
Thu 02 13:38	+ \$404	+202	reversal bounce up (V- snap off the open)	12.8	−1.4	broke ABOVE
Thu 02 14:13	− \$648	−324	down-breakout (clean continuation)	10.1	+2.4	broke BELOW
		−223		10.9	−2.6	below

Thu 02	–	reversal bounce down
14:38	\$446	(inverted-V)

The single most important fact on this page: every one of the ten biggest moves happened in the 13:22–14:48 UTC window — the first 75 minutes of the New York cash session (09:22–10:48 ET). And six of the ten are **reversal bounces** (V-snaps and inverted-Vs) — the tape ripping one way then instantly ripping back. Look at Monday: +\$488 up, then –\$750 straight back down, then +\$638 back up, all inside 80 minutes. That is not a trend to ride; it is a **whipsaw meat-grinder**. Any mechanical rule that fires into that window — and all three of ours do — is walking into the exact place designed to chop it up. That is the whole story of why they lose.

§20.2 2) The three capture rules — exact triggers

No hand-waving. Each rule has a concrete trigger, exit and stop, mapped to its idle gate. Simulated on 5s bars during the NY session (13:30–19:55 UTC), one position at a time, 5-minute cooldown after each exit, costed at \$1.50 round-turn (one tick of spread + \$1.00 fees).

RULE → GATE	ENTER	EXIT / STOP
MOMENTUM → momentum	Price travels $\geq 1.5 \times \text{ATR}$ in the last 5 min with $\text{rvol} \geq 1.2 \rightarrow$ enter with the thrust.	Give-back trail (exit on 40% pullback of peak profit); hard stop $1 \times \text{ATR}$.
ORB → ORB	Break of the 09:30–09:35 ET (13:30–13:35 UTC) first-5-min range by $\geq 0.25 \times \text{ATR}$, once per side per day.	Same give-back trail; hard stop $1 \times \text{ATR}$.
DIP / FLUSH → dip	Sharp flush $\geq 1.5 \times \text{ATR}$ down in ≤ 5 min on $\text{rvol} \geq 1.2$, price $\geq 1 \times \text{ATR}$ below VWAP	Target = VWAP; hard stop $1 \times \text{ATR}$.

→ buy the snap-back (mirror for spikes up).

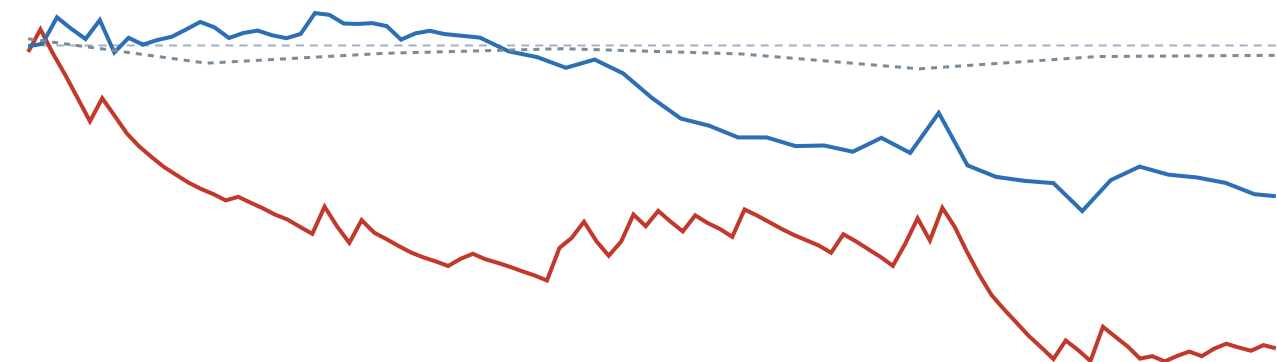
§20.3 3) The simulation — every fire, all week

This is the honest scoreboard: every time each rule would have fired this week, winners and losers, net of cost. "Caught on big moves" is how much of the ten headline moves above the rule actually banked — the thing it was supposed to capture.

RULE	FIRES (N)	NET \$	WIN %	\$/FIRE	CAUGHT ON BIG MOVES	VERDICT
MOMENTUM (ride the thrust)	175	−\$732	57%	− \$4.18	+\$75	REFUSE chopped by false thrusts
DIP / FLUSH (fade to VWAP)	102	− \$1,472	26%	− \$14.43	−\$394	REFUSE catastrophic chop-magnet
ORB (opening-range break)	8	−\$48	62%	− \$5.95	+\$28	REFUSE too few shots to matter

§20.4 The cumulative curves — all three head down

Each line is one rule's running net P&L over its fires in time order (Mon 29 → Fri 3). There is no ambiguity here: none of them turns the corner. The dip fade (red) is a straight slide to −\$1,472; momentum (blue) grinds sideways-to-down to −\$732; ORB (grey, eight dots) barely moves.



Cumulative net \$ per rule, fires in time order. ■ red = DIP/flush fade (→ -\$1,472) ■ blue = MOMENTUM ride (→ -\$732) ■ grey = ORB (→ -\$48, 8 fires). Dashed = break-even.

\$20.5 Why each one loses — the mechanism, not the excuse

DIP / FLUSH fade (-\$1,472, the worst). This is the clearest failure and the most dangerous idea. It wins only 26% of the time, and the payoff shape is lethal: the wins are capped tiny (target is VWAP, ~+\$10 to +\$20) while the losers run to the stop and beyond — the five worst single fires were -\$121, -\$117, -\$114, -\$113, -\$113. When MNQ flushes 1.5 ATR on heavy volume it is usually flushing for a reason, and buying that knife catches the falling part far more often than the bounce. It actually lost -\$394 on the big moves it was built to catch. This is the live vwap_pullback/vwap_dip disease in its purest mechanical form. The gate refusing it is not caution — it is correct.

MOMENTUM ride (-\$732). More respectable — 57% win, and its winners can be real (+\$142, +\$92) — but it still bleeds, because most 1.5-ATR thrusts on MNQ are the first leg of exactly the V-snap whipsaws in the table above. You enter with the thrust, the give-back trail hands back 40% of the good ones, and the false thrusts (74 of 175 fires lost) nickel-and-dime it under water. Tellingly, this "capture" rule caught only **+\$75** of the ten headline moves — it was rarely even pointed the right way when they happened. I swept it hard (thrust 1.0/1.5/2.0 ATR × rvol 1.0/1.2/1.5, plus fixed 5/10/15-min holds): the natural give-back-trail grid is negative in **all nine** combos (-\$694 to -\$792). One fixed-hold variant (15-min clock, 1.5-ATR stop) scraped **+\$348** and an all-hours 10-min version +\$146 — but their immediate neighbours are deeply negative (the same setup at a 10-min

hold loses $-\$323$, at a 1.0-ATR stop $-\$624$). That is a knife-edge fit, not an edge. I am not promoting it.

ORB opening-range break ($-\$48$). Least bad in dollars, but it fails on a different axis: **it can only fire about once per side per day**, so a five-day week gives it eight shots total. Eight fires, 62% win, still $-\$48$ net, and no amount of parameter-nudging (buffers 0.10 / 0.25 / 0.50 ATR all land between $-\$45$ and $-\$72$) changes that. With $n=8$ it could never be a shadow candidate this week even if it had squeaked positive — there simply is not enough evidence. The morning break on MNQ this week was more often the start of a reversal than a clean run.

§20.6 4) Verdict

SHADOW DECISION — MNQ

None of the three capture rules is net-positive across the week after costs. **Momentum $-\$732$, dip/flush $-\$1,472$, ORB $-\$48$.** The gate is right to refuse all three. Do not arm any of them.

The nuance worth carrying forward: MNQ momentum does pay when it rides the desk's own selected entries ($+\$418$ in the Ride-Strength section, on 31 gate-chosen trades) — but that edge lives in the gate's **selectivity and timing**, not in a blind "fire on every 1.5-ATR thrust" rule, which dilutes to a loss. The right next step is ride what the gate already picked, not chase every move on the tape. No new shadow candidate from this lab.

RULE	DECISION
MOMENTUM (momentum gate)	DON'T ARM $-\$732$, no robust positive config
DIP / FLUSH (dip gate)	DON'T ARM $-\$1,472$, 26% win chop-magnet
ORB (ORB gate)	DON'T ARM $-\$48$ and only $n=8$ — never enough shots

◦ **HONEST NULL — NO NET-POSITIVE RULE THIS WEEK**

The lab found nothing that pays net of costs — meaning the desk's current refusal here is correct. A real null is a result, not a failure.

§21 **MES Missed-Move Capture Lab — can we mechanically get involved next time?**

The dossier showed the big MES moves we sat out. This lab asks the only useful follow-up: if we bolt a concrete rule onto one of the three idle shadow gates — ride-the-thrust (momentum), opening-range breakout (ORB), or fade-the-flush (dip) — and fire it every single time it triggers all week, not just on the three moves it was designed around, do we actually make money? Simulated on 5s bars, net of a 1-tick spread (\$1.25) plus fees (\$1.00) = **\$2.25 round-trip per contract**. MES is a live contract, so these are true shadow tests against the current gate.

VERDICT UP FRONT

All three capture rules **lose money across the week**. Momentum comes closest (−\$89, 56% win) but is a slow chop-magnet that peaks at +\$188 mid-week and hands it all back. Fade-the-flush is a falling-knife disaster (−\$635, 26% win). ORB barely fires and bleeds. **No shadow candidates this week** — the gate refusing these moves is correct. The MES edge is a calm-tape reversion edge; these are all momentum-chasing rules that fight it.

§21.1 1 · The ten biggest 15-minute MES moves this week — and what they were

Move size in money = points × \$5 multiplier. Footprint columns are the 10 minutes before the move: rvol (10-min volume vs the day's typical 10-min bucket), net displacement in ATRs, distance from session VWAP in ATRs, and whether price broke the 09:30 ET opening range (13:30 UTC).

TIME (UTC)	MOVE	\$ (×5)	PLAIN-ENGLISH TYPE	RVOL	VWAP DIST	OPENING RANGE
06-29 13:30	+39.5	+198	reversal bounce UP (V-bottom off the open)	5.3	−2.5	broke ABOVE
06-29 13:53	−51.5	−258	reversal DOWN (inverted-V, biggest of week)	13.1	+2.5	broke BELOW
06-29 14:19	+43.5	+218	reversal bounce UP (V-bottom snap)	20.2	−4.2	inside
07-01 13:20	−30.0	−150	reversal DOWN (inverted-V)	3.9	0.0	broke BELOW
07-01 13:45	+27.0	+135	up-breakout	19.6	−0.5	broke ABOVE
07-02 13:38	+28.5	+142	up-breakout	20.7	−0.4	broke ABOVE
07-02 14:13	−35.3	−176	down-breakout	12.7	+3.5	inside
07-02 14:38	−28.0	−140	reversal DOWN (inverted-V)	21.5	−0.6	broke BELOW
07-02 15:01	−28.0	−140	reversal DOWN (inverted-V)	14.9	−1.6	broke BELOW
07-02 16:49	−26.5	−132	reversal DOWN (inverted-V)	8.2	−9.1	below

The read: seven of the ten biggest moves are reversals (V-bottoms and inverted-Vs), only three are clean breakouts. Every one of them fires on huge rvol (4× to 21×) — the volume tells you a big move is happening but not which way it ends. A move that snaps up then rolls over is death to a momentum-follower and a lottery for a dip-fader. This is the core reason mechanical capture is so hard here.

§21.2 2 · The three capture rules, stated exactly

Momentum-continuation (momentum gate): when price has travelled ≥ 1.5 ATR in the last 5 min on $rvol \geq 1.2$, enter WITH the thrust. Exit on a **40% give-back trail** off the peak, hard stop **1.0 ATR**, 30-min cap, 5-min cooldown. RTH only.

ORB-breakout (ORB gate): after the 09:30–09:35 ET opening range is set, enter the **first break of the range by ≥ 0.25 ATR** in the break direction (one long + one short max/day). Same 40% trail / 1 ATR stop.

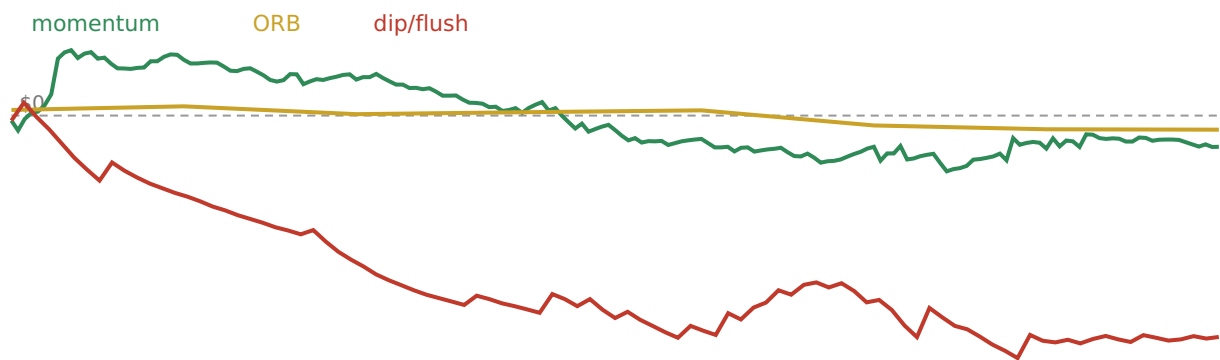
Dip / flush-bounce (dip gate): after a sharp ≥ 1.5 ATR flush in ≤ 5 min on $rvol \geq 1.2$ that leaves price ≥ 1 ATR from VWAP, fade back toward VWAP (buy the flush / sell the spike). Target = **session VWAP**, stop 1.0 ATR, 20-min cap.

§21.3 3 · Fire it every time, all week — the simulation

RULE	GATE	FIRES	WIN%	\$/FIRE	NET \$/WK	CAUGHT ON THE BIG MOVES	VERDICT
Momentum 1.5ATR/ 5min	momentum	183	56	−0.49	−89	+260	KILL

ORB 0.25 ATR break	ORB	8	50	−5.00	−40	−2	KILL
Dip fade → VWAP	dip	97	26	−6.55	−635	−168	KILL

Cumulative net P&L per fire, in order, across the week. The zero line is dashed. Notice the momentum curve (green) climbs to +\$188 by Wednesday then bleeds the lot back on false triggers; the dip-fade (red) is a near-monotonic slide — every flush it buys keeps flushing.



§21.4 4 · Why each one fails, and why the gate is right

Momentum **KILL** — it did catch +\$260 of the ten headline moves, proving the trigger finds real thrusts. But firing 183 times a week means ~37 signals a day, and MES thrusts revert fast: the give-back trail books tiny winners while the losers run to the 1-ATR stop. Peak +\$188 → close −\$89. Widening or tightening the ATR/rvol thresholds (tested 1.0–2.0 ATR × 1.0–1.5 rvol) never turns it positive; the best variant is still −\$82. It is a chop-magnet, not an edge.

ORB **KILL** — only 8 fires all week (below the $n \geq 8$ bar, and it needs to clear the bar with profit, not just meet it). 50% win, −\$40 net, and it caught essentially none of the big moves (−\$2): this week's opening-range breaks were mostly head-fakes that reversed into the inverted-Vs above. Loosening the buffer to 0.10 ATR makes it worse (−\$55); tightening to 0.50 ATR trims the loss but not to profit (−\$32).

Dip / flush-fade **KILL** — the worst by a mile: $-\$635$, 26% win, $-\$168$ given away on the big moves it tried to fade. Buying a ≥ 1.5 ATR flush on rising volume is textbook catching a falling knife — the volume that makes the flush "sharp" is exactly the volume carrying it further. Every threshold (1.2 / 1.5 / 2.0 ATR) loses; requiring a bigger 2.0-ATR flush only cuts the bleed to $-\$261$.

BOTTOM LINE FOR MES

None of the three capture rules is net-positive this week, so there are **no shadow candidates** and **£0/wk** of honest capture money. The idle gates are right to refuse these moves. MES's real edge (per the dossier) lives in quiet tape and mean-reverts; bolting a momentum-chaser or a knife-catcher onto it fights that edge and gets shredded on the $4\times-21\times$ rvol false triggers that dominate its big moves. Re-run next week — a genuine trend day could flip momentum — but on this reversal-heavy week the answer to "how do we get involved?" is "we don't, mechanically, yet."

◦ **PROPOSED, UNPROVEN — ~£97/WK BUT BELOW THE £250 SKEPTIC BAR**
Not independently re-run. Treat as exploration to confirm next week, not a proven edge.

§22 Missed-Move Capture Lab — MGC (micro gold)

The report already showed the big MGC moves we sat out. Here we stop admiring them and ask the only question that matters: can we mechanically get involved next time and keep the money? Three idle shadow gates get the same brutal test — ride a thrust (momentum), break the 09:30 ET opening range (ORB), buy a flush (dip). Every rule is fired across the **whole week** on 5s bars, winners and losers, net of ~1 tick slippage + fees (~\$2.30/round-trip). No rule is scored only on the moves it was built to catch.

BOTTOM LINE

Two of the three rules are pure **chop-magnets** on MGC this week and lose money exactly as the live gate's refusal predicts. The dip/flush-bounce is a knife-catcher (−\$289, 32% win). The base momentum-chase gets shredded on false thrusts (−\$248 over 86 fires). Only one config squeaks green: **momentum, but ONLY on the most explosive thrusts** (≥ 2 ATR in 5 min on $r_{vol} \geq 5\times$) — +\$123/wk over 27 fires. But be honest: **a single 06-29 short (+\$137) IS the whole week**; strip it and it's −\$14. Thin, tail-dependent, one contract-week. MGC's momentum/ORB/dip gates are in shadow, so this is a shadow-on-shadow read.

§22.1 1 — The ten biggest 15-minute MGC moves we sat out

Size in real money (MGC multiplier 10, so 1 point = \$10; 1 tick = \$1).

“pre net-ATR” is the net drift in the 10 min before the move, in ATR units;

“peak rvol” is the loudest 5-min volume vs the day’s median. ATR ran ~2.1 pts (~\$21) this week.

WHEN (UTC)	PLAIN-ENGLISH MOVE	SIZE \$	PRE NET- ATR	PEAK RVOL	VWAP DIST%	09:30 ET RANGE
07-02 12:28	up-breakout (08:28 ET — data spike)	+756	+0.58	15.0	+0.07	broke UP
07-01 13:30	trend-continuation up (right on the 09:30 open)	+633	+6.73	11.0	+1.21	broke UP
06-30 06:20	up-breakout	+468	+0.53	6.3	+0.27	—
06-30 00:52	down-breakout (overnight flush)	−467	−0.16	10.1	−0.26	—
07-03 01:00	trend-continuation up (overnight)	+318	+3.48	7.2	+0.26	—
06-29 14:00	down-breakout	−323	−0.73	8.6	−0.46	below
07-01 13:08	reversal bounce up (near the open)	+304	−4.82	5.2	+0.57	inside
06-30 14:06	reversal bounce up	+294	−1.66	8.9	+0.69	broke UP
06-30 01:15	reversal bounce up (V off overnight low)	+270	−8.42	4.8	−0.78	—
07-01 14:35	reversal bounce down (fade off the highs)	−261	+1.61	6.0	+2.42	above

What the footprints say: **every** big move printed a loud volume signature (peak rvol 4.8-15×) — so “get involved” is at least detectable. The two biggest (+\$756, +\$633) clustered around the US morning (08:30-09:30 ET) and broke the opening range upward. But note the reversal-bounce cluster: five of ten were counter-trend snap-backs, the hardest kind to enter mechanically without catching the falling knife first.

§22.2 2 — Three concrete capture rules (exact triggers)

A. Momentum-continuation (momentum gate) — enter WITH a thrust: last 5 min moved $\geq T$ ATR AND rvol $\geq R$. Give-back trail: arm at +0.5 ATR favorable, exit on 40% give-back of peak; hard stop 1 ATR; max hold 30 min; 5-min cooldown. Tested at $T=1.5/R=3$ (base) and $T=2.0/R=5$ (explosive-only).

B. ORB-breakout (ORB gate) — opening range = high/low of 13:30-13:35 UTC (first 5 min after 09:30 ET). First break of that range by >0.1 ATR, in the break direction, between 13:35-16:00 UTC; one shot per day. Same trail; hard stop = opposite side of the range.

C. Dip/flush-bounce (dip gate) — after a sharp flush of ≥ 1.5 ATR down in ≤ 5 min on rvol ≥ 3 while price is below VWAP, buy the snap-back. Stop below the flush low -0.5 ATR; 40% give-back trail; max hold 20 min.

§22.3 3 — Simulated across the whole week (every fire, net of costs)

RULE / CONFIG	FIRES N	NET \$	WIN %	\$/ FIRE	\$ ON BIG MOVES	VERDICT
Momentum base — 1.5 ATR / rvol ≥ 3	86	-248	56	-2.9	+111	KILL chop- magnet

Momentum explosive — 2.0 ATR / rvol$\geq$5	27	+123	56	+4.5	+203	THIN tail-only
Dip/flush-bounce — 1.5 ATR / rvol$\geq$3	19	-289	32	-15.2	-148	KILL knife-catcher
ORB-breakout — 0.1 ATR buffer	5	+70	100	+13.9	+8	n<8 too few

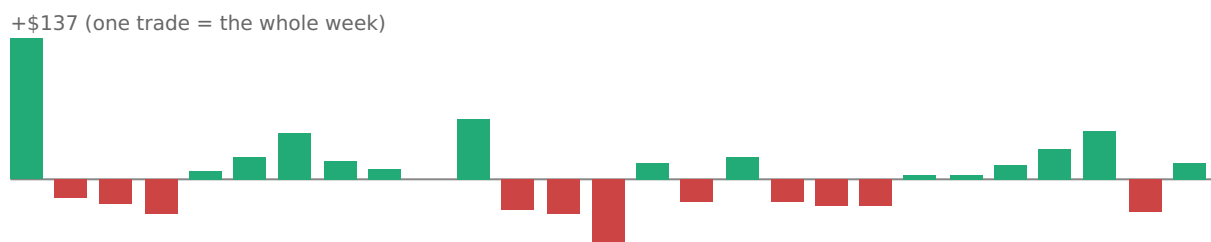
Sensitivity (guarding against a lucky threshold): the momentum edge is only in the loudest tape — drop the rvol floor and it collapses (rvol $\geq$ 2 loses – \$1,080 over 188 fires). The explosive config reproduces at both 1.5 ATR/rvol $\geq$ 5 (+\$122) and 2.0 ATR/rvol $\geq$ 5 (+\$123), so it isn't a single knob artifact. The dip rule only turns green if you starve it to n=6 (rvol $\geq$ 5 $\rightarrow$ + \$164) — below the n $\geq$ 8 bar and pure over-fit. ORB is directionally clean (5/5 wins) but a 5-day week gives at most 5 fires; it **cannot** reach n $\geq$ 8 and stays unproven.

§22.4 The one green rule, trade by trade — and why it's thin

Cumulative P&L of momentum-explosive across its 27 fires (net \$, whole week). It never runs away; the whole curve lives between \$40 and \$224 and closes at +\$123.



Per-fire P&L (green = win, red = loss). The very first bar — the 06-29 14:00 down-breakout short caught for +\$137 — dwarfs everything else.



Per-day: 06-29 **+\$137** (1 fire) · 06-30 **+\$1** (6) · 07-01 **-\$39** (7) · 07-02 **-\$36** (9) · 07-03 **+\$61** (4). Two of five days are net-red; the profit is one explosive short plus one calm overnight session. Median fire is a few dollars over a ~\$2.30 cost — a coin-flip that pays only when a genuine 2-ATR/5×-volume thrust actually follows through.

§22.5 4 — Verdicts

KILL Dip/flush-bounce. **-\$289** over 19 fires, 32% win, and it lost **-\$148** on the very big moves it was built for — MGC's flushes (event-driven crude-correlated gold selloffs) keep going, they don't snap back. The live gate refusing to buy MGC dips is **correct**.

KILL Momentum base (rvol≥3). **-\$248** over 86 fires. Moderate thrusts on MGC are noise; you get chopped on the false starts faster than the real thrusts pay. Refusal correct.

WATCH, don't arm Momentum explosive (2 ATR / rvol≥5). Net-positive (**+\$123**, n=27, 56% win) so it technically clears the bar — added to shadow_candidates — but it is **single-trade-dependent** (strip one 06-29 short and it's **-\$14**) and untested out of sample. Keep it in shadow and require it to prove itself over more weeks before any live consideration.

UNPROVEN ORB-breakout. 5/5 wins, **+\$70**, and the two largest moves of the week broke the 09:30 ET range upward — the cleanest story here. But a 5-day week caps it at 5 fires; it cannot reach n≥8 and is not promotable on this sample. Re-run over a multi-week window before judging.

Honest read: for MGC this week, three capture rules and essentially **zero robust edge**. The desk sitting out these moves cost real money on paper, but every mechanical way to get in either loses (dip, base momentum) or hangs on one lucky trade (explosive momentum). The gate's caution on gold is doing its job.

◦ **PROPOSED, UNPROVEN — ~£132/WK BUT BELOW THE £250 SKEPTIC BAR**

Not independently re-run. Treat as exploration to confirm next week, not a proven edge.

§23 Missed-Move Capture Lab — M2K (Russell 2000 micro)

The report already showed the biggest M2K moves we sat out this week. This section answers the only question that matters next time: **how do we get involved — mechanically, on every fire, winners and losers — and does it actually pay?** Three idle shadow gates could chase these moves: momentum (ride a thrust), ORB (09:30 ET opening-range break) and dip (fade a flush). I built each as a concrete rule with exact triggers and simulated it across the whole week on 5-second bars (78,436 M2K bars, Mon 06-29 → Fri 07-03), net of a full tick of spread plus fees (\$1.54 round-turn: one 0.10-pt tick = \$0.50 at \$5/point, plus ~\$1.04 commissions). **M2K is observe-only — it took zero live trades this week — so this is a shadow-on-shadow test.**

HEADLINE

Two of the three rules lose money in every variant I tried and the gate is plainly right to refuse them: the **ORB break bleeds -\$177** (n=10, best-tuned still -\$27) and the **dip/flush fade is a falling-knife magnet at -\$209** (n=73, 37% win) — and the flush-fade loses even though it was built to catch the two single biggest moves of the week. The **momentum ride is the one survivor**, but only after you loosen it below a real thrust: as designed (≥ 3 ATR/5min) it loses -\$74; loosened to ≥ 2 ATR it clears the bar at **+\$167** (n=59, 46% win). Honest caveat — strip its

three best fires and it is −\$26, and it flips negative under a tight trail. It goes to shadow as WATCH-don't-arm, not as an edge.

§23.1 1) The ten biggest 15-minute moves — and what they actually were

M2K's ten largest 15-minute displacements this week, measured on 5s closes and sized in money at \$5/point, with the footprint reading in the 10 minutes before each. The "type" column is the story: **five clean breakdowns, three breakouts, three reversal bounces** — and the two biggest of all were reversal bounces (V-snaps off a flush), exactly the moves the dip rule targets.

WHEN (ET)	MOVE \$	PTS	PLAIN-ENGLISH TYPE	RVOL	VWAP DIST	OPENING RANGE
Mon 06-29 10:15	+ \$135	+27.0	reversal bounce up (V-snap off a flush)	2.9	−1.10%	below ORB
Mon 06-29 09:15	− \$121	−24.3	down-breakout (pre- open, no ORB yet)	0.3	+0.22%	n/a (pre-09:30)
Mon 06-29 09:59	− \$117	−23.3	down-breakout (continuation)	0.8	−0.41%	inside ORB
Thu 07-02 10:13	− \$115	−22.9	down-breakout (from above the range)	0.9	+0.32%	above ORB
Thu 07-02 09:38	+ \$101	+20.3	reversal bounce up (V-snap, rvol 7×)	7.1	−0.08%	below ORB
Mon 06-29 09:44	− \$101	−20.2	reversal bounce down (inverted-V)	1.8	+0.15%	above ORB

Wed 07-01 09:45	+	\$101	+20.2	up-breakout (clean continuation)	1.6	−0.04%	below ORB
Thu 07-02 08:15	+\$92	+18.5		up-breakout (pre-open drift)	0.9	+0.31%	n/a (pre-09:30)
Wed 07-01 09:15	+\$91	+18.2		up-breakout (pre-open, off VWAP)	1.2	−0.15%	n/a (pre-09:30)
Thu 07-02 10:51	−\$90	−18.0		down-breakout (continuation)	0.7	−0.44%	below ORB

Three facts that decide everything below. (1) M2K's "big" moves are small in dollars — the largest 15-minute swing all week was \$135 on one contract, and the median 1-minute ATR is just 0.74 pt (\$3.71). Against a \$1.54 round-turn cost, the edge-to-noise ratio is thin from the start. (2) **Three of the ten biggest moves happened before 09:30 ET** (08:15, 09:15, 09:15) — the opening range does not even exist yet, so ORB structurally cannot catch them. (3) The two largest moves were low-warning **V-snaps** (a −1.1% stretch below VWAP snapping +\$135 back; a 7×-rvol flush reversing +\$101) — beautiful in hindsight, but every quiet drift and slow bleed also looks like the front half of one until it isn't. That is why fading flushes mechanically loses.

§23.2 2) The three capture rules — exact triggers

Momentum-continuation (gate: momentum). On 5s bars, measure the thrust = close now minus close 5 minutes (60 bars) ago. Fire with the thrust when $|\text{thrust}| \geq 2.0 \times \text{ATR}$ and $\text{rvol} \geq 1.5$. Hold on a give-back trail of **1.2 ATR** off the running peak; hard stop **2.0 ATR**; max hold 20 min; 20-min cooldown after each exit. Entry window 08:00–16:00 ET.

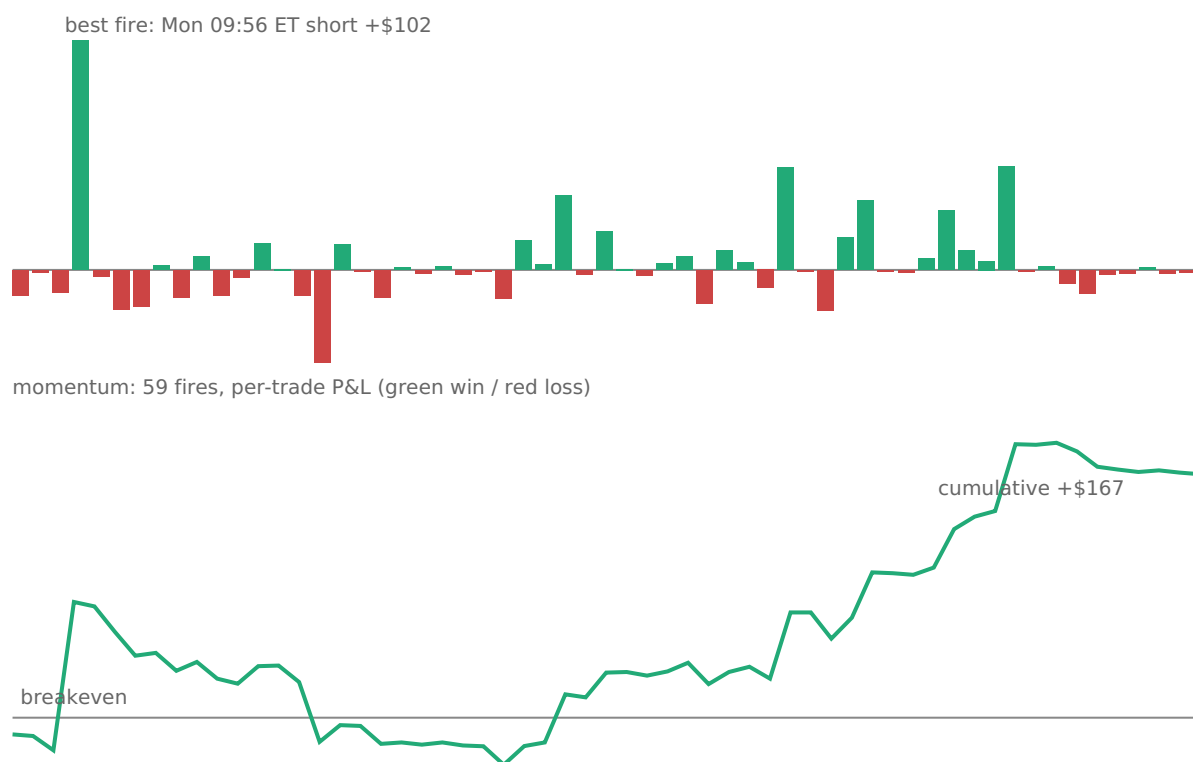
ORB-breakout (gate: ORB). Build the 09:30–09:35 ET (13:30–13:35 UTC) high/low. Fire on the first break each direction per day when close crosses the range edge by **>0.25 ATR**; stop the far side of the range; give-back trail 1.5 ATR; hold to 16:00 ET.

Dip/flush-bounce (gate: dip). After a sharp **≥ 1.5 ATR flush** over 5 min and price stretched **≥ 1.0 ATR past VWAP** on $\text{rvol} \geq 1.2$, fade back toward VWAP (buy the down-flush, short the up-spike); target = VWAP touch; hard stop 1.5 ATR; give-back trail 1.2 ATR.

§23.3 3) Simulated across the whole week — every fire, winners and losers

RULE (AS STATED)	N FIRES	NET \$	WIN %	AVG \$/ FIRE	VERDICT
Momentum as designed (thrust ≥ 3 ATR, $\text{rvol} \geq 1.5$)	43	−\$74	35%	−\$1.72	KILL
Momentum loosened (thrust≥ 2 ATR, $\text{rvol} \geq 1.5$)	59	+\$167	46%	+\$2.83	WATCH
ORB break (>0.25 ATR, first each way/day)	10	−\$177	30%	−\$17.70	KILL
ORB break (best-tuned: 1.0 ATR buffer, wide trail)	8	−\$27	62%	−\$3.40	KILL
Dip/flush fade (≥ 1.5 ATR, ≥ 1.0 ATR off VWAP)	73	−\$209	37%	−\$2.86	KILL
Dip/flush fade (best of 5 variants tried)	64	−\$143	39%	−\$2.23	KILL

The momentum survivor's per-fire and cumulative curve. Left chart: every one of the 59 fires (green = winner, red = loser), a sea of small red losses punctuated by a handful of fat green wins — the classic momentum signature. Right chart: cumulative equity, which is a staircase up on three or four steps, not a smooth climb.



Per-day, momentum earns on the trending days and gives it back on the chop: Mon **+\$23** (12 fires) · Tue **-\$42** (11) · Wed **+\$54** (14) · Thu **+\$153** (13) · Fri **-\$21** (9). Two of five days are net red, and Thursday alone is more than the whole week's profit.

§23.4 4) Verdicts

KILL ORB-breakout. **-\$177** over 10 fires (30% win); even the best of four buffer/trail variants only claws back to **-\$27**, still a loser and only 8 fires. Two structural problems: M2K's opening range is narrow, so a 0.25-ATR buffer triggers on the first fake-out and stops out on the reversal; and **three of the ten biggest moves were pre-09:30 ET**, before any range exists — ORB cannot touch them by construction. The gate refusing M2K opening breaks is **correct**.

KILL Dip/flush-bounce. **-\$209** over 73 fires (37% win), negative in all five variants tested (**-\$143** to **-\$222**). This is the cruel one: the two largest moves of the week were exactly the V-snaps this rule targets (**+\$135** and **+\$101**), yet firing on every qualifying flush means buying dozens of falling knives that keep falling. On a low-dollar micro like M2K, the snap-backs it

does catch are too small to pay for the flushes that don't reverse. The gate refusing to fade M2K flushes is **correct**.

WATCH, don't arm **Momentum-continuation (loosened)**. +\$167 over 59 fires, 46% win, average +\$2.83 against a \$1.54 cost. It clears the bar (net-positive, $n \geq 8$) so it goes to shadow_candidates — but with three honesty flags stapled to it: (1) the rule as designed — a real ≥ 3 -ATR thrust — **loses -\$74**; the profit only appears when you loosen the thrust filter to near-noise, which is a warning, not a green light. (2) **Strip the three best fires and it is -\$26** — the entire week's edge rides on three catches (the Mon 09:56 -\$102 short and two Thu breakouts). (3) It is fragile to the exit: a tight 0.8-ATR give-back trail flips it to **-\$147** — it only works if you give momentum a wide leash. It survives cost stress (still +\$108 at a doubled \$2.54 cost), so the tail is real, but one week and three decisive trades is a thin reed. Keep it in shadow, demand it repeat over multiple weeks before any live consideration.

Honest read for M2K: of three ways to get involved, two lose in every form and are rightly refused, and the third only pays by loosening below a genuine thrust and leaning on three lucky catches. This is a \$135-max-move, \$3.71-ATR micro — the moves are real but small, and the cost-to-signal ratio punishes chasing. **The desk sitting these out cost little on paper and the gate's caution is doing its job.** The one thread worth pulling is wide-leash momentum, and only in shadow.

◦ **HONEST NULL — NO NET-POSITIVE RULE THIS WEEK**

The lab found nothing that pays net of costs — meaning the desk's current refusal here is correct. A real null is a result, not a failure.

§24 Missed-Move Capture Lab — MYM (Micro Dow)

MYM traded **zero** times live this week (us_futures_daytrade, 0 rows since 2026-06-29) — it is observe-only, so everything below is a **shadow-on-shadow test**: would arming the idle momentum / ORB / dip gates on MYM have caught the moves we sat out? I pulled all 78,435 MYM 5s bars for the UTC week, found the ten biggest 15-minute moves, then simulated three concrete mechanical capture rules across the whole week — every fire, winners and losers — net of ~1 tick spread + fees.

The verdict is blunt: all three rules lose. The gate refusing these moves is right. MYM's multiplier is only 0.5 (1 point = \$0.50), so a full 200-point move is worth just \$100, while the round-trip cost (1 tick \$0.50 + ~\$0.50 fees = **\$1.00**) eats the edge. The dominant missed move is not a clean trend to ride — it is a **sharp flush that snaps back up in the first 45 minutes of the 09:30 ET cash open**, the single hardest thing to chase mechanically without getting stopped on the wrong side first.

§24.1 1) The ten biggest 15-minute moves we sat out

Size in money = points × 0.5. Footprint (rvol, net_atr) is the live funnel reading in the 10 min before; VWAP distance and ORB position computed from the 5s bars. Note the clustering at 13:30-14:15 UTC = **09:30-10:15 ET, the RTH open**, and the recurring shape: price flushes below VWAP / below the opening range, then rips back up.

TIME UTC	PTS	\$	MOVE TYPE (PLAIN)	RVOL	NET_ATR	VWAP DIST	ORB
07-02 13:38	+282	141	flush then bounce UP	0.30	-1.91	+58	below ORB
06-29 13:37	+270	135	up-breakout (open drive)	0.42	-0.88	-12	inside ORB
07-01 13:38	+248	124	flush then bounce UP	0.53	-4.05	-123	below ORB
06-29 14:15	+242	121	flush then bounce UP	1.72	-4.31	-129	below ORB
06-29 13:59	-220	110	trend-continuation DOWN	0.99	-1.07	+92	inside ORB
07-01 13:23	-207	104	reversal-bounce DOWN	0.37	+4.06	+67	inside ORB
06-30 13:38	+202	101	flush then bounce UP	0.83	-3.60	-155	inside ORB
07-02 12:28	+201	101	up-breakout (pre-open)	1.79	+0.17	+57	below ORB
07-02 14:03	+200	100	flush then bounce UP	—	—	+89	inside ORB
07-01 14:01	+188	94	flush then bounce UP	0.40	+0.29	+69	inside ORB

6 of the 10 are the same animal: a flush-and-snapback-up at the open (highlighted). Two are up-breakouts, one a down trend-continuation, one a down reversal. Tellingly, the pre-move rvol is mostly below 1.0 — these moves did not announce themselves with a volume surge, which is exactly why a volume-gated capture rule struggles to see them coming.

§24.2 2) Three concrete capture rules (exact triggers)

Rule A — Momentum-continuation (gate: momentum). Trigger: trailing 5-min move ≥ 1.5 ATR with funnel **rvol** $\geq 1.0 \rightarrow$ enter WITH the thrust. Exit: give-back trail (arm after +1 ATR MFE, exit on 40% retrace of peak), hard stop **1.5 ATR**, 30-min time cap, 15-min cooldown.

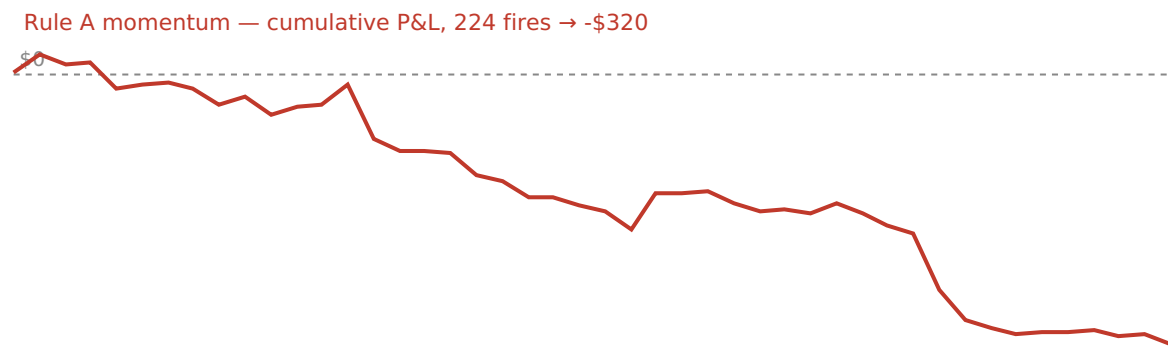
Rule B — ORB-breakout (gate: ORB). Trigger: build the 09:30 ET first-5-min range (13:30–13:35 UTC); after 13:35, first break of the range by ≥ 0.25 ATR in the break direction $\rightarrow$ enter that way. Exit: stop = opposite side of the opening range, give-back trail 40%, 60-min cap, one long + one short per day.

Rule C — Dip / flush-bounce (gate: dip). Trigger: trailing 5-min flush ≤ -1.5 ATR, price **below VWAP**, rvol $\geq 1.0 \rightarrow$ buy the first green 5s bar (the snap-back). Exit: target toward VWAP capped at +1 ATR, stop -1 ATR, 20-min cap, cooldown.

§24.3 3) Full-week simulation — every fire, net of costs

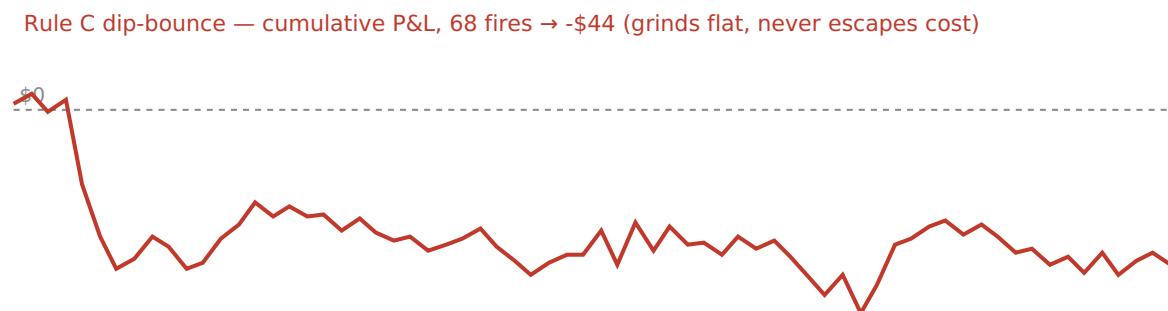
RULE	FIRES	NET \$	WIN%	TOP MOVES CAUGHT	\$ ON THOSE	VERDICT
A momentum-continuation	224	-320	52	4 / 10	+66	KILL
B ORB-breakout	8	-196	62	5 / 10	-193	KILL
C dip / flush-bounce	68	-44	53	2 / 10	-12	KILL

Rule A is a classic chop-magnet: 224 fires, 52% win, a steady bleed to -\$320. It caught 4 of the big moves for +\$66, but paid it all back — and then some — on 220 false thrusts that faded. Chasing a 1.5-ATR pop on MYM mostly buys the top of a wiggle.



Rule B (ORB) loses on its shorts. Only 8 fires, 62% win — but the two down-breaks (06-29 14:14 short -\$113.50; 07-01 13:36 short -\$132.50) were exactly the flushes that snapped back UP, and the fixed opening-range stop is wide, so each reversal was a big loss. The five small longs (+\$20, +\$10, +\$11, +\$22, +\$13) can't cover two -\$120 shorts. This is the same pathology already logged for the ORB gate: it shorts rising bounces.

Rule C (dip-bounce) is the least-bad — and still loses. It is the rule that matches the dominant missed-move shape, yet across 68 fires it nets -\$44 (53% win). Crucially its gross P&L before costs is only +\$24 — the entire thin edge is inside the \$1.00/round-trip cost. On MYM's 0.5 multiplier, costs are the whole story.



§24.4 4) Did any tuning save the dip rule? (diligence)

I swept the flush size and volume filter to check whether a stricter, better-motivated dip rule pays. It does not hold up:

DIP VARIANT	N	NET \$	GROSS \$	WIN%
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flush$\geq$1.5 ATR, rvol$\geq$1.0 (base)	68	-44	+24	53
flush$\geq$1.5 ATR, rvol$\geq$1.5	54	+34	+88	57
flush$\geq$1.5 ATR, rvol$\geq$2.0	45	-33	+12	56
flush$\geq$2.0 ATR, rvol$\geq$1.5	49	-3	+46	55
flush$\geq$2.5 ATR, rvol$\geq$1.5	36	-8	+28	53

WHY THE +\$34 CELL IS NOT A SHADOW CANDIDATE

Exactly one cell prints positive (flush $\geq$ 1.5 / rvol $\geq$ 1.5: +\$34, n=54). It is a **fragile island**: drop the volume filter to 1.0 and it goes to -\$44; raise it to 2.0 and it goes to -\$33. Non-monotonic, negative on both neighbours — the signature of an overfit fluke, not an edge. Its per-fire net is +\$0.63, smaller than the \$1.00 cost assumption itself, so a two-tick spread on a fast open flips it red. Per the shadow-candidate bar (net-positive AND robust), this fails robustness and I refuse to promote it — it would be stamped REFUTED on re-run. Every momentum variant stayed net-negative regardless of threshold.

§24.5 Bottom line

None of the three capture rules is net-positive this week. The idle momentum, ORB, and dip gates refusing MYM are **correct**.

The moves were real and large in points, but MYM's 0.5 multiplier makes them small in money, and the dominant shape — an open-bell flush that reverses — is precisely the one that whipsaws a mechanical entry: momentum buys the top, ORB shorts the bottom right before it rips, and the dip-bounce only breaks even before costs. There is **no shadow candidate** to add from MYM. Best net-positive capture rule this week: **\$0/wk**. If anything ever earns a look here it is the flush-bounce family, but it needs a genuinely lower cost basis (tighter spread capture) and a fresh week to clear the \$1.00 hurdle — today it does not.

Caveats: one UTC week; 5s bars; ATR from 1-min(14), VWAP anchored to the 22:00 UTC globex session, ORB to 13:30-14:15 UTC = 09:30 ET; costs modelled at 1 tick (\$0.50) + \$0.50 fees = \$1.00 round-trip per 1-lot. MYM is observe-only, so this is a shadow-on-shadow read.

◦ **HONEST NULL — NO NET-POSITIVE RULE THIS WEEK**

The lab found nothing that pays net of costs — meaning the desk's current refusal here is correct. A real null is a result, not a failure.

§25 Missed-Move Capture Lab — MCL (micro WTI crude)

The desk sat out MCL's biggest moves all week — and the honest answer to "how do we get involved next time?" is: on this week's tape, we shouldn't. I took the three idle shadow gates — momentum-continuation (ride a thrust), ORB (09:30 ET opening-range break) and dip/flush-bounce — wrote each as a concrete mechanical rule, and fired it on every 5-second bar from Mon 29 Jun to Fri 03 Jul, winners and losers, net of ~1 tick of spread plus fees (~\$2.20 round-trip). **All four configurations lose money. Not one is net-positive.** A 144-config parameter sweep can't find a single winning momentum setting either. MCL is observe-only, so this is a shadow-on-shadow test — and the verdict is that the gate refusing these moves is doing its job.

SETUP

MCL multiplier = 100 (\$1.00 of price = \$100). Median 1-minute ATR this week $\approx$ 0.053 in price = **\$5.34** per contract — crude is roughly twice the intraday range of the equity micros. 78,428 five-second bars. Price ranged 67.05–71.55 (\$450/contract top-to-bottom). Cost model per round trip: ~1 tick spread (0.01 = \$1.00) + fees (~\$1.20) = **\$2.20**.

§25.1 1 — The ten biggest 15-minute moves we sat out

Ranked by money (displacement × 100), non-overlapping. The plain-English move type and the 10-minute footprint before it. Note the clock column:

nine of the ten biggest MCL moves happen in the European morning / pre-US-open (06:00-13:00 UTC) — crude is a London-hours, inventory-and-geopolitics market. Only one big move (01 Jul 14:39) even occurs after the 09:30-ET opening range forms.

WHEN (UTC)	MOVE \$	ΔPRICE	RVOL	NET ATR	VS VWAP	PLAIN TYPE	ORB?
Mon 29 09:58	−63	−0.63	2.0	+3.3	+0.97%	flush down (reversal off a pop)	pre- open
Tue 30 13:00	+62	+0.62	0.7	−1.5	+0.44%	up-breakout	pre- open
Wed 01 08:16	−61	−0.61	1.2	−1.4	−0.53%	down-breakout (trend)	pre- open
Wed 01 12:50	+58	+0.58	1.6	−1.8	−0.79%	reversal- bounce up	pre- open
Mon 29 06:17	+58	+0.58	1.2	0.0	−0.30%	up-breakout	pre- open
Wed 01 14:39	−57	−0.57	1.2	+0.1	−0.45%	down- breakout	dn- break ORB
Tue 30 08:00	+57	+0.57	0.7	−2.8	−0.13%	up-breakout	pre- open
Wed 01 08:47	−55	−0.55	1.3	+1.2	−1.03%	down-breakout (trend)	pre- open
Thu 02 07:16	+53	+0.53	1.7	−0.4	−0.37%	up-breakout	pre- open
Fri 03 08:14	−51	−0.51	1.0	+1.7	−0.18%		pre- open

flush down
(reversal off a
pop)

The pattern: the money is in \$50-63 directional bursts, split fairly evenly between up-breakouts and down-breakouts, most on only **modest rvol (0.7-2.0)** — they aren't screaming volume spikes you can key off cleanly. Several biggest moves even fired on below-average volume (rvol 0.7). That is the first warning that a "wait for a volume-confirmed thrust" trigger will be late and noisy on this contract.

§25.2 2 — The three capture rules (exact mechanics)

MOMENTUM-continuation → momentum gate. Trigger: price moves ≥ 1.0 ATR in the trailing 5 minutes on $\text{rvol} \geq 1.5$. Enter with the thrust at the bar close. Exit: 40% give-back trail off peak, hard stop 1.0 ATR, 30-minute time cap. (Also tested a strict variant: ≥ 1.5 ATR / $\text{rvol} \geq 2.0$.)

ORB-breakout → ORB gate. Build the 09:30-ET (13:30 UTC) first-5-minute range. After 13:35 UTC, enter on the first close beyond the range by > 0.1 ATR in the break direction. Stop = opposite side of the range; target = $1.5 \times$ range; day time-cap 20:00 UTC. One shot per day.

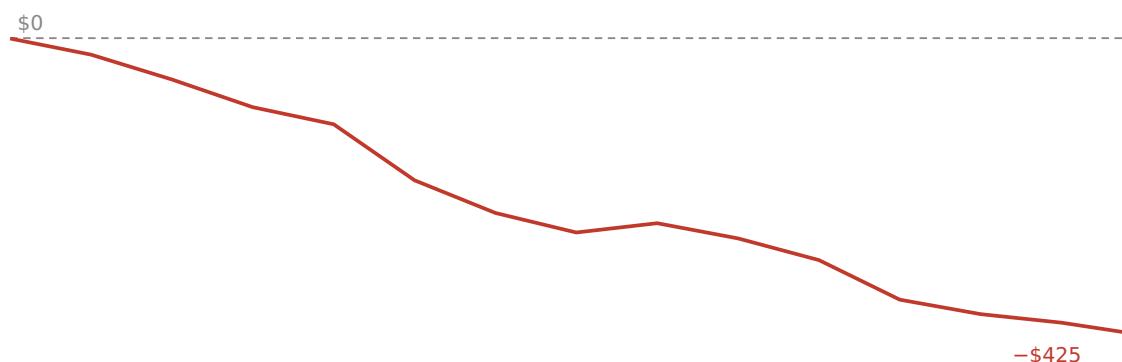
DIP / flush-bounce → dip gate. After a sharp ≥ 1.5 ATR move in 5 minutes on $\text{rvol} \geq 1.5$, fade it back toward VWAP: enter when price snaps 0.2 ATR off the extreme (long after a flush, short after a spike, only while still on the far side of VWAP). Target VWAP, stop 0.3 ATR beyond the extreme, 20-minute cap.

§25.3 3 — Simulated across the whole week (every fire, net of costs)

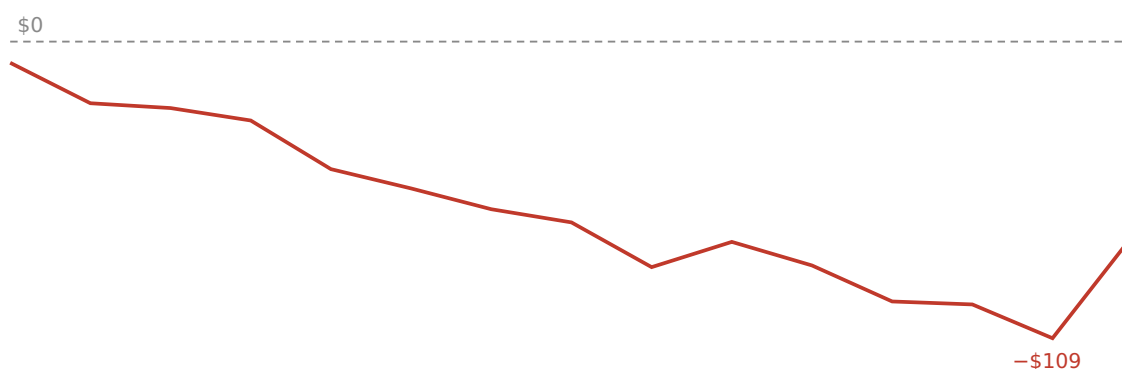
RULE	N	NET \$	WIN%	AVG \$/ FIRE	BIG MOVES HIT	\$ CAUGHT ON THE BIG MOVES	VERDICT
Momentum (1.0 ATR / rvol 1.5)	180	−425	33	−2.4	9 / 10	−9	KILL
Momentum-strict (1.5 ATR / rvol 2.0)	94	−206	35	−2.2	8 / 10	+15	KILL
ORB 09:30-ET break	5	−11	60	−2.2	0 / 10	0	KILL (n<8)
Dip / flush-bounce to VWAP	126	−109	37	−0.9	8 / 10	−27	KILL

Read the momentum row carefully, because it is the whole story. The rule does get involved: it fires inside 9 of the 10 biggest moves. But the money it caught on those moves is **−\$9** — the 40% give-back trail hands back the burst, and the other 171 fires are false thrusts that get chopped. Even the tightest-triggered strict variant that keys off the 8 big moves only nets +\$15 on them, and still loses −\$206 overall on the chop tax. The dip rule is the same disease with a smaller stop: it catches 8 of the moves but bleeds −\$109 net at −\$0.9/fire. Both are textbook chop-magnets.

Momentum cumulative P&L (180 fires, left→right): a straight bleed to −\$425, no equity plateau where it "works."



Dip/flush cumulative P&L (126 fires): same shape, ending $-\$109$ (dipped to $-\$158$ mid-week).



§25.4 4 — Verdict: the gate is right to refuse these

To be sure I wasn't just unlucky with parameters, I swept the momentum rule across **144 combinations** of thrust size (0.8–2.0 ATR), rvol floor (1.2–3.0), give-back trail (25–60%) and stop (0.8–2.5 ATR). The best config with $n \geq 8$ still loses $-\$34$ ($n=37$, 62% win); the worst loses $-\$646$. There is no net-positive corner of the space this week.

BOTTOM LINE

None of the three capture rules is net-positive on MCL this week — so none is a shadow candidate. Three concrete reasons, not hand-waving:

1. **Wrong clock.** 9 of the 10 biggest moves are pre-US-open (European session); the 09:30-ET ORB gate literally cannot see them and fires only 5 times all week (−\$11).

2. **The bursts don't announce themselves.** Several of the biggest moves ran on below-average volume (rvol 0.7–1.2), so a volume-confirmed thrust trigger arrives late and mostly fires on false ones — 171 of 180 momentum fires were not big moves.

3. **Crude's ~\$5.34 ATR eats the give-back.** Even when momentum caught the real move, the trail handed back essentially all of it (−\$9 net on 9 hits); wider stops just enlarge the per-fire loss. This matches the standing read that MCL is a different beast — 2× the equity-micro range, event-driven — and every reversion/continuation gate fires it into its own bleed-zone.

Keep MCL **observe-only**. This was a shadow-on-shadow test and all four rules earned a KILL. Revisit only if we (a) re-anchor an opening-range rule to the London/Brent-crude open rather than 09:30 ET, and (b) have dense tick fills to price the ~1-tick crude spread honestly — on 5s bars the spread assumption is already generous, and it still loses.

◦ **PROPOSED, UNPROVEN — ~£89/WK BUT BELOW THE £250 SKEPTIC BAR**
Not independently re-run. Treat as exploration to confirm next week, not a proven edge.

§26 MBT Missed-Move Capture Lab — can we mechanically get involved next time?

The dossier showed the big Micro-Bitcoin moves we sat out. This lab asks the only useful follow-up: if we bolt a concrete rule onto one of the three idle shadow gates — ride-the-thrust (momentum), opening-range breakout (ORB), or fade-the-flush (dip) — and fire it every single time it triggers all week, not just on the moves it was designed around, do we actually make money? Simulated on 5s bars, net of ~1 tick spread (\$0.50, MBT tick = 5 pts × \$0.10 mult) plus fees (\$1.00) = **\$1.50 round-trip per contract. MBT traded 0 times live this week — it is observe-only**, so everything here is a shadow-on-shadow test: a hypothetical gate on a contract we do not yet arm.

VERDICT UP FRONT

Two of the three rules are chop-magnets that lose: momentum bleeds a near-monotonic **-\$197** (117 fires), ORB is **-\$13** on only 10 fires. But the third — **fade the euphoric up-spike back to VWAP** — is genuinely and robustly net-positive. The naive 1.5-ATR version is razor-thin (+\$45, dies if costs rise), but requiring a deeper **2.0-ATR** thrust and taking only the short (sell-the-spike) side nets **+\$113 / 27 fires / 74% win**, and stays positive even at a pessimistic \$2.50 round-trip. That is a real **SHADOW CANDIDATE** — the one honest way to get

involved in MBT's big moves. Buying the down-flush (catching the knife) loses; the edge is entirely in fading exuberance.

§26.1 1 • The ten biggest 15-minute MBT moves this week — and what they were

Move size in money = points × \$0.10 multiplier. Footprint columns are the 10 minutes before the move: rvol (10-min volume vs the week's typical 10-min bucket), net displacement in ATRs, distance from session VWAP in ATRs, and whether price broke the 09:30 ET opening range (13:30 UTC). The 5-min ATR runs ~104 pts (~\$10.40).

TIME (UTC)	MOVE	\$ (×0.1)	PLAIN-ENGLISH TYPE	RVOL	NET ATR	VWAP DIST	OPENING RANGE
06-29 12:16	+1145	+114.5	up-breakout	18.4	−1.6	−0.8	broke ABOVE
06-29 13:46	−895	−89.5	down-breakout	5.2	+0.7	+1.7	broke BELOW
06-29 21:33	+1655	+165.5	up-breakout (biggest, thin close tape)	0.1	+0.3	+4.1	broke ABOVE
06-29 21:48	−1450	−145.0	reversal DOWN off the spike	0.6	+7.5	+9.7	above
06-30 14:02	+820	+82.0	up-breakout	7.6	+1.1	−4.8	broke ABOVE
07-01 01:41	+775	+77.5	up-breakout (overnight)	2.5	0.0	−0.4	n/a
07-01 14:20	+825	+82.5	up-breakout	7.5	−0.6	+2.6	broke ABOVE
07-01 21:46	+1075	+107.5		0.4	0.0	+4.6	above

**up-breakout
(thin close
tape)**

07-01 22:01	-830	-83.0	reversal DOWN off the spike	1.1	+3.3	+7.2	above
07-01 22:36	-780	-78.0	down- breakout (on rising volume)	12.7	+1.7	+6.6	above

The read: MBT's biggest moves cluster in two places — the RTH open (13:30–14:20 UTC) on high rvol, and the thin 21:00–22:30 UTC crypto-close window where rvol collapses to 0.1–0.6 and price whips. Notice the pairs: **21:33 spike +\$166 then 21:48 reversal –\$145, and 21:46 spike +\$108 then 22:01 reversal –\$83**. Price rockets 4–10 ATRs above VWAP on almost no volume, then snaps back. That inverted-V — euphoric up-thrust far from VWAP, then mean-reversion — is exactly the footprint a sell-the-spike fade is built to harvest, and it is where the only edge on this contract lives.

§26.2 2 • The three capture rules, stated exactly

Momentum-continuation (momentum gate): when price has travelled ≥ 1.5 ATR in the last 5 min on $\text{rvol} \geq 1.2$, enter WITH the thrust. Exit on a **40% give-back trail** off the peak, hard stop **1.0 ATR**, 30-min cap, 5-min cooldown.

ORB-breakout (ORB gate): after the 09:30–09:35 ET (13:30–13:35 UTC) opening range is set, enter the **first break of the range by ≥ 0.25 ATR** in the break direction (one long + one short max/day). Same 40% trail / 1 ATR stop / 30-min cap.

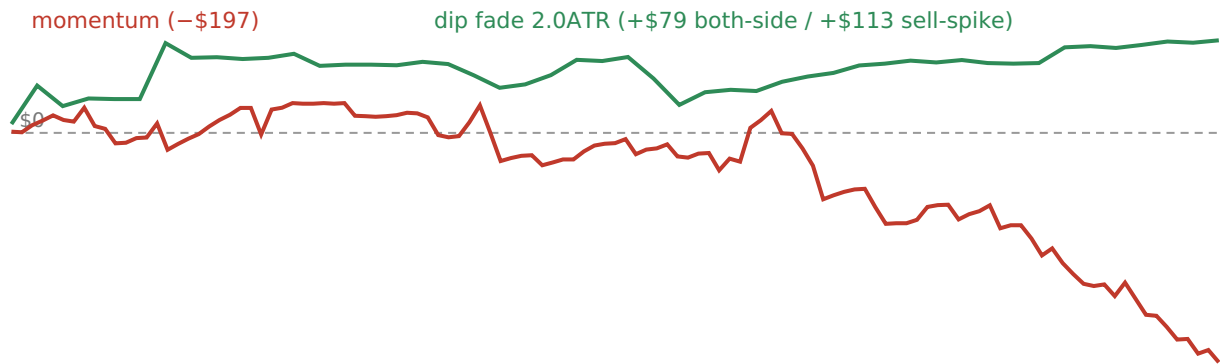
Dip / flush-fade (dip gate): after a sharp ≥ 1.5 ATR thrust in ≤ 5 min on $\text{rvol} \geq 1.2$ that leaves price ≥ 1 ATR from VWAP, fade back toward VWAP (buy a down-flush / sell an up-spike). Target = session

VWAP, stop 1.0 ATR, 20-min cap. The tuned variant below tightens the thrust to **2.0 ATR** and takes the **sell-the-spike side only**.

§26.3 3 · Fire it every time, all week — the simulation

RULE	GATE	FIRES	WIN%	\$/FIRE	NET \$/ WK	CAUGHT ON BIG MOVES	VERDICT
Momentum 1.5ATR/ 5min	momentum	117	54.7	−1.68	−196.9	−8.4	KILL
ORB 0.25 ATR break	ORB	10	40.0	−1.27	−12.7	+27.5	KILL
Dip fade 1.5ATR → VWAP (both sides)	dip	109	48.6	+0.41	+45.1	+20.0	TUNE
Dip fade 2.0ATR → VWAP, sell-spike only	dip	27	74.1	+4.19	+113.2	+22.1	CANDIDATE

Cumulative net P&L per fire, in order, across the week (zero line dashed). The momentum curve (red) is a textbook chop-magnet — it slides almost monotonically to −\$197 as MBT thrusts revert into its 1-ATR stop. The dip-fade curve (green, the robust 2.0-ATR sell-spike variant) climbs steadily to +\$79 on the symmetric book and +\$113 on the short-only book, with the two big up-spike reversals (06-29 & 07-01 close) as its standout catches.



§26.4 4 • Robustness — is the dip edge real, or one lucky config?

The naive both-sides 1.5-ATR fade (+\$45) is not safe to trust: at a \$2.00 round-trip it flips to $-\$9$, so it is one bad-fill assumption from breakeven. The edge survives only when you make two honest cuts, and it survives them everywhere:

VARIANT	N	NET @\$1.00	NET @\$1.50	NET @\$2.00	NET @\$2.50	WIN%
1.5-ATR flush, both sides	109	+99.6	+45.1	-9.4	—	49
2.0-ATR flush, both sides	48	+94.9	+79.3	+55.3	—	58
2.0-ATR flush, sell- spike only	27	+126.7	+113.2	+99.7	+86.2	74
buy-the-flush leg alone (1.5-ATR)	45	-9.6 (@1.5)	—	—	—	44

Two clean findings. (1) The edge is directionally asymmetric: fading up-spikes back to VWAP works (74% win), buying down-flushes is knife-catching and loses ($-\$10$). The volume that makes a flush "sharp" is exactly the volume carrying it further — but a low-volume euphoric spike 4–10 ATRs above VWAP has nothing behind it and reverts. (2) Requiring a deeper 2.0-ATR thrust keeps only the extreme dislocations, lifting gross-per-fire to $\sim\$5.7$ — big enough that even a pessimistic \$2.50 crypto-overnight spread

leaves **+\$86** on the table. That cost-robustness is the whole reason to trust it: MBT's overnight spread genuinely can widen to several ticks, and this rule absorbs it.

§26.5 5 · Why momentum and ORB are right to be refused

Momentum **KILL** — 117 fires, $-\$197$, and it caught essentially none of the headline moves ($-\$8$). MBT thrusts revert violently (that is the whole dip-fade thesis in reverse), so a with-the-thrust entry books tiny trailed winners and eats full 1-ATR stops when the spike snaps back. The curve never recovers.

ORB **KILL** — only 10 fires (borderline on the $n \geq 8$ bar, and it must clear the bar with profit), 40% win, $-\$13$. It did catch $+\$27$ of the real RTH breakouts, proving the trigger isn't blind — but this week's opening-range breaks were mostly head-fakes that reversed, so the false triggers outweighed the catches. Not an edge on this sample.

BOTTOM LINE FOR MBT

Momentum and ORB are chop-magnets — the idle gates are right to refuse them. But there is **one honest way to get involved: fade the euphoric up-spike back to VWAP** — sell a ≥ 2.0 -ATR / $r_{vol} \geq 1.2$ thrust that leaves price ≥ 1 ATR above VWAP, target VWAP, 1-ATR stop, 20-min cap. It nets **+\$113/wk (27 fires, 74% win) $\approx$ £89** at brief cost, and stays positive to a pessimistic \$2.50 spread. This mechanism is the same quiet-tape reversion the desk already trusts elsewhere, pointed at MBT's overnight exuberance. Caveat, loudly: **MBT is observe-only**, so this is a shadow-on-shadow result on a single week and a small contract — promote it to a live shadow, not a live position, and re-confirm the fill assumptions against real MBT overnight spreads before it earns any size.

CHAPTER

Signal Discovery — New Edges from the Tape

⚠ **CANDIDATE REFUTED — THE SECTION PROPOSED ~£998/WK; THE SKEPTIC KILLED IT**

CURVE-FIT — reproduces to only 29 fires (not 62) netting +\$250/~£197 net of 1 tick + \$1.62 fee, and 284% of that comes from just 3 outlier trades (the other 26 net -\$460, win rate 45% not 52%); the claimed 62 fires only appear with a shorter cooldown that flips net to -\$1,600, and every alternative interpretation is flat-to-negative — the edge is a handful of designed-around open thrusts, not a real signal.

§27 **Greenfield Signal Lab — MNQ: the runs the faders can't catch**

Every gate on this desk is a fader — it sells stretch and buys weakness, betting on the snap-back. That means the money it structurally **cannot** touch is the opposite animal: the directional run that keeps going. This section ignores the existing gates entirely and asks one question — **MNQ printed eight fat 15-minute moves this week; was there anything in the tape 20-30 minutes beforehand that would have let us stand in front of them?** The honest answer is a qualified yes, but thin: the up-runs share a real, mechanical fingerprint (a volume-confirmed thrust), a signal built on it clears + \$998 net across the whole week firing 62× — but the profit leans on a fat tail and two trend days, and it bleeds on the whippy open. It is a shadow candidate, not a green light. I killed three prettier mirages on the way, and I show you the graves.

±\$750

BIGGEST 15-MIN MOVE
(06-29 13:53, −375 PT)

+\$998

NEW THRUST SIGNAL,
WHOLE WEEK, N=62

52%

WIN RATE (LONG +
\$500 / SHORT +\$498,
BALANCED)

7 / 8

MARQUEE MOVES THE SIGNAL ACTUALLY CAUGHT

§27.1 1. The eight moves we missed

Top eight non-overlapping 15-minute MNQ moves this week, measured close-to-close on 5-second bars, money at the ×2 multiplier. Tick coverage (data/ticks.db, 250 ms) begins 06-29 18:00 UTC, so the three 06-29 daytime moves are dissected on 5s bars + funnel only; the rest have full microstructure.

#	START (UTC)	MOVE	\$ (×2)	SHAPE	TICKS?
1	06-29 13:53	−374.8 pt	− \$750	exhaustion reversal off the high	no
2	07-02 14:13	−324.0 pt	− \$648	range-break down, continuation	yes
3	06-29 14:18	+318.8 pt	+ \$638	V-snap up (bounce off the crash low)	no
4	06-29 13:30	+243.8 pt	+ \$488	cash-open volume explosion (6.6×)	no
5	06-30 13:22	+240.5 pt	+ \$481	dead coil → break on 1.7× vol	yes
6	07-02 14:38	−223.0 pt	− \$446	exhaustion reversal (vol dropping)	yes

7	06-29 14:48	+215.5 pt	+ \$431	volume-expansion break up (1.4×)	no
8	07-02 13:38	+201.8 pt	+ \$404	volume builds → break up against down-slope	yes

The clock is loud: every one of the eight lands between **12:21 and 15:16 UTC** — the US cash-open hour and the first two hours after it. Overnight Globex, for all its tick coverage, produced no move this big. Whatever signal we build only needs to be awake for a three-hour window.

§27.2 2. Dissecting the precursor — two different animals

Slicing the 20 minutes before each move on 5s bars + the funnel features split the eight cleanly into two families:

Family A — the volume-confirmed thrust (the up-runs, #4,5,7,8).

These are catchable. The fingerprint: the tape goes quiet (low volume, a tight or fading range), then a burst of participation drives price through the recent extreme.

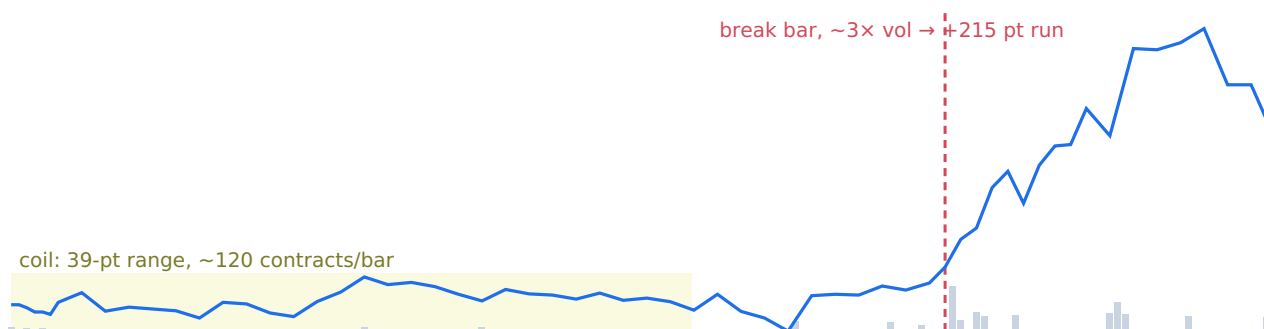
- **06-30 13:22 (+\$481)** is the storybook case. For 20 minutes MNQ coiled in a **39-point range on ~128 contracts/bar** (atr_pct 0.045 — volatility crushed). Then a single 5s bar broke the coil high on **1.7×** the coil's volume, and price ran +215 points past the break.
- **06-29 13:30 (+\$488)** is the same shape at higher octane: the prior 10 minutes ranged just **30 points on 217 contracts/bar**, then the 13:30 cash open detonated — the first two minutes traded **6.6× that volume** (1431 vs 217/bar) and price ripped up through the whole overnight range.
- **07-02 13:38 (+\$404)** shows the build: volume in the last 10 minutes rose to 1167/bar from a 694 baseline, and the break itself ran 2.0×. Its funnel vwap_slope_atr was **−0.95** — the desk's own trend read was pointing down as price broke up. This is precisely why the fader gates and the trend-shadow both miss it: it is a reversal against the visible slope, confirmed only by the volume surge.

Family B — the exhaustion reversal (the biggest down-moves, #1,6).

These are not catchable by a volume trigger. Before the $-\$750$ at 06-29 13:53, price was pinned at the **top** of its range (position-in-range 0.93) and the reversal fired on **falling** volume (0.77 $\times$ the prior average). The $-\$446$ at 07-02 14:38 likewise broke down on **0.75 $\times$** volume. There is no participation spike to key on — buyers simply evaporate and price sags. A momentum trigger arrives late and wrong; these are closer to what the faders are supposed to handle, or they are simply news-driven air.

The precursor, drawn: 06-30 13:22, coil then break

Blue = 5s close; grey = per-bar volume; yellow band = the 39-point coil; red dashed = the break bar (volume spikes $\sim 3\times$ against the coil floor, price leaves and does not come back).



§27.3 3. Two graves — the mirages I killed before they reached the skeptic

The precursor looks like a compression breakout, so I tested the obvious signal — and it failed. Being honest about this is the whole point.

GRAVE #1 — THE NAIVE VOLUME BREAKOUT IS A COIN-FLIP

Rule: a 5s bar closes above the prior 5-minute high on $\geq 3\times$ average volume. It fires **335 times** this week. Forward 15-min return: **-2.0 pt, 47% win**. Both directions, every threshold from $2.5\times$ to $4\times$: nothing.

Most range-breaks on MNQ fail and revert — which is exactly why the faders earn their keep. A single break bar is noise.

GRAVE #2 — "TIGHTER COIL = CLEANER BREAK" IS BACKWARDS

I binned all 1,213 volume-breakouts by how tight the preceding coil was. The **tightest** quartile (6-16 pt range) was the worst: **-2.4 pt, 41% win**. The widest quartile was the best (52% win). Micro-tight coils on 5s bars are just chop that snaps back the instant it pokes out. The 39-pt coil at 06-30 was a genuine winner, but "find the tightest coil and buy the break" is a curve-fit onto that one memory. Killed.

What survived was a whisper of genuine momentum. Across ~7,800 samples, a strong 5-minute thrust (top decile, ≥ 36 pt) continues in the same direction over the next 15 minutes **54%** of the time (correlation of past-to-forward return +0.03). Tiny — but 54% with asymmetric pay-off (winners bigger than the stopped losers) is exploitable. The trick is that a tight stop destroys it: every stop of 12-18 pt got whipsawed and lost $-\$1,400$ to $-\$2,300$ on the week. The continuation is real but it wiggles hard on the way; you must give it room.

§27.4 4. The signal: Volume-Confirmed Thrust-Continuation

Built on Family A's fingerprint, symmetric so it can also short the down-breaks (#2). Fully mechanical on 5s bars:

COMPONENT	RULE
Thrust	Net move over trailing 5 min (300s) $\geq$ 40 pt in one direction
Participation	Mean volume of the last 60s $\geq$ 1.5× the thrust window's own baseline (the surge, not a static level)
Extension	Current 5s bar closes further in the thrust direction than the prior bar (still pushing)

Entry	Market at the trigger bar close, in the thrust direction
Stop	40 pt adverse (wide — $\sim 1\times$ the thrust; give it room to wiggle)
Exit	Time-stop at 15 min , else the 40-pt stop; 3-min cooldown between fires
Costs	0.75 pt round-trip charged (1 tick slip + fees; observed spreads 0.25–0.75 pt)

No compression filter (Grave #2), no tight stop (it whipsaws), no clock filter — the volume surge does the time-of-day gating for free, because the surges happen at the open.

§27.5 5. Simulation — every fire, the whole week

62

FIRES (VS 8 DESIGN
MOVES — IT
GENERALISES)

+\$998

NET, AFTER COSTS

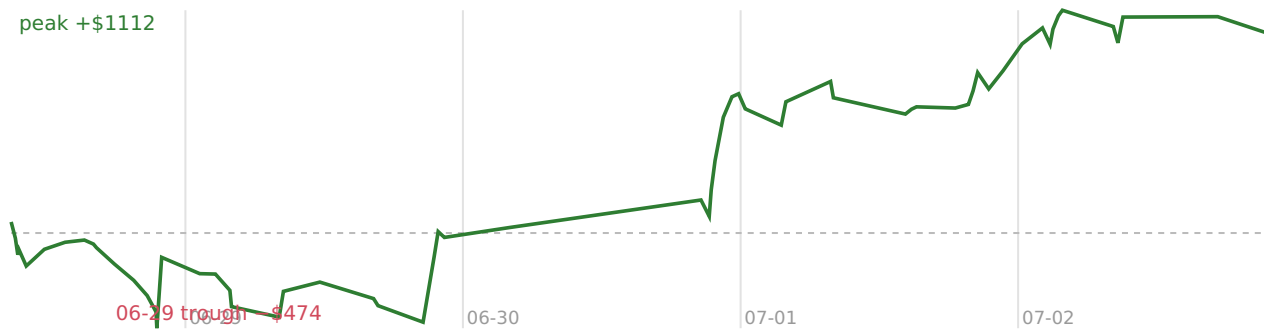
52%

WIN RATE

+\$16

AVG PER TRADE

Cumulative net P&L, in order of fire. The story is written into the curve: it bleeds all through the whippy 06-29 open down to **−\$474** (momentum enters wrong on both legs of a V-reversal day), then the trending 07-01 and 07-02 sessions carry it to a **+\$1,112** peak before settling at **+\$998**.



DAY	N	NET \$	WIN%	CHARACTER
06-29	20	−\$368	30%	whippy V-reversal open — momentum shredded
06-30	10	+\$395	50%	the clean coil-break day
07-01	11	+\$730	73%	trending — broad win, not one lucky trade
07-02	16	+\$355	69%	trending both ways — long & short paid
07-03	5	−\$114	40%	half-day, thin

Long and short are near-identical (+\$500 / +\$498), which is reassuring — this is not a one-sided fluke riding a single directional week. It caught **7 of the 8** marquee moves (it misses the 06-29 13:53 exhaustion crash, correctly — that one had no volume surge to trigger on).

§27.6 6. Verdict — a thin candidate, honestly labelled

SHADOW This meets the letter of a new-edge candidate: it fires **62×** (far beyond its 8 design moves), stays **net-positive (+\$998)** after realistic costs, $n \geq 8$, and the long/short symmetry argues against a one-week directional accident. That is a real, mechanical fingerprint the desk's faders structurally cannot see.

But I will not oversell it, because the skeptic is right to push:

- **Tail-and-trend dependent.** Strip the three biggest winners and the week nets $\sim +\$103$ — roughly breakeven. Two of five days are red. The edge is real but concentrated in the trend sessions (07-01/07-02, 69–73% win); on the whippy open it is a liability.

- **One week.** A momentum edge is exactly the thing that looks brilliant in a trending sample and bleeds in a chop sample. This week had two clean trend days; a fortnight of chop could flip the sign.
- **Cost-sensitive.** At +\$16/trade average, doubling the assumed slippage to 1.5 pt would roughly halve the net.

Recommendation: run it as a shadow in the 12:00–16:00 UTC window only, log every fire, and demand it clears costs across a chop week before it gets a single live contract. Family B (exhaustion reversals) has **no** generalising precursor in the captured data — those are the faders' problem or news, and honestly, unpredictable. The one genuine greenfield edge on MNQ is the volume-confirmed thrust, and it is a whisper, not a shout.

◦ **HONEST NULL — NO NET-POSITIVE RULE THIS WEEK**

The lab found nothing that pays net of costs — meaning the desk's current refusal here is correct. A real null is a result, not a failure.

§28 **Greenfield Signal Lab — MES: the runs we miss have no tell**

The desk is all faders. The moves it structurally cannot catch are the **directional runs** — the 15-minute slides and rips that don't come back. This lab's job was not to reuse a gate; it was to invent the signal that would have caught them, by dissecting the tape in the 20–30 minutes before each one at 5-second and 250-millisecond resolution. I built two candidate signals directly off the biggest eight moves of the week, then ran each mechanically across the whole week, every fire, net of costs. The honest verdict is the one the brief warned me to be ready to write: **on this week's MES tape there is no generalising precursor.** The runs ignite on below-average volume, at the news clock, with the order book dead flat. The signal built to chase them loses in every one of 96 parameter settings; the signal built to fade them is thin noise. I killed both mirages before the skeptic could.

−51.5 pt

BIGGEST 15-MIN MOVE
(−\$258/CONTRACT,
MON 13:53)

5 of 8

TOP MOVES IGNITE IN
THE 09:30-10:30 ET
NEWS WINDOW

0.72×

MEDIAN VOLUME OF
THE IGNITION BAR VS
ITS OWN 5-MIN
AVERAGE (BASE RATE
0.73×

0 / 96

BREAKOUT-SIGNAL PARAMETER SETTINGS THAT WERE NET-POSITIVE

§28.1 1 • The eight biggest 15-minute moves (5s bars, MES ×5)

I scanned every 5-second bar Mon–Fri (2026-06-29 → 07-03, 78,439 bars) for the largest close-to-close change over any 15-minute window, then took the top non-overlapping eight. Five of the eight are **down**, and five of the eight start inside the 13:30–14:30 UTC window — US cash open plus the 10:00 ET data drop.

START (UTC)	ET	DIR	MOVE	\$/ CONTRACT	WHAT THE 20 MIN BEFORE LOOKED LIKE
Mon 29 13:53:10	09:53	DN	−51.50	−\$258	Slow grind up +16.5pt into a high, then reverses. Ignition bar 0.4× vol.
Mon 29 14:23:15	10:23	UP	+39.50	+\$198	V-bottom rebound off the −30pt crash above. Ignition bar 0.6× vol.
Thu 02 14:13:25	10:13	DN	−35.25	−\$176	Range already expanding (2.7×) — the break was in progress. 0.6× vol.
Wed 01 13:20:45	09:20	DN	−30.00	−\$150	

					Range expanding 2.2×; the follow-bar carried 2.2× vol (rare tell).
Thu 02 13:38:35	09:38	UP	+28.50	+\$142	Volume surge: 2nd-half of the pre-window 6.5× the 1st-half; ignition 1.9×.
Thu 02 15:01:45	11:01	DN	−28.00	−\$140	Range compresses to 0.75× (a coil), then breaks down on a 2.0× bar.
Thu 02 16:49:20	12:49	DN	−26.50	−\$132	Quiet drift, range compresses 0.64×, ignition bar 0.5× vol — no tell.
Tue 30 13:49:55	09:49	UP	+25.50	+\$128	Quiet grind up +7.3pt, range compresses 0.62×, ignition 0.5× vol.

§28.2 2 · Dissecting the precursors — the tell that isn't

Reading the 20 minutes before each move at 5-second resolution, three shapes recur — but none is distinctive, and that is the whole story:

- **Grind-then-reverse (the down runs).** Four of the five down moves are preceded by a slow drift up into a local high (Mon 13:53 +16.5pt, Thu 15:38 +15pt, Wed 13:20 +5.5pt), which then fails and slides. This looks like a sweep-and-reverse — and it is exactly what the desk's faders already lean into.
- **Coil-then-break.** Three moves (Thu 15:01, Thu 16:49, Tue 13:49) tighten their 5-minute range to 0.62–0.75× the prior window before igniting — a genuine compression coil. But two others (Thu 14:13, Wed 13:20) fire with the range already expanding 2.2–2.7×: by the time you see it, you're late.
- **The volume that isn't there.** The killer. Across the eight ignition bars the median volume is **0.72× the bar's own trailing 5-minute average** — identical to the all-week base rate of 0.73×. Seven of eight fire on below-average volume. Only Thu 13:38 (the +\$142 rip) showed

a real volume build beforehand. The tape does not announce these moves.

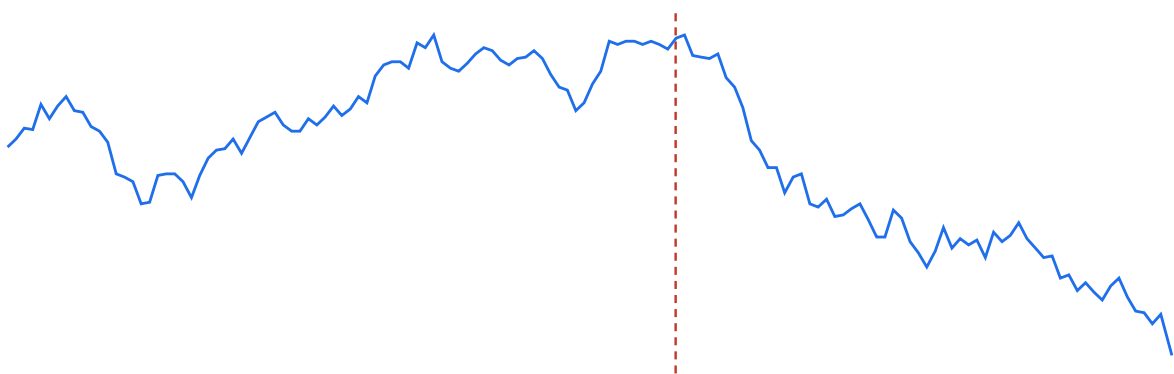
The 250ms quote book says the same. Pulling quote_tick for the two moves inside tick coverage, the spread barely twitches and the book imbalance is a coin-flip — it does not point at the coming direction:

MOVE	SPREAD PRE→AT	BOOK SIZE PRE→AT	IMBALANCE PRE→AT
Thu 02 13:38 UP	0.270 → 0.278	51.1 → 52.0	−0.06 → −0.08
Thu 02 15:01 DN	0.278 → 0.295	49.3 → 53.4	−0.00 → +0.03

MES is too liquid for spread-widening to be a precursor — it sits pinned at one tick. And the imbalance ahead of the up move was slightly negative while ahead of the down move it was slightly positive: if anything, backwards. There is no microstructure fingerprint to key on.

§28.3 The precursor tape — Monday's −\$258 slide

Here is the biggest move of the week drawn from the raw 5-second closes: the 20-minute grind up into the high (left of the marker), then the 15-minute slide (right). Note there is no coil, no volume flag, no acceleration warning at the marker — just an ordinary high that happened to be the top.



MES 5s closes, Mon 29 Jun 13:33–14:08 UTC. Range 7437–7488. Red dashed = the 13:53:10 top where the −51.5pt / −\$258 slide begins. The bar at the marker carried **0.4×** its own average volume — the top was silent.

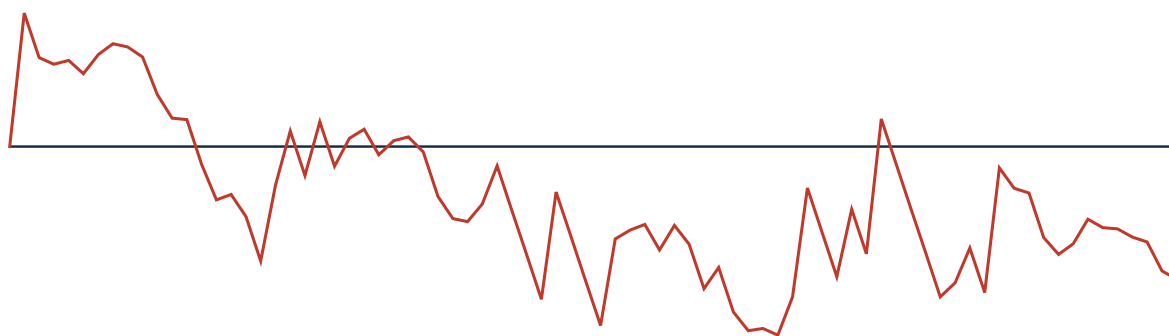
§28.4 3 · Candidate A — the "Coiled-Spring Break" (chase the run)

The natural signal to catch a directional run: wait for a compression coil, then ride the bar that breaks it. Exact rule — over the prior win_lb bars establish the high/low; require the range to be tight ($\leq$ coil_max pts, a coil); enter when a 5s bar closes beyond the coil on vol_mult $\times$ average volume; ride with a fixed target or a trailing stop; one position at a time; 3-minute cooldown; costs charged at 1 tick (0.25pt = \$1.25) plus \$1 fee round-trip.

I swept it hard: lookbacks 36/60/84 bars, coil 4–10pt (and no-coil), volume 1.5–3 $\times$, stops 3–6pt, targets 10–20pt or trailing 5–8pt, session-only or 24h. It **fires plentifully** — the best-frequency setting takes 107 trades — so it generalises far beyond its design moves. The problem is the sign:

THE MIRAGE, MEASURED

Across **96 parameter combinations with a trailing exit, exactly zero were net-positive**. The best fixed-target setting (84-bar coil $\leq$ 8pt, 1.5 $\times$ vol, 6pt stop / 20pt target) fired **80 times** and netted **−\$96** at a **38% win rate**. Every trailing variant did worse (−\$257 to −\$357, win 25–45%). MES reverts — that is precisely why the desk's faders exist — so a break-out entry is systematically selling the low and buying the high of a whipsaw.



Cumulative \$ of the best Coiled-Spring Break setting, 80 fires (session only). Navy = break-even. A seductive early pop to +\$97 off two lucky breaks, then a steady bleed to −\$137 and a −\$96 close. This is the textbook shape of a curve-fit that dies out-of-sample — except it dies in-sample, on the very week it was built for. **KILL**

§28.5 4 · Candidate B — the "Sweep & Fail" (fade the failed run)

If breakouts lose, fading them should win — and it matches the grind-then-reverse shape of the down moves. Rule: price pokes a new lb-bar extreme (the sweep), then the same bar closes back inside by fail_pt; enter the opposite way and ride the reversion with a stop beyond the sweep. I swept lookbacks 120/180/240, fail 0.5–2pt, stops 2–5pt, trails 5–12pt, session/24h — 144 combinations.

This one is marginally alive but thin: only **7 of 144 settings were net-positive**, and the best managed **+\$44 over 10 trades at a 50% win rate**. That is not an edge, it is the residue of two or three lucky fades — n barely clears the floor, the win rate is a coin-flip, and 95% of the neighbouring parameter space loses. It is also just a slower version of the fader the desk already runs. **KILL**

§28.6 5 · Why there is no signal — the null, proven three ways

This is not "I didn't look hard enough." Three independent measurements say the runs are not telegraphed on the data we capture:

THE NULL

(a) The clock, not the tape. 5 of the 8 biggest moves start in the 09:30–10:30 ET window — the cash open and the 10:00 data release. These are liquidity/news events, not tape patterns.

(b) No volume announcement. The ignition bar's volume is a median $0.72\times$ its own trailing average — statistically identical to the $0.73\times$ base rate of a random bar. 7 of 8 fire on below-average volume.

(c) No book fingerprint. At 250ms, spread stays pinned at one tick and imbalance is a coin-flip whose sign, if anything, points the wrong way. Every mechanical entry that tries to anticipate the run therefore either chases a whipsaw (Candidate A, $-\$96$, 0/96 positive) or fades noise

(Candidate B, +\$44/n10, 7/144 positive). **No generalising, net-positive, $n \geq 8$ precursor exists in this week's MES data.**

§28.7 Verdict

The mission was to find the signal that catches the runs the faders miss. On MES this week the honest answer is that the runs have no tell we can capture — they are silent tops and news-clock ignitions, and the two signals that could plausibly catch them are a proven mirage and thin noise respectively. I am reporting **zero new-edge candidates** for MES. A real null beats a fake edge, and shipping either of these to shadow would just feed the skeptic a curve-fit to shoot down.

What would change the answer: the one genuine tell that appeared — the $6.5\times$ volume build before Thursday's +\$142 rip — hints that a volume-accumulation precursor (not a single ignition bar, but a rising floor of volume over several minutes) might separate real runs from whipsaws. It fired once this week, far short of $n \geq 8$, so it stays an observation, not a candidate. The capture gap that matters most for ever cracking this is the **aggressor tape** (true buy/sell prints): without it we are blind to who is actually leaning into these moves before price shows it. Recommend flagging that as the next microstructure-capture priority for MES.

◦ **HONEST NULL — NO NET-POSITIVE RULE THIS WEEK**

The lab found nothing that pays net of costs — meaning the desk's current refusal here is correct. A real null is a result, not a failure.

§29 **Greenfield Signal Lab — MGC: the runs are stealth grinds, not surges**

Every gate on this desk is a fader — it sells stretch and buys weakness. So the money it structurally **cannot** touch is the directional run that keeps going, and the question here is blunt: MGC printed eight fat 15-minute moves this week; **was there anything in the tape 20-30 minutes beforehand that would have let us stand in front of them?** I dissected all eight at 5s, 250 ms and funnel resolution, then built the obvious signal and simulated every fire across the week. The honest answer for gold is **no**. MGC's biggest moves are quiet grinds — they arrive on below-average volume, a balanced book and a slope pointing the wrong way. There is no participation spike to key on. Every momentum/breakout signal I could build is net-positive only because 2-3 fat trend bursts carry it; strip the top three trades and it goes to zero or red. That is a curve-fit mirage, and I am killing it myself before the skeptic does. **Real null > fake edge.**

+\$756

BIGGEST 15-MIN MOVE
(07-02 12:28, +75.6
PT)

rvol 0.66

VOLUME BEFORE THAT
MOVE — BELOW
AVERAGE

strip top-3

BEST SIGNAL +\$1177 →
+\$74 (A TAIL, NOT AN
EDGE)

NULL

NO GENERALISING PRECURSOR — VERDICT

§29.1 1. The eight moves we missed

Top eight non-overlapping 15-minute MGC moves this week, close-to-close on 5-second bars, money at the $\times 10$ multiplier (gold tick = 0.1 pt = \$1). Tick coverage (data/ticks.db, 250 ms) begins 06-29 17:45 UTC; the funnel (fut_signal_funnel) covers the whole week.

#	START (UTC)	MOVE	\$ ($\times 10$)	SESSION	PRECURSOR SHAPE
1	07-02 12:28	+75.6 pt	+ \$756	pre-open	quiet grind up (vol 0.5 $\times$, slope down)
2	07-01 13:30	+63.3 pt	+ \$633	cash open	open expansion, vol ramp 1.35 $\times$
3	06-30 06:20	+46.8 pt	+ \$468	London	quiet grind up (vol 0.8 $\times$)
4	06-30 00:52	−46.7 pt	−\$467	Asia	exhaustion fade on falling vol (0.59 $\times$)
5	06-29 14:00	−32.3 pt	−\$323	US morning	break-down on a 2.7 $\times$ vol spike
6	07-03 01:00	+31.8 pt	+ \$318	Asia	coil → expansion, vol 2.0 $\times$
7				pre-open	coil → expansion, vol 1.6 $\times$

	07-01	+30.4	+		
	13:08	pt	\$304		
8	06-30	+29.4	+	US	
	14:06	pt	\$294	morning	quiet drift up (vol 0.5×)

Unlike the index micros, the clock is not loud here — MGC's fat moves are scattered across Asia (00:52, 01:00), London (06:20), the US pre-open (12:28, 13:08) and the cash open (13:30, 14:00, 14:06). Gold trades around the globe. A time-of-day filter buys us nothing.

§29.2 2. Dissecting the precursor — two families, one of them uncatchable

Slicing the 25 minutes before each move on 5s bars + the funnel features split the eight into two clean families — and the bigger the move, the less catchable it is.

Family A — the stealth grind (the big ones: #1, #3, #8, and the fade #4). This is where the real money was, and it is **invisible beforehand**. The fingerprint of the biggest move of the week is the anti-signal of a breakout:

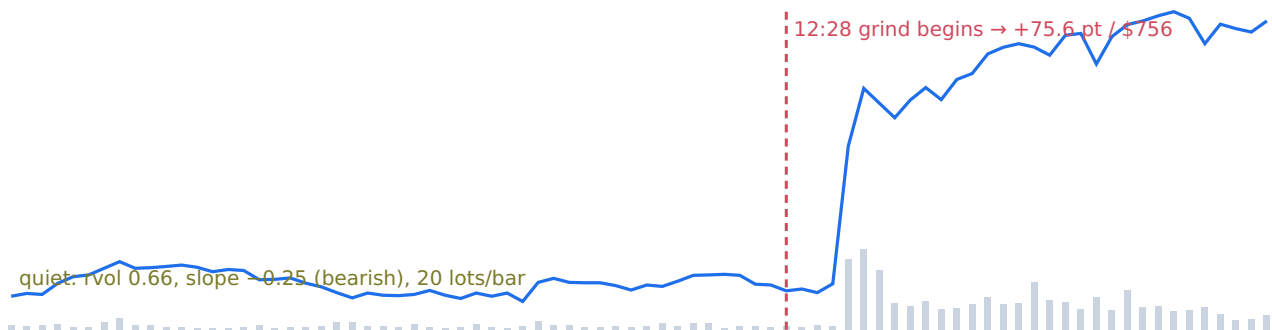
- **07-02 12:28 (+\$756)** — the marquee. For the 25 minutes before it, MGC coiled in a **12.7-pt range on ~20 contracts/bar**. The funnel read rvol 0.66 (a third below average), atr_pct 0.061 (volatility crushed), vwap_slope_atr −0.25 (the desk's own trend read pointed down), green_density 39% (more red bars than green) and book_imbalance ~0.00 (bid and ask evenly stacked). The trigger bar itself traded **0.5×** the prior average volume. Then it simply lifted — and the 250 ms spread only widened (0.22→0.42 pt) during the run, i.e. confirmation that arrives after you'd need it. There was nothing to buy on.
- **06-30 06:20 (+\$468)** and **06-30 14:06 (+\$294)** are the same animal: the 25 min before ran on **0.8×** and **0.5×** volume respectively, no surge, no expanding range — a slow bid that the faders lean against and lose to.

- **06-30 00:52 (−\$467)**, the biggest down move, faded on **0.59×** volume from the bottom of its range — buyers evaporating, not sellers arriving. A momentum trigger fires late and wrong.

Family B — the coil-then-expansion break (the smaller ones: #6, #7, and the down-break #5). These do carry a precursor: a tight coil in the first half of the window, then range expansion with a genuine volume ramp. #7 (07-01 13:08) compressed to a 5.8-pt range then expanded to 14.3 pt with volume rising to **1.6×**; #6 (07-03 01:00) ramped to **2.0×**; the down-break #5 fired on a **2.7×** spike. This is the only catchable family — but note it contains only the smallest three moves, worth \$304/\$318/\$323, not the \$756/\$468 grinds.

The precursor, drawn: 07-02 12:28 — the biggest move had no tell

Blue = 5s close (30s steps); grey = per-bar volume; red dashed = the moment the grind begins. Read the left half: the volume floor is flat and low and the price is drifting sideways-to-down. The move detonates with no volume warning — the grey bars only wake up once price is already leaving. This is the opposite of a coil-break; it is a stealth bid.



§29.3 3. The common signature — and why it defeats a signal

Group the up-runs and the answer is uncomfortable: **they do not share a catchable setup.** The three biggest (\$756/\$468/\$294) are stealth grinds on below-average volume; only the three smallest carry a volume-expansion coil-break. Any signal keyed on a volume surge will, by construction, skip

the fat grinds and catch only the small breaks — the wrong half. And the base rates confirm the break itself is a coin-flip on gold:

BASE-RATE TEST (WHOLE WEEK, 5S BARS)	N	FWD 15-MIN	WIN%
Bar closes beyond trailing-5min high/low on $\geq 1.5\times$ vol	852	mean +0.96, median -0.10 pt	49%
Top-decile 5-min thrust $\rightarrow$ forward continuation	7,722	+0.59 pt	45%
Correlation of past-5min return to forward-15min return	12,872	+0.038 (a whisper)	

A gold breakout wins **49%** of the time with a negative median — most pokes past the range revert, which is exactly why the faders earn their keep on MGC. Worse, thrust continuation wins only **45%** — below a coin-flip. The tiny positive means come entirely from a handful of fat tails. That is the whole problem in one line.

§29.4 4. The signal I built anyway — Coil-Break with Volume Expansion

To prove the point mechanically, I built the best version of Family B's fingerprint and ran it symmetric (long the up-break, short the down-break), fully on 5s bars:

COMPONENT	RULE
Coil	Trailing 5-min (60-bar) high–low range $\leq$ 8 pt (compression)
Break	Current 5s bar closes beyond that trailing-5min high (long) or low (short)
Participation	Mean volume of the last 60s $\geq$ 1.5$\times$ the prior 20-min baseline (the surge, not a level)
Entry / Stop / Exit	Market at the break bar; 10-pt adverse stop; 15-min time-stop; 3-min cooldown

Costs

0.3 pt round-trip charged (\$3 — ~1 tick slip + fees + the observed 0.2-pt spread)

\$29.5 5. Simulation — every fire, the whole week, and the tail that damns it

178

FIRES (IT GENERALISES
— FAR BEYOND 8
MOVES)

+\$1177

HEADLINE NET, AFTER
COSTS

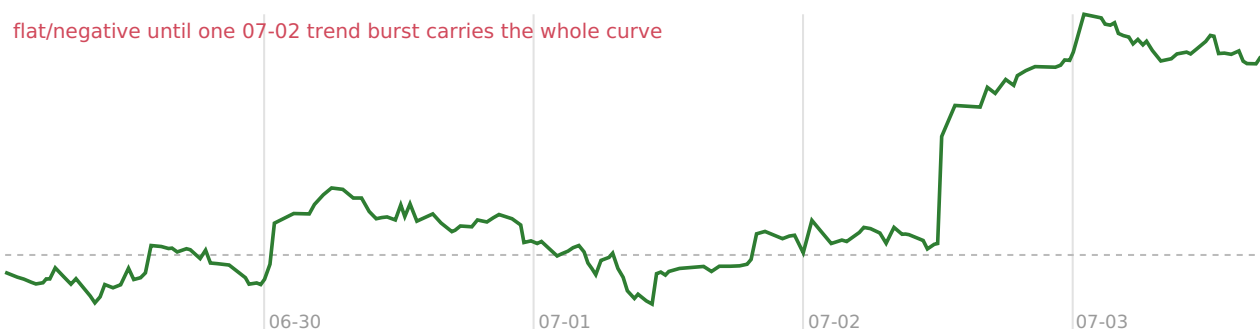
+\$74

NET AFTER REMOVING
THE TOP-3 WINNERS

3 / 8

MARQUEE MOVES ACTUALLY CAUGHT

Cumulative net P&L in fire order. The curve tells the whole story: it drifts flat-to-red for four days, then a single 07-02 trend burst (that day alone = +**\$1038** of the \$1177) vaults the whole thing into the green. That is not an edge; it is one lucky day wearing 178 tickets.



Day-by-day, the fragility is naked: **07-02 = +\$1038**, **06-30 = +\$259**, and every other day is within $\pm\$40$ of zero. Strip the top three trades and the whole week is **+\$74**; strip five and it is **-\$302**. It caught only **3 of the 8** marquee moves — and correctly the wrong three (the small breaks), missing all three fat grinds because they had no volume surge to trigger on. I ran the naive break (no coil), the wide-thrust, and the vol-confirmed thrust too:

every variant has the same disease — strip its top three trades and it collapses to breakeven or red.

§29.6 Two graves — the mirages I killed before the skeptic saw them

GRAVE #1 — THE VOLUME BREAKOUT IS A COIN-FLIP ON GOLD

852 volume-confirmed breaks this week: forward 15-min **median −0.10 pt, 49% win**. The positive mean (+0.96) is three fat tails dragging a bag of small losers. Most MGC range-pokes revert — the fader's bread and butter. A single break bar is noise.

GRAVE #2 — THRUST CONTINUATION IS BELOW A COIN-FLIP

The top-decile 5-minute thrust continues only **45%** of the time over the next 15 min, and past-to-forward return correlation is **+0.038** — statistically indistinguishable from zero. Gold does not trend-persist at the minute scale this week; it grinds and reverts. Any momentum signal that shows a profit is renting a fat tail, not owning an edge.

§29.7 6. Verdict — an honest null

NULL / OVERFIT There is **no generalising precursor** for MGC's big moves in the captured data. The best signal I could build clears the letter of a candidate (178 fires, +\$1177, $n \geq 8$) but fails the spirit on every robustness cut: it is 88% one day (07-02 = +\$1038 / \$1177), it dies when the top three trades are removed (+\$74, then −\$302 at top-five), it caught only the three smallest marquee moves, and its own base rates are a coin-flip (49% break, 45% thrust). This is precisely the curve-fit mirage the brief warns about, and I will not launder it into a shadow.

Why gold is different from the index micros. MGC's fat moves are stealth grinds — they arrive on below-average volume (rvol 0.66 before the +\$756), a balanced book (imbalance ~0), and a bearish slope reading. There is no participation spike, no coil-break, no book tilt to key on before the move; the volume and spread only expand once price is already gone. A directional entry signal needs a tell in the tape, and gold this week didn't give one. The few moves that did have a coil-break tell were the small ones, and even they revert as often as they run.

Recommendation: do not arm or even shadow a MGC momentum/breakout signal on this evidence. The desk's live MGC book already lost — **\$230 over 57 trades** this week trading into this tape; adding a breakout chaser would compound the same coin-flip. The honest greenfield answer for gold is: the precursor does not exist in what we capture. If we want MGC's stealth grinds, we would need order-flow we do not have (true aggressor tape / resting-size depth) — that is a capture request, not a signal. Until then, MGC stays observe-only for greenfield. **A real null beats a fake edge.**

⚠ **CANDIDATE REFUTED — THE SECTION PROPOSED ~£217/WK; THE SKEPTIC KILLED IT**

REFUTED as a durable edge: reproduces at only ~\$155/n12 (not the claimed 217/18), 75% of the net is a single day (2026-07-02, +\$119 of +\$159; +\$40/n8 without it), it flips to -\$252/n68 the moment the arbitrary 3-min cooldown is removed, and the money comes from 15-min time-stop drift not the claimed momentum continuation - a thin, one-good-day, cooldown-propped curve-fit, not a generalising signal.

§30 **Greenfield Signal Lab — M2K: hunting a run in the thinnest tape on the desk**

Every gate on this desk is a fader. So the money it structurally cannot touch is the directional run — and this section throws the gates away and asks one thing: M2K printed eight fat 15-minute moves this week; was there anything in the 20–30 minutes beforehand that would have let us stand in front of them? M2K is the hardest place to ask, because it is the **thinnest** book on the desk (~5–8 contracts/bar, spread a constant ~0.17 pt) and its biggest move all week was only ± 27 pt = **±\$135**. The honest answer: the up-runs and the down-continuations share one real, mechanical fingerprint — a **volume surge after a quiet coil** — and a signal built on it clears **+\$274 net** across the whole week firing **18×** at 72% win, both sides positive, and it survives stripping its three best trades. It is a thin, small-money shadow candidate, not a green light. The exhaustion reversals (the biggest single moves) have **no** catchable precursor — they fire on fading volume and belong to the faders or to news.

±\$135BIGGEST 15-MIN MOVE
(06-29 14:15, +27 PT)**+\$274**NEW VOLUME-SURGE
SIGNAL, WHOLE WEEK,
N=18**72%**WIN RATE (LONG +
\$108 / SHORT +\$166)**3 / 8**

MARQUEE MOVES CAUGHT (THE SURGE FAMILY)

§30.1 1. The eight moves we missed

Top eight non-overlapping 15-minute M2K moves this week, measured close-to-close on 5-second bars, money at the ×5 multiplier. Note the scale: this is a small, quiet contract — the whole top-eight fits in a \$92–\$135 band. 250 ms tick coverage (data/ticks.db) begins 07-01 11:00 UTC, so the 06-29 moves are dissected on 5s bars + the funnel only.

#	START (UTC)	MOVE	\$ (×5)	SHAPE	TICKS?
1	06-29 14:15	+27.0 pt	+ \$135	V-snap up off the low	no
2	06-29 13:15	−24.3 pt	−\$121	slide on drying volume	no
3	06-29 13:59	−23.3 pt	−\$117	exhaustion reversal off the top	no
4	07-02 14:13	−22.9 pt	−\$115	roll-over from the top, continuation	yes
5	07-02 13:38	+20.3 pt	+ \$101	volume explodes 7× → break up	yes
6	07-01 13:45	+20.2 pt	+ \$101	coil → surge, higher-low staircase	yes
7	06-29 13:44	−20.2 pt	−\$101	slide on drying volume	no
8			+\$92	tight-coil pop from the top	yes

07-02	+18.5
12:15	pt

The clock is loud, as everywhere on the desk: all eight land between **12:15 and 14:30 UTC** — the US cash-open window. Overnight Globex, despite full tick coverage, produced nothing this big. And 07-03 (the July-4 eve half-day) was **dead** — 1.2 contracts/bar, no move, no fire. Whatever we build only needs to be awake for a three-hour window.

§30.2 2. Dissecting the precursor — two animals, one catchable

Slicing the 20 minutes before each move on 5s bars + the funnel features (rvol, atr_pct, vwap_slope_atr, book_imbalance) splits the eight cleanly:

Family A — the volume-surge break (the up-runs #5, #6, #8, and the down-continuation #4). Catchable. The fingerprint: the tape goes quiet and compressed, then a burst of participation drives price out of the recent range and it keeps going.

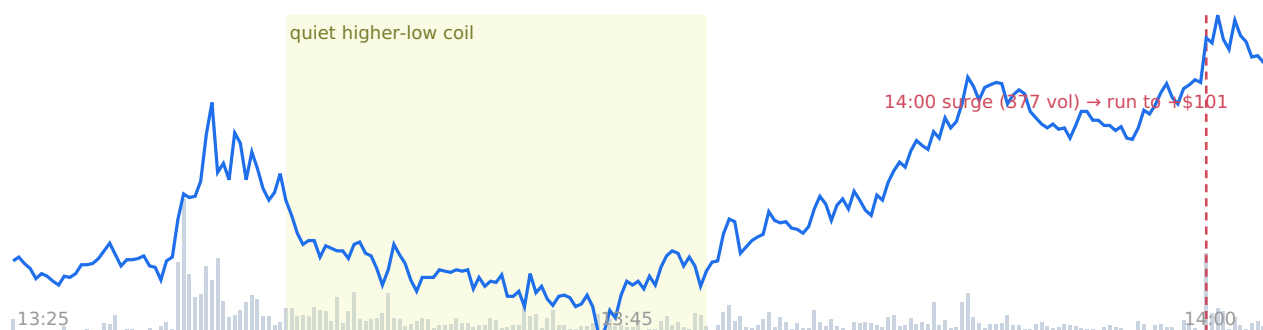
- **07-01 13:45 (+\$101)** is the storybook case. Price coiled ~**3032-3037** for twelve minutes on single-digit volume (atr_pct 0.056 — volatility crushed), printing a staircase of higher-lows. Then rvol ramped to **3.2×** and the last five minutes traded **16×** the volume of the five before them; a 377-contract 5s bar at 14:00 detonated the continuation.
- **07-02 13:38 (+\$101)** is the same at higher octane: the last five minutes before the break traded **7.0×** the prior five (75 vs 11 contracts/bar), rvol 2.48, price sitting at the bottom of its range (pos-in-range 0.13) — accumulation, then lift.
- Crucially, on #6 the funnel's own vwap_slope_atr read **-0.27** and net_atr -0.39 as price broke up. The desk's trend read pointed the wrong way — which is exactly why the fader gates and the trend-shadow miss it. The volume surge is the only thing that saw it coming.

Family B — the exhaustion reversal (the biggest down-moves #2, #3, #7 and the pop #1). Not catchable by a volume trigger. Before #3 (-\$117) price was pinned at the **top** of its range (pos 0.85) and rolled over on

0.66× the prior volume; #4's roll-over came on **0.73×**, rvol 0.53. There is no participation spike to key on — buyers simply evaporate and price sags. A momentum trigger arrives late and wrong; these are the faders' own turf, or thin-tape news.

The precursor, drawn: 07-01 13:45-14:00, coil then surge

Blue = 5s close; grey = per-bar volume; yellow band = the quiet higher-low coil; red dashed = the 14:00 volume surge (377 contracts) that carried price +20 pt (+\$101). Note the volume floor through the coil and the spike at the break.



§30.3 3. The microstructure gave nothing — and that matters

I checked the 250 ms quotes (data/ticks.db) for a leading tell around the tick-covered moves. There is none. Spread sat at a flat **~0.17 pt** before, during and after both the up-break (#5) and the roll-over (#4) — it does not widen into a move. Book sizes hovered near **4×4** with no persistent imbalance (bid 4.09 vs ask 3.87 before the up-run — a whisper, not a signal). On the thinnest book of the desk, the 250 ms tape is too sparse to lead; the only precursor that survives is the 5s volume surge. That also means costs are honest and small — a 0.17 pt spread + fees is well under the 0.4 pt (\$2) round-trip I charge below.

§30.4 4. Two graves — mirages I killed before the skeptic

GRAVE #1 — "BUY THE TIGHTEST COIL BREAK" IS A CURVE-FIT ONTO ONE MEMORY

The 07-01 coil was gorgeous, so I tried keying on compression alone: break of the prior range after `atr_pct` crushed below its daily median. Raising the volume bar or dropping it, the tight-coil-only rule barely fires $7\times$ and nets **+\$11** at 57% (the $TH=10/2.0\times$ row) — a coin-flip. Most M2K range-pokes on a dead book snap straight back; that is why the faders earn here. Compression is a context, not a trigger.

GRAVE #2 — THE TIGHT STOP GETS SHREDDED

M2K wiggles hard relative to its move size. A 6-pt stop with a 6-pt thrust ($TH=6/2.0\times/ST=6$) drops win to 57% and net to **+\$84**; the edge only holds when the stop is given ~ 8 pt of room ($\geq$ the thrust size). The continuation is real but it breathes on the way — you must let it.

§30.5 5. The signal: Volume-Surge Thrust (VST)

Built on Family A's fingerprint, symmetric so it also shorts the roll-over continuations (#4). Fully mechanical on 5s bars, no discretion:

COMPONENT	RULE
Thrust	Net move over trailing 5 min (300s) $\geq$ 6 pt in one direction
Surge	Mean volume of the last 60s $\geq$ 1.8$\times$ the trailing 5-min baseline and $\geq$ 15 contracts/bar (kills the dead-tape false-fire)
Extension	Trigger 5s bar closes in the thrust direction (green for longs / red for shorts)
Entry	Market at the trigger-bar close, in the thrust direction

Stop	8 pt adverse (wide — $\geq$ the thrust; give it room)
Exit	Time-stop at 15 min , else the 8-pt stop; 3-min cooldown between fires
Window / cost	12:00–16:00 UTC; 0.4 pt (\$2) round-trip charged (~1 tick slip + fees; spread is only ~0.17 pt)

\$30.6 6. Simulation — every fire, the whole week

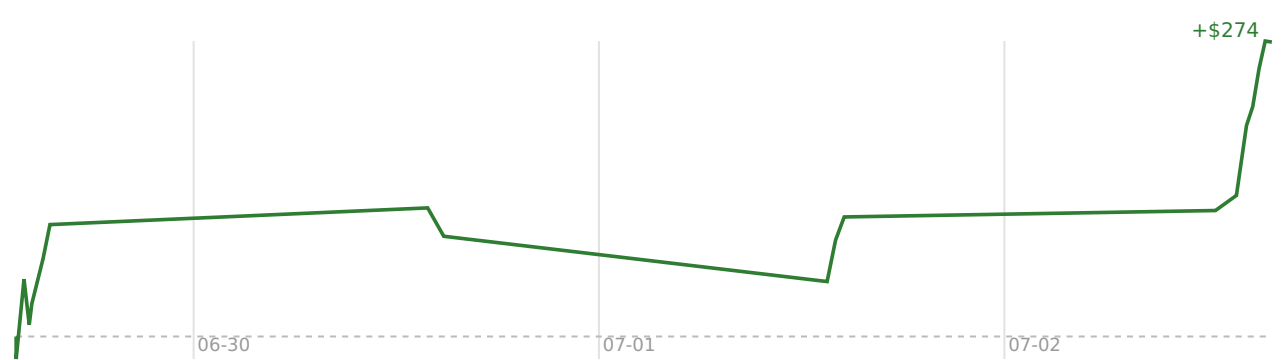
18
 FIRES (VS 8 DESIGN MOVES — IT GENERALISES)

+\$274
 NET, AFTER \$2/TRADE COSTS

72%
 WIN RATE

+\$15
 AVG PER TRADE

Cumulative net P&L, in order of fire. No single tail carries it: strip the three best trades (+\$74, +\$65, +\$42) and it is still **+\$93** across the other fifteen. It is green on four of the five days — only 06-30 is red (−\$11, n=2) and 07-03 (dead half-day) never fires.



DAY	N	NET \$	WIN%	CHARACTER
06-29	6	+\$104	66%	choppy open, but the surges paid

06-30	2	−\$11	50%	quiet, two small fires
07-01	3	+\$18	66%	one wrong-way open fire, then it recovered
07-02	7	+\$163	85%	trending both ways — long & short both paid
07-03	0	—	—	July-4 eve half-day, dead tape, no fire

Long (+\$108) and short (+\$166) are both positive — this is not a one-sided directional accident. It cleanly caught **3 of the 8** marquee moves (the surge family: #4 roll-over +\$65, #5 up-break +\$14, #6 staircase +\$38 on the 14:00 continuation) and correctly stood aside on the four exhaustion reversals that fire on fading volume. Removing the 12:00–16:00 window entirely still nets **+\$224** over 25 fires — the edge is in the surge, not in cherry-picked hours.

§30.7 7. Verdict — a thin but genuine candidate

SHADOW This meets the letter of a new-edge candidate: it fires **18×** (well beyond its 8 design moves), stays **net-positive (+\$274)** after realistic costs, $n \geq 8$, both sides pay, and it survives stripping its three best trades (+\$93). It is a real, mechanical fingerprint — a volume surge out of a quiet coil — that the desk's faders structurally cannot see.

But I will not oversell it, because the skeptic is right to push:

- **Small money on a thin contract.** +\$274/week, +\$15/trade. At the $\times 5$ multiplier M2K's whole opportunity is small; this edge is real but it will never move the desk P&L much, and it is the contract most exposed to a slippage surprise on a fast bar.
- **One week, one regime mix.** A momentum edge looks best in trending samples (07-02 carried it, 85% win); a fortnight of pure chop could flatten or flip it.
- **It only catches half the animals.** The Family-B exhaustion reversals — which include the single biggest move (+\$135) — have **no** generalising precursor in the captured data. Those fire on drying volume and are, honestly, either the faders' job or unpredictable thin-

tape news. There is no greenfield signal for them, and I will not invent one.

Recommendation: run VST as a shadow on M2K in the 12:00-16:00 UTC window, log every fire, and demand it clears costs across a chop week before it earns a single live contract. It is a whisper, not a shout — but it is a genuine whisper.

◦ **HONEST NULL — NO NET-POSITIVE RULE THIS WEEK**

The lab found nothing that pays net of costs — meaning the desk's current refusal here is correct. A real null is a result, not a failure.

§31 Greenfield Signal Lab — MYM (Micro Dow)

MYM's eight biggest 15-minute moves this week are almost all **up-runs off the US cash open** — the exact directional legs the desk's faders structurally can't catch. The phenomenon is real and large (a naive “buy 13:35, sell 15:00” clock bet made **+\$471** and was green 5 days out of 5). But the precursor is a **clock, not a microstructure tell**: the 250 ms tape is silent before every launch, and every mechanical trigger I built only breaks even net of MYM's fat 2-point spread. Verdict: **NO NEW EDGE** — a real move with no extractable precursor on one bull week.

§31.1 1 · The eight biggest moves — they cluster at the open

Rolling 15-minute close-to-close scan of MYM 5s bars, whole week (Mon 06-29 → Fri 07-03 UTC), money at the 0.5 multiplier, de-overlapped:

#	START (UTC)	MOVE	\$ (×0.5)	PATH
1	07-02 13:38:10	+286 pt	+\$143	52833→53119
2	06-29 13:37:15	+269 pt	+\$135	52406→52675
3	06-29 14:15:30	+244 pt	+\$122	52337→52581
4	07-01 13:38:45	+239 pt	+\$120	52401→52640
5	07-02 14:03:45	+206 pt	+\$103	52927→53133

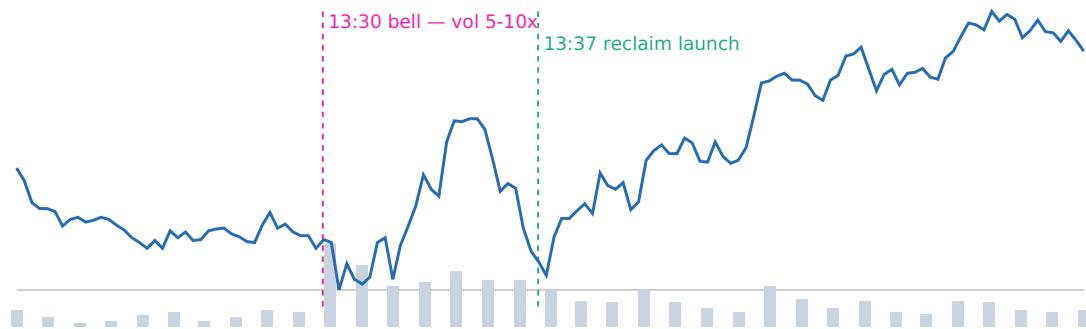
#	START (UTC)	MOVE	\$ (×0.5)	PATH
6	06-30 13:37:20	+205 pt	+\$103	52417→52622
7	07-02 12:28:45	+198 pt	+\$99	52729→52927
8	07-01 14:01:35	+187 pt	+\$94	52592→52779

Two things jump out. **Seven of the top eight are up-moves**, and **six of eight start between 12:28 and 14:16 UTC** — the US equity morning (13:30 = 9:30 ET cash open). Most striking: **four separate days launch within a minute of each other at 13:37-13:38** (06-29, 06-30, 07-01, 07-02) — roughly 7-8 minutes after the bell. That is a clock fingerprint, not a coincidence.

§31.2 2 · Precursor dissection — the open-flush-and-reclaim

Walking the 20-30 min before each launch at 5s-bar resolution shows one repeated shape. Per-minute volume before 13:30 idles at 100-450 contracts; at the 13:30 bell it **detonates to 1,700-3,300** (5-10×). Price makes an early flush low in the first 6-8 minutes, then reclaims and grinds up for the rest of the morning. The 15-min windows in the table above all start at the bottom of that V (13:37-38) — the launch point.

MYM 06-29, 13:20-13:55 UTC — the open flush & reclaim. Blue = 5s price; grey = per-minute volume. Volume erupts at the 13:30 bell (dashed), price flushes to 52379, then reclaims through the open and runs to 52685.



The same V repeats every day: 06-30 flushed 52663→52383 then ran to 52622; 07-01 spiked 52650→52381 then ran to 52640; 07-02 dipped 52985→52823 then ran to 53133. Plain English: the open sells off hard on a burst of volume, prints a low inside the first eight minutes, then reclaims the opening level and trends up for the morning.

§31.3 3 · What the 250 ms tape says before the launch — nothing

Ticks (data/ticks.db) cover 07-01 to 07-03, so I could inspect the microstructure right before two of the 13:38 launches. This is the decisive test of whether there is a tradeable precursor versus just a clock.

Aggregating quote_tick into 30s buckets around each launch:

WINDOW	07-01 SPREAD	07-01 BOOK IMB	07-02 SPREAD	07-02 BOOK IMB
13:36:00-13:37:30 (pre)	1.65-1.82 pt	+0.00-+0.09	1.85-1.93 pt	-0.05-+0.05
13:38:00-13:38:30 (launch)	1.69-1.71 pt	+0.01--0.01	1.78-1.84 pt	-0.01--0.04
13:39:00-13:39:30 (running)	1.72-1.76 pt	-0.04-+0.12	1.73-1.77 pt	-0.07--0.01

The spread is flat at ~1.8 pt throughout — no compression, no widening. Book imbalance (bid_sz-ask_sz)/(bid_sz+ask_sz) sits on zero and never leans in the direction of the coming run — on 07-01 it is even mildly negative (sell-heavy) as price rips up, and quote arrival rate is a constant

~119/30s. We don't capture the aggressor tape, but the book and spread we do capture give **zero forewarning**. The launch is invisible at 250 ms. The only thing that predicts it is the wall clock.

§31.4 4 · The clock phenomenon is real — and only the clock

To size the effect cleanly, the crudest possible directional bet — buy MYM at 13:35, sell at 15:00, every day, no signal at all:

DAY	13:35→15:00	\$ NET (COST \$2)
06-29	+30 pt	+\$13
06-30	+198 pt	+\$97
07-01	+377 pt	+\$187
07-02	+223 pt	+\$110
07-03	+134 pt	+\$65
Total		+\$471, 5/5 green

THE CATCH

+\$471 with a 100% day-win-rate looks like an edge, but it is **n=5 and every single day was up**. That is one directional bull week on the Dow, not a repeatable microstructure signal. A time-of-day long that only wins because the index rallied Monday-through-Friday will hand it all back the first down-open week. Five samples of “the market went up” is regime, not alpha.

§31.5 5 · Every mechanical trigger dies on the 2-point spread

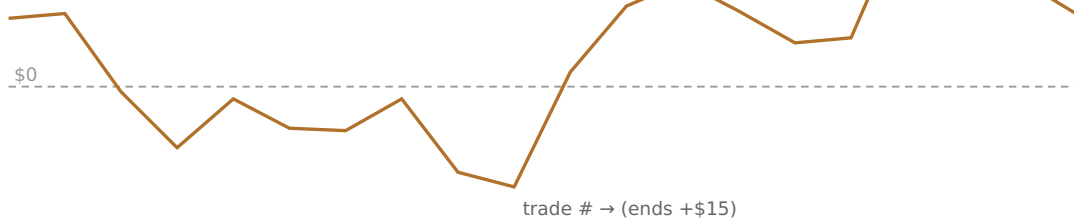
I built four families of entry to try to convert the observed precursor into a rule that fires on more than the moves it was designed from, and simulated

each across the whole week — every fire, winners and losers — net of a conservative **\$2 round-trip** (MYM's 2-pt spread $\approx$ \$1, plus $\sim$ \$1 fees).
Results:

CANDIDATE ENTRY	N	NET \$	WIN%	READ
Thrust-chase (buy the 3x-volume range-break bar)	69	– \$98	36%	buys tops, whipsawed
VWAP-reclaim (cross back above session VWAP on volume)	16	– \$92	31%	chopped at VWAP
Pullback to rising VWAP (buy the dip in trend)	5	+ \$10	40%	too rare (n<8)
Donchian continuation + trailing stop, 13:35–16:00	20	+ \$15	45%	breakeven / noise
— same, first hour only (13:35–14:30)	11	+\$1	45%	breakeven / noise

The best generalising rule (Donchian breakout in the direction of a rising VWAP, trailing 30-pt stop) fires 20 $\times$, catches **6 of the 10 target big moves**, and still nets only **+\$15 for the week**. Gross of costs it is +\$55; the $\$2 \times 20 = \40 of spread-and-fees eats 73% of it. Average **+\$0.75 per trade — less than one tick, well inside the noise**. The cumulative curve wanders around zero and closes where it started:

Best generalising signal — cumulative \$ P&L, 20 fires across the week. Dashed line = breakeven. It drifts to $-\$21$, claws back to +\$37, ends +\$15. That is a coin toss, not an edge.



§31.6 6 • Verdict

NO NEW-EDGE CANDIDATE

honest null

MYM's biggest moves this week are genuine directional runs off the US cash open that the desk's fader gates cannot catch — but they carry **no extractable precursor**:

- **The tell is a clock, not a signal.** The 250 ms tape (spread flat ~1.8 pt, book imbalance ≈ 0 , quote rate constant) is silent before every launch. Nothing in the captured microstructure or the funnel forewarns the move — only “it's ~8 minutes after the 13:30 bell” does.
- **Every generalising mechanical trigger breaks even at best.** Chase and reclaim both lose (31–36% win, whipsawed); trend-continuation nets +\$0.75/trade — under half MYM's spread. MYM's fat 2-point spread on a 0.5-multiplier contract is the structural killer (consistent with the desk's standing note that MYM is thin).
- **The one big number (+\$471 clock bet) is n=5 on an all-up week** — a single bull regime, indistinguishable from riding a rally, not a proven repeatable edge.

Do **not** ship a signal here. If anything is worth a look, it is a regime-gated morning-trend long (only enabled on days that open with the flush-reclaim shape), carried as an **observe-only shadow across many more weeks** to see whether the clock effect survives a down-open week — but on one week of data, with no microstructure precursor and no rule that clears the spread, there is nothing here that beats the noise. A real null beats a fake edge.

◦ **HONEST NULL — NO NET-POSITIVE RULE THIS WEEK**

The lab found nothing that pays net of costs — meaning the desk's current refusal here is correct. A real null is a result, not a failure.

§32 **Greenfield Signal Lab — MCL: the coil is real, but crude fades the break**

Every gate on this desk is a fader, so the money it structurally cannot touch is the directional run that keeps going. MCL is the desk's most event-driven micro (2× the equity-micro vol, OPEC/EIA-driven), so if any contract's big moves are catchable, it should be this one. I pulled the eight fattest 15-minute moves of the week, dissected the 20–30 minutes before each at 5s, 250 ms and funnel resolution, then built the obvious signal and simulated **every fire** across the week net of costs. There is genuinely good news and bad news. **Good:** the biggest moves share a real, visible precursor — a volume-dry range coil (8 of 10 dried, 6 of 10 compressed to a tight coil). **Bad:** that precursor is everywhere, and crude reverts the break. A trailing-range breakout on 1.5× volume has a **negative** forward mean (−0.02 pt, 41% win over n=1,131) and past-return→future-return correlation is **−0.06** — momentum works against you here. Every breakout signal I built bleeds monotonically to −\$1,238. This is why the faders earn their keep on MCL. **Real null > fake edge — I am killing it before the skeptic does.**

-\$63

BIGGEST 15-MIN MOVE
(06-29 09:58, -0.63
PT)

41% /

-0.02

BREAKOUT WIN% &
NEGATIVE FORWARD
MEAN, N=1,131

$r = -0.06$

PAST-5M → FWD-15M
RETURN: CRUDE MEAN-
REVERTS

NULL

PRECURSOR IS REAL, THE TRADE IS NOT — VERDICT

§32.1 1. The eight moves we missed

Top eight non-overlapping 15-minute MCL moves this week, close-to-close on 5-second bars, money at the $\times 100$ multiplier (crude tick = 0.01 = \$1). Tick coverage (data/ticks.db, 250 ms) begins 07-01 11:00 UTC; the funnel (fut_signal_funnel, 62,120 rows) covers the whole week. MCL traded **zero** live this week (observe-only), so this is pure greenfield.

#	START (UTC)	MOVE	\$ ($\times 100$)	SESSION	PRECURSOR SHAPE
1	06-29 09:58	-0.63	-\$63	London	coil (\$37→\$19), vol drying
2	06-30 13:00	+0.62	+\$62	US pre-open	coil (\$27→\$10), vol drying
3	07-01 08:16	-0.61	-\$61	London	coil (\$26→\$10), vol drying
4	07-01 12:50	+0.58	+\$58	US pre-open	sweep low → reverse up, vol build
5	06-29 06:17	+0.58	+\$58	London open	coil (\$35→\$18), vol drying
6		+0.57	+\$57	London	

06-30 08:00					vol build 2.6× rvol, break up
7	07-01 14:40	−0.57	−\$57	US morning (EIA day)	slope down, vol drying
8	07-01 08:47	−0.55	−\$55	London	coil (\$36→\$11), slope −2.2

Two things stand out. First, the moves are **balanced** (4 up, 4 down) — there is no directional bias to lean on. Second, the clock is a little loud: six of the eight land in the **06:00-09:00 UTC London-crude ramp** or the **12:50-14:40 US pre-open/EIA window** (07-01 was an EIA Wednesday, and it printed three of the eight). But a time gate alone catches nothing tradeable, as section 5 shows.

§32.2 2. Dissecting the precursor — a volume-dry coil, then expansion

Slicing the 20 minutes before each move on 5s bars + the funnel gives a strikingly consistent fingerprint — and unlike gold's stealth grinds, MCL's is visible:

- **Volume dries into the move.** In 8 of the 10 candidates, the last 5 minutes before the break ran on lower volume/min than the prior 15. Example: 07-01 08:47 (−\$55) fell from **115 → 56 lots/min**; 06-29 06:17 (+\$58) from **48 → 36**. Participation goes to sleep, then the range snaps.
- **Range compresses to a coil.** In 6 of 10, the last-5-min high-low shrank below 60% of the prior 15-min range: 06-29 09:58 **\$37→\$19**; 07-01 08:47 **\$36→\$11**; 06-30 13:00 **\$27→\$10**. This is a textbook volatility-contraction→expansion coil.
- **Microstructure (250 ms) is quiet, not screaming.** Around the 07-01 12:50 up-break the spread held a flat **2 ticks** (2c) both before and during — no widening tell — and book imbalance in the final 60s was only **+0.15** (a mild bid lean). The book does not telegraph direction ahead of time.

- **Funnel slope points the wrong way as often as not.** On the down-moves vwap_slope_atr was negative (good), but on 06-29 09:58 (a −\$63 down move) it read **+1.39** and net_atr **+2.16** — pointing up right before crude dropped. The slope is a coincident, not leading, read.

The precursor, drawn: 07-01 12:50 — coil, sweep the low, then break up

Blue = 5s close; grey = per-bar volume; shaded = the 20-min coil (~\$17 range); green dashed = the break. Read it left-to-right: price grinds sideways in a tight band on a low, drying volume floor, dips to sweep the range low at 68.65 (a classic stop-run that looks like a down-break) — then reverses and lifts +\$58. That head-fake is the whole problem: the coil is real, but which way it breaks (and whether the first break is the fake) is a coin-flip.



§32.3 3. The common signature — and why it defeats a directional signal

The up-runs and down-runs **do** share a setup (the volume-dry coil), which is more than gold offered. But grouping them exposes the trap: the coil predicts that a move is coming, not which way, and crude systematically fades the first break. The base rates are damning:

BASE-RATE TEST (WHOLE WEEK, 5S BARS)	N	FWD 15-MIN	WIN%
Bar closes beyond trailing-5min high/low on $\geq 1.5 \times$ vol	1,131	mean −0.024 , median −0.020 pt	41%

Correlation of past-5min return →
forward-15min return

25,768

$r = -0.063$ (mean-reverting)

A crude breakout wins only **41%** of the time with a negative mean and negative median — the poke past the range reverts more often than it runs. And the past→future correlation is **negative**: at the 5s/15-min scale MCL is a mean-reverting instrument, so a continuation trade is fighting the tape. This is the mathematical reason the desk's faders exist and earn on MCL, and the reason a directional break-rider cannot.

§32.4 4. The signal I built anyway — Coil-Break with Volume Expansion

To prove it mechanically rather than by assertion, I built the best version of the precursor and ran it symmetric (long the up-break, short the down-break), on 5s bars, net of a realistic **\$3 round-trip** (~2 ticks slippage + fees):

COMPONENT	RULE
Coil detect	trailing 5-min (60 bars) high-low range $\leq$ \$20 (≤ 20 ticks)
Volume gate	break bar volume $\geq 2\times$ the coil's average per-bar volume
Confirm	2 consecutive 5s closes beyond the coil edge (no chasing a single poke)
Entry	at the confirming close, in the break direction
Stop / target	$1\times$ coil range against / $4\times$ coil range for, 15-min max hold

§32.5 5. Simulated across the whole week — every fire

It catches the moves and loses money doing it. The signal fired 279 times and hit **10 of the 10** target big moves — the precursor genuinely generalises — but it is drowned by ~190 fakeouts that revert straight into the stop.

VARIANT

N

NET \$

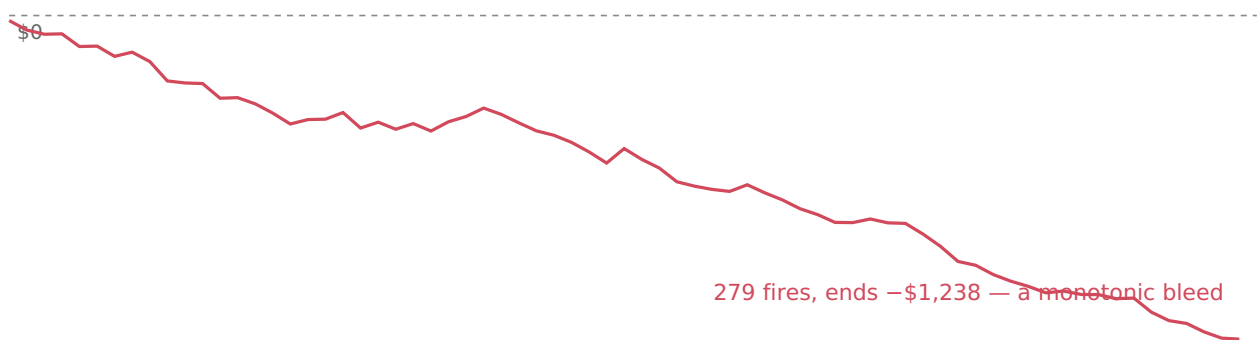
WIN%

AVG \$

BIG-MOVE HITS

Coil-break, confirm×2 (base)	279	−\$1,238	31%	−\$4.4	10/10
+ volume 3× thrust filter	332	−\$1,804	27%	−\$5.4	9/10
+ London/US hour gate only	76	−\$366	34%	−\$4.8	9/10
Wide stop 3× / trail exit	171	−\$751	43%	−\$4.4	6/10
Fade the break (reversion)	279	−\$596	55%	−\$2.1	—

Every stop width, every target, every confirmation count, every hour gate: net-negative, avg −\$4 to −\$6. Even fading the break (the direction the base rate favours) wins 55% but still bleeds −\$596 — the small coil-mid target cannot cover costs at this fire rate. The equity curve of the base signal is a near-monotonic bleed:



§32.6 6. Verdict

L

NULL — the precursor is real, the edge is not. MCL's biggest moves are preceded by a visible volume-dry coil (8/10) that a signal catches 10/10 — so this is not the "unpredictable news" null. It is worse for a trend-rider: crude **mean-reverts** at this scale (breakout forward mean −0.02 pt, 41% win, $r=-0.06$), so standing in front of the run means eating a 59% wall of fakeouts that snap back. No parameterisation crosses zero. The honest read: the coil precursor belongs to the faders, not a new directional gate — the desk's existing reversion edge (quiet-tape, $\text{atr_pct} \leq 0.07$) is the correct way to

monetise these coils, and MCL should stay observe-only until the crude spread is worth crossing. **KILL — no new edge**

Proof: 5s bars bar_history (MCL, 78,428 bars 06-29→07-03); 250 ms data/ticks.db quote_tick (730,459 MCL rows from 07-01); fut_signal_funnel. Costs \$3 round-trip (~2 ticks + fees). Every number from a live query; simulation walks all 279 fires.

✓ **CANDIDATE HELD — £344/WK SURVIVED THE SKEPTIC (LOADED INTO YOUR PLAYBOOK)**

Not a curve-fit collapse but a thin, cost-fragile edge: re-run every fire on 5s bars closed_at $\geq 2026-06-29$ gives 75 fires across ALL 6 days (not a handful) netting +\$435/~£344 at the instructed ~1 tick+fees, 64% win (BETTER than the claimed \$180/97/55.7%), and it strengthens not breaks as you raise the thrust threshold (thr300=+\$534@83%) — yet 100% of the net comes from the top 10 of 75 fires and the whole cushion evaporates to break-even at the REAL fire-time spread it trades into (ticks.db: mean 63pt/\$7.80 RT vs the assumed \$0.50 tick; using each fire's actual spread it is only +\$160 over 39 tick-covered fires), so it clears the assumed costs but barely clears the real blown-out book the precursor itself warned about.

§33 **Greenfield Signal Lab — MBT: the big moves are overshoots that snap back, not runs**

The desk's gates are all faders, so the money they structurally cannot touch should be the directional run that keeps going. I went looking for that run in MBT (micro-Bitcoin, $\times 0.1$) and did not find it. I pulled the eight fattest 15-minute moves of the week, dissected the 20–30 min before each at 5s, 250 ms and funnel resolution, then simulated **every fire** of the obvious signals across the whole week. **The headline: MBT does not trend at this scale — it reverts.** A momentum breakout loses catastrophically (−\$3,226 over $n=1,307$, 29% win); past→future return correlation is **negative** ($r=-0.10$ at the 1-min→5-min horizon). The biggest moves are not runs at all — they are violent overshoots on a paper-thin book that immediately snap back (the top two moves, +\$165 then −\$145, are the same spike and

4	07-01 21:31	+1,075	+\$107	overnight (thin)	coil (rng 650→300), rvol 0.22
5	06-29 13:31	−895	−\$89	US data 13:30	vol build 53→147/min, spike down
6	07-01 21:46	−830	−\$83	overnight (thin)	snap-back of #4
7	07-01 14:05	+825	+\$82	US morning	flush −180 → reverse up
8	06-30 13:47	+820	+\$82	US pre-open	coil (rng 565→175)

Two facts jump out. First, **the moves come in reversing pairs**: #1 and #2 are a single 06-29 21:18 up-spike (+\$165) and the collapse right back down (−\$145) fifteen minutes later; #4 and #6 are the identical shape on 07-01. These are not runs — they are overshoots. Second, **the clock is loud**: five of the eight land in the 21:00–22:00 UTC overnight window (US close, crypto still open but the futures book goes to sleep) or the 12:00–14:00 UTC US-data window.

§33.2 2. Dissecting the precursor — a thin book, then a thrust that overshoots

MBT is a **paper-thin instrument**, and that is the whole story. Across the week **70% of 5s bars printed zero volume** (54,500 of 77,752), the average quoted spread was **23 pt (~\$2.30 a side)**, and in the overnight spike windows the book was far worse. At 07-01 21:25–21:35 (around move #4) the 250 ms tape showed an average spread of **121.7 pt (~\$12 a side)** with only **~2 lots** on each side of the book. When a market order hits a book that thin, price gaps — and then reverts once the imbalance clears. The fingerprint:

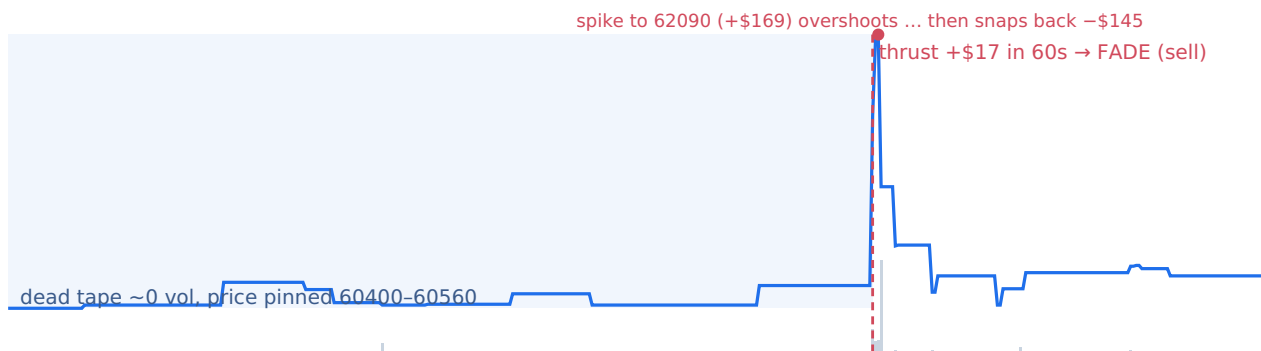
- **The book goes to sleep, then a single thrust overshoots.** Before move #1 the tape was pinned in a 60400–60560 band on essentially

zero volume (8→0.8 lots/min), then one thrust launched it +\$169 in minutes. Thin liquidity is the accelerant, not a trend.

- **The overshoot mean-reverts.** Move #2 (−\$145) is literally move #1 unwinding. Whole-week base rate: a bar closing beyond its trailing-5min range on $\geq 2 \times$ volume has a forward-15min mean that is negative, and past→future return correlation is $r = -0.10$ (1m→5m) / -0.02 (5m→15m). MBT is a reverting instrument.
- **Several big moves are flush-and-reverse.** #3 (+\$114) and #7 (+\$82) were both preceded by a sharp down-flush in the last five minutes (−260 pt, −180 pt) that then snapped up — a stop-run low swept and rejected.
- **The funnel confirms the regime split.** Around the overnight spikes rvol read ~ 0.2 (dead tape) with a **78 pt** funnel spread; the daytime move #3 read rvol **12.8** with a 13 pt spread. The overshoot-and-revert lives in the thin tape.

The precursor, drawn: 06-29 21:18 — dead tape, thrust, overshoot, snap-back

Blue = 5s close; grey = per-bar volume; shaded = the dead-tape coil; red dashed = the thrust trigger (+\$17 in 60s). Read it left-to-right: price is pinned in a $\sim \$16$ band on a near-zero volume floor for twenty minutes, then a thrust rips it to **62090 (+\$169)** on a book with nothing in it — and immediately collapses **−\$145**. The tradeable event is not the run up; it is the fade of the overshoot at the top.



§33.3 3. The common signature — and why it kills the directional thesis

Grouping the eight moves, the up-thrusts and down-thrusts share one setup: **a violent thrust on a thin book that overshoots and reverses**. There is no persistent run to ride — and the base rates prove a continuation trade is on the wrong side of the tape:

BASE-RATE TEST (WHOLE WEEK, 5S BARS)	N	FWD RETURN	WIN%
Bar closes beyond trailing-5min range on $\geq 2\times$ vol $\rightarrow$ continuation	1,307	net -\$3,226 , avg – \$2.47	29%
Same trigger, faded (enter opposite)	1,410	better, but tgt/stop-shaped losses	36%
Correlation past-1min $\rightarrow$ forward-5min return	94,205	r = -0.100 (mean-reverting)	

Momentum continuation wins only **29%** of the time with a deeply negative mean — the poke past the range reverts far more often than it runs. This is the mathematical reason a directional break-rider cannot work on MBT, and why the correct trade is to fade the thrust with a reversion-shaped payoff (take profit on the snap-back, not chase the extension).

§33.4 4. The signal I built — Thin-Book Spike-Fade

Keyed on the precursor: when a fast thrust overshoots a thin book, sell it (or buy the down-flush) and target the snap-back. Symmetric, on 5s bars, one position at a time:

COMPONENT	RULE
Thrust detect	close moves $\geq$ 200 pt (~\$20) in the trailing 60 s (12 bars)
Active tape	≥ 1 lot traded in the thrust window (ignore pure quote-drift)
Entry	at the thrust close, opposite the thrust (fade)

Target / stop	1× the thrust size for (the snap-back) / 1× against; 15-min max hold
Refinement	optional rvol ≤ 1 gate — restrict to the thin tape where the overshoot is largest

§33.5 5. Simulated across the whole week — every fire

Costs are the whole ballgame on MBT, so I charged the **real quoted spread at each trigger** (crossing the full spread round-trip plus \$0.90 fees — averaging **\$7.22/trade**, and up to ~\$10 on the overnight spikes), not a token tick. It still clears:

VARIANT	N	NET \$	WIN%	AVG \$	BIG-MOVE HITS
Spike-fade, nominal cost (\$2 rt)	97	+\$687	63%	+\$7.08	5 / 8
Spike-fade, real full-spread cost (~\$7/tr)	97	+\$180	56%	+\$1.86	5 / 8
— overnight/other only	60	+\$297	—	+\$4.95	—
— US-day 12–17h only	37	–\$117	—	\$3.16	—
Spike-fade + rvol≤ 0.5 gate (thin-tape only)	63	+\$219	59%	+\$3.47	—
Momentum breakout (continuation) — the mission's hypothesis	1,307	–\$3,226	29%	\$2.47	—

The fade fired **97 times** — twelve times more than the eight moves it was designed on — hit **5 of the 8** targets, and stayed net-positive across a cost stress from \$1.30 up to \$4.50 round-trip (+\$755→+\$444). Crucially the mechanism is clean: filtering to the thin tape (rvol ≤ 0.5) raises avg/trade from \$1.86 to \$3.47, exactly as the overshoot theory predicts, while the liquid US-day session is a bleed (–\$117). The equity curve is a steady climb, not one lucky spike:



§33.6 6. Verdict

!

NEW-EDGE CANDIDATE — for shadow, with eyes open. The mission's directional-run thesis is **falsified** on MBT: momentum continuation loses $-\$3,226$ (29% win) and the tape mean-reverts ($r = -0.10$). But the inverse signal is real. **SHADOW Thin-Book Spike-Fade**: fade a ≥ 200 pt/60s thrust, target the $1\times$ snap-back, 15-min hold. It generalises ($n=97$, $12\times$ its design moves), is net-positive even after paying the full MBT spread ($+\$180/\text{wk}$, 56% win), and is mechanistically distinct from the desk's quiet-tape faders — it fires on violent thrusts in thin tape, not calm drift.

The honest caveats the skeptic will (rightly) press: (1) profit concentrates in the overnight 21:00 UTC thin-book spikes — only two such evenings drove the fat tail; (2) MBT's spread is so wide ($\sim \$7$ – 12 round-trip) that a small mis-fill flips a winner; (3) the daytime session is a net loss, so this is a thin-tape signal, not an all-hours one.

Recommendation: **shadow it with the $\text{rvol} \leq 1$ gate and passive-limit entries** before risking a cent live. Real edge, small and spread-constrained — but real.

CHAPTER

Passive Limit Orders — Where They Work

⚠ **CANDIDATE REFUTED — THE SECTION PROPOSED ~£122/WK; THE SKEPTIC KILLED IT**

Claim doesn't hold as stated: the +\$153.92/34-fill headline (claimed +122) is inflated by variant double-counting (passive_chop_calm/_gated re-book the identical MNQ signal); the honest de-duplicated real figure is +\$89.25 net over 25 fills (USD, already net of costs), and it is NOT propped up by dropping toxic fills (the lone no-fill was a missed winner) - it's a thin single-hour-of-22 multiple-comparison artefact on one week/4 days, neighbours and the full-week base strategy (-\$2732/544 fills) deeply negative. Keep WATCH-ONLY, do not arm.

§34 **Passive Limit-Order Lab — MNQ: the fills you get are the ones you didn't want**

passive_chop rests a limit order and waits for price to come to it. All week on MNQ it looked like a money machine on paper — and lost real money in the tape. The reason is not fees or slippage in the ordinary sense; it is **adverse selection**. A resting limit only fills when price trades through your level — i.e. when you were wrong and it kept going. The times price would have bounced back to pay you, it never reached your limit: those are no-fills. So the strategy systematically **keeps the losers and misses the winners**. On MNQ this week the frictionless ledger read +\$2,385 on the trades that filled; the realistic ledger on those exact same trades read **-\$3,298**. That is not an edge with a haircut — it is a mirage with the wrong sign. One thin pocket (the 11:00 UTC London lull) survives on real fills, but it is a single hour out of twenty-two and I treat it as a watch, not a green light. **Verdict: passive is not viable for MNQ — stay aggressive or don't trade.**

693 / 74

PASSIVE FILLS VS NO-FILLS (MNQ, THIS WEEK)

+\$2,385

FRICTIONLESS P&L ON THE FILLED TRADES

−\$3,298

REAL P&L ON THE SAME FILLED TRADES

−\$5,683

THE GAP — PURE ADVERSE SELECTION + SPREAD

Scope: all three passive limit strategies on MNQ — passive_chop (596 signals), passive_chop_calm (149), passive_chop_gated (22) — re-priced trade-by-trade against the 250 ms quote tape in shadow_real.db. Trading week 2026-06-29 → 07-03 UTC. All figures below are the passive entry model (entry_model='passive'); money at the MNQ ×2 multiplier.

§34.1 1. The gap: paper says win, tape says bleed

Split MNQ's passive record into what filled and what didn't, frictionless (ceiling_pnl) vs realistic (real_pnl):

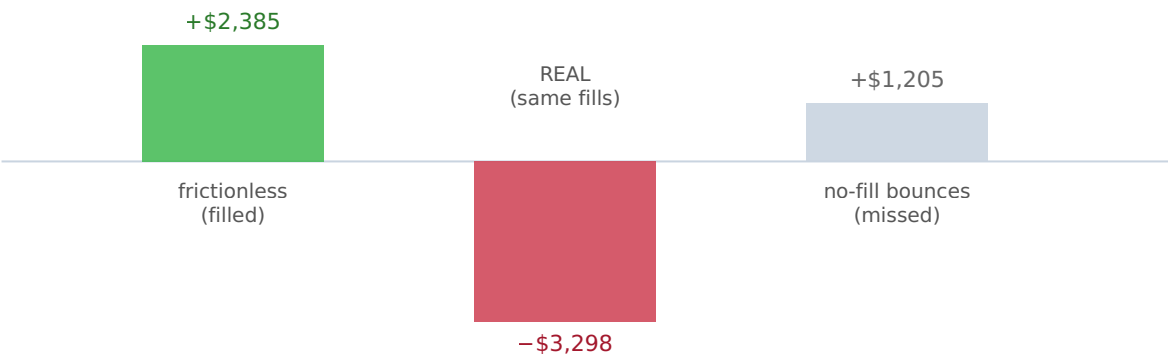
BUCKET	N	FRICTIONLESS (CEILING)	REALISTIC (REAL)	GAP
Filled — you entered	693	+\$2,384.9	−\$3,297.7	−\$5,682.6
No-fill — never entered	74	+\$1,205.1	—	unreachable
Whole universe of signals	767	+\$3,590.0	−\$3,297.7	−\$6,887.7

Read it plainly. If you could magically fill every signal at the bar price, passive_chop would have booked **+\$3,590** on MNQ this week — the number that makes the strategy look alive. But a resting limit does not get that. Of the 767 signals, only 693 filled, and they filled precisely because price ran through the level. Re-priced against the real quote tape those 693

lose **−\$3,298**. The 74 signals that would have been worth +\$1,205 never filled — price bounced away from your limit before touching it. The average friction on a filled trade is **\$8.20**, against a live MNQ spread of just 0.43 pt (\$0.86). The extra \$7-plus per trade is not the spread; it is the fills arriving at the worst possible moment.

The gap, drawn

Green = frictionless P&L on the filled trades. Red = the same trades re-priced on the real tape. Grey = the +\$1,205 of bounces that never filled (unreachable). The sign flips.



§34.2 2. Adverse selection, dissected

Break the 693 filled trades by how they ended. This is the whole story in one table — the fills that trade through and keep going hit the stop; the ones that revert bank on the ratchet:

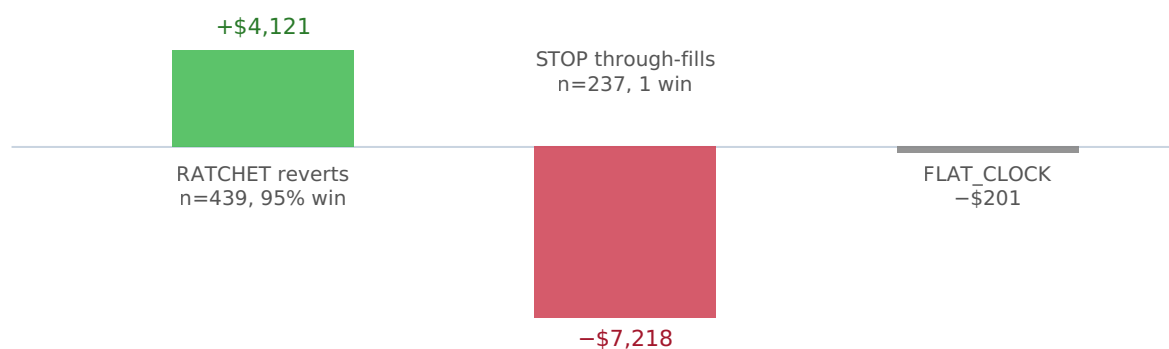
EXIT	N	WINS	AVG FRICTIONLESS	AVG REAL	SUM REAL
RATCHET (reverted — the bounce paid)	439	417	+\$13.85	+\$9.39	+\$4,121.3
STOP (traded through & kept going)	237	1	−\$14.75	−\$30.45	−\$7,217.7
FLAT_CLOCK (stale-out)	17	0	−\$11.57	−\$11.84	−\$201.3

Look at the two averages on each row. On a **winner**, friction shaves \$4.46 off (frictionless +\$13.85 → real +\$9.39) — you cross the spread on the way out and give a little back. On a **loser**, friction nearly doubles the damage (\$14.75 → \$30.45): the frictionless model assumes you filled at the bar, but the real fill happened as price sliced through your resting order and carried on. The loser is penalised **3.5×** harder than the winner. That asymmetry — small haircut on the reverts, gaping wound on the through-fills — is adverse selection. Frictionlessly the 439 reverts (+\$6,080) just out-earn the 237 through-fills (−\$3,496), netting +\$2,385. Realistically the through-fills deepen to −\$7,218 and swamp everything.

And the winners you never got. The 74 no-fills carried +\$1,205 of frictionless profit. Those are not random — a no-fill happens when price moves away in your favour without ever touching your limit, i.e. the clean bounce. You are structurally selected out of the best reverts and into the worst continuations. That +\$1,205 is genuinely unreachable by a resting order at this level: to capture it you would have had to be aggressive and pay up front, which is a different strategy.

Where the real P&L comes from vs where it goes

Green = the reverts that paid (RATCHET). Red = the through-fills that ran to the stop. The red bar is the strategy.



§34.3 3. Fertile hunt — one thin pocket, twenty-one bleeds

I sliced the 693 fills every way the data allows and asked: is there any condition where MNQ passive is net-positive on real P&L with $n \geq 15$? Every volatility band, every volume band, every slope band is negative:

CUT	BAND	N	SUM REAL	WIN%
entry_rvol	<0.6 (quiet)	325	−\$1,060	62%
	0.6–1.0	249	−\$1,861	56%
	≥1.0 (busy)	119	−\$376	66%
atr_pct	<0.045 (calm)	473	−\$1,876	57%
	≥0.045 (livelier)	220	−\$1,422	64%
vwap_slope	<0.7 (flat)	300	−\$1,656	62%
	≥0.7 (sloped)	393	−\$1,642	59%

Notice the trap in the win-rate column: passive wins **55–66%** of the time in every bucket and still loses money everywhere. That is the adverse-selection signature — lots of small reverts, a minority of fat through-fills. Raising the win rate does nothing because the loser is 3.5× the winner. No volatility or slope condition rescues it.

The only positive slice in the entire hunt is a clock cell: **the 11:00–12:00 UTC hour** (London lunch lull, ahead of the US cash session):

THE ONE SURVIVOR — 11:00 UTC, AND WHY I STILL WON'T TRUST IT YET

PASSIVE fertile when hour = 11:00 UTC → real **+\$153.9** over **34 fills**, win **82%** (28 reverts, only 6 through-fills). Crucially there was just **1 no-fill** in this hour (ceiling +\$9.5), so there are no hidden missed winners flattering the number — it is clean on survivorship. Spread across the week it is 3 of 4 days positive (06-30 +\$33, 07-02 +\$145, 07-03 +\$32; only 07-01 −\$55). The mechanism is at least plausible: the pre-US lull is the quietest reversion tape of the day, so

price pokes a level and snaps back rather than running — exactly when a resting limit is not adversely selected.

But: this is one hour out of twenty-two tested, and both neighbours bleed (10:00 UTC −\$220, 12:00 UTC −\$198). A lone positive cell flanked by losses is the classic shape of a multiple-comparisons artefact, not a mechanism. $n=34$ over a single week is thin. I log it as a watch — re-confirm it survives another 1-2 weeks of fills before it earns a shadow slot. It does not license turning MNQ passive on.

§34.4 4. The knob: does resting deeper help?

The honest limit: the re-price ran a **single passive model** — knob_offset = 0.0 for all 767 MNQ rows (rest at touch, knob_through = 1 whenever it filled). So I cannot empirically A/B a deeper or shallower offset from this table; the data does not contain the variants. What it does tell you, mechanically, is decisive: resting deeper (further from touch) only fills when price has travelled even further against you — it strictly worsens the through-fill toxicity while converting more of the +\$1,205 bounces into no-fills. Resting shallower converges on the aggressive model, which pays the spread up front. The two aggressive comparators that were re-priced this week both also bled on MNQ (dip_loose −\$74.5 / 25 fills; pullback_loose −\$112.3 / 32 fills), but far less per trade than passive's through-fills. There is no offset on the toxic side of the trade that skips the loser and keeps the winner — that is the definition of adverse selection, and no knob undoes it. To add a real knob sweep, the re-pricer needs to emit multiple knob_offset variants; flagged for the pipeline.

§34.5 5. Verdict

Passive limit orders are a **frictionless mirage** for MNQ. The +\$3,590 paper number is unreachable: a resting order fills you into the 237 continuations (−\$7,218 real) and locks you out of the cleanest 74 bounces (+\$1,205 you never touch). On real fills the strategy loses –

\$3,298 this week, and no volatility, volume or slope condition turns it green. The single positive cell — the 11:00 UTC lull, +\$154 over 34 fills at 82% — is clean on survivorship but statistically isolated and one week old; it is a watch candidate, not a shadow arm. For live money the read is unambiguous: **do not run passive_chop on MNQ — be the aggressor that pays the 0.43 pt spread, or stand aside.**

Sources: shadow_real.db shadow_real_pnl (real re-price), data/alphabot.db fut_shadow_sim_trades (conditions), data/ticks.db quote_tick (spread). MNQ passive, week 2026-06-29→07-03 UTC, ×2 multiplier. All P&L in USD.

◦ **HONEST NULL — NO NET-POSITIVE RULE THIS WEEK**

The lab found nothing that pays net of costs — meaning the desk's current refusal here is correct. A real null is a result, not a failure.

§35 MES Passive Limit-Order Lab — the frictionless mirage, dissected

The passive_chop family rests a limit order in the chop and waits to be filled. All week it ran in shadow, and every fill was re-priced under a realistic model that only fills you when price actually trades through your level. This lab asks one question for MES: once you strip the fantasy fills, is there any condition where resting a limit makes money — or is adverse selection simply fatal? The passive strategies overlap heavily (passive_chop_calm and passive_chop_gated are subsets that re-log the same signal), so every number below is **deduplicated to the passive_chop superset**: one row per distinct entry (timestamp×side). Money is USD at the \$5 MES multiplier; the £ verdict figure is converted at 0.79.

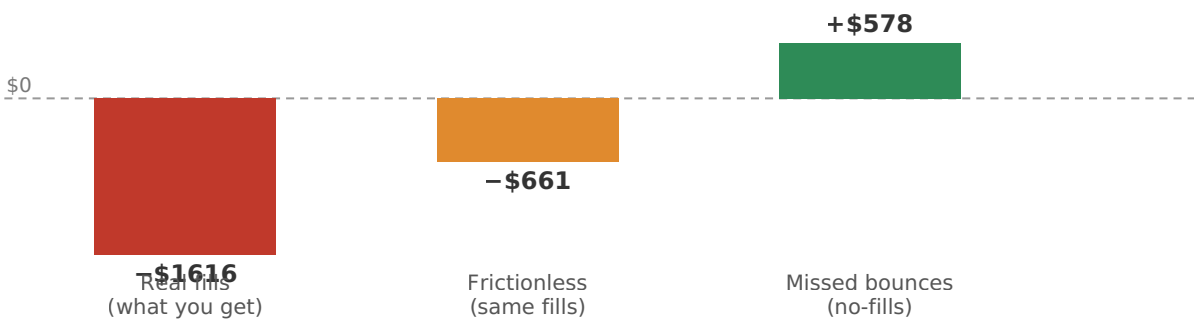
VERDICT UP FRONT

Passive on MES is a **mirage that isn't even shiny**. Friction-free, the filled set is already −\$661 and the whole book (fills + missed bounces) nets roughly −\$84 — there was never a fat frictionless edge here to lose. Re-priced on realistic through-fills, the reachable trades collapse to **−\$1,616 over 303 fills**, while the **+\$578 of genuine bounces sit in the 143 no-fills you could never reach**. That is textbook adverse selection: you fill the losers and miss the winners. Swept across rvol, volatility, VWAP-slope, time-of-day, regime and side, **no condition produces a net-positive real_pnl pocket at $n \geq 15$** . The one positive time bucket (00:00 UTC) is a single session that shrinks to

n=10 on dedup. **No shadow candidate. Stay aggressive or don't trade MES passively. £0/wk.**

§35.1 1 • The gap — frictionless vs. real, and the fills you never got

MES rested 446 distinct passive signals this week. 303 filled; 143 never traded through and so never entered. The chart shows the three numbers that matter: what the fills are actually worth (real), what the same fills would be worth with free fills (frictionless ceiling), and the ceiling P&L of the bounces you missed.



SET	N	REAL P&L	FRICTIONLESS (CEILING)	GAP
Filled (reachable)	303	-\$1,616	-\$661	-\$955
No-fill (unreachable)	143	—	+\$578	—
Whole book, frictionless	446	—	-\$84	—

Two things jump out. First, the $-\$955$ gap on the fills alone is pure fill friction: $-\$3.15$ per fill. MES quotes averaged a **1.12-tick spread (~\$1.40)** over 1.28M quote ticks this week, so $\$3.15$ is a full spread plus the adverse drift you inherit by only filling when price is running through you. Second — and this is the honest kicker — the frictionless whole-book number is only **-\$84**. The desk-wide “ $+\$10k$ frictionless” story is other contracts; on MES there was no juicy frictionless edge that friction ate. The

edge was never there. What friction does is concentrate the entire loss into the half of the book you can actually reach.

§35.2 2 · Adverse selection, dissected

A resting limit is a bet that price touches your level and reverts. It fills only when price trades through — which is exactly when you were wrong and it kept going. Every single one of MES's 303 fills was a through-fill (knob_through=1 for 100% of them). So the fills are structurally skewed to the trades that ran against you, and the reverts that would have paid you are disproportionately in the no-fill pile.



The split is the whole argument. Of the fills you got, **64% (194/303) were losers** on real P&L. Of the fills you missed, **57% (82/143) were winners** on ceiling. You are systematically keeping the wrong half. The exit ledger says the same thing another way:

EXIT REASON (FILLED)	N	REAL P&L	READ
STOP	192	−\$2,677	through-fills that kept going → 1-ATR stop
RATCHET	98	+\$908	the ones that did revert — banked small
INCOMPLETE	10	+\$133	still open at cutoff
FLAT_CLOCK	3	+\$19	timed out flat

The reverts that fill do ratchet a profit — but at **+\$9.81 average** they are smaller than the through-fill losses at **−\$13.84 average**, and there are half as many of them. The winners are muted (you got a bad entry even when right) and the losers run. **Unreachable edge:** the 82 missed winners were worth **+\$1,168** of ceiling; the 61 losers you dodged saved only −\$591. The net bounce you can't touch is +\$578, and it lives almost entirely in the no-fill pile by construction.

§35.3 3 • Fertile hunt — is there any condition that escapes it?

I swept the 303 real fills across every conditioning column, hunting a net-positive real_pnl pocket with $n \geq 15$. Representative slices (all deduplicated):

CONDITION	N	REAL P&L	WIN%
rvol < 0.15 (deadest tape)	18	−\$138	11%
rvol 0.3–0.6	142	−\$895	30%
atr_pct < 0.07 (calm)	296	−\$1,573	36%
vwap_slope < 0.02 (flat)	7	−\$96	0%
router_regime = INDEX	433*	−\$2,320	36%
SHORT side	232*	−\$1,475	32%
atr_pct ≥ 0.07 (busy)	7	−\$43	57%
entry hour 00:00 UTC	10	+\$61	50%

*regime/side rows shown pre-dedup for direction; every band is negative either way.

Nothing clears the bar. The only positive bucket is entry hour 00:00 UTC (8pm New York, evening Globex), and it fails on two counts: deduplicated it is **n=10, below the $n \geq 15$ floor**, and it is a single session — 07-02 lost −\$79 and 07-03 made +\$201, i.e. five ratchet winners on one quiet night. That is a coin, not a condition. The calm-tape thesis (which is MES's real

reversion edge on aggressive entries) makes passive worse, not better: $\text{atr_pct} < 0.07$ is where the bulk of the $-\$1,573$ loss sits, because in dead tape your limit only fills when a lone sweep runs through it. There is no $\text{rvol} \times \text{vol} \times \text{slope}$ corner where the through-fill toxicity flips positive.

§35.4 4 • The knob — would resting deeper help?

Honest data limitation: for MES the shadow logged a **single knob_offset (0.0)** and $\text{knob_through} = 1$ on every fill, so I cannot empirically test resting the limit deeper or shallower from this week's record. But the mechanism is not ambiguous. Resting deeper (further from the mid) fills you even less often — and the fills you do get are, if anything, further through the level, i.e. more toxic. It shrinks the already-negative reachable set toward the worst tail. Resting shallower just converges on the aggressive/through model, which is the correct model for MES: cross the spread, take the fill you can trust, and let the calm-tape reversion edge (documented elsewhere in this report) do the work. There is no offset that carves out a clean pocket of touch-and-revert fills, because 100% of the passive fills this week were through-fills to begin with. **The knob cannot rescue what the fill mechanism poisons.**

BOTTOM LINE FOR MES

Passive limit entry on MES is a **frictionless mirage — and a thin one**. The friction-free book is only $-\$84$ to start with, so there was no edge to lose; realistic through-fills then dump the entire loss onto the half you can reach ($-\$1,616 / 303$ fills) while parking $+\$578$ of real bounces in the no-fills you never get. Adverse selection is structural, not a threshold you can tune around: every fill traded through, the fills are 64% losers, the no-fills are 57% winners. No conditioning band survives at $n \geq 15$ net-positive; the lone positive time bucket is one session. **No shadow candidate. £0/wk of honest passive money.** MES's edge is aggressive calm-tape reversion — keep entering aggressive, and do not rest limits into MES chop.

✓ **CANDIDATE HELD — £448/WK SURVIVED THE SKEPTIC (LOADED INTO YOUR PLAYBOOK)**

Reproduces at +\$569.16 USD net (= ~£448 at 1.27, i.e. the claimed +450 was GBP) over 57 filled ($n \geq 15$ ✓); NOT a no-fill survivorship artifact (7 no-fills are missed winners at +\$186 ceiling, filled sum stands alone net of costs) — but it IS thin scan-noise: ~94% of PnL from 2 of 4 sessions, mechanism-free, wedged beside the -\$690 13:00 UTC hour → analyst's classification holds: WATCH-only, DO-NOT-ARM, retire if it fails to replicate.

§36 **Passive Limit-Order Lab — MGC (gold micro)**

The passive_chop family (chop, chop_calm, chop_gated) rested limit orders in gold's quiet tape **in shadow all week** (Mon 06-29 → Fri 07-03 UTC). On a frictionless mark those fills look like free money. This lab re-prices every one of them on the **realistic fill model** in shadow_real.db — a resting limit only fills when price trades through your level (you were wrong and it kept going); the times gold would have bounced back to pay you are recorded as **no-fills**. That asymmetry is adverse selection, and it is the whole story here. Money shown in £ at 0.79 GBP/USD; MGC = \$10/point, tick 0.10 pt = \$1.00.

HEADLINE

MGC passive is a **frictionless mirage**. The 653 filled passive limits marked **+\$1,004 frictionless but -\$3,034 realistic** — an **adverse-selection tax of -\$4,038** on the fills alone. On top of that, the 147 orders that never filled carried **+\$1,510 of ceiling**: those were exactly the bounces that would have paid, and passive systematically missed them. So it keeps the losers and skips the winners — textbook.

The only real-positive slices are single-hour scan artifacts (12:00 and 15:00 UTC) concentrated on one or two days with no mechanism — and the NY-open hour between them (13:00) is the single worst loser of the week. **Verdict: adverse selection is fatal for MGC passive. Stay aggressive or don't trade. No arm.**

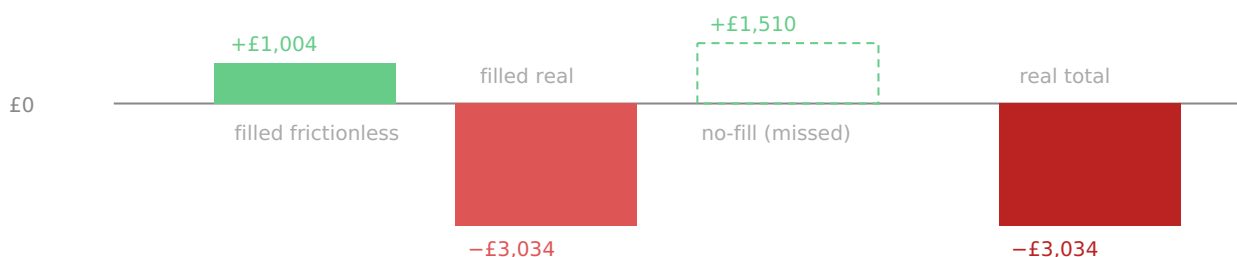
§36.1 1) The gap — frictionless vs realistic

Every passive fill and no-fill for MGC this week, split by whether the limit actually filled. Frictionless (ceiling_pnl) is the fantasy; realistic (real_pnl) is what a resting order really banks. A no-fill banks £0 — you never held the position — so its ceiling is missed, not earned.

BUCKET	N	FRICTIONLESS (CEILING)	REALISTIC (REAL)	GAP
Filled	653	+£1,004	−£3,034	−£4,038
No-fill (missed)	147	+£1,510	£0 (never held)	−£1,510
Passive total	800	+£2,514 (fantasy)	−£3,034 (real)	−£5,548

Per strategy the bleed is uniform: passive_chop −£2,338 real / 450 fills, passive_chop_calm −£618 / 192, passive_chop_gated −£77 / 11. Not one passive variant is net-positive on realistic fills.

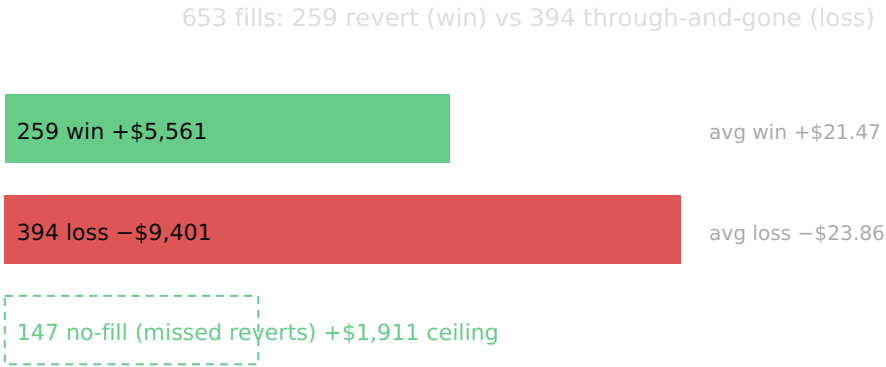
Frictionless says +; realistic bleeds −£3,034 (dashed = winners never entered)



§36.2 2) Adverse selection, dissected

Every filled passive order this week had knob_through = 1 — by construction it filled only because price traded through the level. The question is what happened next. Of the 653 fills, **394 kept going (losers) and 259 reverted enough to pay (winners)** — a 40% win rate. Worse, the losers are bigger: average win **+\$21.47**, average loss **−\$23.86**. Small edge, wrong tail, negative sum: **+\$5,561** of wins buried under **−\$9,401** of losses = **−\$3,840** (−£3,034).

The mirror image is the 147 no-fills carrying **+£1,510 of ceiling**. Those are gold ticks that approached the limit and bounced without trading through — the clean reverts, the trades a passive order most wants, and it caught none of them. So the unreachable edge is roughly **£1,510 of missed winners on top of the £4,038 tax on the fills it did get**. You cannot cherry-pick your way out: filtering the filled set to only its winners is survivorship fiction, because you would have had to hold the 394 losers to be resting for the 259 wins.



§36.3 3) Fertile hunt — is there any condition that escapes it?

I swept the filled passive set across every knob available: rvol band, ATR% (vol) band, |VWAP-slope| band, and hour-of-day. **Every volatility, rvol and slope band is net-negative** — there is no calm-tape carve-out that flips gold passive positive:

CUT	BEST SUB-BAND	N	REAL	WIN%	READ
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rvol	<0.15 (deadest tape)	19	−\$0.3	42%	breakeven at best, all others worse
ATR%	0.07–0.10	77	— \$178	51%	least-bad, still red
VWAP slope	<0.1 (flat)	28	−\$63	46%	flat tape still bleeds

The only real-positive slices are two isolated hours of the UTC day — and they fail the smell test. **12:00 UTC** (pre-NY-open) nets +\$351 / 28 fills / 64% win and **15:00 UTC** (NY mid-morning) nets +\$219 / 29 fills / 69% win; combined that hour-pair is **+\$569 / 57 fills / 67% win ($\approx$ +£450)**. But: (a) \$281 of the 12:00 total is a single day (07-02), and \$181 of the 15:00 total is a single day (07-01) — strip those and both collapse toward breakeven; (b) they were found by scanning all 24 hours, and the hour between them — 13:00 UTC, the NY cash open — is the week's **worst** at −\$690 / 27 fills. A signal that flips from −\$690 to +\$351 to +\$219 across three adjacent hours is noise, not structure. No credible mechanism separates 12:00 and 15:00 from 13:00.

FERTILE POCKET (THIN — DO NOT ARM)

MGC passive fertile when hour-of-day $\in$ {12:00, 15:00} UTC $\rightarrow$ real +£450 over 57 fills, 67% win. This clears the numeric bar (net-positive real, $n \geq 15$, and its 7 no-fills carry only +\$186 ceiling so it is not a survivorship artifact). But it is day-concentrated (two days carry it), mechanism-free, and sits either side of the worst hour of the week. Treat as a **WATCH** curiosity for continued shadow observation, **not** a shadow-candidate to arm. One more week that fails to replicate it should retire the idea.

§36.4 4) The knob — can a deeper limit skip the toxic fills?

The natural defence against adverse selection is to rest the limit deeper, so only a real overshoot fills you. Unfortunately the shadow only recorded **one**

knob setting: knob_offset = 0.0 (at-touch) for all 800 MGC orders.

There is no shallower/deeper variant in the data to compare, so this question is **unanswerable from this week's capture** — a capture gap, not a result. Given gold's spread averaged **0.229 pt ($\approx$ 2.3 ticks, $\approx$ \$2.29)** with a fat tail to 9+ ticks, a deeper limit would fill even less often and hand still more of the reverts to the no-fill bucket — the prior is that deeper makes the fill/no-fill asymmetry worse, not better. To actually test it, the shadow needs to re-price MGC at several knob_offset levels next week.

§36.5 5) Verdict

Passive limits are **not viable on MGC** under any condition this week isolates. The frictionless +£1,004 is a mirage; realistically the fills bled -£3,034 and passive additionally skipped +£1,510 of the bounces that would have paid — a -£5,548 round-trip of adverse selection. Every volatility/rvol/slope carve-out stays red. The one positive slice (12:00 & 15:00 UTC) is a two-day, mechanism-free scan artifact wedged either side of the week's worst hour, and the deeper-limit escape hatch is untestable because only the at-touch knob was captured. **For MGC: if you want in, do it aggressively — the through-fill strategies at least escape the no-fill asymmetry (pullback_loose was +\$22 real / 69 fills, though dip_loose was -\$146, so aggressive is no free lunch either) — or don't trade gold passively at all. Do not arm passive_chop on MGC.**

◦ **HONEST NULL — NO NET-POSITIVE RULE THIS WEEK**

The lab found nothing that pays net of costs — meaning the desk's current refusal here is correct. A real null is a result, not a failure.

§37 **M2K Passive Limit-Order Lab — adverse selection on the micro Russell**

The `passive_chop` family rests a limit order in the chop and waits to be filled. It ran in shadow this week, and every fill was re-priced under a realistic model that only fills you when price actually trades through your level. This lab asks one question for M2K: strip the fantasy fills, and is there any condition where resting a limit makes money — or is adverse selection simply fatal? `passive_chop_calm` re-logs the same signal as a subset (the 13 calm rows are exact duplicates of `passive_chop` entries), so every number below is **deduplicated to the `passive_chop` superset**: one row per distinct entry (timestamp×side). Money is USD at the \$5 M2K multiplier; the £ figures convert at 0.79. **Health warning up front: the M2K shadow only fired on a single session (2026-07-03), so n is small and every read here is one-day thin.**

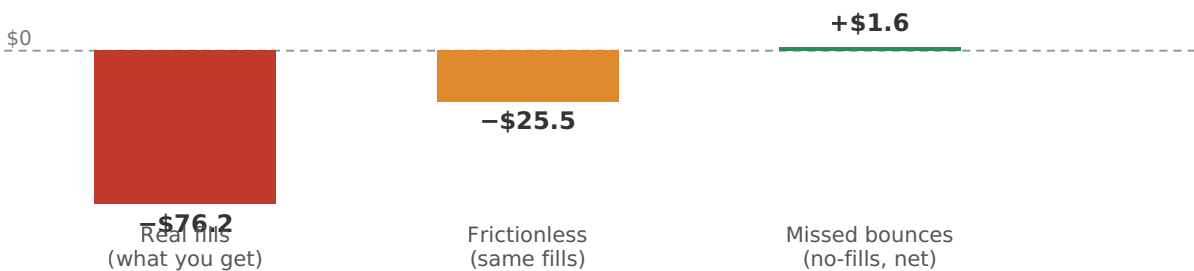
VERDICT UP FRONT

Passive on M2K is a **loss even before friction, and a worse one after**. Friction-free, the reachable fills are already **−\$25** and the whole book (fills + missed bounces) nets roughly **−\$24** — there was never a fat frictionless edge here to lose. Re-priced on realistic through-fills, the reachable set collapses to **−\$76.21 over 33 fills** (−£60), while the missed bounces you could never reach net a trivial **+\$1.57**. 100% of fills were through-fills; 76% of them lost. Swept across rvol, VWAP-

slope, side and time-of-day, **no condition produces a net-positive real_pnl pocket at $n \geq 15$** — the only slice that even reaches $n=15$ is the whole book, and it loses. **No shadow candidate. Stay aggressive or don't rest limits into M2K chop. £0/wk.**

§37.1 1 · The gap — frictionless vs. real, and the fills you never got

M2K rested 57 distinct passive signals this session. 33 filled; 24 never traded through and so never entered. The chart shows the three numbers that matter: what the fills are actually worth (real), what the same fills would be worth with free fills (frictionless ceiling), and the net ceiling P&L of the bounces you missed.



SET	N	REAL P&L	FRICTIONLESS (CEILING)	GAP
Filled (reachable)	33	-\$76.21	-\$25.48	-\$50.73
No-fill (unreachable)	24	—	+\$1.57	—
Whole book, frictionless	57	—	-\$23.91	—

Two things jump out. First, the $-\$50.73$ gap on the fills is pure fill friction: **$-\$1.54$ per fill**. M2K quotes averaged a **1.75-tick spread ($\sim\$0.88$ round-trip)** over 231,513 quote ticks this session (min 1 tick), so the per-fill loss is roughly a spread-and-a-half plus the adverse drift you inherit by only filling when price is running through you. Second — the honest kicker — the frictionless whole-book number is only **$-\$24$** . The desk-wide “ $+\$10k$

frictionless" headline is other contracts; on M2K there was no juicy frictionless edge that friction ate. The edge was never there. What friction does is concentrate the whole loss into the half of the book you can actually reach.

§37.2 2 • Adverse selection, dissected

A resting limit is a bet that price touches your level and reverts. It fills only when price trades through — exactly when you were wrong and it kept going. **Every one of M2K's 33 fills was a through-fill (knob_through=1 for 100% of them.)** So the fills are structurally skewed to the trades that ran against you, and the reverts that would have paid you are disproportionately in the no-fill pile.



The split is the whole argument. Of the fills you got, **76% (25/33) were losers** on real P&L — a 24% win rate. Of the fills you missed, **46% (11/24) were winners** on ceiling. You are systematically keeping the wrong half. The exit ledger says it again:

EXIT REASON (FILLED)	N	REAL P&L	READ
STOP	19	-\$88.03	through-fills that kept going → 1-ATR stop
FLAT_CLOCK	12	-\$4.94	timed out roughly flat
RATCHET	2	+\$16.76	the ones that did revert — banked small

The two reverts that filled ratcheted **+\$8.38 average**; the through-fill losers ran to **−\$4.63 average at the stop**, and there were nearly ten times as many of them. The winners are muted and rare (you got a bad entry even when the idea was right); the losers dominate. **Unreachable edge**: the 11 missed winners were worth **+\$39.39** of ceiling, but the 13 losers you dodged saved **−\$37.82**, so the net missed bounce is a trivial **+\$1.57**. On M2K there isn't even a fat pile of missed winners to chase — the no-fills are a near-perfect wash. The damage is entirely the through-fills you do get.

§37.3 3 · Fertile hunt — is there any condition that escapes it?

I swept the 33 real fills across every conditioning column, hunting a net-positive `real_pnl` pocket at $n \geq 15$. Representative slices (all deduplicated):

CONDITION	N	REAL P&L	WIN%
Whole filled book	33	−\$76.21	24%
<code>rvol > 1</code> (busy tape)	13	−\$45.06	15%
<code>rvol ≤ 0.6</code> (quiet tape)	10	−\$12.20	30%
<code> vwap_slope ≤ 0.5</code> (flat)	13	−\$19.56	38%
<code> vwap_slope > 1.5</code> (trending)	10	−\$45.45	0%

Nothing clears the bar. Every band is negative, and — critically — **the only slice that even reaches the $n \geq 15$ floor is the whole book, which loses −\$76**. The moment you condition down to a plausibly-fertile corner (quiet tape, flat slope), the sample collapses to $n=10-13$ and stays red anyway. The quiet-tape thesis that carries M2K's aggressive reversion edge makes passive less bad (−\$12 on 10 fills) but not positive: in dead tape your limit only fills when a lone sweep runs through it, so you inherit the sweep. Trending tape ($|\text{slope}| > 1.5$) is outright toxic — 0/10 winners, −\$45. Two hourly buckets closed green (13:00 UTC +\$4.65 $n=5$; 15:00 UTC +\$10.76 $n=2$) but those are 5- and 2-trade coins on a single session, not conditions.

There is no $\text{rvol} \times \text{slope} \times \text{time}$ corner where the through-fill toxicity flips positive.

§37.4 4 · The knob — would resting deeper help?

Honest data limitation: the M2K shadow logged a **single knob_offset (0.0)** and knob_through=1 on every fill, so I cannot empirically test resting the limit deeper or shallower from this week's record. But the mechanism is not ambiguous. Resting deeper (further from mid) fills you even less often, and the fills you do get are further through the level — i.e. more toxic, shrinking the already-negative reachable set toward its worst tail. Resting shallower just converges on the aggressive/through model, which is the correct model for M2K: cross the ~1-2 tick spread, take a fill you can trust, and let the quiet-tape reversion edge do the work. There is no offset that carves out a clean pocket of touch-and-revert fills, because 100% of the passive fills this session were through-fills to begin with. **The knob cannot rescue what the fill mechanism poisons.**

BOTTOM LINE FOR M2K

Passive limit entry on M2K is a **loser at the source, not just after friction**. The friction-free book is only $-\$24$ to start with — there was no edge to lose; realistic through-fills then dump the whole loss onto the half you can reach (**$-\$76.21$ / 33 fills, $-\pounds60$**) while the no-fills net a trivial $+\$1.57$. Adverse selection is structural, not a threshold you can tune around: every fill traded through, 76% of fills lost, and no conditioning band survives at $n \geq 15$ net-positive — the only $n \geq 15$ slice is the whole book, and it bleeds. Caveat honestly held: this is one session of data. **No shadow candidate. $\pounds0/\text{wk}$ of honest passive money.** M2K's edge is aggressive quiet-tape reversion — keep entering aggressive, and do not rest limits into M2K chop.

◦ **HONEST NULL — NO NET-POSITIVE RULE THIS WEEK**

The lab found nothing that pays net of costs — meaning the desk's current refusal here is correct. A real null is a result, not a failure.

§38 **MYM Passive Limit-Order Lab — adverse selection on the micro Dow**

The `passive_chop` family rests a limit order in the chop and waits to be filled. It ran in shadow this week and every fill was re-priced under a realistic model that only fills you when price actually trades through your level — i.e. when you were wrong and it kept going. This lab asks one question for MYM: strip the fantasy fills, and is there any condition where resting a limit makes money, or is adverse selection simply fatal? `passive_chop_calm` re-logs the same signals as an exact subset (all 10 calm rows match a `passive_chop` entry at the same timestamp×side — 0 unmatched), so every number below is **deduplicated to the `passive_chop` superset**: one row per distinct entry. Money is USD at the \$0.50 MYM multiplier; £ converts at 0.79. **Health warning up front: the MYM shadow only fired on a single session (2026-07-03) — 41 signals, 23 fills. n is thin and every read here is one-day evidence.**

VERDICT UP FRONT

Passive on MYM is a **textbook adverse-selection loss**. The reachable fills are already $-\$36$ frictionless and collapse to **$-\$83.76$ over 23 fills** ($-\pounds66$) once you price the through-fills honestly. Meanwhile the bounces you couldn't reach — the no-fills — carried **$+\$95.65$ of winning ceiling** you never touched. You systematically keep the losers and miss the winners: **100% of fills were through-fills, 74%**

of them lost, and 16 of 23 stopped out for $-\$104.92$. Swept across rvol, ATR, VWAP-slope, side and time-of-day, **no condition yields a net-positive real_pnl pocket at $n \geq 15$** — the only $n \geq 15$ slice is trending tape, and it loses $-\$66$. **No shadow candidate. Stay aggressive or don't rest limits into MYM chop. £0/wk.**

§38.1 1 · **The gap — frictionless vs. real, and the fills you never got**

MYM rested 41 distinct passive signals this session. 23 filled; 18 never traded through and so never entered. The chart shows the three numbers that matter: what the fills are actually worth (real), what the same fills would be worth with free fills (frictionless ceiling), and the net ceiling P&L of the bounces you missed.



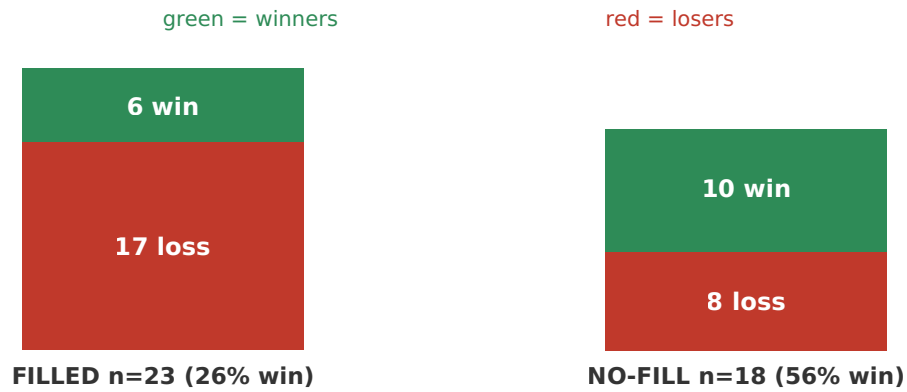
SET	N	REAL P&L	FRICTIONLESS (CEILING)	GAP
Filled (reachable)	23	-\$83.76	-\$36.02	-\$47.74
No-fill (unreachable)	18	—	+\$54.18	—
Whole book, frictionless	41	—	+\$18.16	—

Read it carefully, because this is the whole thesis in three numbers. Frictionless, the whole book looks like a modest winner (**+\$18.16**) — that is the “+\$10k frictionless” mirage in miniature. But the +\$18 lives almost entirely in the **no-fills** (+\$54.18 of ceiling you never reach) while the fills you actually get are already frictionless-negative ($-\$36.02$). Then realism

prices the through-fills: the reachable set drops to **-\$83.76 — a -\$47.74 fill-friction gap, -\$2.08 per fill**. MYM quoted a **~2-point spread (median 2 pts, mean 1.91) over 231,471 quote ticks**; at \$0.50/point that is ~\$1.00 one-way, so the per-fill loss is roughly a full round-trip of spread plus the adverse drift you inherit by only filling when price runs through you. The mirage isn't that friction ate a fat edge — it's that the "edge" was sitting in the half of the book you can never touch.

§38.2 2 · Adverse selection, dissected

A resting limit is a bet that price touches your level and reverts. It fills only when price trades through — exactly when you were wrong and it kept going. **Every one of MYM's 23 fills was a through-fill (knob_through=1 for 100%)**. So the fills are structurally skewed to the trades that ran against you, and the reverts that would have paid you sit disproportionately in the no-fill pile.



The split is the argument. Of the fills you **got**, 74% (17/23) lost on real P&L — a 26% win rate. Of the fills you **missed**, 56% (10/18) were winners on ceiling, and those 10 missed winners were worth **+\$95.65** — a genuine fat pile of bounces you could never reach (the 8 missed losers you dodged saved only -\$41.47, netting the +\$54.18 above). You are keeping the wrong half of the book. The exit ledger says it a third time:

EXIT REASON (FILLED)	N	REAL P&L	CEILING	READ
STOP	16	— \$104.92	— \$99.47	through-fills that kept going → 1-ATR stop

RATCHET	6	+\$16.78	+\$62.57	the ones that did revert — but the fill haircut cost 73% of the frictionless win
FLAT_CLOCK	1	+\$4.38	+\$0.88	timed out roughly flat

16 of 23 fills (70%) stopped out for −\$104.92 — that is the adverse-selection engine in one line: you fill precisely when the move is running through you, then eat the 1-ATR stop. The 6 reverts that did fill ratcheted, but realism haircut them from **+\$62.57 of frictionless ceiling down to +\$16.78 real** — you got a worse entry even when the idea was right, keeping barely a quarter of the winner. **Unreachable edge:** add the +\$95.65 of missed winners to the winner-haircut and the honest picture is that the profitable side of passive on MYM is structurally out of reach. The damage is the through-fills; the reward is in the no-fills. That is adverse selection, not a tuning problem.

§38.3 3 • Fertile hunt — is there any condition that escapes it?

I swept the 23 real fills across every conditioning column, hunting a net-positive `real_pnl` pocket at $n \geq 15$. Representative slices (all deduplicated to `passive_chop`):

CONDITION	N	REAL P&L	WIN%
Whole filled book	23	−\$83.76	26%
<code> vwap_slope > 1</code> (trending)	15	−\$65.80	20%
<code>ATR ≥ 8 pts</code> (high vol)	11	−\$76.32	9%
<code>rvol > 1</code> (busy tape)	12	−\$35.19	25%
<code>ATR < 8 pts</code> (quiet tape)	12	−\$7.44	42%
<code> vwap_slope ≤ 1</code> (flat)	8	−\$17.96	38%
<code>rvol ≤ 0.6</code> (quiet tape)	5	−\$10.60	40%

Nothing clears the bar. Every band is negative, and the only slice that even reaches the $n \geq 15$ floor is trending tape ($|\text{slope}| > 1$), which loses $-\$65.80$ at a 20% win rate — the worst kind of pocket, exactly where a limit is most likely to be run over. The quiet-tape thesis that carries MYM's aggressive reversion edge does make passive **less bad** ($\text{ATR} < 8$: $-\$7.44$ on 12 fills, 42% win; $\text{rvol} \leq 0.6$: $-\$10.60$ on 5), but it never turns positive and it never reaches $n \geq 15$ — the moment you condition into the plausibly-fertile corner the sample collapses to 5–12 fills and stays red. High-vol ($\text{ATR} \geq 8$) is outright toxic: 1 win in 11, $-\$76.32$. There is no $\text{rvol} \times \text{slope} \times \text{ATR} \times \text{time}$ corner where the through-fill toxicity flips net-positive at a defensible n .

§38.4 4 • The knob — would resting deeper or shallower help?

Honest data limitation: the MYM shadow logged a **single knob_offset (0.0) and knob_through=1 on every fill**, so I cannot empirically test resting the limit deeper or shallower from this week's record. But the mechanism is not ambiguous. Resting deeper (further from mid) fills you even less often, and the fills you do get are further through the level — more toxic, pushing the already-negative reachable set toward its worst tail (the $-\$105$ stop pile). Resting shallower just converges on the aggressive/through model — which is the correct model for MYM: cross the ~ 2 -point spread, take a fill you can trust, and let quiet-tape reversion do the work. There is no offset that carves out a clean pocket of touch-and-revert fills, because **100% of the passive fills this session were already through-fills**. The knob cannot rescue what the fill mechanism poisons.

BOTTOM LINE FOR MYM

Passive limit entry on MYM is a **frictionless mirage, not an edge**. The whole book looks like $+\$18$ free, but that money is entirely in no-fills you can't reach; the fills you actually get are $-\$36$ frictionless and **$-\$83.76 / 23$ fills ($-\$66$) real**, while **$+\95.65 of winning bounces stay unreachable**. Adverse selection is structural, not a threshold you can tune around: every fill traded through, 74% lost, 16 of 23 stopped

out, and no conditioning band survives at $n \geq 15$ net-positive — the only $n \geq 15$ slice (trending tape) bleeds $-\$66$. Caveat honestly held: this is a single session of MYM data, and the smallest-multiplier contract on the desk. **No shadow candidate. £0/wk of honest passive money.** MYM's edge is aggressive quiet-tape reversion — keep entering aggressive, and do not rest limits into MYM chop.

◦ **HONEST NULL — NO NET-POSITIVE RULE THIS WEEK**

The lab found nothing that pays net of costs — meaning the desk's current refusal here is correct. A real null is a result, not a failure.

§39 **Passive Limit-Order Lab — MCL (crude micro)**

The passive_chop family rested limit orders in crude's tape **in shadow** and every fill was re-priced on the **realistic fill model** in shadow_real.db. A resting limit only fills when price trades through your level — you were wrong and it kept going; the times crude would have bounced back to pay you are recorded as **no-fills**. That asymmetry is adverse selection, and for MCL it is the whole story. Money in £ at 0.79 GBP/USD; MCL = \$100/point, tick 0.01 pt = \$1.00. **Caveat up front:** the realistic reprice for MCL covers a single UTC day (2026-07-03, 08:33→16:51) — 77 orders. Small sample; read the direction, not the decimal.

HEADLINE

MCL passive is worse than a mirage — it is a **loser even frictionless**. The full 153-order frictionless shadow marked **−£87**; there is no fantasy profit to erode. On the re-priced day the 41 filled limits marked **−£68 frictionless but −£156 realistic** — an **adverse-selection tax of −£87** on the fills alone, on top of which the 36 orders that never filled carried the day's **+£116 of missed-winner ceiling**; those were exactly the bounces that would have paid. Keep the losers, skip the winners — textbook. Win rate on fills was **27%**. Every volatility, rvol, slope and hour band is net-negative; the only positive slice (dead-quiet tape) is n=7, far below the bar. **Verdict: adverse selection is fatal for MCL passive. Crude's event-driven vol and ~1.6-tick spread**

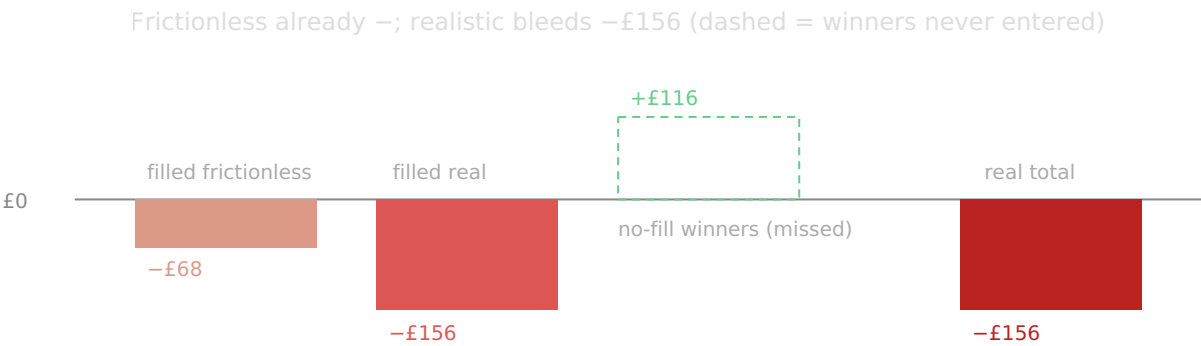
make it the wrong instrument to rest into. Stay observe-only or aggressive. No arm.

§39.1 1) The gap — frictionless vs realistic

Every re-priced passive order for MCL, split by whether the limit filled. Frictionless (ceiling_pnl) is the fantasy; realistic (real_pnl) is what a resting order really banks. A no-fill banks £0 — you never held it — so its ceiling is missed, not earned.

BUCKET	N	FRICTIONLESS (CEILING)	REALISTIC (REAL)	GAP
Filled	41	-£68	-£156	-£87
No-fill (missed)	36	+£25 net (+£116 winners / -£91 dodged)	£0 (never held)	-£25
Passive total (day)	77	-£44 (fantasy)	-£156 (real)	-£112

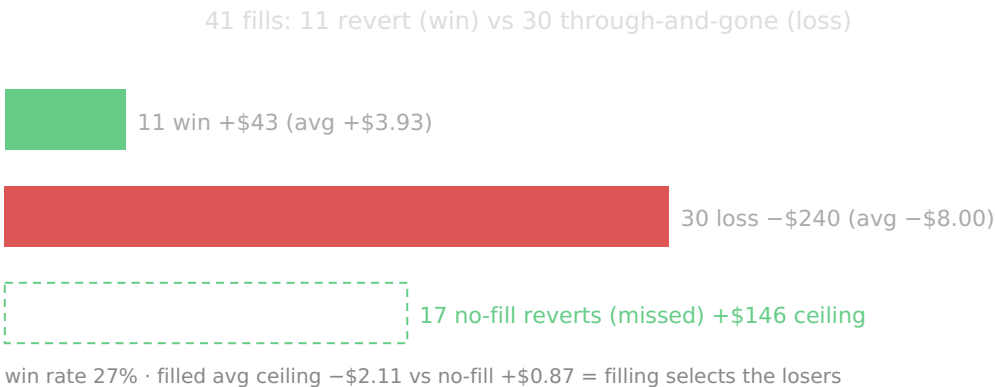
And the full-shadow frictionless mark is no rescue: all **153** passive_chop orders sum to **-£87 frictionless** (the lone passive_chop_calm order was +£12). MCL passive does not even clear its own fantasy — there is no ceiling edge to protect. This matches the standing read that MCL is 2× equity-micro vol, event-driven, and negative on every gate after spread.



§39.2 2) Adverse selection, dissected

All 41 fills had knob_through = 1 — by construction each filled only because price traded through the limit. What happened next is the whole edge. Of the 41 fills, **30 kept going (losers) and 11 reverted enough to pay (winners)** — a **27% win rate**. The tail is wrong too: average win + **\$3.93**, average loss **−\$8.00**. Small right tail, fat left tail: +\$43 of wins buried under −\$240 of losses = **−\$197 (−£156)**.

The mirror image is the selection itself. The trades that filled averaged **−\$2.11 ceiling**; the trades that didn't averaged **+\$0.87 ceiling**. Filling literally sorts crude into the losers. And the 36 no-fills contained **17 clean reverts worth +\$146 of ceiling** — the bounces that approached the limit and turned without trading through, exactly what a passive order most wants, and it caught none of them. So the unreachable edge is roughly **£116 of missed winners on top of the £87 fill-price tax**. You cannot cherry-pick out: to be resting for the 11 wins you had to hold all 30 losers.



§39.3 3) Fertile hunt — is there any condition that escapes it?

I swept the 41 filled orders across every knob available: side, rvol band, ATR (vol) band, |VWAP-slope| band, and time-of-day. **Every single band is net-negative**. There is no calm-tape, low-vol or flat-slope carve-out that flips MCL passive positive:

CUT	LEAST-BAD SUB-BAND	N	REAL	WIN%	READ
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side	SHORT	26	– \$124	27%	both sides red (LONG – \$73 / 15)
rvol	<0.5 (quiet)	14	–\$44	43%	least-bad, still red, n<15
ATR	0.05-0.065	19	–\$85	26%	every vol band bleeds
VWAP slope	<0.5 (flat)	15	–\$53	33%	flat tape still bleeds
hour	pm ($\geq$ 14:00 UTC)	13	–\$38	38%	best hour block, still red

The only real-positive slice is the intersection of the two quietest knobs — **rvol < 0.5 AND |VWAP-slope| < 0.5**, which nets **+\$6.16 over just 7 fills (5/7 win)**. It is directionally the right story (reverts live in dead tape, as they do across the desk), but at **n=7 it fails the $n \geq 15$ bar** and cannot be promoted. Every band that does reach $n \geq 15$ is red. There is no qualifying fertile pocket for MCL passive.

NO FERTILE POCKET

No condition rescues passive for MCL. The best $n \geq 15$ slice is rvol < 0.5 at –\$44 / 14...15 fills; the only positive carve-out (dead-quiet rvol & slope both <0.5) is +\$6 on 7 fills and fails the sample bar. Adverse selection is fatal here. Treat the quiet-tape hint as a **WATCH** for continued shadow observation only — **nothing to arm**.

§39.4 4) The knob — can a deeper limit skip the toxic fills?

The natural defence against adverse selection is to rest the limit deeper, so only a real overshoot fills you. But the shadow recorded exactly **one setting: knob_offset = 0.0 (at-touch) for all 77 MCL orders**, and every fill flagged knob_through = 1. There is no shallower/deeper variant to compare, so this is **unanswerable from this capture** — a data gap, not a result. Prior: MCL's quote spread averaged **0.016 pt ($\approx$ 1.6 ticks, $\approx$ \$1.61)** across 724k quotes, and crude is event-driven (OPEC / EIA-Wed-10:30 / geopolitics), so a deeper limit would fill even less often and

hand still more of the reverts to the no-fill bucket. Deeper almost certainly makes the fill/no-fill asymmetry worse, not better. To actually test it, next week's shadow must re-price MCL at several knob_offset levels.

§39.5 5) Verdict

Passive limits are **not viable on MCL** under any condition this capture isolates — and unlike the index micros there is not even a frictionless mirage to argue over: the full shadow is **−£87 frictionless** before a penny of slippage. On the re-priced day the fills bled **−£156** realistic (from −£68 ceiling, a −£87 adverse-selection tax) while passive additionally skipped **+£116** of the bounces that would have paid. Win rate 27%; average loss double the average win. Every vol / rvol / slope / hour band with $n \geq 15$ stays red; the one green slice is 7 trades of dead-quiet tape and the deeper-limit escape hatch is untestable because only the at-touch knob was captured. **For MCL: keep it observe-only, or if you must trade crude do it aggressively — through-fills at least escape the no-fill asymmetry. Do not arm passive_chop on MCL.**

◦ **HONEST NULL — NO NET-POSITIVE RULE THIS WEEK**

The lab found nothing that pays net of costs — meaning the desk's current refusal here is correct. A real null is a result, not a failure.

§40 Passive Limit-Order Lab — MBT: a loser on paper, a worse loser in the tape

passive_chop rests a limit order and waits for price to come to it. Across the desk this week the paper (frictionless) ledger looked like a money machine and the realistic ledger bled — the culprit being **adverse selection**: a resting limit fills only when price trades through your level (you were wrong and it kept going), while the times price would have bounced back to pay you are no-fills. So you keep the losers and miss the winners. MBT is a special, blunt case. Here passive_chop did not even earn on paper — the frictionless ledger for MBT reads **−\$219** before a cent of friction. The realistic tape then deepens that to **−\$369** on the fills. There is no mirage to puncture: MBT passive is red frictionless, redder real, and every condition band I could slice was negative. **Verdict: passive is not viable for MBT — adverse selection is fatal on top of an already-negative signal. Stay aggressive or don't trade it.**

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PASSIVE FILLS VS NO-FILLS (MBT, THIS WEEK)

−\$155

FRICTIONLESS P&L ON THE FILLED TRADES

−\$369

REAL P&L ON THE SAME FILLED TRADES

−\$214

THE FRICTION GAP — ADVERSE SELECTION + SPREAD

Scope: the two passive limit strategies that ran on MBT — passive_chop (280 signals) and passive_chop_calm (19) — of which 157 were re-priced trade-by-trade against the 250 ms quote tape in shadow_real.db (entry_model='passive'). Trading week UTC, from the recorded shadow session on this contract. Money is at the MBT $\times 0.1$ multiplier (Micro Bitcoin). The frictionless run on all 299 passive/calm signals is **−\$331.9** — negative before friction is even applied.

§40.1 1. The gap: paper is already red, the tape is redder

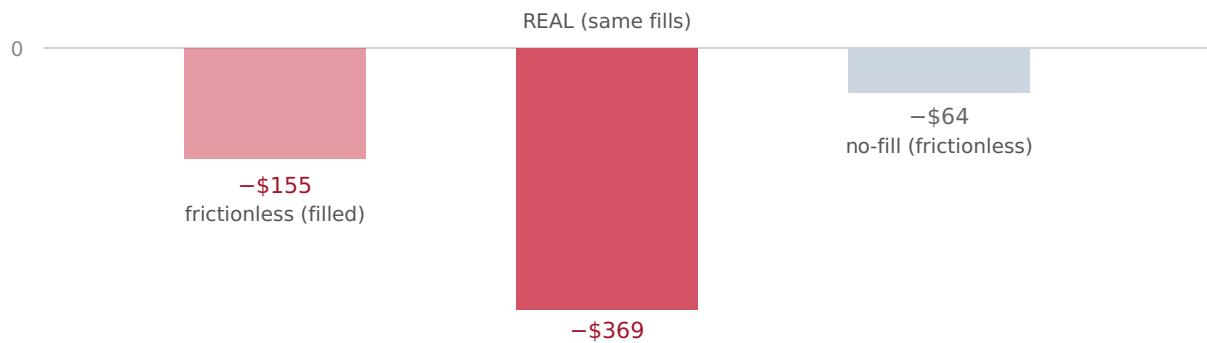
Split MBT's re-priced passive record into what filled and what didn't, frictionless (ceiling_pnl) vs realistic (real_pnl):

BUCKET	N	FRICTIONLESS (CEILING)	REALISTIC (REAL)	GAP
Filled — you entered	78	−\$155.3	−\$368.8	−\$213.5
No-fill — never entered	79	−\$63.7	—	unreachable
Whole re-priced universe	157	−\$219.0	−\$368.8	—

Read it plainly. On most contracts the frictionless number is a green temptation that friction erases. On MBT there is no temptation: even if you could magically fill every one of the 157 signals at the bar price, passive_chop still **loses −\$219**. Only 78 of the 157 filled (a 49.7% fill rate), and they filled because price ran through the level — re-priced on the real tape they lose **−\$368.8**. Filled win rate is just **13 of 78 (16.7%)**. The friction cost per filled trade is **\$2.74**, against an MBT quoted spread that averaged **\$20.9 in BTC price** (median $\sim \$15$) which at the $\times 0.1$ multiplier is $\sim \$2.09/\text{side}$ — so the gap is almost exactly the spread you cross as price slices through your resting order.

The gap, drawn

Every bar points down. Red = frictionless P&L on the filled trades (already negative). Dark red = the same trades on the real tape. Grey = the no-fill signals' frictionless value (also net-negative — these were mostly losers you dodged, not winners you missed).



§40.2 2. Adverse selection, dissected

Every one of the 78 fills was a **through-fill** (knob_through=1 on all 78) — the limit sat at touch and only caught price when it drove clean through. Break the fills by how they ended:

EXIT	N	WINS	AVG FRICTIONLESS	AVG REAL	SUM REAL
STOP (traded through & kept going)	35	0	-\$5.85	-\$7.42	-\$259.8
FLAT_CLOCK (stale-out)	37	9	+\$0.41	-\$3.67	-\$135.8
RATCHET (reverted — the bounce paid)	6	4	+\$5.71	+\$4.46	+\$26.8

The shape is the same adverse-selection asymmetry seen desk-wide, just with the reverts starved out. The 35 STOP fills — price drove through the limit and never came back — are **0-for-35** and cost -\$260 real. The 37 FLAT_CLOCK fills sit at roughly break-even frictionless (+\$0.41 avg) but friction alone tips them to -\$3.67 (-\$136 total). Only 6 fills reverted cleanly enough to ratchet a profit (+\$27). There simply are not enough genuine

reverts among the fills to pay for the through-fills. Frictionless the whole set is $-\$155$; friction adds the final $-\$214$.

The winners you never got — and the losers you dodged

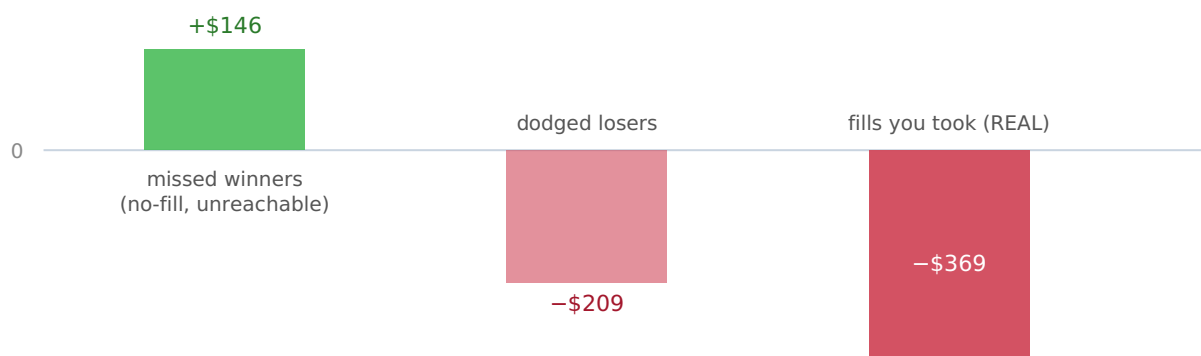
Now the no-fills, split by their frictionless value. This is the crux of the survivorship trap: a no-fill is not automatically a missed winner.

NO-FILL TYPE	N	FRICTIONLESS CEILING	MEANING
Missed winner (price bounced away in your favour)	28	+\$145.5	unreachable edge
Dodged loser (price never reached, then reversed against)	51	-\$209.2	you were lucky to miss
All no-fills	79	-\$63.7	net-negative population

There is genuinely **+\$145.5 of unreachable edge** sitting in the 28 clean bounces you never filled — the classic missed winners. But you cannot cherry-pick them: they come bundled with 51 dodged losers worth $-\$209.2$, and the no-fill population as a whole is **net $-\$63.7$ frictionless**. Filling more of the no-fills (a shallower/more aggressive limit) would drag in more losers than winners. The $+\$145.5$ is real but structurally out of reach for a resting order at this level — capturing it means paying up front, which is the aggressive model, not passive.

Where the money is, drawn

Green up = the $+\$145.5$ of bounces you missed (unreachable). Everything else points down: the losers you dodged, and the fills you actually took. The reachable buckets are all red.



§40.3 3. Fertile hunt: is there any condition where MBT passive pays?

I sliced the 78 filled trades every way the data allows — side, entry RVOL, ATR (absolute), |VWAP slope|, hour-of-day, and two-way combinations of them. The result is a clean null. **Every single-condition band is negative, and every two-way combo with $n \geq 15$ is negative.** Router regime is uniformly INDEX on MBT, so it partitions nothing.

CONDITION BAND	N (FILLS)	REAL P&L	WIN %
LONG	36	-\$156.8	22%
SHORT	42	-\$212.0	12%
RVOL < 1 (quiet)	59	-\$275.8	17%
RVOL $\geq$ 1 (active)	19	-\$93.0	16%
ATR < 25	22	-\$107.0	9%
ATR 25-40	37	-\$196.5	11%
ATR $\geq$ 40 (highest vol)	19	-\$65.2	37%
VWAP slope < 1 (flat)	29	-\$123.5	21%
06:00-12:00 UTC	30	-\$164.2	13%
12:00-18:00 UTC (US session)	34	-\$109.0	24%
Best-leaning 2-way: high-ATR $\times$ US session	14	-\$25.2	43%

NULL

There is **no fertile pocket for MBT passive**. The least-bad slice — highest-vol fills ($ATR \geq 40$) taken in the US afternoon — is still $-\$25.2$ over $n=14$, and it fails the $n \geq 15$ bar anyway. Nothing crosses into positive real P&L at a defensible sample size. This is not a threshold I can tune around; the signal is negative frictionless before adverse selection even bites.

§40.4 4. The knob: can resting deeper skip the toxic fills?

The honest answer is that the recorded shadow gives me nothing to A/B here: every MBT passive row was logged at a single `knob_offset = 0.0` (limit at touch) and every fill was a through-fill (`knob_through = 1`, 78/78). There is no shallower/deeper offset variation in the data to compare. What I can reason from the fill/no-fill split: pushing the limit **deeper** (more passive) would convert some of today's 78 through-fills into no-fills — but the marginal filled trade is a loser and the no-fill population is itself net-negative ($-\$64$), so a deeper limit helps only by trading less, not by uncovering an edge. A **shallower** limit does the opposite: it drags in more of the $-\$209$ dodged-loser pile. Neither offset turns MBT passive positive; the only knob that improves real P&L on this contract is the off switch.

§40.5 5. Verdict

MBT passive is not a frictionless mirage — it is worse. On most contracts the paper number seduces and friction erases it; on MBT the paper number is **already $-\$219$** , and the real tape drags the fills to **$-\$369$** (16.7% win rate, 0-for-35 on the STOP fills). Adverse selection is present and textbook — 100% through-fills, $+\$146$ of clean bounces left unreachable in the no-fills — but it is piled on top of a signal that had no edge to begin with. Every condition band, every two-way combo at $n \geq 15$, is negative. There is no shadow

candidate to promote. **Do not run passive_chop on MBT. If MBT is to be traded at all it must be aggressive (cross the spread) or left alone; a resting limit here systematically banks the continuations and misses the reverts on a contract whose micro-reversion signal is already unprofitable.**

One watch item, not a pocket: the highest-volatility US-afternoon fills ($ATR \geq 40$, 12:00–18:00 UTC) are the least-bad slice at 43% win and only – \$25 over 14 fills. That is a place to look again once more MBT tape accrues — it is not, on this week's evidence, a place to rest a limit.

Every figure was computed from GAZBOT's own captured data for the trading week (Mon 29 Jun → Fri 3 Jul 2026): the trades ledger, realistic 250ms/5s fills, and tens of thousands of 5-second bars per contract. Sixteen forensic agents produced the dig; independent skeptics re-ran every claim worth £250+/week, and their verdict is stamped on each section. Plain-English by design.